ENGLISH FOR ACADEMIC RESEARCH

Adrian Wallwork

English for Writing Research Papers

Third Edition

Springer

English for Academic Research

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Adrian Wallwork, English for Academics SAS
Pisa, Italy

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This series aims to help non-native, English-speaking researchers communicate in English. The books in this series are designed like manuals or user guides to help readers find relevant information quickly, and assimilate it rapidly and effectively. The author has divided each book into short subsections of short paragraphs with many bullet points.

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Introduction

Who is this book for?

This book is part of the *English for Research* series of guides for academics of all disciplines who work in an international field. This volume focuses on how to write a research paper in English.

It is designed both for inexperienced and experienced writers with an upper intermediate (or above) level of English, as well as EAP trainers and scientific editors.

How is this book organized? How should I read it?

Part I: Guidelines on how to improve your writing skills and level of readability.

Part II: Guidelines about what to write in each section (Abstract, Introduction, Methodology etc.), what tenses to use, and how to use a chatbot to correct and improve your English.

I recommend you read all of Part I before you start writing your paper. Then refer to specific chapters in Part II when you write the various sections of your paper.

Chapter 20 concludes the book and contains a checklist of things to consider before sending your manuscript to the journal.

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Each chapter begins with some factoids or quotations that either act as a brief introduction to the topic of the chapter or are just simply interesting in themselves (I hope!). They can be used by EAP teachers as warm-ups for their lessons.

Chapters are divided into short subsections in answer to specific questions or relating to various guidelines. Each chapter ends with a summary.

What is different from the previous editions of this book?

There are two new chapters: Chapters 9 and 10. They cover how to use ChatGPT (and similar tools) and machine translation to write and/or correct your paper. Full details on how to use AI to help you write a paper can be found in the companion volume *AI-Assisted Writing and Presenting in English* (2024). Writing and grammar exercises related to using AI can be found in the 2024 editions of the exercise books.

Many of the chapters in Part I have been totally revised or combined together, and several subsections in Part II have been shortened or removed in order to make room for the new chapters. The chapters on how to write Abstracts, Introductions, Methods etc now include at least one subsection on using ChatGPT to help you write those specific sections of a paper.

Example sentences of good and bad English are now in a blue and red font, respectively.

The *What's The Buzz* sections have been integrated into the main text.

The chapter on hedging has been removed to provide space for new material, but is still available in the 2016 edition.

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Part I

Writing Skills

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Chapter 1

Planning and Preparation



Publications and PhDs

No one knows how many scientific journals there are, but several estimates point to around 30,000, with close to two million articles published each year.

*****

According to Wikipedia, more scientific works are published in China than in any other country, followed by the USA, UK, India, and Germany. However in terms of publications per capita, Switzerland and Scandinavian countries have the highest output.

*****

Approximately 20% of the comments referees make when reviewing papers for possible publication in international journals regard English language issues.

*****

Only 1% of the world's 25-64 year-olds who have been to university, have a PhD – the top five countries are: Slovenia (5%), Switzerland (3%), and Luxembourg, USA, Sweden, UK, Germany and Australia (2%).

*****

In the EU alone there are over 650,000 PhD students.

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1.1 Why should I publish? How do I know whether my research is worth publishing?

Before you start writing you need to have an absolutely clear idea of:

- what your research goal was
- what your most important findings are and how you can demonstrate that they are true
- how these findings differ from, and add to, previous knowledge

You know implicitly what the importance of your findings are – after all, you may have been working for months and years on the project.

But the reader does not know.

You must give the reader a clear message.

Discussing and presenting your findings to colleagues should help you to identify what your key findings are.

Make a list of your key findings and choose the most important ones to fit the space you have available (i.e. the total word count allowed by your chosen journal). For each key finding decide if there is another possible explanation for what you have found. You can do this by looking in the literature again. Make sure you have not inserted any bias in your explanation of your findings. Next, write an explanation saying why you think each key finding is true.

These three quotations should inspire you to write simply and clearly:

From note taking to publishing to teaching, language is the tool that gives sense to scientific activity. Whatever scientists do or observe, everything they come to know or to hypothesize, is mediated through language. Robert Goldbort, Writing for Science

The writing of an accurate, understandable paper is just as important as the research itself. Robert A Day, How to Write and Publish a Scientific Paper

Writing helps you to learn. Writing is not simply a task to be done once research or other preparation is completed – it can be an integral part of the work progress. Nicholas Highman, Handbook of Writing for the Mathematical Sciences

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1.2 Should I write the initial draft in my own language rather than writing it directly in English?

Artificial intelligence (AI) has reached a point where you can write a paper directly in your native language, automatically translate it, and then check / improve it with a chatbot. Full details on how to use AI to help you write a paper in the companion volume *AI-Assisted Writing and Presenting in English* (Springer Nature, 2024).

If you speak a major European language, AND you implement ALL THE GUIDELINES outlined in Chapters 9 and 10 of this book, the result of the above process will be over 95% accurate in terms of English usage – at least according to tests that I have personally carried out with my PhD students. This percentage is very likely higher than if you wrote directly in English without any help from AI.

If you do not speak a major European language, the level of accuracy will be lower. My experiments with Chinese and Arabic, for instance, show that machine translation is not consistent – sometimes it is surprisingly accurate, other times it makes a lot of mistakes. Nevertheless, AI can still help you, as outlined in Chapter 9 and the various subsections in Part II.

The key advantage of writing in your own language is that you have total control over what you write. It is much easier for you, for example, to:

- write quickly with much less mental effort
- reduce the number of words if you find you have exceeded the journal's wordcount, i.e. the maximum of number of words permitted for each section of your paper.
- move sentences around, i.e. change their position within a paragraph or section
- work with a co-author – provided, of course, they speak the same language as you
- show your work to your colleagues, who will be better able to judge its merits if it is written in their first language

If you already have some parts of the paper in English, it does not necessarily mean that you have to write the other parts directly in English. You can upload a document that is made up of parts in your language and parts in English. The automatic translator will automatically output only in English, maintaining the parts that were already in English. However, you need to check that no changes were made by the translation software. For example, the spelling or capitalization of some technical or foreign words may have been altered.

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Another advantage of writing in your own language is that you will save an incredible amount of time that you can then dedicate to

- checking your results
- improving your tables and figures
- making your paper more readable by removing redundancy, replacing generic words with more specific words, avoiding possible ambiguity etc
- checking your final version

Of course there are also good reasons for writing directly in English. English may be

- the language that you habitually use to write up your research
- the only language that you have in common with your co-authors
- a language you enjoy writing in

However, I suggest that you don't see automatic translators and chatbots as a form of 'cheating'. You still have to check that the output of these applications is correct, and by doing so, you will also improve your level of English by comparing what you would have written with what the application has written.

The chatbots only become of form of 'cheating' when you use them to generate totally new work, and pass this new work off as your own work.

Note that I mentioned you can achieve 99% accuracy. This only refers to accuracy of English grammar, syntax and vocabulary. However your paper may

1. still contain redundancy and ambiguity – see Chapter 4 and Chapter 5
2. fail to clarify if a particular study mentioned in your paper was written by you or someone else – see Chapter 6
3. fail to highlight the importance of your work – see Chapter 6
4. be poorly structured at a sentence, paragraph and section level – see Chapter 2 and Chapter 3
5. be full of general statements

Consequently it makes sense to eliminate the five problems above in your original version in your own language.

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Final advice: If you write in your own language, try to use an English style of writing. For example, use shorter sentences and paragraphs than perhaps you normally would, put the subject or key info near the beginning of the sentence, do not separate the subject and the verb, be concise, and remove anything that is not strictly necessary.

This means following all the guidelines in this book.

1.3 Which journal should I choose?

If you have never written a paper before and your supervisor has not indicated a specific journal where they would like you to publish, it is a good idea to ask colleagues in your research group what they read and what sort of publications they aspire to publish in.

Even if you are writing a paper for the first time, it does not mean that it will only be suitable for a marginal or not very well known journal. Your progress in academia very much depends on your ability to publish in journals that have a high impact factor.

An impact factor is a measure of how prestigious a journal is. The higher the impact factor, the more widely read the journal is, and the more likely other researchers will cite your paper. Tables of impact factors which rank all the peer-reviewed journals in the world are available on the internet, you can use Google Scholar to help you find them.

However, given the difficulties of getting published in a high impact journal, you might consider opting for a short article or a 'letter'. A literature review or a methodological text is often publishable. For instance, if you are studying medicine, you could consider writing a clinical review – a 2,500 word article which is essentially a review of the management of important and common problems. Many disciplines have such an equivalent.

When you have chosen three or four possible journals, look at their styles and think about their audience – what do the editors and readers expect from the articles.

You could try to insert your paper into an ongoing discussion that is currently being covered in the journal. This approach may increase the chances of getting your paper approved by the editor.

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One way of improving your writing skills and raising your profile in your area of expertise is to consider writing letters. Journals generally publish letters that offer a short critical review of the research of others. Such letters tend to be about 300 words long, so the same as or a little longer than an abstract. You can also write online rapid responses to letters in print journals.

1.4 How do I know what style and structure to use?

Each journal has its own requirements and style guide. These instructions tend to have different titles, for example: 'instructions for authors', 'notes for authors', 'author guidelines'. They often appear under a page called 'author resources'.

The guidelines include:

- types of titles that are acceptable
- structure of paper – for example, is the review of the literature near the beginning of the article or at the end? Are the Methods after the Introduction, or are they after the bibliography? Are the Results included in the Discussion or in a separate section? Is there a Conclusions section?
- layout (including how the Abstract should be presented – one long paragraph, or 5–6 short paragraphs)
- structure of sections – some journals prescribe exactly how certain sections (most commonly the Discussion) are organized, and what subheadings should be included
- use of passive rather than personal style (*we*, *I*)
- how to make citations
- how to arrange the bibliography
- use of key words
- American or British spelling

It is vital that you rigorously follow your chosen journal's instructions to authors. So download these instructions from the journal's website before you start writing.

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If you opt for a low impact journal, you will still find it very useful to look at the instructions of an equivalent high impact journal. Higher impact journals tend to have better author resources, which are useful for all authors, not just for those in the specific field of the journal itself.

If no journals in your discipline offer such resources, then I suggest that you look at the 'Welcome to resources for authors' page of the website of the British Medical Journal (bmj.com), one of the world's most prestigious journals.

1.5 What preparation do I need to do?

Once you have chosen your journal, look at the most frequently cited papers to see how the authors rationalize the various steps of their research. Try to use papers that you will probably quote in your section on the review of the literature, and which are highly relevant to your topic and/or classic papers in your general field.

For example, you could create a table with some or all of the following headings:

- problem that the research addresses
- background information and relevant references
- elements that validate the level of innovation of the research
- conceptual model, methodology or procedure that the research takes into consideration
- materials, equipment and software used
- method used and steps that the author carried out
- results achieved
- analysis and interpretation of these results
- strengths and weaknesses of the research
- implications for further research

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Then you can fill in your table with brief notes for each of the papers you have analyzed. This analysis should help you to:

1. write your own literature review, because after this analysis you will be very familiar with the literature
2. identify the differences in other researchers' approaches and results compared to your research
3. note down the strengths and weaknesses (including possible bias) in the work of others

These three points should enable you to understand in what ways your research is unique, innovative, interesting and useful, and how it extends what is already in the literature. Your aim is to find a knowledge gap to fill.

If you have done a very thorough literature search, then another publishing opportunity for you is to write a literature review.

1.6 How can I create a template?

Choose one paper that is close to your topic, that is written by a native English speaker, and that you enjoyed reading. Use this paper as a model into which you can 'paste' your own research.

Notice how your model paper is structured:

- how does the author begin?
- what points do they make in each section?
- how do they link paragraphs together?
- how do they connect the Results with the Discussion?
- how do they present the Conclusions?

As you read your model paper, note down some useful English phrases that the author uses. Such phrases will help to increase the readability of your text, as they will be familiar to your readers.

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1.7 Writing style: how do I keep the referees happy?

It is possible to write a paper in completely accurate English, but still have a paper rejected for poor writing skills – which is what happens even to native English-speaking researchers. On the other hand, a paper that is constructed well, and is easy to read, may be accepted (perhaps with some requests for minor revisions) even if the English is not totally accurate.

In my experience native referees tend to be more interested in how the paper flows and how easy it is to read. Non-native referees seem to focus more on grammatical and vocabulary mistakes, so very accurate English is important in order to keep them satisfied too.

All referees will appreciate it if you use simple language.

There are no journals, as far as I know, that are easier to write for in terms of level of English required.

When writing your paper bear the following in mind:

A REFEREE HAS NO OBLIGATION TO REVIEW YOUR PAPER

Referees review manuscripts in their own time and have no direct financial reward for doing so. So do everything you can to make the referee's work easier and more pleasurable – clear English, clear layout, clear tables etc. By doing so you will increase the chances of your paper being accepted.

WRITE IN A WAY THAT A NON-EXPERT OR LESS EXPERIENCED PERSON CAN UNDERSTAND

Research is becoming increasingly more specialized, so that even two people with the same degree may not be able to understand each other's papers. Also, due to the fact that research groups cannot always get the funding they need for research in their specific field, they may have to shift their interests to a related field where funds are available. This entails them reading the literature from this new field. The clearer the literature is, the more they will understand.

This means that when you begin the writing process, you need to bear in mind that your reader may not be as expert as you are.

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MAKE YOUR PAPER INTERESTING ENOUGH FOR AN EXPERT

Try to ensure that your paper has enough scientific substance for the experts. This does not mean you have to write in a more complicated way, but just that you include enough details to get experts interested.

LOOK AT THE FORMS USED IN REFEREES' REPORTS

Every journal has a standard form for use by referees when writing their reports, which the editor then uses to judge whether your paper is suitable for publication or not. Through your professor and colleagues, try to find as many such forms as you can, and preferably the one for your chosen journal.

You can use the following questions as guidelines for your writing.

- Is the research novel and of international relevance?
- Does the article fit the aims and scope of the journal?
- Is the paper written grammatically and clearly?
- Is the writing style succinct and appropriate to the work?
- Is the title appropriate to the content?
- Does the abstract accurately describe the content?
- Are the conclusions borne out by the evidence and arguments?

1.8 In what order should I write the various sections?

There is no standard order in which you should write the various sections of your paper. You should choose the order that suits you best. This may involve writing several sections simultaneously.

Many authors start with the Methods, which is often the easiest section to write because this is the part that will usually be clearest in your mind. Beginning with the Methods will also give you the confidence and impetus you need to move on to the other sections of the paper.

In reality, it is best to start with the Abstract as this will help you to focus / orient your ideas on what are the key aspects of your research. In any case, if you are going to present your work at a conference, the organizers will ask you to submit an abstract before you write the related paper – you can still change the Abstract when you have finished writing the actual paper.

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You might find it useful to look at the scientific study protocol that you wrote when you outlined the aims of your research at the beginning of your PhD or before you began your current project. Here you should have written out your goals very clearly, and this will help you to write your Abstract.

The hardest part for most authors is the Discussion where you have to interpret your results and compare them with other authors' results. While you are writing the Discussion, you may find it useful to draft the Introduction, as some of the authors you mention will appear both in the Introduction and the Discussion.

A typical order for writing the various sections is thus:

Abstract (very rough draft)

Methods

Results

Discussion

Introduction

Conclusions

Abstract (final version)

It is a good idea to write the Results and Discussion before the Introduction. This is because you will only truly understand the significance of what you have done after you have written these two sections. Laying the background foundations on which you can highlight the significance of your research is a major part of the Introduction.

1.9 How can I highlight my key findings?

While you are planning what to put in each section, think of where and how you can highlight your contribution. It may help you to imagine that the reader has asked you these questions:

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1. what problem are you trying to solve / investigate?
2. how did you solve / investigate it?
3. how does your solution / investigation differ from previous approaches?
4. what did you discover?
5. how do your findings differ from what is already in the literature, and what do they mean?

Readers generally read the Title and Abstract of a paper first, followed by the Discussion; though some may just look at your figures and tables! However, you cannot be sure at which section your readers will begin reading, so they need access to the answers to these questions in most or all the sections. Look at other papers in your chosen journal to see how the authors deal with such questions. Clearly, the emphasis you put on answering the questions will vary from section to section, and is likely to be greatest in the Abstract and Discussion, but consider covering it in the other sections too.

When you revise your paper if you think you have done too much highlighting, then you can always remove a few sentences. But while drafting your paper if you constantly try to highlight your contribution, this will give you extra focus.

Think of your paper as a product that you are trying to sell to the referee and journal. The clearer and more convincing you are, the more likely a journal will 'buy' your manuscript.

For more on underlining your contribution see Chapter 6.

1.10 How can I improve the chances of my paper not only being published, but also being read, understood and cited?

One important way of improving the chances of your paper being cited and remembered and cited is to make it as interesting a read as possible. A good start is to think of the following points:

- Why is my research important? Why is it needed? What would happen if my research was not done?
- Who will benefit from my research? Why will they benefit?
- What makes my research exciting? What makes it different from what has been done before?

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When writing a paper, there is often a tendency for authors to simply recount what methods they used, what results they got, and how these results compare to what is already known. Authors often make an effort to paint the big picture, by making big statements such as *Given the climate crisis … Given the prediction that world population will reach 9 billion in 2050 … Given consumer demand for sustainable products …* but they fail to detail how their research will contribute to solving this problem.

It is also very common to use adjectives such as *interesting, intriguing, remarkable,* and *important*. Instead, it would be better to say WHY what you are saying is *interesting, intriguing, remarkable,* and *important*. Why and how is your research *fundamental, pivotal, key, crucial and state-of-the-art*? If your writing is more concise and more incisive, and includes concrete details, it will become much more of a pleasure to read.

Yes, this is hard when you are writing in a foreign language. But in this new edition of this book, I am suggesting that you use AI to help you (see 1.2 and Chapter 9). If you speak one of the world's major languages, you can now write in your own language, make every word of your text meaningful and of value for your reader, and then automatically translate it and further refine it if necessary using tools such as ChatGPT.

The more readers enjoy your paper – as well as finding it useful – the more likely they are to cite it in their own papers. And the more quickly you will climb the academic ladder.

1.11 I know that the recommendations in this book about writing simply and clearly will improve the readability of my paper, but my professor …

Here is an extract from a typical conversation with a PhD student (S) who does my writing courses:

S: I appreciate the rules you have taught us about clear writing etc, but my professor likes to write in a complex way.

Me: Does your professor have any evidence that writing in a complex way will increase their chances of publication, and also increase the chances of having their paper read and cited?

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S: Probably not, but in my field we tend to write in a complex way.

Me: Why?

S: I don't know. Perhaps we would not be taken seriously if we wrote in a simple clear way that everyone could understand immediately.

Me: Have you or your professor ever received a comment from a reviewer saying: "This is paper is well organized and clear, its results are highlighted as well as the implications of the results. The language is simple and straightforward, and at the same time uses all the correct technical terms appropriately. For these reasons I am afraid I will have to reject the paper"?

The problem is that in academia simplicity is a taboo.

Complexity is considered to be a sign of an intelligent author, and simplicity to be an indication of a simple mind.

In reality it is much more difficult to write in a simple way. You really have to think about what you are telling the reader, and then use the most direct and concise way possible. By doing this, you are showing your readers respect. You are not forcing them to decipher your turgid writing.

If writing simply was easy, then everyone would write in a simple way.

Academics often write in a complex way because it is much quicker to do so and requires considerably less mental effort on their part. In some cases, it may also help them to disguise the fact that either they don't actually have much to say, and in some rarer cases that they don't fully understand themselves what they are saying.

Throughout history great minds have been advocating simplicity in writing. I live in Italy, and some Italians love to write exceedingly long, vague, and complex sentences. Yet in 11th-century Italy, the Franciscan monks in their *Regola* recommended short, clear, and immediately understandable sentences. And in the early 20th century, Antonio Gramsci, Marxist philosopher, writer, politician and founder of the communist party in Italy, wrote:

> Comrades please write in a simple way … that stimulates the mental energy of the readers but without forcing them to make too much effort.

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No research has ever shown that writing in a complicated way will help your findings reach a wider audience. In fact, the exact opposite has been proved.

So, if your professor tells you to write in a complex way, show them this section from this book and ask them if they have any RATIONAL reasons for not adopting a simpler style.

I am NOT advocating a paper full of short sentences. What I AM advocating is not to write a paper that only consists of long contorted phrases. Your aim is to make the reader's experience easy and enjoyable, so that they will understand and remember what you have written, and thus be more inclined to cite your work in theirs.

The entire first part of this book is dedicated to teaching you how to write clearly.

1.12 Summary

- Ensure you have 100% clear idea of your research goal, your most important findings, and how these findings differ from, and add to, previous knowledge.
- Consider using AI to help you write and correct your paper.
- Consult with your professor and colleagues about the most appropriate journal where you can publish your research.
- Match your topic to the journal, or vice versa.
- Download the guidelines for authors – these will tell you about the style and structure of your paper.
- Choose frequently cited papers in the journal to see how other authors construct their argumentation, and note down ways in which your research is different and innovative with respect to theirs.
- Choose one paper as a model onto which to map your research, imitating the style and organization. This model should be written by a native English speaker.
- Note down useful / standard phrases from your model paper which you can then use in your own paper.
- Decide on the best order to write the various sections of your paper. It is generally best to start with a very rough draft of the Abstract, and then whichever section is clearest in your head (generally the Materials and Methods).

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- Consider having separate documents for each section. This enables you to work on several sections at the same time.
- Make sure your unique contribution to your community is very clear in every section, not just in the Abstract.
- Write in a way that even a non-expert can understand.
- Referees work for free and often outside working hours – make it worth the time they are dedicating to your work: never submit a carelessly written manuscript.
- Access referees report forms to understand the ways that referees will evaluate your work.
- Learn how search engines index your paper. Key words are essential in order for the search engines to identify your paper.
- No amount of key words is going to help you if readers are not immediately able to understand your paper, and cite it in their own.

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Chapter 2

Word order and sentence length



Factoids

In Old English, the language spoken in English over 1000 years ago, a word could be placed almost anywhere in a sentence, and often with no change in meaning.

*****

Word order differs massively from language to language, even to say a simple concept such as 'I like you': *like to me you* (Croatian), *you like to me* (Estonian), *you are liking to me* (Irish), *I you like* (Korean), *to me you like* (Spanish), *you me I like* (Wolof).

*****

The average length of a sentence in English has become shorter and shorter over the centuries. In Shakespeare's time it was about 45 words, 150 years ago it was about 29 words, and today's experts recommend between 15 and 18 words.

*****

In *The Effective Communicator*, communications expert John Adair reports that approximately 90% of people understand an 8-word sentence on first reading, but only about 4% understand a 27-word sentence first time around, especially if it is poorly punctuated.

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_2

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2.1 Basic word order in English: subject + verb + object + indirect object

The order in which you put information in a sentence (or paragraph) conditions the weight that your reader will give to each element of information.

Native English-speaking readers have a clear expectation regarding the order in which information should be given to them.

English has a strict order in which words can appear in a sentence. Below is an example of this order.

The researchers sent their manuscript to the journal.

This order is rarely altered. It is:

- subject (*the researchers*)
- verb (*sent*)
- direct object (*their manuscript*)
- indirect object (*the journal*)

The key is to keep the subject, verb, direct object and indirect object as close to each other as possible:.

The researchers last week *sent* for the second time *to the journal their manuscript.*

Last week *the researchers sent their manuscript to the journal* for the second time.

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2.2 Place the subject before the verb

In English, place the subject before the verb. In the red sentences below the subject comes after the verb.

In the survey participated *350 subjects*.

Three hundred and fifty subjects participated in the survey.

Were used *several different methods* in the experiments.
Several different methods were used in the experiments.

With these values are associated *a series of measurements*.
A series of measurements are associated with these values.

Once verified *the nature of the residues* …

Once *the nature of the residues* had been verified …

Among the factors that influence the choice of parameters *are time and cost*.

Time and cost are among the factors that influence the choice of parameters.

Of particular interest *was the sugar transporter*, because…

The sugar transporter was of particular interest, because…

When we scan results from a search engine, our eye rapidly goes vertically down the left hand side of the page, before starting again to read horizontally.

This means that you need to think carefully about what information to place at the beginning of the first sentence that begins a new paragraph. If you misplace the key information, there is a strong chance that browsers and readers won't spot it.

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In the following red sentences the parts highlighted in italics occupy the key left-hand position. They delay the subject, with the risk that readers may not even see the subject.

It is interesting to note that x is equal to y.

Note that x is equal to y. // *Interestingly*, x is equal to y.

As a consequence of the preceding observations, x is equal to y.

Consequently, x is equal to y. // X is *thus* equal to Y.

It is probable that this is due to poor performance.

This *may / might / could* be due to poor performance.

It is possible do this with the new system.

This *can* be done with the new system.

It is mandatory to use the new version.

The new version *must* be used.

The blue sentences highlight that to avoid delaying the subject you should:

- delete or reduce the part before the subject
- shift the linking expression to later in the sentence
- try to avoid sentences that start with an impersonal form (*it is possible / it is obligatory etc*)

2.3 Keep the subject and verb close to each other

The verb contains important information: keep it as close as possible to the subject. Anything that comes between the subject and the verb will be read with less attention, and readers will consider it of less importance.

The red sentences force the reader to wait too long to find out what the verb is and thus delay important information. The blue versions shift the verb nearer to the beginning of the sentence and make the meaning / direction of the sentence immediately clear.

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A gradual decline in germinability and vigor of the resultant seedling, a higher sensitivity to stresses upon germination, and possibly a loss of the ability to germinate *are recorded* in the literature [5, 8, 19].

There is generally a gradual decline in germinability and of the resultant seedling, followed by a higher sensitivity to stress upon germination, and possibly a loss of the ability to germinate [5, 8, 19].

People with a high rate of intelligence, an unusual ability to resolve problems, a passion for computers, along with good communication skills *are generally employed* by such companies.

Such companies generally employ people with a high rate of…

People with a high rate of intelligence are generally employed by such companies. They must also have other skills including: an unusual ability to…

2.4 Avoid inserting parenthetical information between the subject and the verb

If you insert more than a couple of words between the subject and the verb this may interrupt the reader's train of thought. Readers may consider this parenthetical information to be of less importance.

Sentences are much easier to read if they flow logically from step to step, without any deviations.

The result, after the calculation has been made, can be used to determine Y.

After the calculation has been made, the result can be used to determine Y.

This sampling method, when it is possible, is useful because it allows….

When this sampling method is possible, it allows us…

These steps, owing to the difficulties in measuring the weight, require some simplifications.

Owing to the difficulties in measuring the weight, these steps require some simplifications.

These steps require some simplifications, owing to the difficulties in measuring the weight.

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This does not mean that you cannot have a series of short clauses within one sentence. In the example below, readers do not have to change their perspective while moving from one clause to the next.

In Old English, the language spoken in English over 1000 years ago, a word could be placed almost anywhere in a sentence, and often with no change in meaning.

Not inserting parenthetical information is just a guideline. Ignore it, if you think that the insertion makes the sentence clearer and flow better.

2.5 Don't separate the verb from its direct object

When a verb is followed by two possible objects, place the direct object (i.e. the thing given or received) before the indirect object (the thing it is given to or received by).

This kind of construction is often found with verbs followed by 'to' and 'with': associate X with Y, apply X to Y, attribute X to Y, consign X to Y, give X to Y (or give Y X), introduce X to Y, send X to Y (or send Y X).

We can *separate*, with this tool, *P and Q*.

We can *separate P and Q* with this tool.

We can *associate* with these values *a high cost*.

We can *associate a high cost* with these values.

In the red sentence below, the direct object is very long and consists of a series of items, so the reader has to wait a long time before discovering what all these items are associated with.

We can *associate* a high cost, higher overheads, a significant increase in man-hours and several other problems *with these values*.

The solution is to put the indirect object after the first item and then use 'along with', or to use a colon.

We can *associate* a high cost *with these values*, *along with* higher overheads, a significant increase in man-hours and several other problems.

We can *associate several factors with these values*: a high cost, higher overheads, a significant increase in man-hours and several other problems.

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You could also use bullets, but bullets are not commonly found in published papers.

The *following can be associated with these values*:

- a high cost
- higher overheads
- a significant increase in man hours

2.6 Put the direct object before the indirect object

Don't put the indirect object (in *italics*) at the beginning of the sentence or main clause.

However, only *for some cases* this operation is defined, these cases are called…

However, this operation is only defined *for some cases*, which are called…

Although *in the above references* one can find algorithms for this kind of processing, the execution of …

Although algorithms for this kind of processing are reported *in the above references*, the execution of…

This occurs when *in the original network* there is a dependent voltage.

This occurs when there is a dependent voltage *in the original network.*

2.7 Don't use a pronoun (it, they) before you introduce the noun that the pronoun refers to

It is OK to use a pronoun at the beginning of the sentence, provided that this pronoun refers back to a noun in a previous sentence. For example:

Beeswax is a very important substance because… In fact, *it* is…

In the sentence above it is clear that *it* refers to *beeswax*. However in the red sentence below, *it* refers to a noun that comes after. The reader does not know what the pronoun refers to and thus has to wait to find out.

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Although *it* is a very stable and chemically inert material, studies have verified that the composition of *beeswax* is …

Although *beeswax* is a very stable and chemically inert material, studies have verified that *its* composition is …

2.8 Locate not before the main verb, but after auxiliary and modal verbs

The word *not* should be placed before the main verb it is associated with.

Patients *seemed not* to be affected by intestinal disorders.

Patients *did not seem* to be affected by intestinal disorders.

When the verbs *to have* and *to be* are used in the present simple or past simple, *not* is located after the verb.

These findings *are not* significant.

Their results *had no* value. // Their results *did not have* any value.

Not is located after modal verbs and auxiliary verbs.

Such patients *should not* be treated with warfarin.

We *have not* encountered such a problem before.

2.9 Locate negations near the beginning of the sentence

The order you put the words in your sentence should be designed to take your reader through a logical progression of thoughts. These thoughts should move forward, never backtracking, never forcing the reader to reconsider or reinterpret what they have just read in the light of what they are reading now.

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In the sentences in red below, readers cannot predict how the sentence might progress. They are forced to wait to the end before being able to understand what they have just read. In fact, the sentences appear to begin in a positive way and then suddenly change direction.

Data regarding the thyroid function and the thyroid antibodies before the beginning of the therapy *were not available*.

All of the spectra of the volatiles *did not* show absorptions in the range …

The blue versions below help the reader to immediately understand the central purpose and driving force of the sentence.

No data were available regarding thyroid function and thyroid antibodies before the beginning of the therapy.

Before the beginning of the therapy, *no data were available* regarding …

None of the spectra of the volatiles showed absorptions in the range …

Negations (*no, do not, does not, none, nothing* etc) are often a key element in a sentence – try to locate them as close as possible to the beginning of a sentence:

The number of times this happens when the user is online is generally *very few*.

This *rarely* happens when the user is online.

Documentation on this particular matter is almost *completely lacking*.

There is *virtually no documentation* on this particular matter.

Consequently *we found* this particular type of service *not* interesting.

Consequently *we did not find* this particular type of service interesting.

As highlighted in the first two red examples, English tends to express negative ideas with a negation. This helps the reader to understand immediately that something negative is being said. The last red example is incorrect English because the verb and the negation (*not*) have been separated. See 17.21 in *English for Research: Grammar, Usage and Style*.

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2.10 Deciding where to locate an adverb

If you are in doubt about where to put the adverb, the following rules apply to most adverbs including *only* and *also*. Locate the adverb:

Immediately before the main verb.

Dying neurons do not *usually* exhibit these biochemical changes.

The mental functions are slowed, and patients are *also* confused.

Immediately before the second auxiliary when there are two auxiliaries.

Language would *never* have arisen as a set of bare arbitrary terms if …

Late complications may not *always* have been notified.

After the present and past tenses of 'to be'

The answer of the machine is *thus* correct.

However other types of adverbs (e.g. certainty, manner, time) follow different rules.

2.11 Put adjectives before the noun they describe

Adjectives normally go before the noun they describe.

This is a paper *particularly interesting* for PhD students.

This paper is particularly interesting for PhD students.

We examined a patient, *30 years old*, to investigate whether …

We examined a 30-year-old patient to investigate whether …

If you want to put the adjective after the noun, you have to use a relative clause (i.e. *which, that, who*).

This is a paper *that* is particularly interesting for PhD students.

We examined a patient, *who* was 30 years old, to investigate whether …

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Never put an adjective before a noun that it does not describe. Generally, you cannot put an adjective between two nouns.

The main document *contribution*
The *main* interface of the editor

The editor *main* interface
The main *contribution* of the document

The algorithm *computational* complexity

The *computational* complexity of the algorithm

2.12 Deciding where to put new and old information within a sentence

The two sentences below begin with the same subject *English*, which is the main topic of the sentence. They then present the same two pieces of information, but in a different order.

English, *which is now studied by 1.35 billion people*, is the international language of communication.

English, *which is the international language of communication*, is now studied by 1.35 billion people.

In both cases if you removed the 'which' clause (in italics) the sentence would still make sense. But if you removed the final clause it wouldn't. This would seem to indicate that the final clause is where we locate the most important information. Thus the relative position of the various parts of the phrase tells the reader the relative importance of the information contained on those parts.

In the blue sentence, the order of the information tells you that the fact that English is *the international language of communication* is old news, but that *1.1 billion people* is new information that the reader probably does not already know. Thus, the order of the information in the red sentence is a little strange because it puts the new information (*1.1. billion people*) before the old information (*international language*).

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Readers tend to focus on the first and last words of a sentence, so avoid placing your most important information in the middle of a long sentence. Readers don't want to make an effort to identify the key points, they want to be told immediately.

Here are some more examples that show how by changing the order of information within a sentence you can achieve a different effect:

English is now studied by 1.35 billion people, though this number is expected to drop with the rise in importance of Chinese.

Although English is now studied by 1.35 billion people, this number is expected to drop with the rise in importance of Chinese.

Although the importance of Chinese is expected to lead to a drop in the numbers of people studying English, 1.35 billion people still study English.

All three sentences contain the same information, but the weight that this information is given varies.

Sometimes, you may want to give equal weight to the two parts. You can do this by presenting the information in two distinct sentences.

English is the international language of communication. It is now studied by 1.35 billion people.

The importance of Chinese is expected to lead to drop in the numbers of people studying English. Despite this, 1.35 billion people still study English.

This device should not be used too often because it can lead to a series of very short sentences, which after a while begin to sound like a list.

2.13 Analyse why and how long sentences are created

Below is a short sentence. It is divided into two parts, each of which contains one idea. You can understand it without having to read it twice.

English owes its origins to the Angles and Saxons, two tribes from what is now northern Germany and Denmark.

The next sentence is more than twice as long. However, its content is simple and it would probably be understood by most readers without too much difficulty.

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Owing its origins to the Anglo Saxons (a tribe who lived in what is now Denmark and Northern Germany), English is the international language of communication, in part due to the importance of the USA, rather than the Queen of England, and is now studied by 1.35 billion people.

Long sentences become a problem when they contain difficult concepts, and when there are several of them in sequence. Although the red sentence above does not contain difficult concepts, it is not the optimal construction. There is not a logical progression of thought. Below is a better solution. The slashes (//) indicate how the red sentence has been divided up into shorter sentences.

English owes its origins to the Anglo Saxons, who were a tribe from what is now Denmark and Northern Germany. // It has become the international language of communication. // This is in part due to the importance of the USA, rather than the Queen of England. // English is now studied by 1.1 billion people.

As a general rule if the first part of a sentence is more than 12–15 words long, don't add a second part that is more than 10–12 words.

Finally, you will notice how using shorter sentences forces you to write much more clearly. It is in fact much easier to write using long sentences - that is, it is easier for you, but not for your reader!

2.14 Learn how to break up a long sentence

If the sentence is less than around 25 words, then you don't need to break it up provided that readers will understand the meaning with no effort.

However, if a long sentence requires too much effort from the reader, you can break up long sentences by following the guidelines below.

BEGIN A NEW SENTENCE WHERE THERE IS A LINK WORD (E.G. *HOWEVER, IN ADDITION, IN CONTRAST, ON THE OTHER HAND*).

The levels of cadmium in Site C were comparable to the levels found in Sites A and B in the previous years, *whereas / on the other hand* the levels for copper were much lower in Site C with respect to the values found in the previous sampling campaigns in 2023 and 2024.

… in the previous years. *On the other hand*, the levels for …

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The levels of cadmium in Site C were comparable to the levels found in Sites A and B in the previous years, *although /however* this was not the case for the levels found in the south-east part of Site C.

… in the previous years. *However*, this was not the case for the levels found in the south-east part of Site C.

BEGIN A NEW SENTENCE WHEN *WHICH* IS USED TO ADD INFORMATION. FOR EXAMPLE:

English is now the world's international language and is studied by more than a billion people in various parts of the world thus giving rise to an industry of English language textbooks and teachers, *which* explains why in so many schools and universities in countries where English is not the mother tongue it is taught as the first foreign language in preference to, *for example*, Spanish or Chinese, *which* are two languages that have more native speakers than English.

… textbooks and teachers. *This* explains why in so … first foreign language. *For example*, English is taught in preference to Spanish or Chinese, *which* are two languages that have more native speakers than English.

However using *which* is fine in cases such as the two sentences below.

English is now the world's international language, *which* is why it is used in scientific papers.

English, *which* has now become the world's international language, is studied by more than a billion people.

However the second sentence is more immediate written like this:

English has now become the world's international language and is studied by more than a billion people.

AVOID THE – *ING* FORM TO LINK PHRASES TOGETHER

Using automatic translation software can considerably ease the work of researchers when they need to translate documents *thus saving* them money (for example the fee they might have otherwise had to pay to a professional translator) and *increasing* the amount of time they have to spend in the laboratory rather than at the PC.

… when they need to translate documents. *Such software saves* them money, for example the fee they might have otherwise had to pay to a professional translator. It *also increases* the amount of time they …

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As can be seen from Table 1, the concentrations were far higher than expected especially in the first set of samples, *indicating* that one cause of pollution was …

… in the first set of samples. *This indicates* that one cause of pollution was …

REPLACE *AND* WITH A PERIOD (.)

The aim of this paper is to confirm that how we speak *and* write generally reflects the way we think *and* that this is true not only at a personal but also at a national level, *and* to this end two European languages were analyzed, English *and* Italian, to verify whether the structure of the language is reflected in the lifestyle of the respective nations.

How we speak and write generally reflects the way we think and act. *This* paper aims to prove that this thesis is true not only at a personal but also at a national level. *Two* European languages were analyzed, English and Italian, to verify ...

LIMIT THE NUMBER OF COMMAS IN THE SAME SENTENCE

When commas are used in lists, they are fine:

Many European countries are now part of the European union, these include France, Germany, Italy, Portugal, Spain, …

However, when commas are used to separate various clauses within a sentence, readers have to constantly adjust their thinking. Also, the more commas there are in a sentence, the longer the sentence is likely to be.

As a preliminary study, in an attempt to establish a relationship between document length and level of bureaucracy, we analyzed the length of 50 European Union documents, written in seven of the official languages of the EU, to confirm whether documents, such as reports regarding legislative and administrative issues, vary substantially in length from one language to another, and whether this could be related, in some way, to the length of time typically needed to carry out daily administrative tasks in those countries (e.g. withdrawing money from a bank account, setting up bill payments with utility providers, understanding the clauses of an insurance contract). The results showed that …

Our aim was to see if there is a direct relationship between the length of documents produced in a country, and the length of time it takes to do simple bureaucratic tasks in that country. Our hypothesis was: the longer the document, the greater the level of bureaucracy.

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In our preliminary study we analyzed translations from English into seven of the official languages of the European Union. We chose 50 documents, mostly regarding legislative and administrative issues. We then looked at the length of time typically needed to carry out daily administrative tasks in those countries. The tasks we selected were withdrawing money from a bank account, setting up bill payments with utility providers, and understanding the clauses of an insurance contract.

The results showed that …

Note that the blue version:

- uses more words in total, but is considerably easier to follow
- rearranges the various subordinate clauses and puts them into a more logical order and in separate sentences
- divides up the information into three paragraphs - the first explains the rationale, the second shows how the investigation was carried out. This makes the connection between ideas much clearer

ONLY USE PARENTHESES WHEN GIVING LISTS

Phrases in parentheses can considerably increase the length of a sentence. Parentheses should be avoided when giving explanations or examples that are not lists.

Using automatic translation software *(e.g. Google Translate, DeepL, and Systran)* can considerably ease the work of researchers when they need to translate documents thus saving them money *(for example the fee they might have otherwise had to pay to a professional translator)* and increasing the amount of time they have to spend in the laboratory rather than at the PC.

… to translate documents. Such software saves them money, *for example the fee they might have otherwise had to pay to a professional translator.* It also increases the amount of time …

In the red version the first use of parentheses is fine, but the second interrupts the flow of the sentence and considerably adds to its length.

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Notes:

- Repeating key words is NOT bad style in scientific writing (e.g. repeating *software* in the blue example above) In fact repetition helps readers to follow your text, and it also helps your co-authors if they need to modify the order of the sentences in your draft (5.4).
- You cannot and should not write a whole paper using short sentences. However, a sequence of short sentences is perfect for highlighting some important point in the Results or the Discussion or when expressing the key points of your research (6.10).
- Manuscripts are often written by several co-authors. Having short sentences in the initial draft means that co-authors can i) add to them without making the resulting sentence too long, ii) change the order in which these sentences follow each other.
- Two short sentences should be combined into a longer sentence if this will reduce the amount of redundancy and improve readability.

2.15 Summary

- Basic English word order is: (1) subject, (2) verb, (3) direct object, (4) indirect object. Keep these four elements in this order and as close to each other as possible.
- If you have a choice of subjects, choose the one that is the most relevant and leads to the shortest construction.
- Put the subject before the verb and as close as possible to it.
- Avoid delaying the subject.
- Avoid inserting parenthetical information between the subject and the verb.
- Don't separate the verb from its direct object.
- Don't separate the verb from its direct object
- Put adjectives before the noun they describe, or use a relative clause. Do not insert an adjective between two nouns or before the wrong noun.
- Do not indiscriminately put nouns in a string.

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- Avoid writing long sentences. If you have a written a particularly long sentence, learn how to divide it up. Make sure that a sentence has a maximum of 30 words, and don't use more than three or four 30-word sentences in the whole paper.
- You don't lose any of the complexity of your thoughts by dividing up a long sentence into shorter ones. The information contained is exactly the same. All you have done is to present that information in a way that is easy for the reader to absorb at a first reading. But do not exclusively use short sentences.

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Chapter 3

Structuring Paragraphs



Students around the world

- Students at the University of York in England drink more alcohol than any other university in the UK – the equivalent of nearly five liters of lager per week.
- In the US, the average college student attends 62 parties per year.
- Brigham Young University in Provo (Utah, USA) holds the record for being the US university where you are most likely to meet your future spouse.
- Undergraduates in the USA often form fraternities and sororities, which are single-sex organizations for male and female students, respectively. The names of these groups are often taken from Greek letters (alpha, beta etc) and their members are sometimes known as 'Greeks'.
- The University of Cambridge in England has over 400 registered clubs and societies for students.
- The Indira Gandhi National Open University located in Delhi India has the highest number of enrolled students – nearly 3.5 million. The top 10 universities by student numbers are located in India, Turkey, Iran, Pakistan, Bangladesh, Indonesia and Nepal.
- Estonia holds the record for the highest proportion of female graduates – more than two thirds are women.
- The Japanese word *ronin* means a 'masterless samurai' and refers to those students who fail the entrance examination to their university of their choice, and then spend an entire year or longer, studying for a second attempt to gain entrance.
- Michael Kearney is the US's youngest ever college student. At the age of 10 he was awarded a bachelor's degree in anthropology from the University of South Alabama.

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_3

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3.1 Elegance vs Effectiveness

Most English-speaking readers today are generally not prepared to read long sentences, such as the one below:

When we reflect on the vast diversity of the plants and animals which have been cultivated, and which have varied during all ages under the most different climates and treatment, I think we are driven to conclude that this greater variability is simply due to our domestic productions having been raised under conditions of life not so uniform as, and somewhat different from, those to which the parent-species have been exposed under nature.

The text above comes from Darwin's *On the Origin of Species*. In 1859, when Darwin's book was published, it was quite common among native English-speaking scientists to use more than 70 words in a sentence. In reality, due to its logical structure the sentence above is not particularly difficult to understand despite its length.

However, the average sentence length today, outside academia, is around 20 words.

No research has ever proved that long sentences and long paragraphs are an aid to reader comprehension.

A lot of research has, however, proved that shorter sentences and shorter paragraphs make comprehension much easier for the reader.

You may think that writing in a simple way with short sentences is not elegant and is superficial.

But the question to ask is: Is this text effective or not? Will my readers be able to understand it easily?

John Kirkman is a British consultant specializing in research and training in scientific and technical communication. In his book *Good Style – Writing for Science and Technology* he says:

To be easy to digest, sentences must be reasonably short and not too complex. The reasons for this are not grammatical: they are connected with the number of items of information the reader can absorb in a single unit or 'thought'.

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In fact, whether they are Nobel Prize winners, Oxford professors, or first-year university students, all readers prefer texts that they:

- only need to read once
- can read quickly because no intense concentration is required
- can process word by word and thus understand the build-up of the author's logic immediately, rather than only being able to reach their interpretation of the whole meaning at the end of the sentence

These goals are much easier to achieve if you write short sentences. In the world of academic writing, I think you should aim for an upper limit of around 25 words.

3.2 Choose the most relevant subject to put it at the beginning of a sentence that opens a new paragraph

Clear English requires that you put the subject at the beginning of the sentence.

Generally you will have a choice of possible subjects.

X was elicited by Y.

Y elicited X.

In the simple example above, your choice will depend on whether you want to emphasize X or Y. The one you want to emphasize should be put as the subject.

As readers, we tend to focus on the areas of a sentence that come immediately before and after a full stop. This is because there is extra white space between one sentence and another, which acts as a restful pause for the eye. Our eyes are also drawn to the capital letter that begins each sentence. These are the moments where you potentially have the reader's attention, so don't waste them.

If the first few words routinely contain no useful or new information, then it can become very tedious.

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So the best solution is to shift 'no value added' phrases to later on in the sentence and preferably reduce them to one word. Otherwise you are encouraging readers to skim i.e. to read very fast and skip words, sentences and even whole paragraphs and sections.

The sentences below contain exactly the same information, but the grammatical subjects are different:

Particularly interesting for *researchers in physics* is the new feature, named X, for calculating velocity.

Physics now has a new feature, named X, for calculating velocity.

Velocity can now be calculated with a new feature, named X, which is particularly interesting for physicists.

X is a new feature for calculating velocity. It is particularly interesting for physicists.

When deciding what the subject is for the first sentence in a paragraph, it is generally best to choose the most recent or newest information. The two red sentences refer to known situations – physics, and physicists – they do not give any new information.

The first blue sentence also begins with a known, in this case *velocity*. This is fine if velocity is the main focus.

However, given that velocity is a common factor for physicists, then the last sentence may be the best solution as it begins with completely new information.

In any case, the choice between the four sentences above will obviously depend on what information you want to highlight for your readers.

3.3 First paragraph of a new section – begin with a mini summary plus an indication of the structure

Readers do not necessarily read the paper from beginning to end. They may begin with any section in the paper.

This means you could consider starting some sections (e.g. Introduction, Discussion, Conclusions) with a one or two-sentence summary of the main aims and/or findings of the paper. This style is also typical if you are writing chapters in a book.

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However, check the general style of papers in your chosen journal. If they do not begin sections in such a way, then don't do it yourself. Instead go for a more direct approach.

Here are four examples of mini summaries at the beginning of a section:

The X Committee has for some years encouraged collaborative clinical trials in X by reporting the results in the medical literature. In this section we describe the first of two unreported results that we believe deserve such publication and which constitute the main contribution of this paper.

As mentioned in the Introduction, a principal concern in the field of X is to understand why… This section attempts to answer the question…

Our aim is to provide a simple alternative to the complex theoretical models that attempt to explain… In this section we present a simplified model, which we believe is…

This section reviews the process of… This process provides the backbone to the system that is at the core of our research.

In addition to this mini summary, some authors also briefly outline what will be contained in the rest of the section. Here are four more examples:

In this section, we briefly review the broad perspectives that have shaped the direction of thinking about …

However**,** particularly in shorter papers, you may not have the space to have mini summaries at the beginning of your section or subsections. In any case, readers often don't have the time or the inclination to read them. In such cases you need a more direct approach.

Being direct does not necessarily entail telling the reader what you did, but telling them what it means. A typical sentence to open the Results section is:

An analysis of the number of words used in English with respect to Italian, showed that the average sentence in English was 25 words long, whereas in Italian was 32 words long (see Table 1). This indicates that when an Italian document is translated into English, there is …

A much more direct approach is to say:

Italian tends to use more words per sentence than English, so when an Italian document is translated into English, there is …

The blue version begins with the main information, and then provides the implications. You do not necessarily need to tell the reader the exact details of what you did (this would be more appropriate in Methods) but just what you found.

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3.4 Deciding where to put new and old information within a paragraph

Known information is traditionally placed at the beginning of a sentence or paragraph. Below are the first three sentences from the Abstract of a fictitious paper entitled 'Readability and Non-Native English Speakers' intended for a journal dedicated to communication in the world of business.

VERSION 1 Readability formulas calculate how readable a text is by determining the level of difficulty of each individual word and the length of sentences. All types of writers can use these formulas in order to understand how difficult or readable their texts would be for the average reader. However, readability formulas are based purely on what is considered difficult for a native English speaker, and do not take into account problems that may be encountered by non-natives. In this paper…

The first word, *readability*, is one of the author's key words. It immediately alerts the reader to the topic of the sentence and of the abstract (and paper) as a whole. However, the information contained in it is not new – readability formulas and their indexes are well established in the literature on business communication.

The role of the first two sentences is thus to set the context and gently guide the reader into the paragraph. The third sentence then introduces the new element, i.e. the fact that readability indexes do not take into account non-native speakers. The third sentence thus highlights the problem that the paper intends to tackle.

However, the abstract could have begun like this:

VERSION 2 Current readability formulas are based purely on what is considered difficult for a native English speaker. They fail take into account problems that may be encountered by non-natives. One thousand five hundred PhD students from 10 countries were asked to evaluate the difficulty of five technical texts from their business discipline written by native English speakers. Three key difficulties were found: unfamiliar vocabulary (typically Anglo-Saxon words), unfamiliar cultural references, and the use of humor. The paper also proposes a new approach to assessing the level of readability of texts to account for such difficulties.

In Version 2, the author still begins with his key word, *readability*. But he precedes it with *current*, which signals to the reader that the author will then probably propose an alternative. The author also assumes that his readers will be aware of what a readability formula is, so he feels he doesn't need to mention it. Thus, in the second sentence he immediately underlines a critical problem with current formulas. In the third sentence he then tells his readers what his research was and then what was found.

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Version 3, below, contains only new information.

VERSION 3 Unfamiliar vocabulary (typically Anglo-Saxon words), unfamiliar cultural references, and the use of humor: these, according to our survey of 1500 PhD students, are the main difficulties non-native speakers have when reading a business text in English. Our results highlight the need to adjust current readability formulas in order to take non-native speakers into account. The paper also proposes a new approach to assessing the level of readability of texts to account for such difficulties.

Version 3 immediately attracts the reader's attention. In contrast, the first 50 words of Version 1 contain no new information at all. Version 2 has 40–50% new information or more, depending on whether readers are familiar with the limitations of readability formulas with regard to non-natives.

So, which version should you use?

The best version to use depends on two factors: i) the section of the paper, ii) what you are trying to achieve

Version 1 would only be appropriate in an Abstract if the journal where it is being published does not usually deal with communication and / or readability indexes. In this case the readers need the context to be set for them. It might be more acceptable in an Introduction in a slightly more specialized journal. In an Introduction the aim is not principally to attract attention. If readers are reading your Introduction, you can presume that you probably already have their attention.

So the information contained in Version 1 would be used in an Introduction just to remind the readers of the context. This is a very typical way to begin an Introduction – it is what readers expect and therefore it is generally a good technique.

Version 2 would be appropriate as an Abstract or Introduction in a specialized journal on business communication.

Version 3 would only be appropriate in an Abstract of a specialized journal. It can only be used if you have clear findings, or a clear new methodology, to report. It works very well because it does not force readers to read background information that they are probably already familiar with.

You might also choose Version 3 as an Abstract for a congress. In such cases you are competing for the attention of the referees who will use your Abstract to decide whether to include your contribution at the congress.

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3.5 Link each sentence by moving from general concepts to increasingly more specific concepts

A key issue when linking up sentences in a paragraph is to decide how to link one sentence to the previous one. The following is an extract from the beginning of a paragraph from a paper on pollution in soil. It fails to make a strong impact because of its lack of logical progression between the third (S3) and fourth (S4) sentences.

(S1) The *soil* is a major source of *pollution*. (S2) Millions of *chemicals* are released into the environment and end up in the soil. (S3) The impact of most of these *chemicals* on human health is still not fully known. (S4). In addition, *in the soil* there are naturally occurring amounts of potentially *toxic substances* whose fate in the terrestrial environment is still *poorly known*.

S1 puts *the soil* as the topic of the sentence. S2 is more specific and talks about the quantity of this pollution – *millions of chemicals*. S3 reports the impact of the chemicals mentioned in S2. But S4 does not continue this logical progression from general to increasingly more specific. Instead, it begins by putting *soil* in the topic position. This breaks the logical progression, because *soil* was the topic of S1.

The following sentence would be a good replacement for S4 because it continues the logical structure developed in S1–S3.

There are also naturally occurring amounts of potentially toxic substances *in the soil* whose fate in the terrestrial environment is still poorly known.

A better structure is thus:

1. S1: main topic (*soil*) introduces subtopic 1 (*pollution*)
2. S2: subtopic 1 is specified by introducing subtopic 2 (*millions of chemicals*).
3. S3: subtopic 2 is specified introducing subtopic 3 (*impact of these chemicals*).
4. S4: a further / related aspect of subtopic 3 is introduced via subtopic 4 (*impact of toxic substances, i.e. chemicals, is poorly understood)*.
5. etc.

Basically each sentence is a link in a chain. A full chain is a paragraph. And a series of linked chains makes up a section.

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3.6 Present and explain ideas in the same (logical) sequence

Readability can be increased massively if you take some time to think about the best way to present information. The red version below is in perfect English, and it may seem fine until you see how the blue version makes the information much easier to assimilate.

Memory can be subdivided into various types: long-term memory, which involves retaining information for over a minute, and short-term memory, in which information is remembered for a minute or less, for example, the memory required to perform a simple calculation such as 5 × 7 × 3. Another type of short-term memory is also recognized: sensory memory, for example we see a video as a continuous scene rather than a series of still images. Research shows sex differences in episodic (i.e. long term) memory: women tend to remember better verbal situations, whereas men have a better recollection of events relating to visuals and space. Long- term memory can be further subdivided into recent memory, which involves new learning, and remote memory, which involves old information.

Memory is the capacity to store and recall new information. It can be subdivided into two main types: short-term and long-term. Short-term memory involves remembering information for a minute or less, for example, the memory required to perform a simple calculation such as 5 × 7 × 3. Another type of short-term memory is sensory memory, for example, we see a video as a continuous scene rather than a series of still images. Long-term memory can be further subdivided into recent memory, which involves new learning, and remote memory, which involves old information. Interestingly, research shows sex differences in remote memory: women tend to remember better verbal situations, whereas men have a better recollection of events relating to visuals and space.

In the red version, the beginning of the first sentence gives the illusion to the reader that the various types of memory will be introduced in a logical order. In reality a rather random selection of information is given, with no clear sequence. This makes it hard for the reader to follow. The blue version uses shorter sentences and follows a much more logical series of steps:

1. definition of memory given
2. clear indication of the number of types of memories (OV various types, RV two main types)
3. short-term memory mentioned first, as later in the paragraph long-term memory will be developed in more detail
4. additional information about short-term memory (the discussion of short-term memory ends here)
5. returns to second topic (long-term memory), which is then subdivided into recent and remote
6. interesting fact about remote memory

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3.7 Break up long paragraphs

Look at the paragraph below. Does it invite you to read it?

The only advantage of a long paragraph is for the writer, not for the reader. It enables writers to save time because they avoid having to think about where they could break the paragraph up to aid reader comprehension. But breaking up long paragraphs is extremely important. Firstly, long blocks of text are visually unappealing for readers, and tiring for their eyes. They fail to meet the basic rule of readability – make things as easy as possible for your reader. Evidence of this can be found in newspapers. If you look at newspapers from 100 years ago, they were basically big blocks of text that took a great deal of effort to read. Today many online newspapers have one sentence per paragraph, with lots of white space between each paragraph. Secondly, your points and the related logical sequence of these points will be much more clearly identifiable for the reader if they are in a separate paragraph. Thirdly, you will find that you will write more clearly if you use shorter paragraphs. This is because it will force you to think about what the main point of your paragraph is and how to express this point in the simplest way. If you just have one long paragraph, the tendency is just to have one long flow of frequently disjointed thoughts. This tendency is known in English as 'rambling'. Fourthly, having shorter paragraphs enables you (and your co-authors) to quickly identify if you need to add extra information, and allows you to do this without having to extend an already long paragraph. Likewise, it enables you to identify paragraphs that could be cut if you find you are short of space. The third and fourth points are also valid reasons for using short sentences. The maximum length of a paragraph in a well-written research paper is about 15 lines. But most paragraphs should be shorter. If you have already written more than 8–12 lines or 4–6 sentences, then you may need to re-read what you have written and think about where you could start a new paragraph. When you begin to talk about something that is even only slightly distinct from what you have mentioned in the previous 4–6 sentences, then this is a good opportunity to begin a new paragraph. For example, when you have been talking about how another author has approached the problem of X, and you then want to make a comparison with your own approach. The topic (i.e. X) is the same, but the focus is different. Likewise, if you have been comparing X and Y, and you have spent a few sentences exclusively on X, then when you start on Y you can use a new paragraph. Basically, there is an opportunity to begin a new paragraph every time there is a change in a focus.

Breaking up long paragraphs is extremely important.

1. Long blocks of text are visually unappealing for readers, and tiring for their eyes.
2. Your points and the related logical sequence of these points will be lost in a long paragraph. They are much more clearly identifiable for the reader if they are in a series of separate paragraphs.

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3. If you have phrases such as *This study shows that / Our findings highlight / These results indicate that* in the middle of a long paragraph, readers may not even notice the sentence. Thus you lose a good opportunity to get the reader to focus on your findings. So whenever you want to highlight the importance of your study or findings, begin a new paragraph.
4. You will find that you will write more clearly if you use shorter paragraphs. If you just have one long paragraph, the tendency is just to have one long flow of frequently disjointed thoughts.
5. Having shorter paragraphs enables you (and your co-authors) to quickly identify if you need to add extra information, and allows you to do this without having to extend an already long paragraph. Likewise, it enables you to identify paragraphs that could be cut if you find you are short of space.

When you begin to talk about something that is even only slightly distinct from what you have mentioned in the previous 4–6 sentences, then this is a good opportunity to begin a new paragraph. For example, when you have been talking about how another author has approached the problem of X, and you then want to make a comparison with your own approach. The topic (i.e. X) is the same, but the focus is different (i.e. Y). Likewise, if you have been comparing X and Y, and you have spent a few sentences exclusively on X, then when you start on Y you can use a new paragraph.

If you print your paper you will be able to see more clearly which paragraphs are long.

Below are some typical words and phrases used to link one sentence to the next in order to create a logical progression of thought (see Chapter 6 Link Words in *English for Research: Grammar, Usage and Style*). These words and phrases also act as markers to indicate that you could begin a new paragraph.

Firstly, secondly, finally; In order to do this / To this end / With this mind; Then / Following this / Afterwards; For example, / An example of this is / In fact, / Unlike / Nevertheless; In addition / Another way to do / An additional feature of; On the other hand / However / In contrast; Due to / Since / Although; Thus / Therefore / Consequently / Because of this; This means that / This highlights that / These considerations imply that / In conclusion / In sum; As can be seen in Table 2 / Figure 1; As far as X is concerned, / In relation to X, In the case of / With regard to / As noted earlier

There is no minimum length to a paragraph. A paragraph can occasionally be just one sentence. However, a series of paragraphs containing only one or two short sentences would be a little strange.

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Where you begin a new paragraph will also depend on which section you are writing. In the review of the literature, you may want to begin a new paragraph when (i) you begin to talk about a different phase in the logical buildup of research in your field, or (ii) you start talking about another author. In the Methods, it may help the reader to identify the various components or understand the various steps, if these components or steps are in separate (probably quite short) paragraphs.

3.8 How to structure a paragraph: an example

In the early 1960s, a senior staff scientist at NASA, Sam Katzoff wrote a 30-page pamphlet entitled 'Clarity in Technical Reporting'. This short document was designed to help his colleagues at NASA to write clearly and to think of better ways to express themselves. His pamphlet is still being read – not just by NASA scientists – but all over the English-speaking world. It is a truly great introduction to writing skills, for native and non-native speakers alike.

I am now going to analyze how Katzoff writes the first paragraph at the beginning of a section entitled 'Organization of a Technical Report'.

Different writers have different methods of organizing their reports, and some seem to have no discernible method at all. Most of the better writers, however, appear to be in remarkably close agreement as to the general approach to organization. This approach consists of stating the problem, describing the method of attack, developing the results, discussing the results, and summarizing the conclusions. You may feel that this type of organization is obvious, logical, and natural. Nevertheless, it is not universally accepted. For example, many writers present results and conclusions near the beginning, and describe the derivation of these results in subsequent sections.

Let's begin with some statistics. The average number of words per sentence is 13.8. The longest sentence has 22 words, the shortest only 6. There are 10 commas, but no semicolons. Five words are repeated up to three times: *approach, method, organization, results, writer*.

Try looking at some of your own work. With respect to Katzoff's paragraph, you will probably notice you use more words, commas and semicolons per sentence. You will have many sentences of around 30–40 words, but also up to 70–80. You will also have fewer full stops and fewer repetitions of key words.

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Sam Katzoff was a top scientist. His document was intended for fellow scientists, who were, like him, native English speakers.

These fellow scientists were also amongst the most brilliant scientists in the world. They could potentially understand even the most complex text.

Yet Dr Katzoff decided to write his document in the simplest and clearest way possible.

And he encouraged his fellows to do the same. According to a fellow colleague:

> He was the kind of person who could look at a paper and tell whether it was a lot of bull. If you were writing a paper and were publishing, he would review it, and that would help a lot of people in the field to come up with a better way of saying what they were trying to get across.

By *bull* the colleague was politely saying *bullshit*, i.e. words, phrases and paragraphs that clearly made no sense, but were just included for effect.

Now let's analyze the structure of Katzoff's paragraph.

(S1) Different writers have different methods of organizing their reports, and some seem to have no discernible method at all. (S2) Most of the better writers, however, appear to be in remarkably close agreement as to the general approach to organization. (S3) This approach consists of stating the problem, describing the method of attack, developing the results, discussing the results, and summarizing the conclusions. (S4) You may feel that this type of organization is obvious, logical, and natural. (S5) Nevertheless, it is not universally accepted. (S6) For example, many writers present results and conclusions near the beginning, and describe the derivation of these results in subsequent sections.

S1 introduces the general topic and summarizes current practice with regard to report writing. S2 qualifies what was said in S1. The reader is warned of this qualification by the link word *however*.

Katzoff repeats the word *writer* from S1 to link it into S2, but precedes it with a different adjective (*different, better*) to show that he is moving from something general (all authors) to something more specific (better authors). The repetition of *approach* in S3 serves a similar linking purpose. It gives readers the feeling that they are being guided step by step along the path on which Katzoff develops his topic.

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In S4 he addresses the reader directly, which is probably something that you would not do in a paper. Instead you would probably phrase such a concept in the passive: *it may be argued that*. Katzoff's idea is to anticipate possible objections to what he is about to say. S5 is only six words long. Such a short sentence is rare in academic work. Yet it is very effective in capturing reader attention. The link word, *nevertheless*, placed prominently at the beginning of the sentence, also catches the reader's eye and helps to underline the importance of what is being said.

In S6 he uses another link word, *for example*. These link words all serve to show how each sentence relates to what has been said before. Without these link words, the reader would be forced to figure out Katzoff's train of thought. However, Katzoff only uses link words when they really serve a purpose.

One constant device Katzoff uses is to repeat words. He uses the word *writer* three times. He could easily have found synonyms, e.g. *author, researcher, technician*. But this might have confused readers who might think that there was a difference in meaning between these four terms.

Another massive aid to helping readers understand, is to have a maximum of two ideas per sentence. S4 and S5 contain just one idea. S6 contains two ideas linked by *and*.

3.9 Summary

- Always think about your readers – order the information you give them in the most logical way and in the simplest form. Your aim is to be effective – forget elegance.
- Don't worry about repeating key words. If dividing up a long sentence into shorter sentences means that you have to repeat key words, this is not a problem. In fact this repetition will increase the clarity of your writing
- Begin each paragraph with a topic sentence, then use the rest of the paragraph to develop this topic. If appropriate have a short concluding sentence at the end of the paragraph.
- Decide whether to begin a new section with a short summary, or whether to go directly to the main points.

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- Put the topic as the subject of the paragraph or sentence, then give known information (context, background) followed by new information. Consider not giving the known information if it will be obvious for your readers.
- Move from the general to the increasingly specific, do not mix the two. Always progress in the most logical and consistent order, do not go backwards and forwards.
- Begin a new paragraph when i) you move on to a new topic (e.g. you move from general background info to discussing a particular case; ii) you have been talking about the literature and now you start talking about your contribution; iii) when you are talking about your contribution and you want to mention a specific gap that your contribution fills; iv) you are discussing your results, and you want to highlight a key finding
- Avoid redundancy in the final paragraph of a section.

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Chapter 4

Being Concise and Removing Redundancy



Expert advice

Read over your compositions, and wherever you meet with a passage which you think is particularly fine, strike it out.

Samuel Johnson, English writer

A good scientific theory should be explicit to a barmaid.

Ernest Rutherford, British / New Zealand chemist and physicist

The ability to simplify means to eliminate the unnecessary so that the necessary can speak.

Hans Hoffman (1880–1966), German-born American abstract expressionist painter

I don't want to bother readers unless I think it is important.

Barbara, American novelist

Human beings are not logical mechanisms into which information can be fed.

Bruce Cooper, British, author of Writing Technical Reports

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_4

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4.1 Being concise is not just an option

Many journals, particularly widely-read ones such as *Science* and *Nature*, have restrictions on the number of words per article. *Nature* states:

Our experience has shown that a paper's impact is maximized if it is as short as is consistent with providing a focused message, with a few crucial figures or tables.

Your aim is NOT to receive a referee's report on your manuscript (MS) like this one (the key points that the referee makes are in italics):

It is the duty of the authors to present their MS in a way that it is readable and to the point. Only then can a reviewer critically evaluate the most essential data on which the conclusions are built. When a MS is written in a highly redundant way it takes *too much time and effort* to judge whether or not all the analyses have been done correctly.

The MS is far too detailed making it *unreadable*. There is *a lot of redundancy* in the text, some parts are written as if this is a chapter in a text book. There are 144 references!!! And 12 pages of discussion!!

The result is that the actual findings that could be interesting are completely lost. There is no focus on what the authors really want to tell the readers.

My suggestion to the editor is to reject this MS and give the authors the opportunity to *resubmit a much more focused and condensed MS.*

Note: These rules in this chapter are designed to help you write in a more concise way. However it is also important to vary the way you write. It is perfectly acceptable to write a long phrase or sentence, or a complicated construction, provided that you only do this occasionally.

Being concise does not always mean using the least number of words. It means using the least number of words that make the meaning 100% clear.

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4.2 Write less and you will make fewer mistakes in English, and your key points will be clearer

The less you write, the fewer opportunities you will have to make mistakes in your English! Look at the two sentences below.

The activity aimed *at / to* the extrapolation of the curve is not trivial.

We did the calculation manually. This *choice / choose* meant that …

Imagine you are not sure if *aimed* should be followed by *at* or *to*, or whether *choice* or *choose* is the correct spelling of the noun.

If you make the sentences more concise by removing the redundancy you will avoid the problem and thus avoid risking making a mistake when using them!

The extrapolation of the curve is not trivial.

We did the calculation manually. This meant that …

By the way, *aimed at* and *choice* would be the correct versions in S1 and S2.

Note how the blue sentences are much more effective in highlighting the key information for the reader. The reader is not distracted by words that add no information.

4.3 Cut any unnecessary generic words

The words in square brackets below could simply be removed without having to make further changes to the sentence.

It was small [in size], round [in shape], yellow [in color] and heavy [in weight].

This will be done in [the month of] December for [a period of] six days.

Our research [activity] initially focused [attention] on [the process of] designing the architecture.

The [task of] analysis is not [a] straightforward [operation] and there is a [serious] danger that [the presence of] errors in the text …

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The analyses [performed in this context] highlighted [among other things] the [fundamental and critical] importance of using the correct methodology in a consistent [and coherent] manner [of conduction].

This was covered in the Materials and Methods [section].

Note how the words that have been cut are more generic than the words that have been left (e.g. *color* is generic, *yellow* is precise). Wherever possible use the most concrete word available.

Avoid using strings of words with little or no difference in meaning. In the sentences below, just one of the words in italics would be sufficient.

So far researchers have failed to solve this equation due to various *issues, problems* and *difficulties*.

This point is *critical* and *fundamental* for our research purposes.

4.4 Consider deleting abstract words and phrases

Give your readers specific information without preceding such information with a generic statement (in *italics* in the three sentences below).

Meetings will be held *twice a year* in June and December.

We investigated *two countries (i.e.* Italy and France*)*, both of which …

The process of registration can take up to ten minutes.

Abstract words and phrases add no value to what you are saying. They are not memorable words for the reader. If you find that your paper is full of the words listed below, first decide if you could cut them, if not try to find a more concise and concrete alternative.

activity, case, character, characteristics, choice, circumstances, condition, consideration, criteria, eventuality, facilities, factor, instance, intervention, nature, observation, operation, phase, phenomenon, problem, procedure, process, purpose, realization, remark, situation, step, task, tendency, undertaking

Ask yourself: What is important about my work? What is new about it? What real contribution am I making?

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You can only write in a concrete way if you know the answers to those questions. And then you can use specific examples to explain the importance.

Not all abstract nouns should be cut. Abstract words that express a clear concept should be retained, e.g. *freedom, love, fear*.

When drawing consequences or introducing the next point that follows on from information given in the previous sentences, avoid redundancy and use *thus*.

From the previous list of properties, it emerges that cooperation with devices is a complex task.

Cooperation with devices is *thus* a complex task.

In this respect, the design of a suitable gateway is necessary in order to guarantee the interoperability between the gateway and other communication protocols.

The design of a suitable gateway is *thus* necessary in order to guarantee the interoperability between the gateway and other communication protocols.

Other types phrases that you could consider deleting:

It is worthwhile noting that …, As a matter of fact …, Experience teaches us that …, It is well known that …, In particular, …

4.5 Prefer verbs to nouns

English tends to use more verbs than nouns. This reduces the number of words needed, makes sentences flow better, and provides variety. Too many nouns make a sentence heavy to read.

X was used in the *calculation* of Y.

X was used to *calculate* Y.

Symbols will be defined in the text *at their first occurrence*.

Symbols will be defined *when they first occur* in the text.

Lipid *identification* in paint samples is based on the *evaluation* of characteristic ratio values of fatty acid amounts and *comparison* with reference samples.

Lipids are generally *identified* in paint samples by *evaluating* the characteristic ratio values of fatty acid amounts and *comparing* them with reference samples.

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Avoid using a verb+noun construction (e.g. to make a decision). Using such a construction entails choosing a 'helper' verb to associate with the noun. For example, should you say *do* or *make* a *comparison* of x and y? If you simply say *to compare x and y*, you avoid choosing the wrong helper verb.

X *showed* a better *performance* than Y.

X *performed* better than Y.

Heating of the probe can be *obtained* in two different ways:

The probe can be heated *in two different ways*:

The *installation* of the system is *done* automatically.

The system is *installed* automatically.

The *evaluation* of this index *has been carried out* by *means* of the correlation function.

This index was *evaluated using* the correlation function.

Other examples:

achieve an improvement (improve), *carry out* a test (test), *cause* a cessation (stop), *conduct* a survey (survey), *effect* a reduction (reduce), *execute* a search (search), *exert* an influence (influence), *exhibit* a performance (perform), *experience* a change (change), *give* an explanation (explain), *implement* a change (change), *make* a prediction (predict), *obtain* an increase (increase), *reach* a conclusion (conclude), *show* an improvement (improve), *subject to* examination (examine).

The above verbs in italics add no value for the reader. The red version below highlights the redundancy that a verb+noun construction causes.

In Figure 2 the curve *exhibits a downward trend* (portion A–B); then it *undergoes a rapid rise* (part B–C), it then *assumes a leveled state* (zone C–D). It *possesses a peak* at point E before displaying a slow decline … On the other hand, the curve in Fig 3 *is characterized by a different behavior*.

In Figure 2 the curve initially *falls* (segment A–B) and then *rises rapidly* (B–C). It then *levels off* (C–D). Finally it *peaks* at point E before falling slowly … On the other hand, the curve in Fig 3 *behaves* differently.

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However, sometimes using the noun is inevitable:

Detection was carried out at 520 nm, using a Waters 2487 dual λ UV-visible detector.

Chromatogram analysis was performed using Millennium 32 (Waters).

4.6 Choose the shortest expressions

Try to use the expression that requires the least characters.

X is large in comparison with Y. (26 characters)

X is larger than Y. (15 characters)

Instead of using an adjective + a generic noun (*way, mode, fashion*), use the adverb form of the adjective.

To do this, the application searches for solutions *in an automatic way / fashion / mode.*

To do this, the application searches for solutions *automatically.*

This should be avoided since *it is generally the case that* it will fail.

This should be avoided since it *generally* fails.

From a financial standpoint, it makes more sense to ...

Financially, it makes more sense to ...

Other examples:

in the normal course of events (normally), on many occasions (often), a good number of times (many times, frequently), from time to time (occasionally), in a rapid manner (rapidly), in a manual mode (manually), in an easy fashion (easily), from a conceptual point of view (conceptually).

Consider replacing impersonal phrases that begin with *it is,* with a modal verb, an adverb or by rearranging the sentence.

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It is advisable to clean the containers.

The containers *should* be cleaned.

It is surprising that no research has been carried out in this area before.

Surprisingly, no research has been carried out in this area before.

It is clear / evident / probable that inflation will rise.

Inflation will *clearly / probably* rise.

It is possible to demonstrate [Kim 1992] that …

Kim [1992] *demonstrated* that …

It may be noticed that … *It is possible to observe* that …

Note that …

A frequent result of reducing the overall number of words is that the subject of the sentence tends to be shifted closer to the beginning of the sentence, as you can see in the blue examples above. This means that the reader gets a much quicker picture of the topic of the sentence. Also, if you use the minimum number of words the importance of what you are saying will stand out more clearly for the reader.

4.7 Cut redundant adjectives

Whenever you use an adjective, decide if it really is necessary. The adjectives in red below could probably be deleted without any change in meaning.

an *acute* dilemma, a *real* challenge, a *complete* victory, a *novel* solution, an *interesting* result, an *appropriate* method

Only use an adjective and adverb if it adds precision to your sentence. If you really think that such adjectives are necessary you should explain why something is *novel, interesting, appropriate.*

Don't use pairs of adjectives or nouns that essentially mean the same thing. What contribution, if any, do the words in square brackets below add to the reader's understanding of the sentence?

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This is [absolutely] necessary as the reader could interpret the sentence in a [completely] different way.

This has made it possible to review the analysis of important [fundamental and practical] problems [and phenomena] of engineering.

Numerical methods have increasingly become quick [and expedient] means of treating such problems.

Equation 1 is [readily] amenable to numerical treatment.

The method lends itself [most amiably] to being solved by …

4.8 Cut pointless introductory phrases and unnecessary link words

Often you can avoid an introductory phrase when it is preceded by a heading. For example, immediately after a heading entitled *Results*, the following phrases would be completely redundant.

The salient results are summarized in the following.

The results of this work may be synthesized as follows.

Let us recapitulate some of the results obtained in this study.

Likewise, under a heading entitled *Conclusions* don't begin by saying:

In conclusion, we can say that …

A link word is an adverb that connects one phrase with another. Many link words are vital for highlighting contrasts (*although, but, whereas, on the contrary* etc), for giving explanations (*owing to, because* etc), and showing consequences (*thus, consequently, thereby* etc) – to learn more about these link words, see Chapter 6 in *English for Academic Research: Grammar, Usage and Style*.

However, there are some link words that are not always necessary, and where often it would be best to delete them and / or begin a new paragraph. Such words are

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1. *in fact, indeed (*quite archaic*), as a matter of fact*
2. *in addition, furthermore, moreover*
3. *as far as x is concerned, with regard to, regarding*
4. *it is worth noting / mentioning that*

The above words are occasionally useful, but generally can be removed. You just need to make sure that the order of your sentences is sufficiently logical not to need such words. The first set is used when you have said something and you then want to show an example that confirms what you have said. Try removing the link word, and decide if it makes any difference without it.

Authors tend to use the second set in order to give the idea that a series of sentences are in some way connected. I suggest you remove them. This will show you that you may need to restructure the paragraph so that the logical thread of what you are saying becomes clearer.

The third set is used to (re)introduce a topic. This is fine. But don't use them in the following way:

As far as artificial intelligence is concerned, it is becoming more and more pervasive.

Artificial intelligence is becoming more and more pervasive.

The fourth set is, in my opinion, totally redundant. If something was not worth mentioning, then surely you would not mention it. If you want your readers to notice something, then the best way is to begin a new paragraph, and put what you are saying in a series of shorter than normal sentences (6.10).

By removing all these link words, you are shifting the subject of the sentence nearer the beginning of the sentence. This is a much more immediate, dynamic, and satisfying for the reader.

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4.9 Be concise when referring to figures and tables

The blue versions below highlight how it is not difficult to be concise when referring to figures and tables.

Figure 1 shows schematically / gives a graphical representation of / diagrammatically presents / pictorially gives a comparison of two components

Figure 1 compares two components.

From the graphic / picture / diagram / drawing / chart / illustration / sketch / plot / scheme that is depicted / displayed / detailed / represented / sketched in Figure 3, we can say that …

Figure 3 shows / highlights / reports that …

The mass spectrum, reproduced in the drawing in Figure 14, proved that …

The mass spectrum (Fig. 14) proved that …

We can observe / As can be seen from Table 3 that … From an analysis of Table 3 it emerges that …

Table 3 highlights that …

If you refer your readers to a figure, you don't need to describe the figure using words like *graphically* or *schematically*. You don't need to use many different synonyms either to describe what kind of figure it is or to say what it shows. If possible use active verbs – *this figures shows x*, rather than *x is shown in this figure*.

4.10 Consider reducing the length of your paper

Bear in mind what Mark Twain, the American author and humorist, once wrote: *I didn't have time to write a short letter, so I wrote a long one instead.*

Ask yourself:

- is my paper 40 pages long, simply because it was easier to report everything rather than taking the time to really consider what was the most important information? Or does it contain 40 pages of meaty essential information?

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- how are my editor and reviewers likely to react to my massively long paper?
- when my 40-page paper is published, will readers naturally want to read it rather than a 10–20 page paper on a similar topic?

Having said that, there is no evidence to prove that a shorter paper is likely to be cited more than a longer paper. However, to be cited your paper:

- has to have been published – excessive length is likely to be an issue for reviewers who may react badly to receiving a very long paper (remember reviewers generally do their work for free)
- contain useful data written in clear understandable way

4.11 Summary

You can be more concise by:

- deleting any words that are not 100% necessary (typically vague and generic words and expressions; typically adjectives, adverbs and abstract nouns)
- cut pointless introductory phrases and unnecessary link words
- finding ways of expressing the same concept with fewer words
- using verbs rather than nouns
- choosing the shortest words and expressions
- avoiding impersonal phrases that begin *it is* …
- being concise when referring to figures and tables

The result is that you will write less, make fewer mistakes, your key results will be clearer, and you will provide your readers with a more enjoyable experience.

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Chapter 5

Avoiding ambiguity, repetition, and vague language

Newspaper headlines

Dealers will hear car talk at noon

Drunk gets nine months in violin case

Enraged cow injures farmer with axe

Juvenile court to try shooting defendant

Kids make nutritious snacks

Killer sentenced to die for second time in 10 years

Local high school dropouts cut in half

Milk drinkers are turning to powder

Miners refuse to work after death

Panda mating fails; vet takes over

Plane too close to the ground, crash probe told

Sex education delayed, teachers request training

Stolen painting found by tree

Two sisters reunited after 18 years at checkout counter

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_5

5.1 Use a simple and accessible style

The author resources section of Nature highlights the importance of writing in a 'simple and accessible style':

> Many papers submitted for publication in a Nature journal contain unnecessary technical terminology, unreadable descriptions of the work that has been done, and convoluted figure legends. Our journal subeditors and copyeditors edit the manuscript so that it is grammatically correct, logical, clear and concise, uses consistent search terms, and so that the terminology is consistent with that used in previous papers published in the journal. Of course, this process is assisted greatly if the authors have written the manuscript in a simple and accessible style, as the author is the best person to convey the message of the paper and to persuade readers that it is important enough to spend time on.

A sentence or phrase is ambiguous or vague when it has more than one interpretation or its interpretation is not obvious. Below are some examples (not from the world of science!).

For those of you who have small children and don't know it we are now serving ice cream.

I plan to mow the lawn with my husband.

You should never crumble your bread or roll in your soup.

In the first it seems that the parents don't know that they have a child; in the second that the wife is using her husband to cut the grass; in the third, *roll* has two meanings – as noun it means a type of bread, but as a verb it means to move in a particular direction by turning over and over on an axis (it seems like the guest is swimming inside the soup!)

If referees are not clear about what you are saying in a particular sentence, this may affect their overall understanding of the contribution of the paper. They may thus feel that they are not in a position to judge the merits of your paper. Just two or three ambiguous sentences are enough for referees to recommend delaying publication until 'the English has been revised by a native speaking expert'.

Ambiguity arises when a phrase can be interpreted in more than one way, as highlighted by the two red sentences below.

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Professors like annoying students.

Professors like to annoy their students.

Professors like students who are annoying.

I spoke to the professor with a microphone.

Using a microphone, I spoke to the professor.

I spoke to the professor who was holding a microphone.

Here is another example where poor word order can create confusion:

To obtain red colors, insects and plant roots were used by our ancestors.

Readers may initially think that *red colors* and *insects* are part of the same list. Readers will only understand that *insects and plant roots* is the subject of the verb when they get to the end of the sentence. To avoid this problem there are two possible solutions.

Insect and plant roots were once used to obtain red colors.

To obtain red colors, our ancestors used insects and plant roots.

The first blue sentence puts *insects and plant roots* as the main subject, and the second *primitive people*. The choice of which sentence will probably depend on whether the ancestors have already been mentioned or not.

We tend to read words in small groups. Often we think that if two or three words immediately follow each other they must be related in some way. Consequently, the red sentence below is initially confusing.

The European Union (EU) adopted various measures to combat these phenomena. This resulted in smog and pollution levels reduction.

When we read *resulted in smog and pollution*, our initial interpretation is that the smog and pollution are the result of the EU's measures. Then when we move on and read *levels* we have to reprocess the information. This is not important if readers have to change their interpretation only once or twice in a paper. But if they have to do it many times, the cumulative effort required becomes too much. Some readers will stop trying to guess the meaning and stop reading. In your case, it may mean that your paper could be initially rejected. Below is a clearer version.

The European Union adopted various measures to combat these phenomena. This resulted in a reduction in smog and pollution .

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5.2 Beware that pronouns are probably the greatest source of ambiguity

Some sentences that would not be ambiguous in your language may become ambiguous in English. For example:

I put the book in the car and then I left *it* there all day.

In English we do not know whether *it* refers to the *book* or the *car*. Some languages have a case system or a gender for nouns. Thus if your word for *book* is – for instance – masculine, and your word for *car* is feminine, you will use a different form of *it* to indicate whether the noun *it* refers to is masculine or feminine, and this will make it clear for your reader. In English *it* can refer to all nouns (apart from those that refer to human beings).

In any case, if you use *it* in one sentence to refer to a noun you have mentioned in a previous sentence, you may be forcing to the reader to re-read the previous sentence to remember what *it* refers to.

So if you think that there could be possible ambiguity or that the reader may have forgotten the subject, then simply repeat the key word:

I put the book in the car and then I left the book there all day.

You may think this is not very elegant, but it is much clearer for your reader and is not considered bad style in technical English.

In the red sentence below, does *they* refer to all three locations, to Canada and the Netherlands, or just to the Netherlands?

We could go to Australia, Canada or the Netherlands, but *they* are a long way from here.

To avoid misunderstandings, be more specific:

… Australia, Canada or the Netherlands, all of which are a long way from here.

… Australia, Canada or the Netherlands. But Canada and the Netherlands are a long way from here.

… Australia, Canada or the Netherlands. But the Netherlands are a long way from here.

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In the red sentences below, what do *one* / *this* / *these* refer to? (a) user names (b) passwords?

No user names or passwords are required, unless the system administrator decides that *one* is necessary.

… decides that *this* is necessary.

… decides that *these* are necessary.

Interpretations (a) and (b) are much clearer rewritten as the two blue sentences below, respectively.

… unless the system administrator decides that a user name is necessary.

… unless the system administrator decides that a password is necessary.

In the red sentence below, what does *this* refer to?

There are two ways to learn a language: take private lessons or learn it in the country where the language is spoken but *this* entails spending a lot of money.

Does *this* refer to the cost of private lessons, the cost of living in the country where the language is spoken, or both? To clarify, you just need to repeat the key concept.

There are two ways to learn a language: take private lessons or learn it in the country where the language is spoken. However, *living in a foreign country* entails spending a lot of money.

There are two ways to learn a language: take private lessons or learn it in the country where the language is spoken. However, *both these solutions* entail spending a lot of money.

In the first blue sentence above it is now clear the cost only refers to living in a foreign country. The second blue sentence clarifies that lessons plus living in a foreign country have a high cost. Note that *solutions* has been used to replace *ways* in the first part of the sentence – using synonyms for non-key words is fine.

In the red sentence below does *them* refer to just the branches or the logs as well?

We cut the trees into sectors, then separated the logs from the branches, and then burnt *them*.

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To make it clear, you just need to replace *them* with *branches* (if it is just the branches that were burnt), or with *both of them* (if both branches and logs were burnt).

In my experience as an editor, pronouns cause more ambiguity than all the other sources of ambiguity combined.

This is also due to the way we read.

You are the author and you expect your readers to read every word, sentence and paragraph.

However, few readers will have the time or energy to do this. Thus if you refer to something, let's call it X, that you mentioned a few sentences before, or in the paragraph before, and you refer to X using *this, these, it, them, which, the former etc*, then you risk losing your reader who simply may not have read the original instance of X.

It makes life much simpler for everyone if, where ambiguity could arise, you replace pronouns with the noun that they refer to. You make consider this inelegant, but elegance should NEVER be considered more important than clarity and ease of reading.

5.3 Avoid replacing key words with synonyms and clarify ambiguity introduced by generic words

When you were at school learning your own language, your teachers probably encouraged you not to use the same word in the same sentence more than once, and maybe not even in the same paragraph. Finding synonyms was good. Consequently, like many researchers you probably now suffer from monologophobia – the fear of using the same word twice!

Monologophobia can cause ambiguity or confusion for the reader. For example, do the three words in italics below have a different meaning?

Companies have to pay many taxes. In fact, occasionally *enterprises* fail because of over-taxation. Some *firms* resolve this problem by moving their headquarters to countries where the tax rate is lower.

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For the author, the words probably have the same meaning. But the reader cannot be sure, and may try to work out what the difference between the three terms is. The author is thus forcing the reader to make an unnecessary mental effort.

If you decide to use words that have similar – but slightly different – meanings, then you should define these differences for the reader. In the red sentence above you would need to define the difference between a *company*, an *enterprise* and a *firm*.

A very important rule in scientific English is: never find synonyms for key words – avoid synonymomania!

So a better version would be:

Companies have to pay many taxes and occasionally may fail because of over-taxation. Some *[companies]* resolve this problem by moving their headquarters to countries where the tax rate is lower.

This problem is accentuated when authors use different words to express the same concept over several paragraphs. For instance, let's imagine that in one paragraph the author uses the word *test*, in another paragraph *experiment*, and in another paragraph *trial*. The reader cannot be sure if *test*, *experiment* and *trial* all refer to the same scientific procedure or to two/three different procedures.

Authors come up with ingenious solutions for not repeating the same word. One device is to replace the key word with a generic description of it.

The maximum solubility of *mercury* occurs in an oxygenated environment, which is the typical condition found in soil. The principle forms that are found in soil are $Hg(OH)_2$ and $HgCl_2$. With these ions, this *metal* can form soluble complexes that are …

Readers will probably understand that *metal* refers to *mercury*. But it will help readers if you repeat the word for them:

The maximum solubility of *mercury* occurs … With these ions, *mercury* can form soluble …

This is particularly important when the generic word (*metal*) appears several lines later than the original concrete word (*mercury*).

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So, be careful when you use words such as *process, parameter, element, feature, function* to refer to a key word – can you be sure that your readers will associate these generic words with the key word?

Another typical device to avoid repetition is to use *one* or *that* as in the two red examples below, which both sound strange to a native speaker.

This can be done by using either a *chromatographic pump or a peristaltic one.*

This can be done by using either a *chromatographic or peristaltic pump.*

With regard to the *TTC* output the *arbitrariness* of a g_{pk} parameter can be exploited by starting from *that* of g_{pa}.

With regard to the *TTC* output the *arbitrariness* of a g_{pk} parameter can be exploited by starting from the *arbitrariness* of g_{pa}.

5.4 Restrict the use of synonyms to non-key words

Synonyms are helpful for replacing repetitive usage of adjectives and verbs. Examples:

We would like to stress / underline / emphasize / highlight that x = y.

We performed / carried out / did several experiments.

This is a critical / very important / fundamental issue.

Another typical case where synonyms are useful is to avoid repetition of the same generic word (or derivatives). For example:

The identification is *mainly* based on three *main* strategies.

This function has three main *aims* all *aimed* at reducing stress.

The *use* of synonyms is *useful* to replace *overuse* of the same adjective.

Such unnecessary repetition may irritate some readers, so you could replace *mainly* with *generally*, or alternatively *main* with *principal*, and *aims* with *objectives* or *aimed* with *targeted*. In the last case (*use … useful … overuse*) you can simply delete *the use of* and replace *useful* with *helpful*.

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Note how in the sentence immediately preceding this one I have specified what *the last case* refers to by putting the example words in brackets. Otherwise you might have initially thought that *the last case* referred to the previously mentioned case (i.e. *aimed with targeted*). Putting an explanation in brackets or using *i.e.* is a great way of clarifying what you mean, particularly when you are forced to use a generic phrase that could be open to ambiguity.

Finally, don't be worried if a sentence contains two instances or more of the same preposition (typically *of*). Most prepositions don't have exact synonyms, so don't be creative in trying to find them! This is particularly true when deciding on the title for your paper (11.4).

However, do check that you have used the preposition correctly, for example we say 'an increase *of* 10%' but 'an increase *in* inflation' (*of* plus a percentage, *in* plus a noun – see 2.11 in *English for Research: Grammar, Usage and Style*).

Synonyms are also essential when you need to paraphrase the work of others or your own work (15.9–15.12).

5.5 Be as precise as possible

Aim for precision. Instead of saying something happened *in a number of cases*, be more exact: *this happened in 11 cases*.

A common mistake by authors is in making assumptions about what the reader will understand. This is because you as the author know your topic extremely well. This means that you may use words and expressions which to you are clear, but to the reader may not be. Below are a few examples of words and expressions that could be interpreted in many different ways. In all cases you need to be more specific:

in the short term, in the near future

a relatively short / long duration

[quite a] high / low number of

recently, recent

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In the last case, for example, the reader may be reading your paper several years after its publication, thus it is no longer *recent*.

Referees often criticize authors for sentences such as:

Usually the samples were cooled to room temperature. [So what happens or happened in other cases?]

It was necessary to study the problem with *attention*. [What exactly does *attention* mean?]

Subjects performed *fairly* well and their results were *substantially* better than their counterparts.

With reference to the last example, *fairly* and *substantially* mean different things to different people.

Other examples of potentially ambiguous adjectives and adverbs are: *adequate, appreciable, appropriate, comparatively, considerable, practically, quite, rather, real, relatively, several, somewhat, suitable, tentative*, and *very*. These adjectives and adverbs do not have a single unequivocal meaning. They can be open to interpretation by the reader.

5.6 Choose the least generic word

Another way to be more precise is to choose the least abstract / generic word. In the two red sentences below a generic word is followed by specific definition – this type of construction is often an unnecessary repetition.

This kind of investigation, i.e. the analysis of the AS profiles, also aims to find sets of nodes which behave similarly and …

Climatic conditions (i.e. temperature, rainfall) were also checked.

In such cases, delete the preceding phrase and just use the definition.

By analyzing AS profiles we can also find sets of nodes that behave similarly and …

Temperature and rainfall were also checked.

For more on specific vs generic, see 5.4.

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5.7 Use punctuation to show how words and concepts are related to each other

Punctuation in English is used exclusively to show how words and concepts are related to each other. This is not true of all languages.

COMMAS BEFORE *AND*

Here is an example from an email:

I will be free the whole of Monday *and* Tuesday *and* Thursday morning unless one of the professors decides to arrange an extra class.

Does this mean that I will be free (a) all Monday and Tuesday, or (b) all Monday and also Tuesday and Thursday mornings? This can be clarified by using a comma:

a) I will be free the whole of Monday and Tuesday, and (also) Thursday morning.

b) I will be free the whole of Monday, and (also) Tuesday and Thursday morning.

SEMICOLONS TO DIVIDE UP LISTS

If you have lists of items, you need to show how the various items relate to each other. In some cases semicolons can be useful:

The languages were grouped as follows: Spanish, Italian and Romanian; German and Dutch; and Swedish and Norwegian.

However, the above could be written more clearly as:

The languages were allocated to three groups: 1) Spanish, Italian and Romanian; 2) German and Dutch; and 3) Swedish and Norwegian.

HYPHENS

Hyphens are used in English to show the relationship between words in a sequence, which without a hyphen would be ambiguous. For example, what does the following sentence mean?

We have a little used car in the garage.

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Does it mean that we don't use the car very often, or that we bought the car second-hand? Hyphens can be used to make this differentiation clear.

We have a *little-used* car in the garage.

We have a little *used-car* in the garage.

5.8 Defining vs non defining clauses: *that* vs *which / who*

Look at the two sentences below – in which case do I have more than one sister?

My sister, *who* lives in Paris, is a researcher.

My sister *that* lives in Paris is a researcher.

In the first sentence, the information contained between the two commas is not essential. I have only one sister and she is a researcher – the fact that she lives in Paris is just additional information. I could simply say: *My sister is a researcher*.

In the second sentence, I am telling you that I have more than one sister, and that the sister that lives in Paris is a researcher. Perhaps my other sister is a doctor and I am using Paris to distinguish between my two sisters.

This difference between *who* and *that* is the same as the difference between *which* and *that*.

In scientific English, *which* and *that* have distinct uses. For example, imagine you are instructed to do the following:

Correct the sentences below which contain grammatical mistakes.

Does this mean (i) that all the sentences contain grammatical mistakes, or (ii) that you should correct only those sentences that contain mistakes?

i) Correct the sentences *below, which* contain grammatical mistakes.

ii) Correct the sentences *below that* contain grammatical mistakes.

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The rule is that if you are simply adding extra information (sentence i) then use *which* (things) or *who* (people) preceded by a comma (,). If you are defining the previous noun then use *that* (sentence ii).

Given that not many people are aware of this distinction, it is probably better to rewrite the sentences more explicitly:

i) Correct the sentences *below, all of which* contain grammatical mistakes.

ii) Correct *only those sentences below that* contain grammatical mistakes.

Here is another ambiguous example:

The table below gives details of the parameters which are not self-explanatory.

The reader does not know if the writer

- has forgotten to put a comma after *parameters* and thus means that none of the parameters are self-explanatory
- should have put *that* instead of *which* and thus means that the table only gives details of those parameters that need to be explained

A similar problem arises when the author does not use either *which* or *that*, as in the red sentence below.

This is followed by a characterization of the states *poorly represented* at atmospheric pressure.

This is followed by a characterization of the *states, which* are poorly represented at atmospheric pressure.

This is followed by a characterization of *all those states that* are poorly represented at atmospheric pressure.

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5.9 Clarifying which noun you are referring to when *which, that, who* and the *-ing* form

Which, that and *who* should only refer to the noun that immediately precedes them.

A group of patients was compiled using this procedure, as proposed by Smith and Jones [2024], *who* had died under surgery.

A group of patients *who* had died under surgery was compiled using this procedure, as proposed by Smith and Jones [2024].

An initial reading of the red sentence above gives the impression that Smith and Jones died under surgery! This ambiguity arises because the subject (*patients*) has been separated from its verb (*had died*) by a subordinate clause (*as proposed* …). The solution is to keep the subject and verb as close as possible to each other.

Here is another example:

Each scheduling service is characterized by a mandatory set of QoS parameters, as reported in Table 1, *which* describes the guarantees of the applications.

Each scheduling service is characterized by a mandatory *set* of QoS parameters, as reported in *Table 1*. *This set* describes the guarantees of the applications.

The blue version splits the sentence in two and repeats the key word (*set*).

Be careful when using the *-ing* form.

Professor Rossi teaches the students *having* a good level of English.

Professor Rossi teaches the students *that have* a good level of English.

Professor Rossi teaches the students *since he has* a good level of English.

Professor Rossi, *who has* a good level of English, teaches the students.

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5.10 *-ing* form vs. subject + verb

In clear unambiguous writing, verbs should be immediately preceded by their subject.

If you take your child in the car, don't let her put her head out of the window *while driving.*

If you take your child in the car, don't let her put her head out of the window *while you are driving.*

After consuming twenty bottles of wine, the conference chair presented the awards to the fifty best PhD students.

After the fifty best PhD *students had consumed* twenty bottles of wine, the conference chair presented them with the awards.

In the first red sentence above it initially seems that the child is *driving*. This is because the person located in the phrase nearest to the word *driving* is the child not *you*. In the second red sentence above, it seems that the conference chairperson consumed 20 bottles of wine, whereas presumably it was the students who did the drinking!

In the red sentence below there is an initial ambiguity: the order of words makes it seem that psocoptera read books! (*Psocoptera* are an order of wingless insects that attack paper).

We cannot understand how psocoptera survive by reading books alone. Instead we need to …

If we rephrase the sentence by putting the *-ing* form at the beginning, the true meaning is a little clearer:

By reading books alone, we cannot understand how psocoptera survive. Instead we need to …

However the clearest way is to avoid the *-ing* form completely and replace it with a subject + verb construction:

If we limit ourselves to reading books, we cannot understand how psocoptera survive. Instead we need to …

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So, beginning a sentence with the *-ing* form can be dangerous. This is because the reader doesn't know who or what is carrying out the activity introduced by the *-ing* form.

In the red sentence below, it initially seems that the muscles are watching television, though this is clearly absurd.

By sitting and watching too much television, our muscles become weaker.

The solution is to put a subject (*we*) in front of the verb:

When we sit and watch too much television, our muscles become weaker.

Ending a sentence with the *-ing* form can also be dangerous. This is again because the reader doesn't necessarily know who or what the *-ing* form refers to.

Students were taught French, German, and Arabic using a new method.

What was the *new method* used with: i) Arabic only, ii) all three languages

i) Students were taught French and German, as well as Arabic by using a new method.
i) Students were taught French German, and Arabic. A new method was used to teach Arabic.
ii) Students were taught French, German, and Arabic – all with a new method.
ii) We used a new method to teach French, German, and Arabic.
ii) Using a new method students were taught French, German, and Arabic.

5.11 Avoiding ambiguity with the *-ing form*: use *by* and *thus*

The red sentence below is ambiguous – why?

This will improve performance *keeping* clients satisfied.

Does it mean: (a) the way to improve performance is if clients are kept satisfied? or (b) as a consequence of improving performance clients will be satisfied?

We can show the true meaning if, before the *-ing* form, we insert *thus* or *by*:

This will improve performance *thus keeping* clients satisfied.

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If performance improves, clients will be satisfied – *thus* means *as a consequence*.

This will improve performance *by keeping* clients satisfied.

The way to improve performance is through client satisfaction – *by* indicates how something is done.

Often it is a good idea simply to break up the sentence or use *and*.

This will improve performance *and* clients will (*thus*) be satisfied.

Below in red is another ambiguous sentence. It can be disambiguated as in the two blue sentences, both of which have the same meaning.

The Euro indirectly raised prices, *causing* inflation.

… raised prices. This *consequently / subsequently caused* inflation.

… raised prices *and so / thus caused* inflation.

It is best to replace the *-ing* form with *and* when you are simply giving additional information.

This section focuses on the reasons for selecting these *parameters, trying* to explain the background to these choices.

This section focuses on the reasons for selecting these *parameters, and tries* to explain the background to these choices.

Finally, note the difference between these three sentences.

To paraphrase something you can use QuillBot or ChatGPT.

Paraphrasing your work now takes only a few seconds.

By paraphrasing someone else's work you may be guilty of plagiarism.

The infinitive (*to paraphrase*) means *If you want to / If your aim is to …*

The *-ing* form with no preceding preposition (*Paraphrasing*) refers to the activity of paraphrasing, it acts as the subject of the sentence.

Finally, *by paraphrasing* means *If you paraphrase …*

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5.12 Uncountable nouns

A countable noun is something you can count – *one apple, two apples*. An uncountable noun is something that, at least in English, you cannot count. You cannot say *an information, these informations* etc. *Information* is considered a mass, and for English speakers it is not easily split into different parts.

Spinach leaves can be clearly separated and counted, but when cooked they become one big mass. You cannot clearly and easily identify cooked spinach as separate parts – so you cannot say *these spinaches taste very good*, but only *this spinach tastes very good*. Similarly, you can count *cars* but not *traffic*, *steps forward* but not *progress*, *comments* but not *feedback*.

These kinds of subtleties do not normally cause problems. But when an uncountable noun (i.e. a noun that is always singular) is referred to in a later phrase with a plural pronoun (*they, these, those*) or adjective (*many, few*) it can create confusion for readers.

For example, be careful of using a word like *feedback*. *Feedback* is uncountable, so it has no plural form (at least not in common English).

Such *feedbacks are* vital when analyzing the queries. At subsequent stages in the procedure, for instance after steps 3 and 4, *they* are also useful for assessing …

Such *feedback is* vital when … At subsequent stages *it* is also useful for …

Such *feedbacks are* vital when analyzing the queries. At subsequent stages in the procedure, for instance after steps 3 and 4, *many of them* are also useful for assessing …

Such *feedback is* vital when … At subsequent stages *much of it* is also useful for …

There is ambiguity in the first red sentence above because a native speaker would think that *they* must refer to *queries*, since *queries* is plural. Likewise in the second red sentence, the reader would be totally confused and would probably be unable to understand what *many of them* refers to.

Pronouns are in any case a constant source of ambiguity in English, so the best solution is to repeat the noun that the pronoun refers to.

Such feedback is vital when … At subsequent stages (a lot of) this feedback is also …

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5.13 Definite and indefinite articles

The usage of articles is very complex in English – for full details see Chapter 3 in *English for Research: Grammar, Usage and Style*.

In brief, here are the differences in meaning.

A researcher spends many days in the lab.

A generic researcher, who we have not mentioned before.

One researcher spends many days in the lab.

We have already mentioned a group of researchers, now we are focusing on one individual who spends many days in the lab, unlike the others in her group who are rarely in the lab.

Researchers spend many days in the lab.

Researchers in general, i.e. 'all researchers', so the non-use of *the* is correct.

The researcher spends / *The researchers* spend many days in the lab.

The researcher has (or the researchers have) already been mentioned before and that the reader knows which researcher we are talking about.

If the last sentence appeared at the beginning of a new section in a paper, you as the reader might be confused and would be forced to look back to earlier sections to see if you could find a previous reference to the researcher/s. In fact, if *the* is used with a countable noun it implies that this noun has already been mentioned before.

Below is an example of the *a* versus *one* rule. The two sentences thus have very different meanings.

We made *one* experiment before the equipment exploded.

We are implying that we had planned a series of experiments (at least two), but that these were interrupted by the explosion.

We made *an* experiment before the equipment exploded.

No series of experiments had been planned.

So where is the possible ambiguity?

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5.14 Referring backwards: the dangers of *the former, the latter*

When you refer back to something you mentioned before, it is often not immediately clear what *the former* and *the latter* refer to.

Africa has a greater population than the combined populations of Russia, Canada and the United States. In *the latter*, the number of people is only …

Does *the latter* refer just to the US alone, or to the US and Canada?

The simplest and clearest solution is to replace *the latter* with the exact word or words it refers to:

Africa has a greater population than the combined populations of Russia, Canada and the United States. *In the USA* the number of people is only …

Africa has a greater population than the combined populations of Russia, Canada and the United States. *In Canada and the USA* the number of people is only …

It is not a problem to repeat words if the result is that the reader will be clear about what you want to say. This is particularly true if the word that *the former / the latter* refers to is some distance away. For example:

Smith was the first to introduce the concept of readability in websites. In his seminal paper, written in 2018, he realized that the way we read pages on the web is totally different from the way we read a printed document. Five years later in 2023, Jones studied the differences between the way that people of different languages, whose scripts are written left right (e.g. English), right left (e.g. Arabic) and top down (e.g. Japanese), read texts on the web. The former author then wrote another paper …

By saying *the former author* you are forcing the reader to go back four or five lines in order to remember which author you are talking about. By simply saying *Smith then wrote* you save the reader time and frustration.

Clearly there are some occasions when using *the former* and *the latter* is not ambiguous:

Water organisms can be contaminated directly or indirectly. *The former* occurs by contact or ingestion of the substance dissolved in water, whereas *the latter* happens when the contaminant is accumulated in the food chain.

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In the sentence above there is no ambiguity, but it would be better rewritten as:

Water organisms can be contaminated directly or indirectly. *Direct contamination* occurs by contact or ingestion of the substance dissolved in water, whereas *indirect contamination* happens when the contaminant is accumulated in the food chain.

Specific words (*contamination*) are more readily absorbed and memorable than generic words (*the former*).

Another reason for NOT using *the former … the latter* is that generally you will be writing with a co-author. This co-author might decide to change the order of something in the initial sentence, which means that *the former* and *the latter* no longer refer to the same words.

For instance you might write: *Italy and France are key players in Europe. The former is …* And then your co-author might decide to put *France* before *Italy*, but forget to rearrange the second sentence.

5.15 Referring backwards and forwards: the dangers of *above*, *below, previously, earlier, later*

When making reference to things that are mentioned earlier or later in your documents do not give readers generic locations.

As mentioned *above / before / earlier / previously*, these values are important when …

These points are dealt with in detail *below / later* …

If readers are interested in these things, then they need an exact location, for example: *see Sect. 1.1 / see the above paragraph / see points 4–5 below*.

The term *previously* is often ambiguous because the reader may be not sure if you mean at some point earlier in this paper, in another of your papers, or in someone else's paper

It is acceptable to say *as mentioned above* or *as mentioned before* when you don't want the reader to go back to what you said before, but simply to reassure them that you are aware that you are saying the same thing again. However, ask yourself if the reader really does need such reassurance.

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5.16 Use of *respectively* to disambiguate

Respectively is a very useful word for clarifying how items are related to each other. In the sentence below, a basic knowledge of geography makes it clear that London is associated with England, and Paris with France.

London and Paris are the capitals of England and France.

But such connections are not always so obvious:

… where X is the function for Y, and f1 and f2 are the constant functions for P and Q.

Are f1 and f2 constant functions for both P and Q? If so:

… and f1 and f2 are the constant functions for *both* P and Q.

Or is f1 for P and f2 for Q? If so, use *respectively*:

… and f1 and f2 are the constant functions for P and Q, *respectively*.

Most style books recommend placing *respectively* at the end of the phrase. It is best to put a comma before *respectively*.

5.17 Distinguishing between *both … and,* and *either … or*

Both … and is inclusive. *Either … or* is exclusive.

We studied both English and Spanish.

You can study either English or Spanish.

You cannot study both Russian and Korean.

You cannot study either Russian or Korean.

The first sentence above means that we studied English and we studied Spanish. The second means that you can only study one language. You cannot study English and Spanish. You can study English or you can study Spanish.

The third means that you have to choose between Russian and Korean. You can only study one of the two languages.

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And the fourth means that these two languages are not offered. Neither of them can be studied.

The position of *both* can change the meaning. Note the difference between these two pairs of sentences:

This is true both for the students and the professors.

This is true for both the students and the professors.

We had fun in both the parks we visited and also the museums.

We had fun both in the parks and the museums.

In the first there are several students (and professors) involved, in the second there are only two students and an undefined number of professors.

In the third just two parks are being referred to, whereas in the last sentence there is an unknown number of parks.

5.18 Talking about similarities: *as, like, unlike*

Be careful when you are comparing your methods and results with those of another author. In the red sentence below it is not 100% clear whether you are or are not in agreement with Walker's suggestion.

We also demonstrated that x does not equal y as suggested by Walker (2025).

Does this mean that Walker suggested that x is equal to y and is thus in contrast to what you are saying? If so, it should be re-written as:

Unlike what was suggested by Walker (2025), we demonstrated that x does not equal y.

Our findings do not concur with Walker (2025). In fact, we have clearly demonstrated that x does not equal y.

Or does the red sentence mean that Walker, like you, found that x does not equal y:

In agreement with Walker (2025), we demonstrated that x does not equal y.

Ambiguity affects readability. If you force your reader to constantly interpret what you are writing, the reader will soon want to stop reading.

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5.19 Differentiating between *from* and *by*

These two words have different meanings: *from* (origin) and *by* (agent).

This paper was drafted *by* several different authors *from* three different universities.

We received an email *from* Professor Southern written *by* her secretary.

In the above two sentences if you used *by* instead of *from*, or vice versa, this would not lead to ambiguity – though it would be ungrammatical.

But in some cases there could be ambiguity if you use the wrong one.

This product was created *from* smartcon.

This product was created *by* smartcon.

In the first case *created from* would mean that *smartcon* is a material. In the second case, *created by* means that *smartcon* is the creator. These are two very different meanings.

5.20 Be careful with Latin words

The problem with many Latin expressions is that you may know what they mean, but your reader may not. Notice the difference between *e.g.* and *i.e.* in the following two sentences.

Great Britain, *i.e.* England, Scotland and Wales, is the ninth biggest island in the world and the third most populated.

Some EU members, *e.g.* Spain, Italy and France, are not in agreement with this policy.

i.e. is used to define Great Britain, which contains only those three countries.

e.g. is used to give examples. Spain, Italy and France are just some examples of countries in the European Union that do not agree with the policy – but the implication is that there are other countries involved as well.

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If you are not short of space then it is generally better to use alternative versions. Another way to say *i.e.* is *that is to say*. Other ways to say *e.g.* are: *for example, such as*, and *for instance*.

Unless they are commonly used in your chosen journal, try to avoid other Latin expressions such as *a priori, a posteriori, ex ante, in itinere, ex-post,* and *ceteris paribus*. Some readers, including native English speakers, may not know what they mean.

This argument holds, *a fortiori*, in mergers, where the reduction of the number of firms in the market is an explicit objective.

This argument holds *for similar but even more convincing reasons* in mergers, where the reduction of the number of firms in the market is an explicit objective.

If you use Latin expressions, check with your journal whether they should be in italics or not.

5.21 False friends

False friends are words from two different languages that look very similar but have different meanings. If you speak a European language, then there is a good chance that there are several false friends between your language and English. The most common of these is *actually*, which in English means *in reality*, but its false friend in other languages means *currently / at the moment*.

Another false friend, which frequently appears in research, is to *control* which does not mean *verify*. Here is the difference:

A thermostat is used to *control* the temperature. [i.e. adjust, act on]

We *checked* the patient's temperature with a thermometer. [i.e. verify without any intervention]

In scientific papers, only a limited number of false friends tend to cause problems:

actual (real) vs *effective* (successful in producing desired effect); *alternately* (first one, then the other) vs *alternatively* (another option); *coherent* (intelligible) *consistent* (not contradictory, always acting in same way over a period of time); *comprehensive* (including everything) vs *understanding* (sympathetic awareness); *eventually* (at the end of series of difficulties) vs *if necessary / if any*; *occur* (happen) vs *need*; *sensible* (reasonable) vs *sensitive* (quick to respond to slight changes)

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5.22 Be careful of typos

What impression would a referee have if they read the following?

There are three solutions to *asses*.

A solution of lead was added to the mixture. Note: this *addiction* is likely to cause health problems.

Acknowledgements: We would like to offer our *tanks* to the following people:

The author meant to write *assess* (asses = donkeys), *addition* (addiction = pathological dependence), and *thanks* (tanks = armored vehicles).

No spell checking system currently available is likely to spot such mistakes.

See 11.9 in *English for Research: Grammar, Usage and Style* for a list of typical typos of this kind.

5.23 Summary

Your writing will be much clearer if you take into account the following:

- ensure that the order of the words in a sentence is not ambiguous
- beware that pronouns are probably the greatest source of ambiguity
- do not replace key words with synonyms – always use the same word
- if possible replace generic words with precise words
- *which* is used for adding information about the preceding noun, *that* defines the preceding noun
- *which, that* and *who* should only refer to the noun immediately preceding them
- Ensure it is clear what the subject of the *-ing* form is.
- clarify whether something is a consequence of doing something or a means to do something by using *thus* (consequence) and *by* (means) before the *-ing* form
- use the definite article (*the*) before a noun only if you refer to a specific example of that noun. If you are giving a generic idea, do not use the article
- learn the most frequent uncountable nouns and false friends in your field

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- be very careful when you use pronouns (*this, that, them, it* etc.) – make sure it is clear what they refer to and don't be afraid of repeating the same word many times (if this will improve clarity)
- avoid using *the former … the latter*, simply repeat the related noun
- if necessary specify exact locations, when using *above* and *below*
- use *respectively* when it is not 100% clear how items are related to each other
- be careful of punctuation with *which* and – punctuation must help the reader understand the relationships between the various parts of the sentence
- don't confuse *both … and* (inclusive) with *either … or* (exclusive); *i.e.* (definitions) and *e.g.* (examples), by (agent) and from (origin, and be careful when you use *as* to mean 'in a similar way'
- never use synonyms for key words, only for generic verbs and adjectives
- use the most precise word possible

There are other serious cases of ambiguity. These are dealt with in separate chapters of this book.

1. strings of nouns and adjectives (8.9)
2. misusage of tenses – using the present instead of the past, and vice versa, can create considerable confusion, particularly in the Introduction and Discussion sections
3. active vs passive (6.4, 18.3)
4. poor or incorrect word order (Chapter 2, 16.10).
5. when it is not clear if you are referring to your own work or other people's work (Chapter 6, 18.5)
6. cut&pastes that have not been checked (20.6)

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Chapter 6

Clarifying and Highlighting



Clarifying who invented what

- Cocktails: Not the Americans – their first cocktail, the *Sazerac*, was invented by a Frenchman in New Orleans in 1838. The British were the first to mix drinks – in India in around 1750.
- Electricity: Not Thomas Edison. The type of electricity we use today was proposed by Tesla, though Edison was the first to produce a long-lasting electric lightbulb.
- Guillotine: Proposed (but not invented) by Joseph-Ignace Guillotin as a less painful method of execution, but a similar apparatus had been around since the 14th century,
- Planes and helicopters: not the Wright brothers, but Leonardo Da Vinci.
- Printing: Not Johannes Gutenberg – the ancient Chinese were the first to print on block.
- Radio: Not Guglielmo Marconi. Tesla discovered a number of years before Marconi that radio signals could be transmitted, but never filed the patent on his discovery (Marconi patented his 'wireless telegraph').
- Sparkling wine: not Dom Pérignon (1638–1715), but fellow Benedictine monks in the Abbey of Saint-Hilaire, near Carcassonne, in 1531.
- The unconscious mind: Not Sigmund Freud (1836–1939), though he was responsible for making it popular. It was 'invented' by early psychology experimenters and thinkers, including the American, Charles Sanders Peirce (1839–1914).

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6.1 Why is it so important highlight and differentiate my findings in relation to the findings of other research groups?

In various sections of your paper, you need to compare your methodology or results with what has already been established in the literature. You must make it 100% clear to the reader whose methodology or results you are talking about.

Look at this extract from a fictitious paper entitled: *Do we talk more to people who are far from us than to those people who are next to us? How mobile phones have changed the way we communicate*, written by Joe Bloggs.

Imagine that your eye fell on this paragraph from the Results section and that this was the first paragraph in the paper that you had read so far.

Subjects sitting in train compartments on a 60-minute journey were found to spend an average of 55 minutes either talking on their cell phone or sending messages, watching videos etc. Only 2% of passengers talked to other people, in such cases merely to say 'sorry' or 'is this seat taken'? This contrasts with research conducted in 1989 on the same train journey. Cell phones were very rare at the time, and the study found that around 58% of passengers spoke to each other in a meaningful way for 10 minutes or more, with the prevalence being woman to woman, or man to woman. Conversations between two men were also found to be rare on the line between Copenhagen and Malmö.

Can you answer with 100% certainty the following questions: i) Who found the data regarding the 55 minutes talking? a) Bloggs b) someone else; ii) Who conducted the 1989 research? a) Bloggs b) someone else; Who conducted the Copenhagen / Malmö study? a) Bloggs b) the same author who conducted the 1989 research, in fact it was part of the same study c) another author.

The problem is that readers cannot answer the three questions above. If the referees cannot immediately understand who did what, and why what you did is important then this will make it difficult for the referee to:

- identify your contribution
- decide how useful the contribution is
- make a decision as to whether this contribution is worth recommending for publication

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For example, if you say *It was found that X = 1*, the referee needs to know whether you found that X = 1, or whether another author made this finding.

Your findings may be extremely valid and important. However, if the referees are not able to see or understand your findings because you have neither highlighted nor described them clearly enough, then your paper may not be published. Your contribution to the community may thus vanish into oblivion.

In the words of English botanist, Sir Francis Darwin: *In science the credit goes to the man who convinces the world, not to the man to whom the idea first occurred.*

6.2 Check your journal's style – first person or passive

Check your journal's 'guidelines to authors' to see whether you are permitted to use *we*. If you can use 'we' then it is relatively easy for you to distinguish between your work and others. Some journals, particularly those regarding Physics, tend to opt for an impersonal form in the belief that science is independent of the person writing about it. This entails adopting a lower profile and using the passive form.

If your journal insists on the passive form, you need to be extremely careful.

The most important point to remember is that YOU know which is your work and which is someone else's. But the readers do not! You must make it clear for THEM.

6.3 How to form the passive and when to use it

Active: We ***performed*** two tests. Blake et al. ***carried out*** one replication.

Passive (*is / was / will be* etc. + past participle): Two tests ***were performed*** (by us). One replication ***was carried*** out by Blake.

The passive is particularly useful when you describe a process, for example in the Methods. This is because it puts the equipment, chemicals, procedures etc. that you used in the first position in the phrase. In review papers, and in other sections of research papers, for example the Introduction and the Discussion, you may want to use the passive to describe what other authors have done, or what is already established knowledge in your domain. In such cases you can say:

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It has been demonstrated / It is believed that bilingual children adapt…

Bilingual children *have been demonstrated / are believed* to adapt better to new situations than monolingual children.

The advantage of the blue sentence is that the subject of the sentence (*bilingual children*) is at the head of the phrase, whereas it is delayed in the red sentence.

6.4 Use the active form when the passive might be ambiguous

Look at the two sentences below, is it clear who is the subject of the verbs in italics?

[In the Conclusions of a paper] This is a limitation of our data assimilation system, which *should be changed* in the near future.

Is the author suggesting that other researchers in the community should make these changes, or is the author going to do it herself?

The same lack of regard for others was present amongst subjects who used their mobile phone while driving their car, in agreement with the higher values of selfish behavior *observed* in cigarette smokers [17].

Who made the observation, the author of the paper or the author of reference 17?

If the author is the subject of the verbs, then the two red sentences above could be rewritten, using the active form, as follows:

This is a limitation of our data assimilation system, which *we plan* to change in the near future.

The same lack of regard for others was present amongst subjects who used their mobile phone while driving their car, in agreement with the higher values of selfish behavior that *we observed* in cigarette smokers and which has also been found by other authors [17].

If the author is not the subject of the verbs, then the two red sentences could be rewritten as follows:

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This is a limitation of our data assimilation system, and *we invite others in the community to suggest possible remedies*.

The same lack of regard for others was present amongst subjects who used their mobile phone while driving their car, in agreement with the higher values of selfish behavior that *have been found by other authors* [17].

Often a passive form can be disambiguated by using an adverb of frequency.

Using the x methodology *is considered* the same as using the y methodology.

Using the x methodology is *generally / usually / often* considered the same as using the y methodology.

In the red sentence it is not clear who has made the 'consideration'. The use of *generally / usually / often* resolves this problem – as it now clear that this is a general consideration rather than one just made by the author.

The following verbs when used in the passive are often ambiguous: *acknowledge, conceive, consider, describe, design, develop, find, observe, propose, suggest*.

Obviously, there are occasions when the passive form contains no ambiguity and is perfectly acceptable. For example:

It is well known that smoking causes cancer.

Mobile phone usage during meetings is often criticized.

In the first sentence, this fact is known by everyone. In the second sentence, it is clear that the criticism comes from other attendees at the meeting.

For more on this critical point see 16.4 in this book, and 14.3, 15.9 and 15.11 in *Grammar, Usage and Style*.

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6.5 Consider starting a new paragraph to distinguish between your work and the literature

Throughout the Discussion, and sometimes during your Introduction, you will need to switch from talking about your work to discussing other authors' work. Each time you begin a new area of comparison, begin a new paragraph. This makes it much easier for the reader to follow.

Also consider using one paragraph to describe other authors' work, and a new paragraph to describe your own. Constantly switching within the same paragraph from your work to other authors' can be quite hard for readers as they may not be totally clear if you are talking about your work or another author's.

The switch from one topic (your work) to another topic (the literature) is much clearer if it is also visual, i.e. if there is a paragraph break.

For some good examples of how to effectively differentiate your work from others, see 16.6.

6.6 Ensure you use the right tenses to differentiate your work from others, particularly when your journal prohibits the use of *we*

The following extract is the first paragraph of a Discussion (though something very similar might also be found in an Introduction). It is poorly written because often it is difficult to understand if the verb refers to what the writer of THIS paper (hereafter, Smith) found or to something ANOTHER author (hereafter, Another) did or found in another paper.

Bilingual children *(1) were found* to show a greater adaptability to new situations (e.g. change of school, change of diet) and demonstrated a greater ease in communicating confidently with adults [Blake, 2024]. As result of an extensive search for bilingual children in ten European countries, 149 children *(2) were identified* (Table 1). One hundred and twenty two children with parents of different nationalities *(3) were assigned* to a group (hereafter Group A). It *(4) has been found* that those children with parents of the same nationality but who live in a foreign country (for example, a child with English parents living in Italy) *(5) have* a greater level of adaptability than those children with parents of different nationalities living in the native country of one of the parents.

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1) The use of the past tense (*were found*) seems to indicate that this is Smith's finding. But when I get to the end of the sentence I see the reference, so I now realize that this is Another's finding.
2) Reading the first part of this sentence I am not sure if Smith is adding more information about Blake's findings or if Smith is now going to talk about Smith's own results. When I reach the end I see a reference to a Table, so I now assume that Smith made the identification.
3) There is no ambiguity here. Smith is talking about what he did.

4+5) The change in tense from the past simple (*were assigned* in 3) to the present perfect (*has been found* in 4) followed by the present tense (*have* in 5) suggests that I am reading about Another's findings. But in reality, I suspect that these are Smith's findings.

Below is a clearer version. The changes with respect to the red version are underlined.

Bilingual children *show (1)* a greater adaptability to new situations (e.g. change of school, change of diet) and *demonstrate* a greater ease in communicating confidently with adults [Blake, 2024]. *Blake investigated children from the US and Canada. As mentioned previously, the focus of our study was Europe and* a result of an extensive search for bilingual children in ten European countries, 149 children *were identified (2)* (Table 1). One hundred and twenty two children with parents of different nationalities *were assigned (3)* to a group (hereafter Group A). It *was found (4)* that those children with parents of the same nationality but who lived in a foreign country (for example, a child with English parents living in Italy) *had (5)* a greater level of adaptability …

1) The use of the present tense (*show*) indicates to the reader that this is general knowledge, i.e. this is Blake's finding and not Smith's. An alternative here would be to write *Blake [1995] showed that*… One problem in the red version was that there was no real connection between the first and second sentences, and this added to the confusion about whose work was being discussed. In the blue version a new sentence has been added to explain the connection and to introduce Smith's work. Because of the addition of the extra sentence, it is clear that *were identified* and *were assigned* are Wordsworth's findings.
4) In the previous two sentences, Smith has been talking about what he did, so the reader can assume that *It was found* refers to Smith's work.
5) The use of the PAST SIMPLE (*had*) rather than the PRESENT SIMPLE (*have*) makes it clearer for the reader that these are Smith's findings. The general convention (but not rule) of tense usage in Results and Discussions sections is that you use the PRESENT SIMPLE, PRESENT PERFECT OR PAST SIMPLE to refer to other authors but only the PAST SIMPLE to refer to your work. The PRESENT PERFECT should not be used to refer to work that you have carried out.

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6.7 For journals that allow personal forms, use *we* to distinguish yourself from other authors

The simplest way to make a distinction between your results and those of other authors is to use *we* – provided that your journal allows you to do this. Using *we* would make the text in 6.6 much clearer for the reader.

Blake investigated children from the US and Canada, *whereas we studied* children in Europe. *We conducted* an extensive search for bilingual children in ten European countries and identified 149 children (Table 1). One hundred and twenty two children with parents of different nationalities *were assigned* to a group (hereafter Group A). *We found* that those children with parents of the same…

The above text highlights how making contrasts between what you did and what others did is much simpler when you use *we*. However, you don't want to begin each sentence with *we*, as this would be monotonous for your readers. So you can use a mixture of active (*we found*) and passive (*it was found*).

Only use the passive to describe your work if you have clearly established that now you are talking about your work. You can do this by using *we* or *in our study* at the beginning of a paragraph – this alerts the reader that you are going to discuss your work, so even if you then use the passive the reader still knows that it is your work. If you then introduce someone else's work, make sure that the next time you talk about your own work again you begin the sentence with *we* or *in our study.*

6.8 Make good use of references

Read the red text below. Think about the answers to the questions.

1) Who made the measurements, the author of this paper or the author of another paper?

2) Who carried out the 'similar tests'?

3–6) Is the writer talking about what their research group found, or what others have found before?

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Measurements (1) *were made* of the speed with which bilingual adults performed simultaneous translations of politicians' speeches because politicians tend to use formal language [Anderson and Wordsworth, 2023]. (2) *Similar tests* with Nobel prize winners' acceptance speeches gave similar values of speed. This finding strongly suggests that formal language represents an easier element for translation than informal language. The performance of teenagers (3) *in analogous situations* also confirms the above finding. Considering that informal language, in particular slang, (4) *intensifies* the stress levels of subjects undertaking simultaneous translation (5) *the lack of changes in stress levels* of the bilingual adults with respect to bilingual teenagers when simultaneously translating extracts from a teenage soap opera, would seem to indicate that experience plays an important role. Consequently, stress levels in bilingual subjects *tend* (6) to decrease with age.

It is impossible to answer the questions above. The person who wrote the text undoubtedly knows the answers, but the reader doesn't. Now read the blue text. It is now clear who (the author of the text or another author) carried out the various actions.

In a previous paper [Anderson and Wordsworth, 2008] we made measurements of the speed with which bilingual adults performed simultaneous translations of politicians' speeches. We chose politicians because *it is well known that* they tend to use formal language. *In the same study [Anderson and Wordsworth, 2023] we* conducted similar tests with Nobel prize winners' acceptance speeches, which gave similar values of speed. *These two findings* strongly suggest that formal language represents an easier element for translation than informal language. The performance of teenagers in analogous situations also confirms the above finding [*Williams*, 2019].*Williams* found that informal language, in particular slang, intensifies the stress levels of subjects undertaking simultaneous translation.

Therefore the lack of changes *that we found in our present research* in the stress levels of bilingual adults with respect to bilingual teenagers when simultaneously translating extracts from a teenage soap opera, would seem to indicate that experience plays an important role. *As a consequence of our latest findings, we conclude* that stress levels in bilingual subjects tend to decrease with age.

The blue version clarifies *who* did *what* and *when*. It also divides the original red version into two paragraphs: one describing previous work, and the other describing the current work. The reader is carefully guided through various studies before reaching the conclusions for his present paper.

This results in an increase in the number of words you will need to use – but clarity is more important than conciseness.

I cannot overstress how important it is for you to make such differentiations between your work and that of others. Lack of such a differentiation is one of the most common and serious mistakes made in research papers.

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6.9 Avoid long blocks of text

To be able to read your key findings and to understand the contribution of your paper, readers need to be able to easily find these key findings on the page.

If your key findings are buried in the middle of a paragraph, there is less chance that readers will see them and read them. Readers tend to concentrate at the beginning and ending of paragraphs, rather than the middle.

Read the two versions below. Which is easier to read? Why?

The results showed that tourists in front of important monuments who take selfies using selfie sticks and those who drop litter have an equivalent negative empathy value suggesting that such people should be considered under the category of 'majorly selfish'. Additional observations support our view: i) subjects of the selfie group had a mean lag time of 30.3 seconds between arriving at the monument and the onset of the need to take a photograph of themselves. ii) The mean time of the litter group between arrival and dropping cans and food packages was aligned with the expected response from the selfie group to being given a warning by the monument guards. iii) The MEMEME ego ratio in the selfie group was compatible with a destructive form of graffiti writing, and not significantly different from that found in the can't-see-the-writing-on-the-wall group. iv) No significant differences in the recurrence rate of Kudnt Givadam Syndrome (KGS) were observed between the groups.

*****

The results showed that tourists in front of important monuments who take selfies using selfie sticks and those who drop litter have an equivalent negative empathy value, thus suggesting that such people should be considered under the category of 'majorly selfish'.

Four additional observations support our view.

Firstly, subjects in the selfie group had a mean lag time of 30.3 seconds between arriving at the monument and the onset of the need to take a photograph of themselves.

Secondly, the mean time of the litter group between arrival and dropping cans and food packages was aligned with the expected response from the selfie group to being given a warning by the monument guards.

Thirdly, the MEMEME ego ratio in the selfie group was compatible with a destructive form of graffiti writing, and not significantly different from that found in the can't-see-the-writing-on-the-wall group.

Fourthly, no significant differences in the recurrence rate of Kudnt Givadam Syndrome (KGS) were observed between the groups.

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The blue version clearly takes up more space, but readers are far more likely to notice it and consequently read it:

It is six paragraphs long, making it very easy to follow with each key observation contained in its own dedicated paragraph, which each begin with an ordinal number.

The right-hand margin is not justified, i.e. it does not fill all the white space. This is much easier to read, though may not be part of your journal's style.

6.10 When you have something really important to say, make your sentences shorter than normal

Readers' eyes tend to be attracted most to the white space between sentences and to the capital letter that begins each sentence (try testing this out for yourself). This means that shorter sentences are noticed more, and of course they are generally easier to follow and understand.

This visual factor is critical to the impact of your paper. It is very similar to a good oral presentation. When presenters have something important to say, they slow down the speed of their voice, speak a little louder or more emphatically, use much shorter sentences, and use particular adverbs (e.g. *importantly, interestingly, remarkably*) to attract attention. Presenters do this to (i) attract the audience's attention, (ii) to underline the importance of what they are saying, (iii) to help the audience understand what is being said.

Here is an example from a Discussion. The red version is one long sentence. The italics in the blue version highlight where each new sentence begins.

The method developed in this work relies on a sample pre-treatment that allows a low final dilution, *guaranteeing, on the other hand*, a negligible shift of pH with regard to different specimens to be tested (±0.15 units from 23 samples tested)*; however, the slight shifts* of pH do not alter the response of the test, *as shown* by the overlapping of standard curves obtained by spiking buffers at different pH with IGF-1.

Our method relies on a sample pre-treatment that only requires a minimal level of dilution. *In addition, it guarantees* a negligible shift in pH with regard to the different specimens to be tested (±0.15 units from 23 samples tested). *Importantly*, the slight shifts in pH do not alter the response of the test. *This is revealed* by the overlapping of standard curves obtained by spiking buffers at different pH with IGF-1.

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In the blue version it is much easier for readers to quickly identify where the innovation in the author's method lies, what the results are, and how these results reveal themselves. Note the replacement of *however* with *importantly*. The link word *however* seems to suggest that something negative will follow, whereas the use of *importantly* shows that in fact it is something very positive.

To learn how to break up long sentences see 2.14.

6.11 Other means of attracting the reader's eye and keeping their attention

WRITE CLEARLY AND SIMPLY

Just because you are an academic, it does not mean that you have to write in a complicated way. Be as simple as possible, so that both experts and non-experts will immediately be able to see the value of your work.

Notice how the blue version is much quicker and easier to read because it is immediately understandable.

Results obtained have management implications and suggest that there is a high degree of improbability that sound fruit will be subject to an infection process by Panama disease and wounds have an inherent tendency towards a phenomenon of infection susceptibility with regard to bananas, therefore, necessary steps should be taken to set in place various guarantees so that bananas are handled in an adequately careful manner in order to undertake a strategy of lesion prevention.

Our results highlight that Panama disease is unlikely to infect sound fruit, but rather it is wounds that make fruit susceptible to infection. Thus the best way to avoid infection is by ensuring that the fruit is handled carefully and not wounded. This is clearly critical for those involved in picking, packing, transporting and displaying bananas.

To learn more about how to write clearly see Chapter 8.

REMOVE REDUNDANCY

One of the most effective ways to gain your reader's attention is to remove words that serve no purpose.

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To learn how to remove redundancy see Chapter 4.

UNDERSTAND WHAT KINDS OF WORDS ATTRACT ATTENTION

Nouns and abbreviations like *ANOVA, spectrometry, equation*, i.e. words specific to particular disciplines, are key words and will always attract the reader's attention.

Nouns like *process, characterization, phase* are commonly used in science but do not attract attention and can often be deleted (4.4).

Nouns like *speed, brightness* and *lightness* are concrete words, but are often less effective than their adjectival equivalents

Oriental lacquers have been used since ancient times in East Asia as coatings for every kind of surfaces, because of their *brightness, toughness and durability*.

Oriental lacquers are *bright, tough and durable*. They have thus been used since ancient times in East Asia as coatings for all kinds of surfaces.

The blue version also highlights how changing the structure of a sentence can lead to a more dramatic impact.

Readers are more interested in reading specifics than general concepts (5.4). Particularly when you give your key findings, you need to use the most concrete and specific words and phrases possible. If you don't, you are in danger of losing the attention of the reader.

PLACE TABLES AND FIGURES IN STRATEGIC POSITIONS

The readers' eyes will inevitably be attracted to any non-textual information, such as graphs and tables. Placing tables and figures strategically throughout the paper is thus another visual way of attracting attention. The next thing their eyes will focus on will probably be the legend to the figures, and then the paragraph immediately following the legend. So use this paragraph to make an important point.

Some readers may begin your paper by looking at the figures and tables plus their legends. Thus both the figures/tables and the legends must be immediately understandable.

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6.12 Show your paper to a non-expert and get them to underline your key findings

It is imperative that you check through every sentence in which you report a finding, and make it 100% clear to the reader who is responsible for the finding.

The best way to do this to get another person outside of your field of research to tell you who (i.e. you or another author) is the subject of each verb.

6.13 Summary

- One of the biggest causes of confusion for readers is not being clear about which findings are yours and which are another author's, and also not really understanding the importance of your findings.
- Follow the journal's instructions regarding whether you can use *we,* as this will clearly differentiate your findings from that of other authors.
- You may have the impression that the passive form is considered to be more elegant in scientific papers. Whether this impression is true or not, be aware that the passive inevitably creates problems for your readers because it may be difficult for them to know immediately and with certainty whether you or another author made a particular finding.
- Do not rely on a reference to a figure or a table, or a reference to the bibliography to distinguish your new data from those in the literature. Make sure the reference clearly indicates it is another author's work and not a previous paper by you.
- Be aware that if you make mistakes in the usage of tenses when you are comparing your work with other authors' work, you could really confuse your readers. Make sure you consistently use the correct tenses and remember that in English there is a real difference between the simple past (finished actions with time indication) and the present perfect (past to present actions, finished actions with no time indication)
- Help readers to distinguish between your work and others by using a series of short paragraphs, rather than one long paragraph.
- If you mention another author's paper, make sure that the reader understands why you are mentioning that paper and how it relates to your own work.

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- Be aware of how the layout of your paper can affect where readers focus their eyes – break up long blocks of text using shorter paragraphs, headings, bullets, and figures / tables etc
- Begin a new paragraph when highlighting something important
- Use shorter sentences and paragraphs to make your key points

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Chapter 7

Discussing your limitations



Mistaken predictions

- Cars: *That the automobile has practically reached the limit of its development is suggested by the fact that during the past year no improvements of a radical nature have been introduced.* Scientific American, January 1909
- Computers: *Where a calculator on the ENIAC is equipped with 19,000 vacuum tubes and weighs 30 tons, computers in the future may have only 1,000 vacuum tubes and perhaps only weigh 1.5 tons.* Popular Mechanics, March 1947.
- Electric light: *When the Paris exhibition closes, electric light will close with it and no more will be heard of it.* Erasmus Wilson, English surgeon and dermatologist, 1878
- Intelligence: *The most important fact about intelligence is that we can measure it.* Arthur Jensen, US psychology professor, 1969
- Inventions: *Everything that can be invented has been invented*. Charles H. Duell, Commissioner of the US Patent Office, 1899
- Nicotine: *It is my conviction that nicotine is a very remarkable, beneficient drug that both helps the body to resist external stress and also can as a result show a pronounced tranquilizing effect.* Charles Ellis, Senior Scientist, British American Tobacco company, 1962
- Nuclear Energy: *There is not the slightest indication that [nuclear energy] will ever be obtainable. It would mean that the atom would have to be shattered at will*. Albert Einstein, 1932.
- Planes: *Heavier-than-air flying machines are impossible*. William Thomson (Lord Kelvin), British mathematician, physicist and engineer, 1895
- Surgery: *The abolishment of pain in surgery is a chimera. It is absurd to go on seeking it. … Knife and pain are two words in surgery that must forever be associated in the consciousness of the patient*. Dr. Alfred Velpeau, French surgeon, 1839

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7.1 What are my limitations? Should I mention them?

Every researcher has limits:

Even the most impressive minds are not flawless, they merely pave the way for the next level of understanding. Mario Livio (author of 'Brilliant Blunders' on scientists' breakthrough mistakes).

You research is likely to be affected by difficulties regarding one or more of the following: old data, poor weather, insufficient funds, too small sample sizes, no collaborations due to pandemics, no access to certain tools, substances, equipment, inability to conduct in vivo tests, same tests but different results each time, inability to get authorization to do the tests, literature and studies in another language.

Once we accept our limits, we go beyond them. Albert Einstein

But out of limitations, sometimes there are breakthroughs. For instance, Post-it notes were created Spencer Silver, a researcher in 3M Laboratories while trying to make a strong adhesive. He actually inadvertently created something that was weaker than the adhesives available at the time. Ink-jet printers were invented by an engineer at Canon who mistakenly put a hot iron on a pen, and notices that ink was injected from the pen a few moments later. Alexander Fleming noticed that mold on a contaminated Petri was dissolving all the bacteria around it. He then grew the mold by itself and discovered penicillin. The inventors of the following products initially had their ideas rejected by potential manufacturers: Barbie doll, the hovercraft, the board games Monopoly and Trivial Pursuit, the safety razor, the vacuum cleaner. And even octogenarians can produce quality work. At the age of 87, Francis Rous was awarded the Nobel Prize for Medicine, and at 88 Michelangelo was still painting.

The British Medical Journal (BMJ) has an extremely useful author resources section on its website. These resources are not only useful for those undertaking medical research, but can be applied to any kind of research writing. I strongly recommend accessing their site: http://www.bmj.com/about-bmj/resources-authors/article-types/research

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The following extract is from the BMJ site:

Please do not use the term "negative" to describe studies that have not found statistically significant differences, perhaps because they were too small. There will always be some uncertainty, and we hope you will be as explicit as possible in reporting what you have found in your study. Using wording such as "our results are compatible with a decrease of this much or an increase of this much" or "this study found no effect" is more accurate and helpful to readers than "there was no effect/no difference".

7.2 Recognize the importance of 'bad data'

To enable referees to judge whether you have made a mistake or not, you should not hide any negative results. Be upfront (clear and honest) about the limitations of your methods and approach.

In *Why People Believe Weird Things*, author Professor Michael Shermer writes:

In science, the value of negative findings – failures – cannot be overemphasized. Usually they are not wanted, and often they are not published. But most of the time failures are how we get close to truth. Honest scientists will readily admit their errors, but all scientists are kept in line by the fact that their fellow scientist will publicize any attempt to fudge. Not pseudo scientists. They ignore or rationalize failures, especially when exposed.

Dr. Donald Dearborn, of Bates College, comments:

Your results may be of importance to others even though they did not support your hypothesis. Do not fall into the trap of thinking that results contrary to what you expected are necessarily "bad data". If you carried out the work well, they are simply your results and need interpretation. Many important discoveries can be traced to "bad data".

And finally, Linus Pauling, winner of two Nobel prizes and some of whose findings were later found to be majorly flawed by other scientists, is reported to have said:

Mistakes do no harm in science because there are lots of smart people out there who will immediately spot a mistake and correct it. You can only make a fool of yourself and that does no harm, except to your pride.

Negative data are frequently commented on in the Results (17.5) and particularly in the Discussion (18.10).

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7.3 How to avoid losing credibility

Dr Maggie Charles is a Tutor in English for Academic Purposes at the Oxford University Language Centre. She explains the importance of admitting limitations, but doing so in a way that does not undermine your credibility:

As a young researcher you want your scientific community to see you as credible, professional and honest, and also reasonably modest. This means that you can, and should, draw attention to limitations in your research. The community needs to know what went wrong in your research, not just for ethical reasons, but also so that others can learn from your 'mistakes'. It also means that others will see you as a reliable and honest researcher. In fact, because you have drawn attention to the problems you have had in your research, the community is more likely to accept the validity and reliability of what you describe in your paper.

However, you can present these limitations in such a way that you do not have to take direct responsibility for them. You can do this by using impersonal forms. These impersonal forms distance you from the limitations of your study and at the same time they highlight for the community that you can evaluate your 'performance' in accordance with the standards of that community.

The passive form is very useful when you don't want to assume complete responsibility for what you are saying. This is because no agent is necessary with a passive.

It was found that the containers for the samples had become contaminated.

This fraction *is assumed* to originate from…

It *might be speculated* that…

Impersonal phrases beginning with *it* have the same function:

It is regrettable that the containers had become contaminated as this meant that …

It is reasonable to hypothesize that…

It appears possible that…

These tactics give the reader the impression that the responsibility for the contamination does not rest entirely with the author. The author does not explicitly state who is doing the assuming, speculating, hypothesizing etc. This means that you can avoid losing face and so not be perceived as being incompetent.

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7.4 Be constructive in how you present your limitations

When you discuss any limitations and failures, try to do so in a constructive way so that other researchers can learn from your experiences.

However, you don't want to present your limitations in a negative light. Your results may be 'negative' for you, but for the scientific community they are not negative, rather they are helpful indicators essential for the progress of knowledge (7.2).

This means that although the results themselves may have been unexpected or appear disappointing to you, the actual way you present them should not be formulated in negative language as this might produce a negative reaction in your readers. The idea is to report everything in a neutral subjective way.

The limitation of this paper is that the two surveys were unfortunately not conducted in the same period. This will affect our results in terms of …

The above sentence is extremely honest, but it could be expressed in a way that sounds less negative:

Although the two surveys were not conducted in the same period, this will only affect our results in terms of …

The blue version reduces the negative impact of the red version by:

- removing the words *limitation* and *unfortunately*. In reality, *limitation* is not a bad word to use, but if you use it more than once or twice, the reader may go away thinking that your work has more negative aspects than positive ones. If you have to refer to several limitations, another solution to reduce the possible negative effect on the reader is to use synonyms: *shortfall, shortcoming, pitfall, drawback, disadvantage* etc.
- introducing *although* and *only* – these adverbs qualify what you are saying. In this particular case, *although* immediately tells your reader that you are going to say something negative, but that something positive will immediately follow. *Only* implies a limited number of cases, thus it lessens the level of seriousness of the shortcoming
- combining two sentences into one sentence – this gives the reader less time to ponder on the negative content

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Other words to avoid are adverbs such as *regrettably* and *unfortunately*, and the link word *moreover*. *Moreover* tends to be used when you have said something negative, and then add further negative details. On the other hand, *in addition* (*further, furthermore, also* etc) are used to add to any already positive or neutral comment. So I could say to my students:

You are the *worst* class I have ever had. *Moreover*, you appear to understand absolutely *nothing*.

You are the *best* class I have ever had. *In addition*, you appear to understand absolutely *everything*.

7.5 Clarify exactly what your limitations are

When you outline the limitations, you need to be clear what these limitations are and what exactly the implications are. The red sentences below fail to do this. The first does not explain why and in what way the sample size was too small, nor what the consequences of this were. The second does not explain why or how the samples were contaminated, nor to what extent the conclusions are misleading.

One limitation of our research was the sample size, which was too small.

The unfortunate contamination of a few of our samples may mean that some of our conclusions are somewhat misleading.

The limitations would be much better described as follows:

One limitation of our research was the sample size. Clearly 200 Xs are not enough to make generalizations about Y. However, from the results of those limited number of Xs, a clear pattern emerged which …

Two of our samples were contaminated. This occurred because … We thus plan to repeat our experiments in future work. However, our analysis of the uncontaminated samples (24 in total) supported our initial hypothesis that …

The important thing is to be (i) honest, (ii) clear, and, if appropriate, (iii) discuss possible remedies.

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7.6 Anticipate alternative interpretations of your data

If you want the referee and readers to accept your specific interpretation of your data, you will be more convincing if you also provide alternative interpretations. Basically you are anticipating any objections that they might have – you are playing the devil's advocate with yourself.

Let's imagine that you have stated that 'Our findings show that dogs are more intelligent than cats'. Below are some ways to hedge your claim (i.e. to protect yourself form possible criticism) by setting out an alternative interpretation.

Of course, the *opposite* may also be possible. *In fact, it cannot be ruled out* that certain species of cats, for example, Siamese, show intelligence traits that are remarkably similar to those of dogs.

Other factors besides intelligence *could be involved*, such as the visual and olfactory senses. *This implies that*, in a restricted number of cases, cats could be considered as being more intelligent…

We do not know the exact reasons for the discrepancy between our findings and those of Santac [2025], but *it might reflect*… Feeding habits may favor intelligence, *or* they may simply be…, *or* they may result from… Future work will be devoted to investigating these three *alternative possibilities*.

Despite this apparently clear evidence of the superiority of dogs, our findings are in contrast with those of Karaja [2023] and Thanhbinh [2024], whose experiments with Singapura and Sokoke cats apparently showed that both these species were superior to Rottweilers in terms of emotional intelligence. *However*, we believe that the species of cats involved are quite rare, and that Rottweilers were not a good choice of comparison.

7.7 Refer to other authors who experienced similar problems

Another way to lessen the impact of the limitations of your 'bad data' is to say that other authors have experienced similar problems, as illustrated in the extract below:

Analytic expressions for the density (1) were not derived, (2) because their interaction depends on the relative orientation of the spheres, (3) thus making integration considerably more complex. (4) Similar complications in the analytical determination of the density, using the same approach that we used, were experienced by Burgess [2018].

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The strategy used in the above extract is: 1) explain the pitfall (i.e. the limitation in your work); 2) give a reason for the pitfall; 3) outline the consequence of the pitfall; 4) refer to a similar pitfall experienced by another author

However, be careful how you refer to the literature.

The statistical tool is not able to describe all the variables involved. The same tool was used for conducting a similar research with an American sample, and the results were reliable and representative.

In the red sentence the reference to the literature is very vague and is thus not convincing.

This vagueness is resolved by being much more precise:

The statistical tool may not be optimal for describing some of the variables involved. However it is optimal for x, y and z. In addition, exactly the same tool was used for conducting a similar research with an American sample [Williams, 2025]. Williams' results were reliable and representative and were in fact used by the US government.

7.8 Tell the reader that with the current state-of-the-art this problem is not solvable

Your limitations may be due to the fact that current knowledge (theories, models, technologies etc.) is unable to resolve the problems you have encountered.

(1) A full treatment of our problem using Gabbertas's theory (GT) is complicated to handle in our case, (2) *given* the complex geometry. (3) *In fact*, the expressions derived by GT are only available for a few simple geometries [Refs]. (4) *Moreover*, GT is not well suited to describing the upper regions. (5) *An additional problem* is that a theoretical description of X is still the target of active experimental and theoretical research. (6) There is little experimental or theoretical information available for the properties of X [Refs]. (7) *At the same time*, the properties of Y can be described by Burgess's model, (8) *however* its ability to account for X is still under investigation.

The strategy adopted in the above case is: 1) say that current theories (models etc.) cannot deal with your problem, i.e. this is your limitation; 2) explain why they can't deal with your problem; 3+4) support what you have just said.

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Note how (5–8) follow the same pattern as (1–4). The author uses link words (highlighted in italics) to give emphasis and logic to her argumentation and she provides variety by using different link words.

When discussing your limitations, be consistent. Say either *this worked in 75% of cases* (affirmative approach) or *this did not work in 25%* (negative approach), then stick with just one of the two approaches. Otherwise you are in danger of confusing the reader.

Finally, only attribute your limitations to a current lack of knowledge if this really is the case – don't just use it as an excuse!

7.9 Explain why you did not study certain data

Other limitations relating to data are that i) you did not study the most recent data, ii) you did not study a sufficient amount of data. You can deal with these two limitations by writing in your Discussion or Conclusions:

Even though the data were collected two years ago, the stability of this sector means that such data have not changed significantly. In fact, in the last two years the percentage of x has remained exactly the same [Wang 2017, Chu Wa 2018]. In addition, more recent data are not currently available.

Our data only refer to one kind of sector. However, as far as we know there are no similar studies for this sector in South Korea. Thus we believe that this project opens the way for …

In first example above, you justify old data by saying that nothing has changed since the time that data was collected. In the second example, you say that in the country that is the subject of your research (in this case South Korea), such data are not available, and you protect yourself by saying as far as we know.

7.10 Don't end your paper by talking about your limitations

Don't end your Discussion (18.5, 18.6) or Conclusions (19.9) by talking about your limitations. End with something positive – this will be the reader's final impression. Possible endings are:

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- talking about other applications for your findings
- suggesting other avenues of research for the future
- re-highlighting the benefits of your findings

7.11 Summary

- Talk about your weaknesses not just your strengths; do not make the referees suspect any bias in your work
- **Recognize the importance of 'bad data'**
- Always mention your limitations
- Present your limitations in a constructive way using positive language
- Justify your limitations
- Ensure the final words of your paper are not about your limitations

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Chapter 8

Readability



Best writing practices

The KISS principle (*Keep It Simple, Stupid*) was invented during the Second World War by an aeronautical engineer at Lockheed Martin, and was a design principle used by the US navy. In terms of writing, KISS is also used to mean *Keep It Short and Simple* – in fact, no studies on academic writing have proved that long sentences and complex writing are an effective form of communication.

*****

The results of two research projects - the Poynter Institute's Eyetrack survey, and an analysis by Jakob Nielsen (a Danish web usability expert) - show that only half of readers who begin an article, will actually finish it, and if the article is read online, only a fifth of readers will finish it.

*****

A survey carried out at Stanford University revealed that 86.4% of students admitted that in order to appear more intelligent they used complex language in their essays, theses and dissertations.

*****

You will lose more readers in the first 50 words than you will in the next 250.

*****

The Viennese art historian, Ernst Gombrich wrote many of his books in English rather than in his native German. His *Story of Art*, first published in 1950, is one of the most widely accessible art history books ever published, precisely because it is written in a clear, simple, unpretentious style.

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8.1 You are responsible for enabling your readers to understand what you have written

Nobel Prize Winner in Physics, Tony Leggett, notes that:

> In Japanese it seems that it is often legitimate to state a number of thoughts in such a way that the connection between them, or the meaning of any given one, only becomes clear when one has read the whole paragraph or even the whole paper.

Tracy Seeley, an English professor at the University of San Francisco, noted that after a conversation with some of her students she discovered that:

> Most can't concentrate on reading a text for more than 30 seconds or a minute at a time. We're being trained away from slow reading by new technology.

Most of your readers will not be reading your paper with a high level of concentration. This means that your meaning must be clear from the first word of a sentence and remain clear throughout the sentence.

Write from a reader's perspective. Make it easy for them so they can use their brainpower not to decipher your meaning, but to ponder and reflect on your meaning.

This is not the case in what is considered good written English, where the meaning should become clear very quickly.

You may come from a receiver-oriented culture where it is the responsibility of the listener or reader to understand what is being said or written, rather than the speaker or writer. Anglo cultures have also traditionally been receiver-oriented, but have shifted in recent years to place more emphasis on the speaker or writer's responsibility to ensure their audience understands.

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8.2 Basic rules of readability

Below are some of the key guidelines to good writing. These guidelines are detailed in the other chapters of Part I of this book.

- Use the minimum number of words possible, but always prioritize clarity over conciseness. See Chapter 4.
- Follow this word order – subject, verb, object. See Chapter 2.
- Learn how to write clear sentences and paragraphs. See Chapter 3.
- Only use long sentences if readers will have no difficulty understanding them. Otherwise split a long sentence into a series of shorter ones. See 2.14.
- Avoid ambiguity. See Chapter 5.
- Ensure that your readers are clear about: i) what you did as opposed to what another research group did, ii) what your important findings are and what they mean. See Chapter 6.
- Think about the visual aspect of your paper. Readers do not find blocks of text attractive. You need to break up long sections and paragraphs. Use headings, figures and tables to attract the reader's eyes. See 3.7.

8.3 Place the various elements in your sentence in the most logical order possible: don't force the reader to have to change their perspective

When we scan results from a search engine, our eye rapidly goes vertically down the left hand side of the page, before starting again to read horizontally. This means that you need to think carefully about what grammatical subject to place at the beginning of the first sentence that begins a new paragraph, otherwise there is a chance that browsers and readers won't spot the key information that you want to give them.

Readers expect words/phrases that are closely related to each other, to appear next to each other within the sentence.

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Several authors have evaluated the possibility to minimize the levels of background compounds, both those released from the bag material and those from the previous sample collection *using a cleaning procedure*.

Several authors have evaluated the possibility *of using a cleaning procedure* to minimize the levels of background compounds, both those released from the bag material and those from the previous sample collection.

Overall the match between the aggressiveness of season-based inoculations and the capacity of the fungus to be active in vitro as a function of the temperature, *appears close*.

Overall *there seems to be a close* match between the aggressiveness of season- based inoculations and the capacity of the fungus to be active in vitro as a function of the temperature.

In the red versions, the information in italics is key to the readers' overall understanding and should be placed earlier in the sentence closer to the elements it refers to.

8.4 Don't force readers to hold a lot of preliminary information in their head before giving them the main information

The sentence below imposes a lot of effort on the reader.

Considering that peach skin is particularly rich in antioxidants (Figs. 1A, 1B, 2A, 2B), positively reacts to UV-B radiation at the end of postharvest by increasing antioxidant activity (Fig. 3A), and, differently from flesh, is directly exposed to UV-B radiation under natural conditions, the study of free radical generation was performed specifically on this tissue.

The blue version below resolves the problem by splitting the sentence into three. This means that the reader can absorb the information in manageable chunks, i.e. you feed the reader small pieces of information at a time. This enables readers to progress forwards without having to re-read anything.

Peach skin is particularly rich in antioxidants (Figs. 1A, 1B, 2A, 2B) and reacts positively to UV-B radiation at the end of postharvest by increasing antioxidant activity (Fig. 3A). Unlike the flesh, the skin is directly exposed to UV-B radiation under natural conditions. Consequently, the study of free radical generation was performed on peach skin.

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Words such as *since* and *although* are often used in a subordinate clause at the beginning of a sentence, as in red sentence below. Readers are thus forced to carry an idea in their head before they understand how this idea relates to the idea in the main clause (in italics).

Since English is now spoken by 1.35 billion people around the world and is used as a lingua franca in many international business and tourism scenarios between people of different languages and between native English speakers and non-native speakers, *the learning of foreign languages in the United Kingdom has suffered a huge decline*.

The solution is to split the sentence in two.

English is now spoken by 1.35 billion people around the world and is used as a lingua franca in many international business and tourism scenarios between people of different languages and between native English speakers and non-native speakers. The consequence is that the learning of foreign languages in the United Kingdom has suffered a huge decline.

For more on splitting up sentences see 2.14.

8.5 Try to be as concrete as possible as soon as possible

Compare these two texts:

Smart devices may have to manage sensitive information that, often, must be protected against unauthorized diffusion or from malicious attacks. Some notable examples of sensitive information are data concerning the health conditions of a patient or data gathered from caregivers about the status an elderly person.

Smart devices may have to manage sensitive information, for example the health conditions of a patient or data gathered from caregivers on the status an elderly person. Clearly, such data must be protected against unauthorized diffusion or from malicious attacks.

In the red version, readers have to wait to understand exactly what sensitive information is and why it has to be protected.

The blue version tells readers immediately what sensitive information is, and therefore enables readers to understand why it should be protected. The blue versions is also uses fewer words.

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I am not suggesting that you should always follow the order of information given in the blue version: just be aware that the same information can be presented in a different order. Your aim is to choose the most effective order.

In addition, avoid words and phrases that add no value for the reader. Typically:

- words that are not concrete, particularly abstract nouns
- expressions that are vague

Adjectives, too, can create problems - *interesting, challenging, vital, fundamental, innovative, cutting-edge*. What exactly do these words means? Can you be sure that the referee will understand why something is *interesting* and thus agree that is *interesting*? Instead, clearly demonstrate why and how something is interesting or challenging.

If you read on a CV that a candidate describes herself as having 'excellent communication skills', do you believe her even if she provides no evidence in her CV of such excellence? No, you don't. You simply think to yourself "everyone says they have good communication skills, it means nothing". The same is true of any claims you make in your paper - don't simply tell your readers that something is "fundamental", show them how it is fundamental, convince them.

8.6 When drawing the reader's attention to something use the least number of words possible

Occasionally, you may want to draw the reader's attention to an important point. You will do this more effectively if you use two words rather than ten. This will produce a short sentence. Short sentences tend to stand out from the rest of the text, and thus get noted more.

All the phrases below could be replaced by *Note that …*

It must be emphasized / stressed / noted / remarked / underlined …; It is interesting to observe that … ; It is worthwhile bearing in mind / noting / mentioning that … ; It is important to recall that … ; As the reader will no doubt be aware … ; We have to point out that …

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8.7 State your aim before giving the reasons for it

When you explain a new game to someone, do you tell them the rules/strategies and then the objective, or vice versa? Which sounds more logical to you the red or the blue version?

You need to develop a strategy, make decisions as to whether to collaborate or not with the other players, also keep an eye on the progress of the other players, and finally make the most money *in order to win the game*.

In order to win the game you need to make the most money. To do this, you need to develop ...

Game players and readers have the same expectations: they want to know the aim of the game before learning how to carry it out - i) aim ii) means (i.e. how).

In the red version above, you are forcing the reader to wait for the key information, which only appears 38 words into the sentence. In the blue version, the aim is immediately established.

However, if the sentence is short, it does not make too much difference which element (aim or means) you put first:

In order to win the game you need to make the most money.

You need to make the most money in order to win the game.

8.8 Be as specific as possible

Generic statements are often redundant.

The last decade has been characterized by advances in device manufacturing. Devices are becoming increasingly miniaturized, powerful, cheap and have become part of our daily lives. Notable examples include smart phones and smart watches equipped with a plethora of sensors, home appliances and general purpose devices such as tablets and ultra-thin notebooks. We are surrounded by all these devices daily in a pervasive way, at home, work and also in public spaces - as anticipated in Mark Weiser's visionary observation: "The most profound technologies are those that disappear."

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Introductory sentences (in *italics* in the example above) at the beginning a new section or paragraph alert the reader about what is coming next ... but they need to be eye-catching.

The first red sentence contains very obvious information that adds not value to the reader. The second sentence is slightly more specific, but still does not attract attention.

Remember that your readers will not be reading every word, sentence or paragraph of your paper. Their eyes will be skimming down the page (rather than slowly reading across the page horizontally). Their eyes are waiting to rest on something that attracts their attention.

So a better start would be:

In 1991 Mark Weiser observed that "The most profound technologies are those that disappear." In fact, increasingly miniaturized, powerful, and cheap devices have become part of our daily lives, for example smart phones and smart watches, home appliances, tablets and ultra-thin notebooks. We are surrounded by all these devices: at home, work and also in public spaces.

Note how the blue version attracts more attention by

- including a date and a name at the beginning of the sentence. Numbers and names (with their initial capital letters) stand out in a text
- removing all the generic phrases that add no real value: *notable examples include ... equipped with a plethora of sensors ... general purpose devices such as ... in a pervasive way*
- concluding with a sentence that recalls the introductory sentence, thus giving the paragraph a sense of cohesiveness

Clearly, I have been quite extreme in the number of generic phrases I have removed, but this is to show you how often we tend to write too many phrases that give no new information and simply fill the text unnecessarily and thus make the reader waste time.

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8.9 Avoid creating strings of nouns that describe other nouns

A series of nouns in a sentence makes your writing very heavy.

An innovative engineering PhD student scientific English and communication course didactic methodology

If possible, replace some nouns with verbs and add some prepositions.

A didactic methodology *for teaching* a scientific English and communication course *for* PhD students *in* engineering

You cannot indiscriminately put nouns in front of each other in a string. For example, you cannot say *art state technology* (state-of-the-art technology).

Native speakers do tend to string nouns together, but they intuitively know how to do it. In fact, you can say *a software program* or *an aluminum tube, a breakout room, a digital covid certificate,* and *social distancing rules.*

The inventors of these terms were not following any written rules, but they were basing themselves on examples that already exist.

If you are a non-native speaker I strongly recommend that you verify on Google Scholar that your proposed string of nouns already exists and has been used by native English-speaking authors.

If your string of nouns does not exist, it will sound very strange to any native English-speaking referees, and more than one occurrence of such structures could cause the referee to recommend that your English be revised.

If it has not been used by native English-speaking authors, then you need to change the order of the words, which normally entails inserting some prepositions.

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8.10 Be careful how you use personal pronouns and avoid stereotyping

Only use *he* and *she* (and *his, him, her, hers*) when they are specifically used in relation to a male or female subject, respectively.

In 2023, US President Biden claimed in **his** speech that **he** ... whereas in **her** speech the Italian premier, Giorgia Meloni, reaffirmed that **she** ...

The patient (**female**, 46 years old) had lost **her** arm in an accident.

Consider rephrasing sentences which contain *he/she* or *him/her*. Firstly, constantly writing *he/she* or *him/her* in a text can become quite heavy for the reader. Secondly, not everyone associates themself with one gender. Instead, use *they, them, their*.

When **a user** has connection problems, the system tries to reconnect **him/her** automatically. This means that **he/she** may have problems with **his/her** modem, and consequently **he/she** may need to ... In such case **he/she** ...

When **a user has / users have** connection problems, the system tries to reconnect **them** automatically.

When **users have** connection problems, the system tries to reconnect **them** automatically.

When a child is born, **he/she** is often a great source of joy for the parents.

When a child is born, **they** are usually a great source of joy for the parents.

Be aware that machine translators and editing tools tend to be male-focused in terms of their translations with reference to people, e.g. doctors will tend to be associated with the pronoun *he*, and nurses with *she*, professors with *he*, and teachers with *she*.

It was recommended that the patient should visit a doctor and inform him of the illness.

In such cases, use *they/them/their* or simply repeat the key word.

It was recommended that the patient should visit a doctor and inform them of the illness.

It was recommended that the patient should visit *their* doctor and inform the doctor of the illness.

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In some cases you need to be careful to ensure that readers can understand what *they* and *them* refer to.

In our survey of 25,000 patients, most interviewees said their doctor only spent a few minutes with them and that *he* rarely took any notes.

... only spent a few minutes with *them* and that *they* rarely took any notes.

... said their doctor only spent a few minutes with *them* and that *the doctor* rarely took any notes.

In the first red sentence, *he* is wrong given that doctors can be of any gender.

In the second red sentence, *they* is confusing because it could refer to the patients or their doctor.

The blue sentence resolves the issue by repeating the word *doctor*. Never be worried about repeating key words (5.4, 5.5).

When referring to people and their country, make sure you know which country they are from. Do not make generalizations based on color, race, or so-called *level of development*. It will help if you avoid:

- expressing any stereotypical opinions
- bracketing people together who in reality have no or few common traits
- making generalizations about people, groups, countries, religions
- showing stereotypical photographs or harmful depictions of marginalized communities.

8.11 Summary

- Most of your readers will not be reading your paper with a high level of concentration. This means that your meaning must be clear from the first word of a sentence and remain clear throughout the sentence.
- Write from a reader's perspective. Make it easy for them so they can use their brainpower not to decipher your meaning, but to ponder and reflect on your meaning.
- Re-read your paper to check that it fulfills the basic rules of readability (8.2).

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- Place the various elements in your sentence in the most logical order possible: don't force the reader to have to change their perspective.
- Don't force readers to hold a lot of preliminary information in their head before giving them the main information.
- Be as concrete and specific as possible, using the least number of words possible.
- Avoid creating strings of nouns that describe other nouns.
- Never use *he* as a generic pronoun. Try to use plural forms of nouns so that you can use *they/them/their* to refer back to these nouns.
- Avoid stereotypes and generalizations when referring to other parts of the world.

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Chapter 9

Chatbots



From the press in the three months following ChatGPT's release in 2022

ChatGPT ... was launched by the team at AI development company OpenAI on November 30; by December 5, more than one million users had tested it out.

*****

Meet ChatGPT, the scarily intelligent robot who can do your job better than you: OpenAI's chatbot proves the gap between computers and humans is rapidly narrowing

*****

The College Essay Is Dead: Nobody is prepared for how AI will transform academia.

*****

It can write songs! But it can't remember its own name.

*****

Essays produced by a generative-based AI app can have various falsehoods embedded, including manifestly untrue facts, facts that are misleadingly portrayed, and apparent facts that are entirely fabricated. Those fabricated aspects are often referred to as a form of AI *hallucinations*.

*****

But no matter how hard it tries, ChatGPT will never be human.

*****

AI tools present an opportunity to teach and learn in new ways.

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_9

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Preface to this chapter

The initial controversy about ChatGPT, which you can see in the quotations on the previous page, focused on those aspects that affected native English speakers: the biggest ones being the fact that the bot doesn't always return reliable information and that it can be used by college students to generate essays without the students having to study at all.

However, no one seemed to focus on its massive benefits for the non-native English speaking community..

> In my opinion, ChatGPT and Google Translate, and other AI applications, mean that essentially anyone in the academic world with access to the internet can now write a paper in reasonably accurate and generally appropriate English without having to spend part of their research budget on getting their work translated into English. Obviously, it is NOT perfect, and is not a substitute for a professional editor.
>
> To achieve the accuracy and appropriacy mentioned above, you must apply the guidelines suggested in this chapter.
>
> Using AI applications without any pre-editing or post-editing will produce imperfect texts.

Of course editing services will still be required, as AI does have limits in terms of its ability to edit a scientific paper (9.6, 9.7).

What I have tried to do in this chapter and the next, is to show you ways of being more autonomous when writing papers.

This chapter deals with using AI throughout your paper, but without looking at how it could be used in specific sections.

To learn how to use bots to correct your English in specific sections, consult the following: Titles (11.17), Abstracts (12.13, 13.1, 13.9), Introduction (14.10), Literature Review (15.11), Methods (16.2), Results (17.9), Discussion (18.12, 18.13), Conclusions (19.3, 19.10). For full details on how to use AI to help you write a paper, see the companion volume *AI-Assisted Writing and Presenting in English* (2024). Writing and grammar exercises related to using AI can be found in the 2024 editions of the exercise books. For a regularly updated list of prompts to use with ChatGPT: https://tefldiscussions.com/ai-prompts/.

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This chapter is about using bots to improve your English, NOT to write a paper for you. There are ethical issues involved.

9.1 What are the key things I need to know about chatbots?

There are many AI tools available. Key things to know are:

- As with most new AI technologies, they were NOT specifically designed for non-native speakers. Many of the writing functions available are for business (e.g. marketing) not for academia. In any case, these bots can be used with very good results to correct the English of your academic texts.
- These tools are NOT totally 100% accurate. However, their English is probably more accurate than yours (at least according to my tests with PhD students of various nationalities).
- The more precise your instructions, the better the output.
- The 'regenerate' button is very useful as it enables you to see (and then compare by yourself) different versions of a text.

The vast majority of the examples given in this chapter were derived from using ChatGPT (https://openai.com/blog/chatgpt/). Although the bots will be constantly upgraded, I believe that most of the guidelines in this chapter will still be valid in the future.

9.2 What prompts (instructions) can I use to get a chatbot to fulfill my request?

The more precise your instructions to the bot, the greater the chance of an effective response. For a regularly updated list of prompts to use with ChatGPT: https://tefldiscussions.com/ai-prompts/. Below are some possible prompts, varying from short to long.

Improve this ...

Add relevant details to this ...

What could I delete in this ... ?

Is this a good abstract, title etc?

What is your opinion of the style of this ...?

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Where appropriate, convert the active verbs into the passive form in the following abstract from a scientific paper:

Please write a better version of this abstract from a scientific paper. The keywords that define the article are: [insert keywords] Do not delete the sentences in which the keywords are present, but rework them. The abstract is: [insert abstract]

You can ask very specific questions:

What is the difference between the two phrases? "I could finish the manuscript on time" AND ""I was able to finish the manuscript on time"

You can request to see the changes that the bot has made – see 9.3.

Improve the English of this abstract and list the changes you have made.

9.3 How can I see the changes that the bot has made? What can I learn from the list of changes generated?

Some bots, such as QuillBot, will show your original text and their modified text in adjacent windows. They also highlight in different colors the modifications they have made. This is a very useful feature enabling you to:

- check that the bot hasn't changed anything you didn't want changing
- learn different ways of saying the same thing (QuillBot has an excellent paraphrasing feature)
- see how your text can be reduced
- see how your text has been corrected and thus improve your English

At the time of writing, ChatGPT does not have the adjacent window feature. However, when prompted (*list the changes you have made*), ChatGPT will list the changes it has made.

You can use the list of 'Changes Made' to make a quick check that Chat hasn't made any changes that are inappropriate.

Warning: Chat does NOT necessarily list ALL the changes it made. The ones it lists seem entirely random. Pressing 'regenerate' may add more of the changes made, but again, not all of them.

However, it is worth studying the explanations (in SMALL CAPS below) for the changes that the bot makes.

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REVISED SENTENCE STRUCTURE AND PHRASING FOR CLARITY AND CONCISENESS

This means that the bot may have changed:

- the order of the phrases/clauses within a sentence, in order to shift the key info to nearer the beginning
- deleted some words that it thought added no value (e.g. it changed *in the case of the final use of the park being a green space* to *if the park is eventually used as a green space*)
- a phrase to make it sound more dynamic (e.g. it changed *It should also be noted that* ... to *These results show*)
- a phrase to make it more specific (e.g. it changed *In the case under study* ... to *Despite the decrease in bioavailable cadmium*)

CHANGED INDIVIDUAL WORDS

When making vocabulary changes, the bot does not give an explanation. The reasons could be:

- to avoid repetition of non-key words
- to use a more common or more simple word

Below is an example of where a researcher first asked the bot to improve her text, and then asked the bot to explain its changes.

1. *Changed "proven to be" to "emerged as" in the first sentence for a more precise language.*
2. *Added "although" in the second sentence to better contrast the methods proposed for analyzing multivariate time series with the limitations of such methods.*
3. *Added "In this work" to start the third sentence and clarify the focus of the proposed framework.*
4. *Changed "robustly differentiates" to "effectively differentiates" in the third sentence to avoid redundancy.*
5. *Reordered "chaotic dynamical phases and various types of synchronization" to "different types of synchronization and chaotic dynamical phases" in the fourth sentence for better flow.*
6. *Added "By utilizing" at the beginning of the fifth sentence to better connect this sentence to the previous one.*
7. *Changed "sheds light on" to "provides valuable insights into" in the final sentence for a more precise language.*

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Not all the explanations are clear (e.g. 4, 5, 7), but in any case may be good changes.

However, some explanations are really useful to show you how you can write in a clearer and more logical way (2, 3, 6). So, in theory, you can learn from the bot's explanations.

If you want to see ALL the changes made, then use the 'Compare Documents' feature under 'Track Changes' in Microsoft Word. This will show all the changes tracked.

9.4 What good corrections do chatbots make?

Chat makes a lot of sensible (i.e. reasoned) changes.

Note: It does not make the changes listed below automatically. The list just shows examples of the kind of changes that it makes.

The list is in approximate order of importance in terms of making the text more readable as well as more accurate from an English language point of view.

CONCISENESS

- reduces the number of words in a phrase by eliminating the phrase completely or by finding a more concise way to express the same concept.

SENTENCE STRUCTURE

- shifts the key information to the beginning of the sentence. This is good because the reader is more likely to notice this information.
- at the beginning of a sentence it may add a phrase such as *In contrast*, when the sentence is saying something that is very different from the previous one.
- restructures a string of nouns and adjectives in sequence by putting them in a logical order and using prepositions
- divides one long sentence into two parts in cases where the long sentence is not easy to understand immediately at first read
- combines two sentences into one sentence so that the order of the concepts presented is more logical

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AVOIDS AMBIGUITY

- replaces *one* (and other words such as *this, that, which, it, them*) with the word *one* refers to

STYLE

- uses the active rather than passive, when the passive form appears at the end of a long sentence.
- converts contractions into full forms e.g. *can't* to *cannot*
- replaces *his* (used as a generic pronoun, i.e. not in association with a specific man) with *their*

VOCABULARY

- chooses a more suitable or more common word (but is still sufficiently specific)
- corrects spelling mistakes of technical words. Such words may not be in your word processor, e.g. Microsoft Word. Note that it cannot correct all spelling mistakes, especially non-typical ones such as mistakes between these pairs of words: *implantation/implementation, paired/parried, reveled/revealed, series/serious.*

GRAMMAR

- corrects misusage of tenses
- respects the rules of concordance e.g. *this table shows* instead of *this table show*
- understands when *more* should be *most* and vice versa

A bot does not automatically make the changes listed above – sometimes it does and sometimes it doesn't. It is generally accurate, but not always. For instance it may change the tense of a verb when the verb was already in the correct tense.

So you always need to carefully check the bot's output.

However, if you use a bot effectively it should reduce the number of times that reviewers reject your paper for poor use of English.

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9.5 What can't GPT do? Is it a good idea to pre-edit my original text that I wrote in English?

Professional human editors make certain changes to texts that a bot, for the moment, is unable to make. The changes that editors make involve improving the readability and flow of a paper.

In reality, you could make these changes yourself BEFORE you submit your text to a bot. You can find a list of such 'pre-edits' in 10.3. In any case, here are some suggestions for improving the final output of a bot. Before you submit your text:

- Remove any redundancy – cut as much as you can
- Follow the advice given in Chapter 2 on word and sentence order
- Try to analyse key sentences to check that they are not ambiguous
- Replace pronouns with the words they refer to. This is essential when you have two consecutive sentences and the first sentence contains two nouns and the pronoun in the second sentence could in theory refer to either of these two nouns. In such cases, replace the pronoun with the exact noun it refers to.
- Delete unnecessary linkers and begin a new paragraph. If you are using a lot of words and phrases such as *in addition*, *furthermore*, and *it is worth noting* it is probably best to eliminate one of them and begin a new paragraph.
- Replace vague nouns with specific nouns, or simply delete the vague noun.
- Replace a vague adjective with an explanation. So instead of saying something is *reliable* or *important*, explain why it is *reliable* or *important*.
- Ensure there are no informal expressions or constructions.

Ensure that you know the difference between *which* and *that* and the related use of punctuation. See 6.17 and 6.18 in *English for Academic Research: Grammar, Usage and Style*.

- Try to relocate negative information to the beginning of your sentence.

If you apply the above changes, the bot will then just focus on correcting your English – something it tends to do very well. It also makes sense to use a bot in conjunction with your word processor's editing functions – this way you get a double check.

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9.6 What kinds of grammar mistakes does a bot currently fail to correct?

Grammar areas where bots tend to fail to make corrections, include when the past continuous or the past perfect should be used instead of the past simple.

At the time of the study, patients underwent treatment. past simple

At the time of the study, patients were already undergoing treatment. past continuous

At the time of the study, patients had already undergone treatment. past perfect

Past simple: patients had not had any treatment before the study commenced. Past continuous: treatment started before the study and continued during the study. Past perfect: patients received and terminated treatment before the study started.

Bots can fail to spot prepositions that have been used erroneously, particularly in cases where in theory two prepositions are possible.

Potatoes *were introduced from* sailors who brought them home from the new world.

Potatoes *were introduced by* sailors who brought them home from the new world.

Potatoes *were introduced from* the new world to the old world.

Bots are essentially looking for existing combinations of words. Although *were introduced by* is around 30 times more common (on a normal Google search) than *were introduced from,* the latter is still very common and on Google Scholar is only 20% less common. A very different case is where a preposition grammatically does not follow a particular verb. For example, a bot is very likely to correct *associate to* with *associate with*, given that *associate to* is simply not correct.

There are also issues when grammar seems to be sector/discipline specific. For example, nouns are used adjectivally in some cases and with the genitive in others, but without any underlying logic, e.g. *goat's milk* and *cow's milk* are correct and common, but *donkey milk* is much more common than *donkey's milk*.

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9.7 What kinds of errors will a bot probably never be able to correct?

There is a principle in computer science known as GIGO. This stands for *garbage in, garbage out*, meaning the result of data processing can only be as good as the data that have been inputted. The same is true for all artificial intelligence applications that are used to translate or improve a text in English.

Unlike a good scientific editor, no bot can, or is ever likely, to tell or warn you that:

- you have far too much background information
- something that is currently in the Abstract should be in the Introduction
- your Conclusions do not match what you have said in your Abstract
- there are many redundant and repeated phrases in captions to figures
- the meanings of two consecutive sentences apparently contradict each other
- your paragraphs are far too long and/or need headings
- you have not been consistent in your use of certain terms or in terms of style
- using an expression such as 'recently' or 'in a recent paper' makes no sense because if someone reads the paper in five years' time, it will no longer be 'recent'
- you need to mention your limitations
- you have used the same paragraph twice in the same paper (i.e. you shifted something to another position, but forgot to delete it in the original position)
- a word that you have invented yourself will make no sense to a reader unless you give it a clear explanation
- you have made a literal translation from your own language and although the result in English is grammatically correct, it doesn't mean the same thing as it meant in your own language
- a concept you have expressed is totally tied to your own country, language or culture and will make little sense to other readers
- your paper is long, tedious, and of little or no value

Moral of the story: Even if your paper is written in perfectly grammatical English, this does not mean that it will automatically be accepted for publication. A combination of the points listed above could mean that your paper is rejected. Clearly, native English speakers also have their papers rejected.

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9.8 What the most dangerous errors that chatbots make when revising a text in English?

Chatbots make mistakes. Under the 'cons' of using its software copy ai (the company behind ChatGPT) states "errors and unrelated words can pop up". Reviewers of some AI tools have commented:

I usually get a lot of unrelated sentences in the middle of some paragraphs which is very annoying.

There are a lot of errors in the content. Some parts of the content is copied exactly from other articles.

The two errors above are annoying, but are likely to be spotted quickly.

For me, the two most dangerous mistakes are when the bot changes:

- your specific technical word with a generic word
- the pronoun *we* or *our* with the definite / indefinite article or something else. For instance, a bot may change *in our previous paper* to *in a previous paper*. This can be very confusing for the reader as the pronoun *our* unambiguously indicates that this is your paper, whereas *a previous paper* gives the idea that you are talking about another author

To deal with the above two problems, highlight in yellow in your original text all those words (or phrases) that it is essential that the bot does NOT change. Then check the bot's version against your yellow words.

Another serious issue is when the bot changes the words in an acronym or adds words to it.

A chatbot may also break with stylistic conventions. For example, sentences do not begin with a number as a numeral. Instead the word form should be used:

... Smith et al 2025]. 20 pigs were selected for ...

... Smith et al 2025]. Twenty pigs / A total of 20 pigs were selected for ...

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9.9 What are the dangers of modifying the chatbot's version? Is there a solution?

When you check the bot's version, there are various modifications to the output that you may wish to make. For example, you may decide to modify the bot's version by:

- replacing a generic word with a more specific word
- reinserting (from your original version) a technical word that Chat has mistakenly changed
- changing the tense when you are 100% sure that a particular tense is required (e.g. when you are following a guideline that you have found in this book)

However, you need to be careful when you introduce a totally new phrase or radically modify the bot's version – you might also be inadvertently introducing a grammar or spelling mistake.

In such cases, resubmit your new phrase. You should provide the bot with some context by also inserting the sentences that precede and follow it.

9.10 What decisions do I need to make before rejecting a change made by GPT?

Be careful before deciding to ignore changes that a bot has made.

A very banal example is when the bot has inserted a comma before *and* in a case where there are more than two items in a list:

We did research in Zimbabwe, *Zanzibar, and* New Zealand.

In your language a comma before *and* may be considered bad style, but not in modern English. The comma helps to make a visual separation, so that readers can quickly see how items are divided up. In such cases, you should trust the bot.

A much more serious example is when the bot appears to have randomly added a word, or changed the word order, or erroneously divided up a sentence. Take this as a warning that there was probably something wrong with your original version. So before simply rejecting the bot's change, try to understand why the bot had difficulty in understanding your intended meaning.

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Likewise, if the bot has considerably shortened the original. This may not necessarily be a way to be more concise, but could be an attempt to make sense of a phrase that in English doesn't seem to mean anything. Again, this should be a warning to you that something is perhaps not right.

9.11 Will a chatbot negatively affect my writing style?

I am going to use the extract from Sam Katzoff's NASA doc: 'Clarity in Technical Reporting' that I reported in 3.8. Let's look at it sentence by sentence. Sam's version is in blue, and Chat's has a grey background.

Different writers have different methods of organizing their reports, and some seem to have no discernible method at all.

Technical writing requires a clear and organized approach to effectively convey information to the reader.

Comment: Sam focuses on the human element - 'writers'. Chat focuses on the action: 'technical writing. Sam introduces humor - 'no discernible method' - which has been totally removed by Chat's dispassionate approach.

Most of the better writers, however, appear to be in remarkably close agreement as to the general approach to organization.

While some writers may have unique methods of organizing their reports, most successful writers tend to follow a similar structure.

Comment: Sam links his second sentence back to his first by repeating the word 'writer'. As a reader you feel that Sam is moving you gently forward with his logic. Chat's version is also logical, but there is no attempt at making a connection with the reader.

This approach consists of stating the problem, describing the method of attack, developing the results, discussing the results, and summarizing the conclusions.

This structure involves outlining the problem, presenting the methodology, discussing the results, analyzing the results, and concluding with a summary.

Comment: You can feel that there is a human behind Sam's writing: he uses 'method of attack' rather than 'methodology' – this makes his writing more enjoyable to read.

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You may feel that this type of organization is obvious, logical, and natural. Nevertheless, it is not universally accepted.

This approach is widely recognized as being logical and straightforward, yet not all writers follow it.

Comment: Sam addresses the reader directly 'you may feel'. The reader is engaged. Sam also uses a very short sentence (only six words) that begins with 'Nevertheless' – he wants to warn readers that he is going to say something important that is in contrast with what he has said before. This warning and this contrast are lost in Chat's version, which joins the two sentences together, thus losing the opportunity to get the reader's attention by using a short sentence.

For example, many writers present results and conclusions near the beginning, and describe the derivation of these results in subsequent sections.

Some choose to present results and conclusions early on, followed by an explanation of how they were derived in subsequent sections.

Comment: Both Sam's and Chat's version are equally good here.

Essentially, a chatbot has no soul. Not surprising really!

9.12 How can I assess whether the bot's version is actually better than my original version?

It is not easy for you to assess:

1. the quality of a text you yourself have written directly in English
2. a bot's correction of your English
3. an automatic translation from your language into English

If you have implemented the pre-editing strategies outlined in 9.5 and 10.3, and if you also speak one of Europe's major languages, there is a very good chance that an AI tool will produce a better version than you can if you wrote directly in English.

Obviously you cannot, and should not, trust AI 100%. But, do not dismiss it either (see 10.9).

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9.13 Using AI to generate a rebuttal letter and other types of email

A bot can help you in another crucial area of your paper writing experience – the rebuttal letter. The rebuttal letter is a point-by-point response to the issues brought up by the reviewers who have read and assessed your paper.

Imagine that you have received the following comments on a paper you have submitted.

> "The English is poor" - *but the reviewer gives no examples. In your opinion the English is fine.*
>
> "The figures are not clear" - *you think they are very clear.*
>
> "The bibliography does not include two important references: Smith et al [2025] and Smith et al [2026]" *the two references that the referee gives you are irrelevant in your opinion.*

If you give AI the above information and ask it to generate a reply to the editor, the output may be something like this:

Dear Editor,

We would like to thank you and the reviewers for taking the time to evaluate our paper.

We appreciate the valuable feedback provided by the referee who recommended revisions, and we have carefully considered their comments. We would like to provide a formal response to the issues raised.

In regards to the concern regarding the English language quality, we understand the importance of clear and concise language in scientific writing. However, we are surprised that the referee did not provide specific examples of areas that require improvement. We kindly request that the referee provides detailed feedback on the areas of the manuscript that require attention so that we can address them appropriately in a revised version.

Regarding the clarity of the figures, we appreciate the referee's feedback, but we respectfully disagree with their assessment. We believe that the figures are clearly presented, and we believe that they effectively illustrate the key findings in our research. Nevertheless, we will review the figures again and take any necessary steps to further improve their clarity.

Finally, the referee has pointed out that two important references were missing from the bibliography. We appreciate this feedback and have carefully reviewed the references to determine their relevance to our research. After careful consideration, we believe that the references are not pertinent to our work. However, we will review the references again and make any necessary adjustments to ensure that our bibliography is accurate and comprehensive.

We are grateful for the opportunity to submit our work to your esteemed journal and would be honored if you would consider our revised manuscript for publication.

Thank you again for your time and consideration.

Sincerely,

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What AI has done is to produce a response that is essentially error free (though *in regards to* sounds strange to me, I would say *regarding* or *with regard to*). It also has a conventional structure, which you could possibly use as a template.

However:

- The response is very long and full of phrases that mean very little – this is very typical of AI-generated texts.
- It is unnecessarily formal (e.g. *esteemed journal ... would be honored*).
- The final salutation (*Sincerely*) is somewhat archaic; much more common is *Best regards.*

Moreover, the bot cannot tell you that:

- In this particular case, your strategy of contesting what the reviewers have said is not actually going to benefit you as it will inevitably delay the publication of your paper. This is particular important if you are giving groundbreaking results. If you delay publication, another research group may publish their similar findings before you.
- Given that the three requests are relatively simple to manage, the cost of not doing what the reviewers ask is greater than the cost of simply doing them (even if you are not 100% convinced).

Of course, in some cases it is entirely appropriate to question what the reviewers have stated, and to diplomatically explain why you have decided not to do as requested. But the three points in our case are unlikely to affect the purpose, methods or results of your paper.

For more on writing effective emails to journal editors see Chapters 12 and 13 in *English for Academic Correspondence*.

9.14 When not to use a bot to generate / correct an email

I do not recommend using a chatbot when you want to generate an email expressing your own personal views of something, or when thanking someone, or in any cases where you are trying to establish or maintain some kind of relationship.

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I conducted a test with PhD students which entailed writing me an email at the end of the course. Many students took the opportunity to complete the task either by getting the bot to generate the email on the basis of a few prompts, or using the bot to correct their versions in English.

Their answers were error-free, but also totally impersonal.

Students who I had grown to know well during the course produced emails that were full of platitudes and devoid of any emotion: they were literally robotic. It was as if we had never met.

> The effect on the reader of a bot-generated / bot-corrected email is one of being distanced from the writer and having the spontaneity, idiosyncrasy and joy of a relationship sucked out. It is like the email has been censored for any signs of personality and sincerity.

9.15 How easy is it to tell that a text has been generated by artificial intelligence?

Massive amounts of money are being spent on developing algorithms to identify whether a text has been generated by a bot.

In reality it is not that difficult to spot.

If you are a non-native speaker and you have written an assignment in English that contains no mistakes, it is going to be pretty clear to an English teacher that you did not write the text yourself!

But there are other key signs that a text has possibly not been generated by a human.

AI texts are often formulaic, i.e. they always begin in the same way, follow the same structure, and end in the same way. This is true of both academic texts (e.g. Abstracts and Conclusions) and emails relating to your everyday academic work.

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AI tends to use sentences that are similar in length – none are very long and none are very short. This makes the text less interesting to read, and the key points don't always stand out as much as they could.

AI texts are soulless and emotionless. Obviously, when reading them you don't feel there is a personality behind them. There is never anything unusual or quirky, and rarely anything counterintuitive.

Automatically generated emails are, in my opinion, the kinds of texts where it is clearest that there has been no human intervention. Sometimes they generate inappropriate (in the sense of strange) comments and questions. They are often too formal, or too informal.

9.16 Is using a chatbot an act of plagiarism?

I asked ChatGPT: *Is using you an act of plagiarism?*

No, using me, a language model, is not an act of plagiarism. Plagiarism involves using someone else's work, such as their words, ideas, or creative expression, without giving them proper credit.

However, it is important to keep in mind that if you use information or text generated by me, you should still cite the source of the information as "OpenAI's language model" or "ChatGPT". This is especially important if you are using the information or text for academic or professional purposes, as proper citation helps to give credit to the original source and demonstrate academic integrity.

I agree with ChatGPT's explanation ... but only to a certain extent: To write its output, a chatbot by necessity has to utilize other people's works.

Back in the 1980s when I was at university, you could be thrown out if it could be proved that you had copied someone else's work. This ethic is deeply ingrained into Anglo societies. It is considered a form of deception (and laziness).

By all means, use a chatbot to improve, reduce, enhance, correct your English. But remember that, unlike what is reported in the media, it is relatively easy to spot (9.14) whether a text has been generated by a human (especially a non-native English speaking human).

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Consequently, I suggest that you always acknowledge that you have used a chatbot 'to improve the style and correctness of my English writing'.

For more details on what constitutes plagiarism see Grammarly's tutorial entitled *7 Common Types of Plagiarism, With Examples*, available here: https://www.grammarly.com/blog/types-of-plagiarism/#:~:text=Plagiarism%20is%20the%20act%20of,likely%20has%20a%20plagiarism%20policy.

> NEVER use AI generate a totally new text and pass this text off as your own. This is a form of plagiarism.
>
> In my opinion, you should ALWAYS inform the person who you are sending your text to that all or parts of the text were generated by a bot.

9.17 Summary

- When you use AI to help you write or correct a text in English, you still have to do a lot of preparatory work on your original text, and you then have to correct the version that AI produces.
- Learn the best prompts to get a chatbot to fulfill your request: the more specific and detailed the better. For a regularly updated list of prompts to use with ChatGPT: https://tefldiscussions.com/ai-prompts/.
- Check the changes that the chatbot makes, and learn from them.
- Remember that AI has both benefits (9.4) and limitations (9.6, 9.7, 9.8).
- If the bot makes changes that you don't agree with, don't automatically dismiss them. They probably reflect a lack of clarity in your original version.
- Be very careful when using a bot to generate an email. Ensure that it says what you really intended and that it has not changed the level of formality or introduced any bizarre questions or statements.
- Never use a bot to paraphrase the work of another author without acknowledging that author by referencing their work.
- Finally, if you use a bot for work that you later to submit to your professor or a journal, ensure that you inform the prof / editor.

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EXERCISES

For exercises on the use of AI for improving, correcting and paraphrasing your texts, see the 2024 editions of the following exercise books in this series.

English for Academic Research: Grammar Exercises (Chapter 29)

English for Academic Research: Vocabulary Exercises (Chapter 11)

English for Academic Research: Writing Exercises (Chapters 11 and 12)

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Chapter 10

Automatic translation



A bit of history

- Esperanto (meaning one who hopes) was a language devised by Dr Lazar Zamenhof, who published his work in 1887, during a period in which another 53 artificial universal languages were created.
- In 1980, when a hospital interpreter translated the Spanish *intoxicado* (meaning suffering from food poisoning) as intoxicated, the doctors treated the patient as if he had taken a drug overdose. The patient was left with permanent quadriplegia, and the malpractice apparently cost the hospital $71 million in compensation.
- The European Commission's Directorate-General for Translation (DGT) produces more than 2.75 million translated pages per year, 91% from English, 2% from French, and under 1% from Spanish and German.
- At lower secondary level schools in the EU, English is studied by nearly 86.8% of pupils, followed by French (19.4%), German (18.3%) and Spanish (17.5%).
- Google Translate was introduced in 2007, initially with translations from French, German, and Spanish into English, and vice versa. ChatGPT and Bard were introduced in 2022 and 2023, respectively.
- Some politicians' disjointed syntax, run-on sentences, and limited vocabulary pose an ethical dilemma for simultaneous translators. According to one translator, if Trump's words were rendered into smoother intelligible speech this might make him sound like "an ordinary politician who speaks properly."

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_10

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10.1 How does this chapter differ from the chapter on translation in the companion volume on *Grammar, Usage and Style*?

Chapter 16 of the companion volume – *English for Academic Research: Grammar, Usage and Style* – contains various points not mentioned in the book you are reading now. 16.1: lists the main translation services available. 16.2 covers pre-editing, which is dealt with more at length in this book (10.3). 16.3 and 16.4 show how comparing the output of an automatic translation with the original can improve your English (also dealt with here, 10.9). 16.5 lists typical mistakes made by machine translators – this is dealt with more at length in this book (10.4-10.7). Finally, 16.6 outlines mistakes made in the machine translations of emails, which is not dealt with in this book.

On the other hand, the chapter you are reading now is much more specific to writing papers.

*****

This chapter distinguishes between *major* and *minor* languages. For the purposes of this book, by *major* I mean a language pair that is very commonly used by machine translation applications, essentially the following five European languages: Spanish, French, German, Portuguese and Italian.

Although the following languages are major in the sense that they have more than 100 million speakers, they are more complex than western European languages and most have been translated less frequently: Mandarin Chinese, Hindi, Bengali, Russian, Indonesia and Urdu.

By *minor* language, I mean a language that is spoken by less than one hundred million people – the world has almost 7000 languages, but only twelve are spoken by one hundred million or more.

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10.2 Should I use machine translation? Which application?

The following reasons are NOT good reasons for NOT using machine translation.

"It's too easy if I use a machine translator. It's like cheating".

"I won't learn and improve my English by using machine translation".

"I already have parts of my paper that are written in English".

"Machine translators make mistakes".

"I used Google Translate a few years ago and it was a disaster!"

Being 'easy' does not equate with cheating. Using machine translation means that you can write in your own language, yet produce a very good first draft in English and save a lot of time which you can use for improving your manuscript – see 1.2.

You can learn a lot of English using machine translation – see 10.9.

Machine translation software will merge together your language and English into a final English output.

Machine translators do make mistakes, but probably far fewer than you do.

Google Translate (GT) is no longer a disaster. It has improved massively over the years and for some languages, for example, French to English academic texts, it is close to 99% accurate. Machine translations did sometimes make massive mistakes that were then exaggerated and ridiculed in the press – this was true in the 2010s, but very rarely now.

Good reasons for NOT using machine translation are:

"My language is a minor language and machine translation at the moment produces erratic results".

"I love writing in English".

"I have trained myself to write academic texts in English. It would not be natural for me to write in my own language".

My own studies with PhD students have shown that GT is more accurate than the average student. In fact, I now encourage students to write in their own language (modifying it as suggested in 10.3) and then use GT to do the translation.

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This does not mean that you have to write the entire paper in your language. If there are some sections that you have already written in English or feel confident about writing directly in English – for example the Methods section (i.e. the least complicated section from a linguistic point of view) – then write in English. GT will then merge the texts in your language and the texts already in English into a final output entirely in English.

You can also use a mixture of your own language and English even within the same phrase. So if you are certain of a particular word or phrase in English, then you can insert this word or phrase directly into a sentence that is otherwise written in your own language. GT will then convert everything into English.

See 1.2 for more reasons why writing your first draft in your own language might be a good idea.

Which machine translator should you use? I personally don't see much difference in the quality between, for example, Google Translate, Deep L and Baidu; but everyone has their preferences. The examples given in this chapter were generated by Google Translate, which is free and which accepts entire papers. Many other applications claim they are more accurate and thus charge a fee, and may only allow you to upload a certain number of characters.

Note that the success of machine translation (MT) very much depends on the source language. In any case, I suggest that you at least experiment with MT – and try using different machine translators to see if there is any difference in accuracy. It is a good idea to show the results of your experiments to an English teacher who also speaks your language – they are likely to be a better judge than you of the accuracy of the MT.

10.3 Pre-editing: how to improve the chances of getting an accurate automatic translation

The success level of a machine translation (MT) depends to a large extent on how similar the construction of your language is with respect to the normal structure of English.

One solution is to modify the version in your own language before you submit it to translation. To get the most accurate results from MT, make the text in your language follow some of the key rules of English and of good writing in general. You have much more control over your own language than you do over English, so the following recommendations should not be too difficult to implement.

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In brackets are the relevant chapters in this book where you will learn how to implement the suggested pre-edits.

1. Make your text as clear and concise as possible. Remember that you are writing for an incredibly wide range of readers. Don't try to impress them with 'elegant' writing skills (Chapter 8).
2. Reduce the length of any long sentences. Limit the number of ideas in each sentence to two ideas (Chapter 2)
3. Divide up any long paragraphs – this will not help MT, but it will help your readers (Chapter 3).
4. Remove redundancy. Imagine that the editor has told you to cut the paper by 25%, try to find ways to do this, but without removing key content (Chapter 4).
5. Adopt an English-like word order. This may entail altering the word order of your own language, typically by putting the subject at the beginning of the sentence (Chapter 2).
6. Check that you haven't used any synonyms for your key words. You should use the same key word throughout the paper. If you use synonyms, the reader may think you are referring to two different entities. Also, MT will make more sense of your text if a key word is always written the same (Chapter 5).
7. Where there might be ambiguity, replace words such as *it, them, these, those, which* (i.e. the equivalents of these words in your own language) with the words that they refer to. This will help your readers immediately understand what you are referring to (Chapter 5)
8. Look for any vague words or phrases and replace them with specific words and phrases. For example, if you are a medical researcher and you write *Patients all received assistance for their health issues,* then your reader might be interested to know what kind of *assistance*. So it would be better to write *Patients all received x, y and z for their health problem.* Obviously, avoid expressions such as *for a certain amount of time* (be exact), *several countries* (state how many or which ones) etc (8.8).
9. Don't use masculine pronouns as generic pronouns, but only to refer to men and male animals. Instead try using the plural form of the noun so that you can use the third person plural pronoun (8.10)
10. Where possible use active rather than passive forms (16.4). This helps both MT and your readers to understand who carried out the action.
11. If possible try to punctuate your text in an English way. Keep the sentences short, replace semi colons with full stops, and where appropriate use commas to break up the various parts of the sentence. See *English for Academic Research: Grammar, Usage and Style* sections 1.4-1.18 to learn how to punctuate your manuscript.

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12. If you already know the English version of a specialized a term, write this term directly into English. This will avoid any mistranslations by MT. Similarly, if you are sure of a particular phrase, write it directly in English. MT will usually leave any English words as they are and produce a uniform text all in English. You can even write an entire section in English, and MT will just translate the other sections.
13. See *English for Academic Research: Grammar, Usage and Style* section 16.2 for more checks to carry out on your text before submitting it to MT.

It is much easier to carry out the above modifications in your own language than it is for you to do it in English.

Then check the machine-generated translation very carefully, just like you would do if you had translated your text manually.

10.4 Grammar and vocabulary areas where MT may be more accurate than you

Unlike non-native speakers, machine translators of major languages do not make the following types of mistakes. Note: this list is by no means exhaustive, it merely gives you an idea of the utility of using MT.

ADJECTIVE BEFORE NOUN

We used a *doctor experienced* to carry out the diagnosis.

We used an *experienced doctor* to carry out the diagnosis.

ARTICLES

We used this as standard / *in* laboratory.

We used this as *a* standard / *in the* laboratory.

EVEN IF VS *EVEN THOUGH*

Even if the instructions to the author stipulate a maximum of 250 words, we used 300.

Even though / Although the instructions to the author stipulate a maximum of 250 words, we used 300.

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FALSE FRIENDS

This is the *actual* situation today.

This is the *current* situation.

We found that x = 1. *On the contrary*, Smith et al. found that x = 2.

We found that x = 1. *In contrast / On the other hand*,, Smith et al. found that x = 2.

NOUN STRINGS

An internal strains collection

A collection of internal strains

PREPOSITION + ING VS INF

... without to receive any financial compensation.

... without receiving any financial compensation.

SPELLING / TYPOS

First, we *weighted* the samples.

First, we *weighed* the samples.

.. we did x. In *addiction*, we did y.

... we did x. In *addition*, we did y.

This was purchased form IBM.

This was purchased *from* IBM.

SUGGEST, RECOMMEND: CONSTRUCTION WITH

The referee suggested / recommended *us to rewrite* the first section.

The referee suggested / recommended ***that we (should) rewrite*** the first section.

VERB / NOUN + PREPOSITION SPELLING

X depends *by* Y and the association *to* Z.

X depends *on* Y and the association *with* Z.

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10.5 Typical areas where automatic translators may make mistakes in English if your language is not a major language

The mistakes that machine translators make very much depend on the source language, i.e. your mother tongue. Below are some typical mistakes. However, once you have done a machine translation, you can check the accuracy of English usage with a chatbot (Chapter 9, and 10.8).

TENSES

MT sometimes changes the tense from the original. For example, you may use the future tense and GT will translate it into the present tense. In some cases, MT may be correct. For example, if in your language you say 'I will tell him when I *will see* him', MT will correctly translate this as 'I will tell him when I *see* him'. This is because in a time clause, *when* takes the present and not the future. However, MT makes mistakes when it changes tenses, so it is wise to check very carefully.

UNCOUNTABLE NOUNS

An uncountable noun is a noun that cannot be made plural and which cannot be preceded by *a/an* or *one*. For example *information* is uncountable. This means you cannot say *an information, one information, two informations, several informations*.

The problem with uncountable nouns is that the surrounding words (i.e. articles, pronouns and verbs) must also be singular. This means that if, for example, *information* is countable in your language, MT will probably make errors with the surrounding words, as highlighted in this example (note: the example is NOT in correct English):

These information *are* vital in order to understand xyz. In fact, *they are* so crucial that ...

There are two solutions. You can modify your own language so that you put the surrounding words into the singular. Or you can check the English version. Whichever solution you use, the aim is to produce one of the following two correct sentences:

This information *is* vital in order to understand xyz. In fact, *it is* so crucial that ...

This information *is* vital in order to understand xyz. In fact, *such information is* so crucial that ...

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WORDS WITH MORE THAN ONE MEANING

MT generally manages to guess the right meaning when translating into English because it looks at the surrounding words (i.e. how words are collocated together). In any case, you need to check carefully that the machine has translated your text with the meaning you intended.

NAMES OF PEOPLE

The first MT applications tended to translate people's first names and sometimes surnames. This no longer happens with major languages, but if your languages is only spoken by a few million people, MT may mistakenly translate names.

Below are just some examples of the kind of mistakes that MT might make if your language is NOT a major language.

ALLOW : CONSTRUCTION

This allows to reduce x.

This allows x to be reduced.

This reduces x.

This allows you * to reduce x.

This allows us * to reduce x.

* GT tends to write *you* rather than *us*. The use of *you* is not appropriate in a paper.

CAPITALIZATION

a european project

a European project

in figure 1

in Figure 1

We calculated the Odds Ratio by ...

We calculated the odds ratio by ...

We used a Life Cycle Assessment (LCA) in order to ...

We used a life cycle assessment (LCA) in order to ...

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HYPHENS

The decision making task is ...

The decision-making task is ...

Decision making is ...

NUMBERS 1-10 IN MAIN TEXT

Each test was replicated 3 times.

Each test was replicated three times.

POSITION OF *FINALLY*

The samples were finally incubated in ...

Finally, the samples were incubated in ...

10.6 Areas where machine translation will not help you even if your language is a major language

There are some areas where AI (machine translators and chatbots) is unlikely to be able help you, even if your language is a 'major' language (see 10.1).

REDUNDANCY For example, if in your original text you use a verb+noun construction, such as *this causes the reduction of x*, or *after the drying of the samples*, MT does not usually translate as *this reduces x* or *after drying the samples*. Another example: *As we can see from Figure 5, ... = Figure 5 shows ...*

WORD ORDER Good writing in English entails putting the main subject as close to the beginning of the sentence as possible, and then putting the verb and object immediately after. MT is a translator not an editor, so it rarely makes major adjustments in word order.

LONG SENTENCES MT will not divide up long sentences into shorter sentences.

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VAGUE PHRASES MT will not make a vague expression more specific. So, it will not make changes like the ones below:

This analysis ... The xyz analysis ...

These considerations ... The influence of xyz ...

The method mentioned above ... The xyz method (see Section 2.1) ...

PUNCTUATION MISTAKES For example, when you have a list that contains many commas, it is useful to group certain elements in the list together and separate them with a semicolon. MT will not do this for you!

SPELLING MISTAKES IN YOUR LANGUAGE MT cannot deal with spelling mistakes in your original language that then cause the wrong word to be used in the English translation. Before you submit your original text, ensure that the spelling is 100% correct. This includes making sure that the right accent is on a word, and that an accent is used rather than an apostrophe: e.g. *più* not *piu'*.

CAPITALIZATION MISTAKES Stylistic rules differ from language to language particularly with reference to names of authors and formulas, indices, tests etc.; software; manufacturers etc. Double check that these are correct in your original version.

MISTAKES WITH ACRONYMS. Look for all acronyms in your original text and ensure that they are written in the English way. MT will not automatically translate acronyms. This means that if, in your language you write UE to indicate the European Union (EU), MT will not necessarily convert UE into EU.

LEVEL OF FORMALITY. Research papers are written in formal English. If you use informal words in your original language, MT is likely to adopt a similar level of informality in its translation. In any case, be aware that the first word in the following pairs is generally the most appropriate (note: these are just some examples, there are several others): *reported/stated vs said, thus/consequently/therefore vs so, excellent vs brilliant/fantastic, until vs till, let us vs let's, a little vs a bit, such as vs like.*

PLURAL ACRONYMS In English, we say *one PC* but *two PCs*. Most other languages do not have a plural form for acronyms, and thus say *two PC*. MT is able to recognize this for common acronyms such as *gif*, *pdf, SME* (small to medium enterprise), but not for very technical acronyms.

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10.7 The dangers of using machine translation

The abstract below was translated from Italian into English. It highlights some of the types of mistakes that you need to check for. The words that should be used to replace the deleted words are shown in brackets and are in red.

A method for the production of renewable X-polyolefins ~~has been~~ (was) developed. Monomer X was obtained from functionalized sugar industry waste extracts. Monomer X was ~~used for~~ (employed in) reversible deactivation ~~free~~ radical polymerizations (RDRP) under mild and environmentally sustainable conditions to produce X-polyolefin. Molecular weight distributions were measured by size exclusion chromatography (SEC) and show little control over chain length and dispersity. The chain transfer phenomenon was studied by multiple angle light scattering (MALS), highlighting a high branching density and therefore explaining the poor control over the weight parameters. The polymer ~~life~~ (livingness) was studied by NMR, showing that the retention of the terminal functional groups is 20% at long reaction times. The ~~scaling~~ (upscaling) of the procedure was carried out allowing to obtain 20 g of monomer X without significant alterations of the relative yield.

GT has chosen the wrong words in three cases:

- It has added *free* in the third line (you can see from the acronym - *RDRP* - that *free* is not required).
- The verb *used* is incorrect, it should be *employed in* (meaning it was only a part of the deactivation, rather than being the sole deactivator)
- In your field you may need to use very specific terms. GT has used the wrong terms twice (*life, scaling*). GT has in fact made an intelligent guess, and in most other cases *life* and *scaling* would probably have been correct, but in this particular case the correct terms are *livingness* and *scaling*. One solution to this problem is to write your key words in English even though the rest of the text is in your mother tongue. GT will automatically merge the two languages together. In any case, to ensure that GT doesn't change your key words, highlight them in yellow in your original text. If you upload your entire document, GT will keep the yellow formatting and then you can check against your original to see if it has made changes that it should not have made.

So, when you are reviewing an automatic translation, double check that the terminology is correct.

GT has also changed the tense in the first line: *has been developed* is ambiguous as it is not clear who developed it. The convention in an abstract is to use the simple past (*was developed*) to indicate that you did something. However, given the passive form, it is still ambiguous. It would be better to write: *We developed a method* or *We have developed a method*. Using *we* avoids any ambiguity.

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10.8 How can I combine my use of machine translation with a chatbot?

To increase your chances of getting an accurate text in English, you can use an automatic translator and then check the output with a chatbot. Here is a possible list of steps:

1. Write your text in your own language.
2. Rigorously implement all the modifications suggested in 10.3.
3. Submit your text to machine translation (MT).
4. Check that the MT version hasn't altered any of your key words or added extraneous words.
5. Submit the corrected MT version to a chatbot. Give the bot this prompt (for a regularly updated list of prompts to use with ChatGPT: https://tefl-discussions.com/ai-prompts/): *Please improve the English of this Abstract (Introduction, Conclusion etc). Do not change any technical words.*
6. Check the version that the bot produced (see 9.3, 9.6-9.10).
7. Go through the version again looking for any examples of possible readability issues – see 9.5 for the types of mistakes a bot cannot find.
8. If you make any changes of more than two or three words in sequence, check them again with the bot (9.9).
9. Do a final comparison between your original version (Step 1) and the final version in English. Ensure, if you can, that there are no differences in meaning.
10. Do a final check of the English (20.3, 20.6, 20.7, 20.8, 20.9).
11. Use your word processor's Editor function to check for spelling and grammar mistakes. This is the final check that you MUST make immediately before you send your paper to the journal, particularly if you have made any last-minute modifications at Points 8-10.

At this point, if you speak one of the world's major languages as defined in 10.1, you can assume that the English of your paper is close to being perfect. But remember that just because the English is perfect, does not mean that your paper will automatically be accepted for publication. And even if it is accepted for publication, perfect English is not guarantee that your paper will be read or cited (20.1).

As an alternative to Point 3 above, you can submit your text in your own language to a bot, and ask the bot to improve it. The advantage is that because the text is still in your own language you have more control over it, and you

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can identify more easily the effect that the bot's changes will have on the meaning and effectiveness of your text. Then, when you are happy with the text in your own language, you can submit it to MT.

Note that translating the final English version back into your own language is only effective if the machine translator works well when translating into your language – which is very often not the case. MT works best from your language into English, and not the other way round. This is simply because more texts are translated *into* English than *from* English, so MT has more data to work with when translating *into* English.

10.9 Will I still improve my level of English if I use machine translators and chatbots?

The red sentences below are part of an abstract produced by one of my students. She then submitted her abstract to ChatGPT, which produced the sentences in blue. She told me that she thought that 'ChatGPT didn't improve much on the original'. We then made the following analysis together and she realized that in fact the bot had improved her original version. And in the process she also learned / revised some rules of English.

Opinion *dynamics, which* studies how people's *opinion* on a particular topic *evolves* over *time, deals* with key phenomena such as polarization and social influence.

Opinion *dynamics analyses* how people's *opinions* on a particular topic *evolve* over time *and* deals with key phenomena such as polarization and social influence.

The red version splits the subject from its noun (2.4), whereas the blue keeps them together thus making the sentence flow better.

It has become ever more important *given the central role that social media* has gained in our lives.

Given the central role that social media plays in our lives, opinion dynamics has become increasingly important.

The blue version puts the key information at the beginning of the sentence (Chapter 8).

Its main tool is agent-based *modeling, a type of model* that encodes the causal mechanisms that drive the opinion formation process, and is thus easy to interpret.

Its main tool is agent-based *modeling, which* encodes the causal mechanisms that drive the opinion formation process and is thus easy to interpret.

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The blue version avoids the repetition of *model/modeling*.

A proxy-based model (PBM) comprises a set of *proxies which* interact according to pre-determined mechanisms and is simulated to derive outcomes.

A proxy-based model (PBM) comprises a set of *proxies that* interact according to predetermined mechanisms and is simulated to derive outcomes.

The red version is ambiguous. If you are adding extra information about the *proxies*, then *which* should be preceded by a comma (*proxies, which*). If you are defining what the proxies are, then *that* should be used (as in the blue version). See 6.17 and 6.18 in *English for Academic Research: Grammar, Usage and Style.*

Unfortunately, and *differently from* their statistical counterparts, there is no direct, simple, and principled way to make use of data in conjunction with PBMs.

Unfortunately, *unlike* their statistical counterparts, there is no direct, simple, and principled way to use data in conjunction with PBMs.

The red version contains the only mistake in English in the entire abstract: *differently from* is not correct English.

Indeed, the lack of a formal probability function for PBMs makes it impossible to fit a model to a dataset via optimization.

The lack of a formal probability function for PBMs makes it impossible to fit a model to a dataset via optimization.

The blue version removes the unnecessary link word (*indeed*) – see 4.8.

The analysis above highlights two factors:

1. there is a good chance that the bot writes better English than you do
2. you can learn a lot by comparing your original version (either in English or in your own language) with the version produced by a bot or by machine translation

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10.10 Do NOT use an automatic translator to check your English

If you write something directly into English, you may think that you can use Google Translate to check your English by translating it back into your own language to see if it makes sense. Unfortunately this does NOT work.

When you write in English you are naturally translating directly from your own language. So, if you submit your English text into GT and translate it back into your own language, the translated text in your own language will probably be very good because the structure of your English sentence is based on the structure of the same sentence in your language.

However, this does NOT indicate that your English version is correct. It only indicates that the resulting text in your own language is what you wanted to say in English.

10.11 A note for EAP teachers

In my own experiments with PhD students, I found that students make up to 30% fewer mistakes by using Google Translate, and that Google Translate tends to be at least 90% accurate (even 95% accurate if the source text is written well and is GT-ready).

My experiments were conducted over two courses. In the first course, students were taught how to write a scientific paper, but with no mention of using Google Translate (GT). As part of this course, students had to send me assignments for me to correct – I noted the number of mistakes that were made. In the second course, I introduced GT. In their initial assignments they sent me texts in their own language, Italian, and I suggested ways of modifying the Italian so that it reflects the style and rules of English. They then submitted their modified texts to GT, and finally corrected GT's version. Students also worked together by suggesting changes both to their partner's Italian and to their partner's final version in English.

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I compared the accuracy of the GT texts that my students produced in the second course with previous texts that they had sent me in the first course and with full papers for publication that they and their professors had sent me for editing in previous years. My simple (and for the moment limited) experiments have convinced me that GT does a better job than most non-professional translators, provided of course that the GT-generated text is checked for mistakes.

If you are a teacher, I recommend that you test out GT. Don't be discouraged by bad Google translations that you have seen in the past, or by your own experiences of translating from English into another language. GT works best when translating technical texts from European languages into English. However, I have also tested it with Arabic, Chinese and Japanese. If the source texts are written simply and clearly, GT does a good job. But, I repeat, GT works best when translating into English, not from English. Also, the more technical the text, the better the translation. I would not advise using GT to translate colloquial English and complex emails.

10.12 Summary

- Using machine translation (MT) is not a form of cheating. Instead, it will save you time and you will learn a lot if you compare its output with your original version.
- MT currently works best with Europe's five principal languages, translating from those languages into English. Be aware of the mistakes that MT might make if you language is not a major language (10.5)
- To ensure an accurate translation, you should work first on the version in your own language by making it 'look and sound' more like English, and also by removing redundancy and ambiguity. This is known as pre-editing (10.3) – it is essential.
- MT is not a perfect tool – it cannot help you write the perfect paper. There are other considerations to take into account – sentence length, redundancy, ambiguity, vague phrases, word order, punctuation and capitalization, level of formality etc.
- Always check the output of MT, above all checking that MT has not changed any of your key words.

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EXERCISES

For exercises on the use of machine translation, see the 2024 editions of the following exercise books in this series.

English for Academic Research: Grammar Exercises (Chapter 30)

English for Academic Research: Vocabulary Exercises (Chapter 12)

English for Academic Research: Writing Exercises (Chapter 13)

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Part II

Sections of a Paper

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Chapter 11

Titles



Great titles of real papers

- Describing the Relationship between Cat Bites and Human Depression Using Data from an Electronic Health Record
- Dogs are sensitive to small variations of the Earth's magnetic field
- Aesthetic value of paintings affects pain thresholds
- Response Behaviors of Svalbard Reindeer towards Humans and Humans Disguised as Polar Bears on Edgeøya
- 10 = 6 + 4
- A minus sign that used to annoy me but now I know why it is there
- Can One Hear the Shape of a Drum?
- A Midsummer Knot's Dream
- College Admissions and the Stability of Marriage
- On what I do not understand (and have something to say)
- A Smaller Sleeping Bag for a Baby Snake

11.1 How important is my title?

Browsers on the Internet looking for a paper may read hundreds of titles before they select an Abstract to read. According to one of Britain's top editors, writing good headlines represents about 50% of the skills vital to article writing. For this reason the gurus of research writing tend to dedicate more pages to discussing the importance of the title than they do to any section in the paper itself.

Every word in your title is important. So the key is to devise a title that:

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_11

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1 will immediately make sense to the referee
2 will easily be found by a search engine or indexing system
3 will attract the right kind of readers rather than discouraging them, and will also catch the attention of browsers. Note 'attraction' does not mean resorting to newspaper-like headlines, but simply containing those words that readers in your field would expect to find
4 does not consist of a string of nouns and will be immediately comprehensible to anyone in your general field
5 is reasonably short
6 has a definite and concise indication of what is written in the paper itself. It is neither unjustifiably specific nor too vague or generic

The rules for writing good titles reflect the rules on writing skills in Part 1 of this book.

Note that all the rules relating to titles given in this chapter are also valid for headings, subheadings, and legends / captions. They are also valid for book titles and chapter titles.

11.2 How can I generate a title?

With the advent of chatbots it may seem that you can simply insert your title into the chat window and let the bot generate a title for you. It is certainly worth getting the bot's help, but before you do, think about the following questions:

- Which of my findings will attract attention?
- What is new, different and interesting about my findings?
- What are the 3–5 key words that highlight what makes my research and my findings unique?

On the basis of your answers you should be able to formulate a title.

Now you need to review your title. First underline all the key words. Then in your head put the key words in order of priority. Finally, find a way of shifting the top priority key words to nearer the beginning of the title.

When you have a final title, then I suggest you use a chatbot to see if there might be a better title. To learn how to do this, see 11.17.

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If your paper is not about results but proposes a particular methodology, then your title should encapsulate why your methodology is novel and useful.

The rest of this chapter looks at some tips i) for making your titles more readable, ii) to punctuate it correctly, iii) to ensure that it is grammatically correct.

11.3 Should I try to include some verbs in my title?

Yes, titles without verbs are generally more difficult to understand.

The red title below is almost incomprehensible for a reader.

Educational software specification definitions trends

Trends in defining the specifications for educational software

The blue title contains a verb, prepositions (11.4) and the definite article (11.5), which help to make the meaning much clearer for readers.

Below are some more examples.

Examining narrative cinema fiction and fact boundaries

Examining the boundaries between fiction and fact in narrative cinema

New archaeological research and teaching technologies

New technologies for research and teaching in archaeology

Cultural heritage audiovisual material multilingual search gathering requirements

Gathering requirements for multilingual searches for audiovisual materials in the cultural heritage

Where possible use the *-ing* form of verbs rather than abstract nouns. This will make your title more readable as well as making it 2–3 words shorter.

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The *Specification* and the *Evaluation* of Educational Software in Primary Schools

Specifying and *Evaluating* Educational Software in Primary Schools

Methods for the *Comparison* of Indian and British Governmental Systems in the 19th century

Methods for *Comparing* Indian and British Governmental Systems in the 19th century

A Natural Language for Problem *Solution* in Cross Cultural Communication

A Natural Language for *Solving* Problems in Cross Cultural Communication

Silicon Wafer Mechanical Strength *Measurement* for Surface Damage *Quantification*

Quantifying Surface Damage by *Measuring* the Mechanical Strength of Silicon Wafers

However the above titles would be better if the key words were nearer the beginning of the title, for example:

Educational Software in Primary Schools: a Critical Review

Indian and British Governmental Systems in the 19th century: Methods for Comparison

11.4 How will prepositions help to make my title clearer?

Most titles require prepositions. Prepositions help readers understand the relationships between the other words in the title.

Fast computing machines equation of state calculations

Equation of state calculations *by* fast computing machines [*by* – how something is done]

Depression measuring inventory

An inventory *for* measuring depression [*for* – indicates purpose]

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Antonio Gramsci prison notebooks selections

Selections *from* the prison notebooks of Antonio Gramsci [*from* – the origin of]

Vertical flux of ocean particles

Classical theory of elasticity crack problems

Vertical flux of particles *in* the ocean [*in* – location]

Crack problems *in* the classical theory of elasticity

Reality social construction

The social construction *of* reality [*of* – regarding]

Model dimension estimation

Estimating the dimension *of* a model [*of* – belonging to]

Cancer causes: cancer avoidable risks quantitative estimates

The causes *of* cancer: quantitative estimates *of* avoidable risks *of* cancer

Even if you don't understand the exact meaning of the above titles, the important thing to note is that the use of prepositions helps the reader to understand how the various elements in the title are related to each other. Also note that rewriting a title so that it contains prepositions may involve adding *a/an* or *the*. Such cases are underlined in the titles above.

I have given more examples of the preposition *of* than for the other prepositions because the non-use of tends to create more difficulties for the reader than any other preposition.

Do not worry if you use the same preposition more than once in the same title. For example *of* is used three times in the last title above. This is perfectly acceptable and is not considered bad style in English.

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11.5 Are articles (*a / an, the*) necessary?

Although a title is not generally a complete sentence, it does have to be grammatically correct. This means that it must have articles where necessary, even though this will increase the length of the title.

Survey of importance of improving design of internal systems

A survey of the importance of improving the design of internal systems

A general rule of English is that a countable noun that is in the singular must be preceded by an article. *Survey* is a singular countable noun, so it must be preceded by either *a* or *the*. Another general grammatical feature of English is that if you have the following sequence of words: *noun1* + *of* + *noun2*, then noun1 is preceded by *the*. This is because noun1 is used to specify noun2. This means that we need *the* before *importance* and before *design*.

The last noun in the example above is countable but it is plural (*systems*) and unspecified (we know that the systems are *internal*, but we do not know which *internal systems* they are). In such cases, no article is required.

No *the* is required for uncountable nouns (i.e. *lack, feedback* and *equipment*). In the following three titles *the* should thus be removed.

The lack of protective immunity against reinfection with hepatitis C virus

The feedback and optimal sensitivity

Vibration analysis for the electronic equipment

There are some cases where the use or non-use of *the* changes the meaning of the title.

The factors that determine depression

Factors that determine depression

The first title (*The factors*) gives the idea that the author has made a comprehensive survey of depression and has identified all those factors that lead to depression. This makes the paper sound like the final word on depression, i.e. this is the definitive article on depression.

The second title (*Factors*) is not all-inclusive. The reader will expect to learn about some factors. This makes the paper sound much more modest.

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The difference between *a* and *an* in a title follows normal usage. Use *a* before all consonants, before *eu*, and before *u* when *u* has the sound as in *university* and *unit*.

Use *an* before *a, i* and *o*. It should also be used before *e* except before *eu*, and before *u* when *u* has the sound as in *unusual* and *understanding*. Use *an* before *h* only in the following cases: *hour, honest, honor, heir*. Some authors use *an* before *historical* too.

These rules mean that the following are wrong:

An hybrid approach to X.

An unique approach to Y.

The first title should be *a hybrid* (the *h* in *hybrid* is aspirated). The second title should be *a unique*, because the *u* in *unique* is pronounced like *you*.

Note also the words in italics in the following titles:

GNRA tetraloops make *a U-turn*

The evacuation of the Machault, *an 18th*-century French frigate

An NLP application with a multi-paradigm architecture

u as a separate letter is pronounced *you*, *18th* stands for *eighteenth* (i.e. beginning with an *e*), and *N* is pronounced *en*.

11.6 Will adjectives such as *innovative* and *novel* attract attention?

No. The problem with *novel* and *innovative* is that they give no indication as to how something is novel. For example, what does *novel* mean in the following title?

A novel method for learning English

No one is likely to include the words *novel* or *innovative* when Googling papers in their field. In any case, if your research is not novel then no one would want to read about it anyway. You need to explain to readers what makes it novel. More explicit adjectives to replace *novel* could be: *computerized, guaranteed, high-performance, low-cost, minimal-stress, no-cost, pain-free.*

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Generally speaking, choose adjectives that indicate the unique features of your work, e.g. *low cost, scalable, robust, powerful*. Adjectives such as *reliable* should only be used if work in your field has so far only produced an unreliable system or unreliable results.

11.7 What other criteria should I use to decide whether to include certain words or not?

You can use an Advanced Google Scholar Search to check how frequently a word in your title is used. Under the form 'Find articles' insert your word or combination of words into the 'with the exact phrase' field. Then in the 'where my words occur' field, choose 'in the title of the article'.

If the word you choose gets less than a few thousand returns and it is not a specifically technical then you should check whether the authors:

- are native speakers
- use the word in the same way and in the same kind of context as you do

If the answer to either of the above is 'no', then you need to think of another word.

For example, the title below may make sense in the native language of the author, but when translated into English it sounds rather strange:

A study on the use of oils and colorants in Roman cosmetics: a witness of make-up preparation

The problem word is *witness*, which is here being used to mean *evidence* or *example*. A search on Google Scholar for "a witness" only gives 1,300 returns, which is very low given that the concept of evidence and examples is very common in research. Also, a quick look at the titles in which the term *witness* appears quickly indicates that *witness* is generally confined to a legal context meaning someone who sees something, it thus refers to a human subject whereas *make-up* is inanimate. You can also see words in context on wordnik.com.

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11.8 How should I punctuate my title? What words should I capitalize?

There are two common standards for punctuating titles. Both standards are exemplified in the Factoids at the beginning of this chapter. Check which system is used in your chosen journal.

You can capitalize the first letter of every word except for articles (*a, the, an*), and prepositions and conjunctions:

Describing the Relationship between Cat Bites and Human Depression Using Data from an Electronic Health Record

College Admissions and the Stability of Marriage

Or you can just capitalize the first word.

Aesthetic value of paintings affects pain thresholds

On what I do not understand (and have something to say)

Note that some words require an initial capital letter if they are proper nouns:

Does the fact that there are three different electric plug sizes in Italy indicate the level of chaos in the Italian government or economy?

The above title also indicates that the title can end with a question mark. However no other punctuation is required at the end of the title.

Two part titles may contain a colon:

The causes of cancer: quantitative estimates of avoidable risks of cancer

11.9 How can I make my title shorter?

Titles are often constricted by the number of characters that can be used (check with your journal to see how many words or characters you can use). In some cases you can keep your title as it is but reduce it in length simply by replacing the non-key words with shorter synonyms.

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VERBS: achieve vs gain, calculate, evaluate vs assess, rate; demonstrate, display, exhibit vs show; determine vs fix; facilitate vs ease; guarantee vs ensure; prohibit vs block; require vs need; support vs aid; utilize vs use

NOUNS: advantage vs gain, benefit; examination, investigation vs study; modification vs change

adjectives: accurate vs exact; fundamental vs basic; important vs key; necessary vs needed; primary vs main

- remove redundant words (4.3)
- use verbs rather than nouns (4.5)
- choose the shortest word (4.6)

11.10 How can I make my title sound more dynamic?

Every word (apart from articles and prepositions) included in the title should add significance. The words in *italics* in the titles below rarely add value.

A study of the factors affecting the trihyroxyindole procedure for the analysis of deoxyribonucleic acid

An investigation into some psychological aspects of English pronunciation

In the above titles, the first seven and four words, respectively, give the reader no information. They would be more dynamic and more concise if the initial redundant words were removed.

Factors affecting the trihyroxyindole procedure for the analysis of deoxyribonucleic acid

Some psychological aspects of English pronunciation

Similar words that are often redundant are: *inquiry, analysis, evaluation*, and *assessment.*

If you feel that words such as *study* and *investigation* are essential, then relocate them to the end of the title.

Old age: A study of diversity among men and women

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However, the above title might have more impact in one of the following forms:

Old age: diversity among men and women

What factors affect diversity among men and women in old age?

Will women always live longer than men?

11.11 Can I use my title to make a claim?

Some referees and journals editors do not appreciate authors who use the title to present their major conclusion and thus perhaps overstate the importance of their findings. For example:

The consumption of one apple per day precludes the necessity of using medical services

The above is what is known as a declarative title. It summarizes the author's most important findings, as a complete sentence (i.e. with subject – verb - object). It does so in a way that there seems to be no element of doubt. However, if the author's conclusions are only speculations, then such declarative titles are dangerous. This is because they give readers the initial idea that the issue has been settled and that what the author asserts is now scientific fact.

Such titles are increasingly common in medicine and biology, and may be acceptable if well documented. Such titles also get your paper noticed and potential readers may thus become stimulated into reading your paper. The important thing is to ensure that the title reflects the truth and is supported by the rest of the paper.

Before using a declarative title check with other titles in your chosen journal.

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11.12 Are questions in titles a good way to attract attention?

The titles below highlight that a question can be formulated using an auxiliary (e.g. *does, would, can, will*) and using question words (e.g. *why, when, what, which, who*).

Does the ocean-atmosphere system have more than one stable mode of operation?

If homo economicus could choose his own utility function, would he want one with a conscience?

Why Do Some Countries Produce So Much More Output Per Worker Than Others?

When do foreign-language readers look up the meaning of unfamiliar words? The influence of task and learner variables

What do bosses do? The origins and functions of hierarchy in capitalist production

Who would have thought it? An operation proves to be the most effective therapy for adult-onset diabetes mellitus.

Titles with questions also work particularly well for abstracts submitted to conferences. They are generally much more informal and because of their question form they immediately get readers thinking about what the answer might be. They can also be original and fun, as highlighted by the last title. They thus tend to stand out from other titles and are more likely to attract attention.

11.13 When is a two-part title a good idea?

In some cases the first part poses a question, expressed in an informal manner to attract the attention of readers. The second part gives a more technical description of the content of the paper. For example:

What do bosses do? The origins and functions of hierarchy in capitalist production

Who would have thought it? An operation proves to be the most effective therapy for adult-onset diabetes mellitus

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In other cases the second part acts as an explanation for the first part:

Consequences of erudite vernacular utilized irrespective of necessity: problems of using long words needlessly

The role of medicine: dream, mirage or nemesis

Telling more than we can know: Verbal reports on mental processes

Given that two-part titles are much less common than other titles they generally attract more attention, and like questions work well for abstracts submitted to conferences.

11.14 How should I write a title for a conference?

Try to ensure that your abstract will not just be enticing for the editorial board, but also that it will be suitable for publishing in the conference handbook / proceedings. Your title should be interesting but not too obscure or too colloquial / witty. It can be less 'technical' than a title for a journal, and may contain a question (11.12) or two parts (11.13) - the first part is technical, and the second part contains a more informal interpretation of the first part. Or vice versa - the first part is more fun, and the second more serious, as in the example below.

PhD students have miniscule brains. Extremely low IQ levels in PhD student brains as an indicator of wasted tax-payer funding of pointless research

11.15 What is a running title?

Many journals require that a running title, also known as a running head or short title, be included in submitted manuscripts. This shortened form of the main title, usually cited at the top of each published page of an article, serves to guide readers browsing a print journal, shuffling loose printed pages, or toggling between multiple papers in PDF form. The running head may also be used in RSS feeds and mobile applications instead of the frequently more unwieldy main title.

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Requirements for running titles vary between journals, but generally, they must be 50-60 characters long at most, often including spaces. To achieve brevity, these titles typically include abbreviations, even if the main title does not (or cannot, based on journal guidelines). Articles (the, a, an) may also be omitted to conserve characters, and wordy phrasing, including filler phrases, should be minimized. However, if the main title is brief enough, it can function as the running head as well.

Unlike for the manuscript title itself, being catchy is not a priority for a running title. Rather, because it is so abbreviated, clarity and accuracy should be priorities. Some also suggest that as much content as possible should be preserved from the main title, although in practice, this approach is not widespread; authors instead tend to include only what they deem most important to highlight.

The following is an example of effective title abridgment, drawn from a recently published article (Lambert et al., 2013):

Manuscript title: *Dendritic Cell Immunoreceptor Is a New Target for Anti-AIDS Drug Development: Identification of DCIR/HIV-1 Inhibitors* (117 ch with spaces, 103 ch without spaces)

Running title: *Inhibitors of DCIR Limit HIV-1 Infection* (40 ch with spaces, 35 ch without spaces)

The authors combined a few strategies here to reduce the title length by two thirds, applying the abbreviation "DCIR," omitting articles ("DCIR" instead of "the DCIR"), and focusing on what they judged to be the central concept (the limitation of HIV-1 infection by the inhibitors, rather than the novelty of the target, the application in drug development, or the identification process). Of course, this task may be easier for papers with a narrower and more descriptive focus, such as review articles.

I would like to acknowledge the professional manuscript information of American Journal Experts and their excellent online resource (https://www.aje.com/en/author-resource) from which this entire subsection, written by Michaela Panter, was taken verbatim.

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11.16 Is using an automatic spell check enough?

No, it isn't! The following titles contain spelling mistakes and typos (e.g. inverted or missing letters) that spell checkers may not always find. In the examples below, the misspelt words are in italics.

Incidence of *Hearth* Attacks and *Alzeimer's* Disease among Women *form* East Asia

An atmospheric tape *reorder*: rainfall analysis *trough* sequence *weighing*

For example, in the above examples Microsoft Word did not flag *form*, *trough* and *weighing*, which should have been *form*, *through* and *weighting*. However, Word did flag *hearth* (*heart*), *reorder* (*recorder*) and *Alzeimer's* (*Alzheimer's*).

Given that you may be unable to see your own spelling mistakes, it is a good idea to show your title to several other people, firstly to get them to check the spelling but more importantly to get some feedback on whether your title is clear and explicit enough.

In a research paper, poor spelling gives the idea that you did not make the effort to check your paper. By extension, if you did not check your spelling there is a chance you did not check your data. Perhaps for this reason referees seem obsessed with finding and reporting spelling mistakes. If they find more than one or two this may cause them to recommend that publication of your paper should be delayed until the paper has been thoroughly proof read.

Another major reason for checking the spelling in your title, is that if a key word (e.g. Alzheimer's) is misspelled or not punctuated correctly (note the apostrophe before the *s*), then search engines will not be able to find it.

11.17 Using a chatbot to generate or improve your title

You can use a chatbot to improve a title that you have already formulated, using the following prompt: *Improve this title for an academic paper: ...* . For a regularly updated list of prompts to use with ChatGPT: https://tefldiscussions.com/ai-prompts/. The titles in red below are ones that I have cited earlier in this chapter and which I have explained are NOT good titles. The bot's versions (in blue) are considerably better, but could possibly be shortened.

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Cancer causes: cancer avoidable risks quantitative estimates

Quantifying Avoidable Risk Factors for Cancer: A Comprehensive Assessment of Cancer Etiology

Examining narrative cinema fiction and fact boundaries

Exploring the Boundaries Between Fiction and Reality in Narrative Cinema: A Critical Analysis

Educational software specification definitions trends

Emerging Trends in Defining Educational Software Specifications: A Comprehensive Review

Bots can also generate a title from scratch. You simply insert your abstract, and give the bot the following instruction: *Generate a title for this academic abstract*.

Below is the abstract for a fictitious paper that was originally entitled: *Six key strategies to a meaningful life: the non-believer's worldview.*

With no hope of an afterlife, atheists may have difficulty rationalizing their purpose on earth. With the aim of understanding the coping mechanisms of non-believers, we interviewed 150 UK-born couples (125 mixed, 25 same sex; average age 46) who had happily cohabited for more than 15 years. Interviewees were asked ten simple questions regarding their attitudes to the meaning of life. Our results revealed that there are six key strategies in an atheist's pursuit of a happy and meaningful existence: keep everything simple, have fun, cultivate a sense of community, delight in the wonder of nature, find time for creativity, and help other people through frequent acts of kindness. Atheists that implement a combination of these six strategies were found to be as well-equipped as other non-believers to deal with the death of close ones, health problems, financial difficulties, and bad luck.

Below are the various titles the bot generated. Note: You can press 'regenerate' and the bot generates another title.

Making Meaningful Lives: Coping Strategies of Long-term Cohabiting Atheist Couples in the UK

Strategies for a Meaningful Existence: A Qualitative Study of Coping Mechanisms Among Atheist Couples in the UK

Living without the Afterlife: Coping Strategies and Pursuit of Happiness Among Long-term Cohabiting Atheist Couples in the UK

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You can also insert your Conclusions and see what title the bot will generate. This might help you to further improve your title, by incorporating elements from both the abstract- and the conclusions-generated titles. Here is what the bot generated from the Conclusions to the above abstract (19.5):

Living Satisfying Lives as Atheists: The Key Strategies and Implications for Education and Interfaith Dialogue

Note that the bot:

- generally generates a title with more detail than your original, and also tends to use a two-part structure
- suggests various different permutations using the same or different words
- suggests titles that you might otherwise not have thought of
- can generate several titles from which you can choose the 'best' parts and reassemble them into a new title
- may use information that you have previously inserted in the same chat – so begin a new chat if you want the bot not to be influenced by what it has already 'learned' from you

However, the bot will NOT tell you that:

- you have inserted vague words (particularly adjectives)
- your key words are not in the most effective order
- your title goes against the style required by your journal

Finally, the title generated by the bot very much depends on the quality of the abstract that you insert.

11.18 Summary: How can I assess the quality of my title?

- You need to check that your title is:
  - in correct English - in terms of syntax, vocabulary, spelling and capitalization
  - understandable (no strings of nouns)
  - eye-catching and dynamic (through effective use of vocabulary and even punctuation)
  - sufficiently and appropriately specific
  - reflects the content of your paper
  - expressed in a form that is acceptable for a journal

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- Consider using a chatbot to help you generate a more specific and understandable title.
- You can check the syntax and the level of understandability by consulting with a native speaker. Generally speaking titles that contain at least one verb and one or more prepositions tend to be much easier to understand.
- You can check the vocabulary and spelling using Google Scholar. Remember that an automatic spell check is not enough.
- The best way to decide whether it is eye-catching and sufficiently specific is to prepare several titles (including ones in two parts, and in the form of a question) with various levels of specificity and ask colleagues to choose their favorite.
- Unless you get someone to read the whole paper for you, you are probably the best judge of whether your title reflects the actual content of your paper. If it doesn't, the referees will probably tell you.

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Chapter 12

Abstracts: Standard types

Unusual abstracts

In their paper *Can apparent superluminal neutrino speeds be explained as a quantum weak measurement*? the authors wrote possibly the shortest abstract ever: *Probably not*.

*****

Not quite as short as the Abstract above is this*: A zipper-entrapped penis is a painful predicament that can be made worse by overzealous intervention. Described is a simple, basic approach to release, that is the least traumatic to both patient and provider*. The key words to the paper are: *zipper; foreskin/penile skin; bone cutter*.

*****

A paper published in 2014 was co-authored by four economists all with the same surname. Their 3-sentence abstract was: *We explore the phenomenon of coauthorship by economists who share a surname. Prior research has included at most three economist coauthors who share a surname. Ours is the first paper to have four economist coauthors who share a surname, as well as the first where such coauthors are unrelated by marriage, blood or current campus.*

*****

God does not play dice. He flips coins instead. So began Carlos Mochon's abstract to his paper entitled *Quantum weak coin flipping with arbitrarily small bias.*

*****

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_12

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The Socceral Force by Norbert Bátfia began: *We have an audacious dream; we would like to develop a simulation and virtual reality system to support the decision making in European football (soccer).*

*****

The abstract to a paper entitled *Pressures produced when penguins pooh–Calculations on avian defecation* ends as follows: *Whether a bird chooses the direction into which it decides to expel its faeces, and what role the wind plays in this, remain unknown.*

*****

The last line of the abstract to a paper entitled *Fractal Analysis of Deep Sea Topography* is: *No attempt has been made to understand this result.*

12.1 What is an abstract? When should I write it?

An Abstract is like a mini paper. It summarizes all the sections of your paper. It will be judged in isolation from the accompanying paper. Abstracts are sometimes called Summaries.

Abstracts are found before a full article in a journal, standalone in databases of abstracts, and in conference programs.

The structure of abstract and its length will depend on the journal or conference, as well as on your field of research. Make sure you read their instructions to authors (12.6) before you begin writing.

An Abstract generally answers at last the first three of the following questions, and generally in the following order. You can use the answers to these questions to structure your Abstract.

- Why did I carry out this project? Why am I writing this paper?
- What did you I, and how?
- What were my results? What was new compared to previous research?
- What are the implications of my findings? What are my conclusions and/or recommendations?

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Although most Abstracts shouldn't concentrate on the methods but more on the results, some scientists (e.g. chemists, physicists, biologists) who are presenting some new instrumentation may want to focus not on what they found, but on what the benefits of their apparatus are and how well it performs.

To decide what to include in your Abstract, go through your paper and highlight what you consider to be the most important points in each section.

Finally, remember that an Abstract is NOT an Introduction to your paper, it is a summary of ALL your paper. This means that the Abstract (and Title) should be written or rewritten when you have completed the main text.

12.2 How important is the Abstract?

Incredibly important.

Editors may decide whether or not to send your paper for review exclusively on the basis of your Abstract.

Reviewers will probably read your Abstract first before reading any other parts your paper. Ensuring that they have a positive reaction is essential. If they don't like your Abstract they may simply stop reading and reject the paper, rather than wasting their time reading and evaluating the rest of the paper. In fact, a poor Abstract is very often the sign of a poor paper.

Research has proved that what you experience first will condition how you perceive what comes after - we tend to judge everything in comparison with something else, i.e. something that came before. This means that your Abstract is clearly written, you will set up a positive expectation amongst your readers - they will think that if the Abstract is easy to understand then the rest of the paper is likely to be easily understood too. This will certainly encourage them to read on.

Your title and your Abstract will generally be the only parts of your paper that are available online at no cost. So when a potential reader finds your paper, they will use your Abstract to help them decide whether to buy the full version of your paper and / or read the rest of the paper.

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Note: The journal's "instructions to authors" will tell you whether:

- there is a limit on the word count
- you can use the active or passive, and whether you can use *we*
- references are allowed
- you can use a note-like form in some items of a structured abstract (12.5), but full sentences are required in the results and conclusions)
- you can change and/or delete the headings in a structured abstracts

12.3 Where is the Abstract located? What are 'highlights'?

A typical first page of a research paper for publication in an international journal contains the following headings, generally in this order:

1. Title
2. Abstract
3. Highlights
4. Key words

Some journals require you to write between three and five bullet points reporting the core findings of your paper. These are known as 'Highlights'. The 'instructions to the author' (1.4) will tell you how many bullets and how many characters per bullet. For example, the highlights of the paper in 12.5 could be:

► We surveyed grammatical, structural and lexical errors in 120 medical articles.

► Badly written articles correlated with a high rejection rate.

► On equal scientific merit, a badly written article will have less chance of being accepted.

► Rejection may be due to language errors, though editors may not identify this as the cause.

► We suggest the introduction of standardized guidelines in scientific writing.

The Highlights are generally located immediately below the Abstract and immediately above the Key Words.

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12.4 How should I select my key words?

In most journals, directly below the Abstract there is a list of key words. These are for indexing purposes and will help your paper be identified more easily and thus cited more frequently.

Ensure you check with your journal's 'instructions to authors' to see how many key words to include, and whether or not these can also be words that appear in the title of your paper.

You can follow the steps below to help you select your key words.

1. Read through your paper and underline the 'technical' terms that you've used most frequently.
2. Check that the terms you've listed in 1) match the key technical terms used in your specific field
3. Consider including variants / alternatives of some of the terms, and also their acronyms
4. Include common abbreviations of terms (e.g., HIV).
5. Meet the criteria of indexing and abstracting services (see en.wikipedia.org/wiki/Indexing_and_abstracting_service)
6. Type your chosen keywords into Google Scholar (or a similar search engine). Do the results match your topic?

There is a lot of mystery around how Google and other search engines use key words when indexing websites and articles. In any case it makes sense to have key words in your abstract (and title too) because it forces you, the author, to decide what words in your paper really are important. The key words are also the words that readers are looking for in their initial search and then when they actually scan your abstract. General consensus seems to be to not repeat the key words more than three times in the abstract. This can be tedious for the reader. More importantly, 'keyword spamming' may lead to the web page being rejected by the search engine.

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12.5 What is a structured abstract?

Below is a structured abstract entitled *Language and publication in Cardiovascular Research articles* and was published in the international journal *Cardiovascular Research*. The author, Robert Coates, talks about the reasons why papers are and are not accepted for publication.

Read the abstract and think about:

- how it is structured and what information is contained in each of the four parts
- how relevant Dr Coates' research is in terms of you writing a manuscript for publication in an international journal
- what kinds of English language errors you tend to make when writing papers and how you could avoid them

BACKGROUND: The acceptance rate of non-mother English tongue authors is generally a lot lower than for native English tongue authors. Obviously the scientific quality of an article is the principal reason for publication. However, is editorial rejection *purely* on scientific grounds? English mother tongue writers publish *more* than non-mother-tongue writers—so are editors discriminating linguistically? We therefore decided to survey language errors in manuscripts submitted for publication to *Cardiovascular Research* (CVR).

METHOD: We surveyed language errors in 120 medical articles which had been submitted for publication in 1999 and 2000. The language 'error' categories were divided into three principal groups: grammatical, structural and lexical which were then further sub-divided into key areas. The articles were corrected without any knowledge of the author's nationality or the corrections made by other language researchers. After an initial correction, a sample of the papers were cross-checked to verify reliability.

RESULTS: The control groups of US and UK authors had an almost identical acceptance rate and overall 'error' rate indicating that the language categories were objective categories also for the other nationalities. Although there was not a direct relationship between the acceptance rate and the amount of language errors, there was a clear indication that badly written articles correlated with a high rejection rate. The US/UK acceptance rate of 30.4% was higher than for all the other countries. The lowest acceptance rate of 9% (Italian) also had the highest error rate.

DISCUSSION: Many factors could influence the rejection of an article. However, we found clear indications that carelessly written articles could often have either a direct or subliminal influence on whether a paper was accepted or rejected. On equal scientific merit, a badly written article will have less chance of being accepted. This is even if the editor involved in rejecting a paper does not necessarily identify language problems as a motive for rejection. A more detailed look at the types and categories of language errors is needed. Furthermore we suggest the introduction of standardized guidelines in scientific writing.

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As highlighted on the website of the U.S. National Library of Medicine:

> Structured abstracts have several advantages for authors and readers. ... ... They assist health professionals in selecting clinically relevant and methodologically valid journal articles. They also guide authors in summarizing the content of their manuscripts precisely, facilitate the peer-review process for manuscripts submitted for publication, and enhance computerized literature searching.

If you are a medical researcher you can learn more about the different types of abstract from the British Medical Journal's website: www.bmj.com/about-bmj/resources-authors/article-types/researchStructured abstract

Below are some more typical headings for structured abstracts in the medical field:

Background / Context / Purpose - Methods - Results / Findings - Conclusions

Context - Aim / Objective - Design - Setting - Patients (or Participants) - Interventions / Treatment - Main Outcome Measure(s) - Results - Conclusions

Context - Objective - Data Sources - Study Selection - Data Extraction - Results - Conclusions

And here are some from other disciplines.

From a journal of vegetation sciences:

Question - Location - Methods - Results - Conclusions

From an economics journal:

Purpose - Design / Methodology / Approach - Findings - Practical implications - Originality / value - Keywords - Paper type

A structured abstract could really be used for any piece of research, given that all research should have 1) a context, 2) an aim, 3) a method, 4) some results, 5) an interpretation of the overall meaning, possible applications, and ideas for how the research might be continued.

So even if your journal does not require a structured abstract, you will certainly write a much more effective abstract if you at least include the same information in the same order (though you can invert points 1 and 2)

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12.6 What style should I use: personal or impersonal?

There are four possible styles for writing abstracts and papers:

STYLE 1	I found that x = y.
STYLE 2	We found that x = y.
STYLE 3	It was found that x = y.
STYLE 4	The authors found that x = y.

The style you use will depend on your discipline and on the requirements of the journal. Using the first person singular (Style 1), is generally only found in humanistic fields where the author's opinions are often outlined. Here is an example - an abstract from a paper entitled *International scientific English: Some thoughts on science, language and ownership.*

STYLE 1 The intention of this paper is to raise some questions about the 'ownership' of scientific English. Its author is a native speaker of English and a teacher of scientific English, but it aims its arguments at the international scientific community communicating in English. The paper is deliberately somewhat provocative in parts in an attempt to raise some questions about 'scientific English' which *I think* are important but which have not been faced to date.

Style 2 is found in all fields. Here is an example of the beginning of an abstract from a physics paper entitled *Tumbling toast, Murphy's Law and the fundamental constants*.

STYLE 2 *We investigate* the dynamics of toast tumbling from a table to the floor. Popular opinion is that the final state is usually butter-side down, and constitutes prima facie evidence of Murphy's Law ('If it can go wrong, it will'). The orthodox view, in contrast, is that the phenomenon is essentially random, with a 50/50 split of possible outcomes. *We show* that toast does indeed have an inherent tendency to land butter-side down for a wide range of conditions.

[to tumble = to fall and turn; butter-side down = people in Britain often put butter on one side of their toast]

Style 3 is also very common and many journals insist on this style. For an example of this style see the abstract in 13.2. For the problems associated with this style see 6.4.

Style 4 is the least common style. Here is an example of the beginning of an abstract from a fascinating psychology paper entitled *Unskilled and unaware of it: How difficulties in recognizing one's own incompetence lead to inflated self-assessments*

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STYLE 4 People tend to hold overly favorable views of their abilities in many social and intellectual domains. *The authors* suggest that this overestimation occurs, in part, because people who are unskilled in these domains suffer a dual burden: Not only do these people reach erroneous conclusions and make unfortunate choices, but their incompetence robs them of the metacognitive ability to realize it. Across 4 studies, the authors found that ...

To give you an idea of the different effects that using different styles can have, below is an abstract written in a personal style. See 12.8 for a version of the same abstract with an impersonal style.

We have developed an analytical model which predicts the relationship between the number of times a 5G cellular phone battery is recharged, the length of time of each individual recharge, and the duration of the battery. We validated this model by comparison with both experimental measurements and finite element analyses, and it shows strong agreement for all three parameters. The results for the proposed model are more accurate than results for previous analytical models reported in the literature for 5G cell phones. The new model can be used to design longer lasting batteries. It can also lead towards further models that can predict battery failure.

The benefits of using *we* rather than the impersonal passive form are:

- it is more reader friendly and easier to follow
- it is more direct and dynamic
- the word count is reduced and sentences tend to be shorter
- it is much easier for you, the author, to write

I suggest that if given the choice you opt to use *we / our*.

12.7 How should I begin my Abstract?

When you read an advertisement for a product it never begins *The objective of this advertisement is to convince you to buy ...* Instead advertisers go straight to the point. Abstracts are like advertisements for your paper.

You want your abstract to stand out so that there will be a better chance someone will notice it and read it. If you begin your abstract with commonly used phrases (by both native and non-native English speakers) such as *This paper deals with ... The aim of this paper ... This article explores ... We report ...* you are not differentiating yourself from the others. In fact, some journals advise against using such expressions.

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Below are four examples taken from abstracts (the first and third are fictitious) in very different fields. Readers of the red versions have to wait for up to 15 words before reaching a key word that enables them to understand the potential relevance of the topic. They are forced to read words and expressions that they have read thousands of times before and which add absolutely no value to the abstract. In the blue versions, the reader learns either immediately or very quickly how the author has filled the knowledge gap.

In this paper we present the design and development of a *highly innovative* software application, Transpeach, which allows *mobile phone users* to use their own native language when speaking to someone of another native language. The prototype version enables a Japanese mobile phone user ...

Transpeach extends automatic translation from written to oral communication. This software allows, for instance, a *Japanese mobile phone* user to talk to a Greek counterpart in Greek, likewise the Greek's words are automatically translated into Japanese.

In the first blue sentence the author manages to combine both the background (automatic written translation) with the new information (automatic oral translation). The words *highly innovative* have been removed. More concrete examples are given, which reflect what the prototype does.

We present a procedure for the analysis of the content of organic materials present in archeological samples. The procedure allows the identification of a *wide variety* of materials within the same micro sample.

Archeological samples used for identifying organic materials are by necessity extremely small. We have found a way, which *we believe* is the first of its kind, to accurately identify *glycerolipids, natural waxes, proteinaceous, resinous and polysaccharide* materials within the same micro sample.

In the blue version, the abstract now begins with *archeological samples*, so that the reader can immediately understand the general topic of the paper. The vague phrase *a wide variety of materials* has been replaced with concrete examples of these materials. This makes the blue version slightly longer, but it has a much stronger impact.

In this article we conduct an exploration of the crucial role of the invention of the steam engine in the Industrial Revolution, and specifically the modified version created by James Watt, the Scottish inventor born in 1736. However, *we contend that the merit* for the success of the steam engine should be ...

James Watt's modified steam engine is widely acknowledged as paving the road to the Industrial Revolution. But was this Scottish inventor really the brains behind the steam engine? We *contend that Henry Wallwork*, a little-known Mancunian foundry entrepreneur, should be given more credit for ...

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The revised version gets straight to the point without the initial redundancy of the red version, including the birth date of James Watt that serves no purpose for the reader and has thus been removed.

Several authors have highlighted the high yields and low environmental impacts associated with the cultivation of perennial rhizomatous grasses (PRGs).

Perennial rhizomatous grasses (PRGs) tend to have a high yield combined with a low environmental impact.

In the blue version, redundancy has been removed. The abstract now begins with the key word (PRGs), which are also the main topic of the paper. This leads to an opening sentence - describing the background to the research - that is both clear and incisive.

It is essential that the first sentence of your Abstract attracts a reader's attention.

In *recent* years, there has been *increasing attention* on the *influence* of social networks on society, and a *considerable output* of research has focused on investigating the *underlying dynamics*. To further understand this topic, we have developed a model that simulates how information is exchanged on social networks, with a particular focus on how users and algorithms handle information from external sources.

The first sentence above has nothing new to say and is very vague, particularly the words in italics. It would be much better to begin the abstract with *We have developed a model* and then provide the background information either later in the Abstract, or better, in the Introduction.

Here is a similar example, again the first sentence is just frustrating for the reader – it is a series of abstract words containing no concrete information.

One of the key important challenges in conducting research on how information spreads is to understand how the mechanisms that support the diffusion of information affect emergent phenomena of the opinion evolution process of individuals belong to a network. In this work, we conduct a computational investigation of how the competition of posts for users' attention influences the ...

Moral of the story: re-read the first sentence of your abstract and either i) delete it, ii) make it much more specific. If your first sentence adds no value for the reader, they may simply stop reading. This issue is covered in detail in the next subsection.

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12.8 How much background information should I give?

An Abstract is NOT an introduction to your paper. This means that context setting should NEVER take up more than 25% of the whole abstract, as it probably contains information that the reader already knows. The background information in the abstract (fictitious) below represents about four fifths of the total abstract - this is too much.

In the last few years 5G cellular batteries have become increasingly popular in the telecommunications and computer industries. Many authors have studied the various features of such batteries and noted that the lifetime of a 5G cellular battery, in particular those used in the most recent generations of mobile phones, may be subject to the number of times the battery is recharged and how long it is charged for. In addition, it has been found that there is no adequate analytical model to predict this lifetime. Such an accurate model is necessary in order for producers and consumers alike to be able to predict how long the batteries will last and also, in some cases, how they can be recycled. In this work, an analytical model is developed which describes the relationship between the number of times a battery is recharged, the length of time of each individual recharge, and the duration of the battery.

Your readers want new information, not old information. Remember that the reader may be a referee who has to read hundreds of abstracts to decide which to include for a conference or in a journal. He / She wants to know immediately what the topic is and will be negatively affected if forced to wait several lines before understanding this. Of course, you can (and should) give more background details in the Introduction.

Also, the extract above does not describe the methodology or the results, nor what can be concluded from the model the authors have developed. A much better solution is given below:

(1) The lifetime of a 5G cellular phone battery may be subject to the number of times the battery is recharged and how long it is charged for. To date, there has not been an adequate analytical model to predict this lifetime. (2) In this work an analytical model is developed which describes the relationship between the number of times a battery is recharged, the length of time of each individual recharge, and the duration of the battery. (3) This model has been validated by comparison with both experimental measurements and finite element analyses, and shows strong agreement for all three parameters. (4) The results for the proposed model are more accurate than results for previous analytical models reported in the literature for 5G cell phones. (5) The new model can be used to design longer lasting batteries.

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Below is the structure of the above abstract and the questions it aims to answer. The numbers refer to the numbers in the abstract.

1. The problem that this paper is trying to resolve set in the context of the current situation. Why did you carry out your project and why are you writing this paper? What gap in the current knowledge do you hope to fill?
2. New solution given by authors of the paper. What is the innovative contribution of your work? What did you do and achieve? What makes it different from previous research?
3. Validity of the model. Does it really do what you say it does?
4. Results. What is new compared to previous results?
5. Implications and future work. What does this all mean? What are your conclusions and recommendations? What do you plan to do next?

This abstract only has a minimal amount of background information (two lines). This background information is given so that reader can understand the context of the author's research.

12.9 What tenses should I use?

The most commonly used tenses in all kinds of abstracts are the PRESENT SIMPLE (*we show*) and the PAST SIMPLE (*we showed*).

The author of the "tumbling toast" abstract (Style 2) uses the PRESENT SIMPLE to:

- describe the contents of his paper (*we investigate, we show*).
- describe the common opinion that he is trying to question (*the phenomenon is essentially random*)
- refer to what he did during his experiments (*We show that toast does indeed have an inherent tendency*)
- give his conclusions - not shown here - (*Murphy's Law appears to be an ineluctable feature of our universe*)

In fact he uses only the PRESENT SIMPLE . Even though his research has already been done (thus the investigation is complete), he uses the PRESENT SIMPLE because he wants to make his abstract sound more dynamic and his conclusions more convincing. However, in the paper itself he uses the PAST SIMPLE to describe what he did and found.

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In the "incompetence" abstract (Style 4), the authors use the PRESENT SIMPLE to:

- talk about a well-known situation (*people tend to hold overly favorable views*)
- explain their opinion on this well-known situation (*the authors suggest that ...*)

They then use the PAST SIMPLE to describe what they did / achieved and what conclusions they reached (*the authors found that ..*). This is the standard way to use tenses in abstracts.

The author of the "scientific English" abstract (Style 1) ends his abstract by using the PRESENT PERFECT (*which have not been faced to date*). You can use the PRESENT PERFECT and the PRESENT PERFECT CONTINUOUS when you describe a situation that began in the past and is still true now. This is typical when you are giving the context / background.

In the last few years there *has been* considerable interest in ...

Since 2024 attention *has focused* on ...

To date, there *has not been* an adequate analytical model ...

For more than a decade data analysts *have been developing* new ways to ...

Note: the underlined parts highlight the past-to-present timeframe. For example, *in the last few years* means a situation or action that began a few years ago and is still true today. *To date* means so far in the history of this particular branch of study.

Some authors also use the PRESENT PERFECT (in the active or passive) to describe what they achieved during their research.

We *have found / devised / developed* a new approach to X. We *have demonstrated / proved / validated* the effectiveness of this approach by ...

A new approach to X *has been devised.* The effectiveness of the approach *has been demonstrated ...*

Important note: Beginning your Abstract with a phrase such as *In the last few years* or *Recently there has been* is both uninspiring and unnecessary.

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12.10 Why and how should I be concise?

The abstract below comes from a paper that was rejected by the reviewers because it contained a lot of redundant words (NOT redundant information).

Tomato (*Solanum lycopersicum L.*) is a worldwide-cultivated vegetable crop which is affected by many viruses that cause significant economic losses whose detection and identification is of critical importance to plant virologists in general and, in particular, to scientists and others involved in plant protection activities and quarantine and certification programs.

The reviewer informed the author that 'the paper must be rewritten completely'. The reviewer gave an example of what kind of 'rewriting' was required - basically he wanted all the redundancy removed (he managed to remove 40% of what the author had written!):

Tomato (Solanum lycopersicum L.) is affected by many viruses that cause significant economic losses. Their detection and identification is of critical importance in plant protection and quarantine, and in certification programs.

In addition to being heavily redundant, the red version contains one long 50-word sentence. The blue version has the same number of words but contains three sentences.

Moral of the story:

- If the reader or reviewer sees a lot of redundancy in the abstract, then they will probably stop reading.
- In your abstract every word must add VALUE.
- Show respect for your readers by not forcing them to read words and phrases that are the result of your lazy writing.
- Increase the chances of your reviewers accepting your manuscript by simplifying the review process.
- Be obsessed about removing redundancy. I guarantee that redundancy has a very negative impact on all types of writing.

To learn how to reduce redundancy, read the summary of Chapter 4.

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12.11 What should I not mention in my Abstract?

You should try to avoid:

- background information that is too generalist for your readers
- claims that are not supported in the paper
- terms that are too technical or too generic - this will depend on your audience
- definitions of key terms
- mathematical equations (unless the whole paper revolves around these equations)
- generic quantifications (e.g. *many, several, few, a wide variety*) and the overuse or unjustified use of subjective adjectives (e.g. *innovative, interesting, fundamental*).
- unnecessary details that would be better located in your Introduction, such as the name of your institute, place names that readers will not have heard of
- references to other papers. However, if your whole paper is based on an extending or refuting a finding given by one specific author, then you will need to mention this author's name.

12.12 How can I ensure that my Abstract has maximum impact?

There are three main ways to do this. Firstly, put the information in the best possible order (see Chapter 2 and Chapter 3). Secondly, highlight the importance of what you are saying (Chapter 6). And thirdly, be as concise as possible (Chapter 4). Although you should certainly mention the limitations of your research at some point in the paper, Abstract is designed to 'sell' your research, so considering delaying mentioning the limitations until the Discussion (18.10).

12.13 How can chatbots help me improve my Abstract?

The use of chatbots is dealt with in Chapter 9 and in full detail in the the companion volume *AI-Assisted Writing and Presenting in English* (2024). Below are just some of the advantages of using the chatbot. For a regularly updated list of prompts to use with ChatGPT: https://tefldiscussions.com/ai-prompts/.

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You can prompt the bot to:

- expand one or two sentences into an entire abstract
- correct your English
- reduce the number of words
- change style (e.g. from personal to impersonal)
- paraphrase all or parts of the text
- rewrite it from a different point of view (you would probably only do this for fun, e.g. *rewrite this abstract as if it was written by Shakespeare*)
- suggest a Title based on the Abstract

Note: you should NEVER use a chatbot to generate a text for you, and then submit it to your professor or for publication without making any changes and without acknowledging that a bot was involved in creating the text.

12.14 Summary: How can I assess the quality of my Abstract?

To make a self-assessment of your Abstract, you can ask yourself the following questions.

- Have I followed the journal's instructions to authors? Have I followed the right structure (i.e. structured, unstructured) and style (*we* vs passive)?
- Have I covered the background / context, the research problem / aim - the gap I plan to fill, the methods, the results, and possibly the implications.
- Is everything mentioned in the Abstract also mentioned in the main text? Is the information consistent with what is presented in the paper?
- Whenever I have given my readers information, will it be 100% clear to them why they are being given this information? (You know why, but they don't.)
- Can I make my Abstract less redundant? If I tried to reduce it by 25% would I really lose any key content?
- Have I used tenses correctly? present simple (established knowledge), present perfect (past to present background information), past simple (my contribution)?

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- Have I checked the spelling? Have I shown it to other people so that they can find any typos that I may have missed?
- Have I chosen my keywords carefully so that readers can locate my Abstract?
- Have I shown it to a colleague who is not familiar with the details of my research to see how much they can understand and can identify the value of the research?

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Chapter 13

Abstracts: Particular types

Factoids

- Around a billion people travel across borders every year. Around 75% travel from a non-English speaking country to another non-English speaking, yet communicate in English while they are there.
- Non-native English-speakers now outnumber native ones 3 to 1. There are more Chinese children studying English—about 100 million—than there are Britons. Around two billion people study English, and three billion people speak it. Eighty percent of the electronically stored information in the world is in English.
- Probably the biggest selling book in English language teaching history is Raymond Murphy's *English Grammar in Use*, first published in 1985 and subsequently making its author a multi-millionaire.
- Nearly one-third of the citizens of the USA feel that it is not too important or not important at all to speak a second language. The number of children and students studying languages in the UK has dropped considerably, and less money is being invested in language research. Apparently, this lack of language skills has led to a lack of interpreters and is making it harder for international anti-terrorism squads, for example, to work together.
- Misunderstandings are as common with people who share the same language as they are in exchanges between people who are talking in a second language. A study in the UK revealed that as many as 6 out of 10 conversations in the workplace lead to a significant degree of misunderstanding.
- Native speakers often listen at only 25% of their potential. The other 75% is made up of ignoring, or forgetting, distorting, or misunderstanding what the other person has said. Concentration rises above 25% if the interlocutor thinks what he / she is hearing is important or interesting. But in any case it never reaches 100%.

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_13

13.1 I have never written an Abstract before. I don't know where to start. Could I chatbot help me?

If you are a humanities student, you may find it problematic to write an abstract in the standard ways outlined in Chapter 12. In fact, what a humanities abstract should contain does not appear at first sight to be as straightforward as writing an abstract for a science journal. Perhaps you did not follow any recognized method for conducting your research, or you got not results in a strict sense (you were not actually aiming to get 'results'), or you were not pursuing a particular objective.

But a bot may be able to help you.

You can ask the bot to generate an abstract based on only your first two sentences. I gave a chatbot this instruction:

Please expand this Abstract for a psychology review journal: "We studied three types of selfish acts that are characteristic of certain types of citizens. Investigations into selfish behavior are essential to understand why for example, people do not differentiate their trash, or insist on driving large high-fuel consumption cars".

The bot generated an entire abstract:

This study aimed to investigate three specific types of selfish acts commonly observed among citizens, and their implications for environmental sustainability. Selfish behaviors, such as failing to differentiate trash or driving high-fuel consumption cars, have become increasingly prevalent, and it is critical to understand the underlying motivations driving such behaviors. By exploring the reasons behind these actions, we can develop effective strategies to encourage more sustainable practices among the general population. The findings of this study can contribute to the larger discourse on sustainable living and inform policy decisions aimed at promoting more environmentally conscious behavior.

You will NOT want to use the abstract produced, because i) it is unethical to submit work that you did not create yourself, ii) it is very generic.

However, it may give you an idea of how it could be constructed. For instance, the example above implies that you need to detail what the results and wider implications of your study might be.

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However, you should NOT just accept what the bot thinks are the implications. Chatbots are infamous for generating erroneous information. You MUST also check with your professor and colleagues that the bot's suggestions are in some way useful.

In any case, you should NEVER use a chatbot to generate a text for you, and then submit it to your professor or for publication without making any changes and without acknowledging that a bot was involved in creating the text.

13.2 Social and behavioral sciences. How should I structure my abstract? How much background information?

Here is an abstract from a fictitious social sciences paper entitled *Is it Time to Leave Him?* written several years ago by one of my PhD students, Estrella Garcia Gonzalez from Spain. This abstract is designed to prove to you that even if you don't work in a purely scientific field, you can still write an abstract that would fulfill the criteria expected by journals in most formal sciences and natural sciences (as defined by Wikipedia - http://en.wikipedia.org/wiki/Science).

By *sitting-zapping sessions* Estrella means sitting like a zombie in front of the television and constantly changing channels. Note: the numbers in brackets are mine.

(1) Three red flags were identified that indicate that the time to leave him has come. These red flags are: five burps per day, two sitting-zapping sessions per day, and five games on the PlayStation with friends per week. (2) A large number of women have doubts about the right moment for leaving their partner. Often women wait in hope for a change in their partner's habits. (3) One hundred couples were analyzed, recording their daily life for six months. Women were provided with a form to mark the moments of annoyance recorded during the day. Burps, sitting-zapping sessions and games on the PlayStation with friends produced the highest index of annoyance. (4) The probability of eliminating these habits was found to be significantly low when the three red flags had been operative for more than three months. (5) Thus, these numbers provide a good indication of when the time to leave him has come. With these red flags, women will no longer have to waste their time waiting for the right moment.

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Below is a series of instructions for writing an abstract based on Estrella's structure. Again, the numbers refer to the numbers in the abstract.

1. Begin the abstract with one or two sentences saying what you did plus one key result, i.e. begin with information that the reader does NOT already know
2. Introduce the background by connecting in some way to what you said in your introductory sentence. The concept of leaving him is introduced in (1) and then referred to again in (2)
3. Use the background information (which the reader may or not already know) to justify what you did, and outline your methodology (and materials where appropriate)
4. Provide some more information on your results
5. Tell the reader the implications of your results

13.3 I am a historian. We don't necessarily get 'results' or follow a specific methodology. What should I do?

If you analyze history abstracts, and other abstracts from humanistic disciplines, they still have a structure that is similar to a scientific abstract.

You have a primary objective (e.g. a theory or perspective that you would like to share, test, analyze or question), a design to your research, some methods and procedures that you used, some outcomes from your research that support your theory/perspective, and some conclusions or implications derived from these outcomes.

Abstracts from social and behavioral sciences tend to devote more space to background issues and context setting. The 'thesis' is often formulated as a series of questions that inform the reader about what issues will be dealt with in the paper.

In any case your abstract should include the following:

- background information
- your aim and its importance
- your contribution and its value
- what you looked at
- your conclusions and implications

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Here is a fictitious abstract from a researcher interested in the history and evolution of languages.

(1) The Quaker movement was founded in the mid 17th century by George Fox. One of the practices used by this rebellious religious group was the use of 'plain speech' and 'simplicity'. This involved addressing all people with the same second person pronoun, in the words of Fox: 'without any respect to rich or poor, great or small'. The modern use of 'you' in the English language (in 10th century England there were 12 forms of 'you') is thus attributed to Fox's egalitarian movement. (2) Was this use of 'you' for addressing all kinds of people, regardless of their social status, specifically initiated by Fox? Or was it simply a part of an organic unplanned process in the English language of ridding itself of unnecessary devices and formalities? Are some languages more dynamic than others? And does this depend on how 'controlled' they are by official prescriptions? (3) By analyzing 50 English texts from 1012 to 2012, I show that English has successfully eliminated all accents on words, simplified punctuation use, virtually made the subjunctive redundant, and reduced the average sentence length by more than half from around 35 in the convoluted style of the 18th century to 14 words today. (4) Our findings show that English has the potential for being democratic, concise yet profound, and simple to understand. (5) I believe that this has implications for those languages, such as French, Italian, Korean and Turkish, that have conservative academies for safeguarding the 'purity' of their language.

The above abstract covers the following elements, which typically appear in humanistic abstracts. The numbers below refer to the numbers in the abstract.

1. Background information - there tends to be more context setting in humanistic than in scientific abstracts, and this may take up even 50% of the text.
2. Gap in the knowledge - here the author challenges the accepted view on the topic. Using the question format, the author tells the reader what areas of the topic he plans to address. Questions create variety in an abstract.
3. Methodology and results - the author provides some brief information on the data he used to get his findings.
4. Conclusions
5. Implications - having implications in some way justifies why the author did his work, it gives the work relevance, it shows that the work makes a real contribution and was not just carried out for the author's own personal interest

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13.4 I am writing an abstract for a presentation at a conference. What do I need to be aware of?

What do organizers of conferences want to see in an abstract? To answer this question you need to think about why conferences are organized. As an academic, you may think that the primary aim is to bring people from the scientific community together to talk about the state of the art. But conference organizers also want to the conference to make money so that they will be able to propose new editions in future years. They will use any profits made (from accommodation, food, services etc) to finance other events or research, and very importantly they are aware that if the conference is a success it will look good on their CV and further their academic career.

To ensure that all these aims are achieved, they want:

- to appeal to as many people as possible (not just researchers in one very specific field) by inviting presenters who can make their work relevant to a broad audience (however for workshops the participants are likely to be much more homogeneous)
- results to be original and interesting
- speakers to be of a high quality

This means that your Abstract should:

- really fit the conference theme, which in some cases might have a broad spectrum
- highlight the level of innovation
- contain interesting results; thus proposals and future plans tend to be less interesting unless they are already attracting a large community
- be quickly understood and appreciated by reviewers
- be understandable by non-experts - plenary talks in particular have to be more general - the audience can follow the parallel sessions if they wish to be informed of more technical details

The fifth point is becoming increasingly important. A lot of public money is spent on funding research. Those who allocate such funds need to see some kind of ‘return’ on their ‘investment’. For example, imagine you have been given funds for conducting high-energy physics, funders will want you to publicize your results not only in highly specialized physics journals and conferences, but also in related areas such as electronics. Funders want to see how your methods and findings can be applied in other areas.

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Conferences are the perfect opportunity to present your work to a wider audience, thus your abstract must:

- not be too technical (i.e. not take for granted that all readers will understand the importance of the work or the technical jargon used to describe it) - note that certain words have different meanings for scientists with different backgrounds
- give examples that the majority of your audience will understand even if they don't have the same background info as you have
- highlight how your innovations and results could potentially be transferred to other disciplines to solve other problems

In high profile conferences you need to write your abstract as if you were targeting a journal paper. The review process is strict and it's a real privilege if you are accepted.

If the conference that you plan to go to is not in its first edition, you can look at abstracts from the previous editions to see their style and tone. In any case, the rules for writing the abstract itself are the same as for a journal, though your style may be slightly more informal.

Finally, show your abstract to someone outside your research group - if they can't understand it, it probably means you need to rewrite it more clearly.

13.5 How do I write an abstract for a work in progress that will be presented at a conference?

Conferences are generally planned up to two years in advance. When you answer the call for papers, your research may not yet be complete, but nevertheless you think that the conference would be a good way to get feedback on your progress.

Ensure that your abstract contains such words and phrases *currently, will explore, seem to indicate*, and *we hope*, i.e. you need to highlight that this is ongoing research NOT research that has already been included.

You should include your initial results in a way that will make the audience curious to read your future paper to find out whether these results were later confirmed.

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Your abstract should encourage conference attendees to come and hear you rather than going to a parallel session. If you don't have any results at all, you should either consider going to a later conference when you have something more conclusive to say, or recount what you expect your results to show.

13.6 What is an Extended Abstract?

An extended abstract is like a mini research paper, whose ideas and significance can be understood, according to William Pugh, professor of Computer Science at the University of Maryland, "in less than an hour". Professor Pugh says:

> An ideal submission should have a reviewer intrigued within the first 5 minutes of reading, excited within 15 minutes and satisfied within 45 minutes. If your abstract fails any of these tests, it might be rejected no matter how good the research is. Committee members may spend more than 30-45 minutes on your abstract, but you shouldn't rely on it.

Some journals will not require an extended abstract to contain complete details of methodology / proofs and future work. However, they will expect comparisons with related work, the key aspects of the methodology, results and some kind of discussion of the results.

If you are short of space, then you may be allowed to use an appendix, but bear in mind that many reviewers simply won't have the time to read appendices.

13.7 What is a video abstract? How can I make one?

A video abstract, as explained by expert Karen McKee, is a brief description of a technical paper in which you explain your work on camera, physically demonstrate your methods, use animations or simulations to illustrate concepts, and/or discuss the implications of your findings. By using video and other multimedia, you can explain your work in a way that the print article cannot. This approach provides a richer, more accessible and more diverse experience for the readership.

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There are several reasons for using a video abstract:

- have more flexibility to describe and explain your work
- gain greater visibility for you and your research, particularly if you post your video on the internet (e.g. YouTube)
- increase readership for your printed article
- exploit search engines which tend to rank video high in relation to text-based descriptions, so a video abstract can make your work more visible and accessible to fellow researchers searching for papers on that topic

The British Medical Journal (BMJ) encourages authors to create video abstracts lasting up to four minutes to accompany accepted research articles. The BMJ's website (www.bmj.com/about-bmj/resources-authors/article-types/research) gives some great advice:

Video abstracts enable authors to go beyond the constraints of their written article to personally explain the importance of their work to *The BMJ*'s global audience. ... In the simplest kind of video abstract the author(s) talks directly into the camera and, perhaps, presents a slideshow. In the interest of maximizing engagement and visibility, however, we encourage authors, where appropriate, to combine footage of themselves with other relevant visual and audio material (such as animations, video clips showing how the study was conducted and any intervention was delivered, audio, still photographs, figures, infographics).

When using materials from previously published work ensure you have the relevant permissions.

13.8 My aim is to have my paper published in *Nature*. Is a *Nature* abstract different from abstracts in other journals?

In *Nature* an abstract is called a 'summary paragraph', which is structured as follows. You can also use such a summary paragraph if you are submitting your manuscript to another journal:

- One or two sentences providing a basic introduction to the field, comprehensible to a scientist in any discipline.
- Two to three sentences of more detailed background, comprehensible to scientists in related disciplines.
- One sentence clearly stating the general problem being addressed by this particular study.

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- One sentence summarizing the main result (with the words "here we show" or their equivalent).
- Two or three sentences explaining what the main result reveals in direct comparison to what was thought to be the case previously, or how the main result adds to previous knowledge.
- One or two sentences to put the results into a more general context.
- Two or three sentences to provide a broader perspective, readily comprehensible to a scientist in any discipline, may be included in the first paragraph if the editor considers that the accessibility of the paper is significantly enhanced by their inclusion. Under these circumstances, the length of the paragraph can be up to 300 words.

13.9 I know I need to end my Abstract with a view of the big picture. How can I use a chatbot to suggest to me what the implications of my research are?

I have taught many PhD students how to write a scientific paper in English. Very often I am surprised that the students don't seem to have a clear idea of what their research might mean to the wider community, i.e. they don't really understand the importance of what they are doing.

Look back to the abstract on selfishness in 13.1. Based on an input of only two sentences, a chatbot managed to generate the following:

Our results shed light on the complex interplay between individual beliefs, social norms, and environmental attitudes, which influence the likelihood of engaging in selfish behaviors. By providing a more nuanced understanding of these dynamics, our findings can inform interventions aimed at promoting more sustainable behavior among individuals. Ultimately, our study underscores the importance of addressing the psychological dimensions of selfish behavior to achieve meaningful progress towards sustainability goals.

The bot, fortunately, has not generated any false information, but has a provided a template that you can use to formulate your ending for your Abstract.

Another alternative is to feed the chatbot your key results and to request the bot to generate some conclusions.

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Below is my input to the chatbot. Note this is a totally fictitious work based on observations of what color people use when highlighting a text as an indicator of how selfish they are: fuchsia – not very selfish, grey – very selfish.

Write a text to conclude an abstract based on the following information. "Of the 765,854 people videoed, approximately two thirds demonstrated no selfish behaviors. The majority of these used fuchsia when highlighting a text. Of the remaining one third, who predominantly opted for grey when highlighting , 70% were caught on camera dropping litter; 15% double parking, 10% playing loud music, but only 5% queue jumping. An equal number of men and women were shown to drop litter, but in the three other cases there were four times as many men than women adopting the selfish behaviors".

Here is the bot's output:

In conclusion, this study reveals that while queue jumping is commonly seen as a selfish behavior, other actions such as dropping litter, double parking, and playing loud music in public places are far more indicative of selfishness. Among the 765,854 people observed in public spaces, two-thirds demonstrated no selfish behavior, with a majority of them using fuchsia to highlight their texts. However, approximately one third of the individuals, who primarily used the color grey, were observed displaying various selfish behaviors. The most common of these were dropping litter and double parking, with a higher proportion of men than women engaging in these behaviors. These findings provide insight into the prevalence of different selfish behaviors in public settings and can be used to inform interventions aimed at promoting pro-social behavior.

If you use a bot to generate a structure like the one above you should find it considerably easier to write the last part of your Abstract.

> Note: you should NEVER use a chatbot to generate a text for you, and then submit it to your professor or for publication without making any changes and without acknowledging that a bot was involved in creating the text.

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13.10 How do journal editors and conference review committees assess the abstracts that they receive?

Apart from investigating the scientific merit and validity of the research that is the topic of your abstract, reviewers will be answering the questions below. These questions, along with the answers, were either lifted or adapted from an excellent document by Professor Bill Pugh of Maryland University.

Some of the answers may not be strictly related to your field but are nevertheless still likely to be valid.

Is the work a significant advance over previous work in the area, by the same authors or others? The abstract should give a clear description of the advantages offered by the new technique over previous techniques. Simply describing an interesting new way of doing something that could be done as simply and efficiently by previous techniques won't get an abstract accepted. The best abstracts give a clear description of what their results allow that couldn't be done previously and why that is significant. Examples and measurements are great for this.

If the work involves a specialized application, does it make a more general contribution? Would the resulting paper be useful to people not interested in your specific application?

Does the abstract address the obvious questions raised by the research? For example, if an abstract claims to describe "an efficient, practical algorithm" for something, it should give empirical timings, asymptotic analysis or both. If the techniques described require solving a problem that is NP-Complete or undecidable in general, the abstract should discuss the difficultly of solving the problem. It may be that in practice the problems that arise in the author's application can be solved efficiently; but if the abstract doesn't discuss it, the committee doesn't know if the author is even aware of the potential problem.

Is the abstract well presented and understandable? Is the abstract too long? The word limit is to encourage authors to write abstracts that can be absorbed quickly, not to save trees.

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13.11 Summary

- Consider using a chatbot to help you write the first draft of your abstract.
- Remember that even if you are a humanities student your abstract should still follow one of the structures outlined in Chapter 12.
- Before submitting an abstract for a conference, think about how such an abstract should differ from a standard abstract: more appealing and original, must fit conference theme, and be understandable by non-experts.
- Consider using the structure suggested by Nature.
- When assessing the quality of your abstract, make sure it answers the following questions:
  - Is the work a significant advance over previous work in the area, by the same authors or others?
  - If the work involves a specialized application, does it make a more general contribution?
  - Does the abstract address the obvious questions raised by the research?
  - Is the abstract well-presented and understandable?
  - Is the abstract too long?

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Chapter 14

Introduction



Far cited

The measure of a paper's importance is often seen in terms of the number of times it is cited in the works of other scientists. In 1964 the Science Citation Index (SCI) was set up. The index is now owned by Thomson Reuters and the expanded version covers 6,500 top journals in 150 disciplines, from 1900 to the present.

*****

To rank in the top 100 of the Thomson Reuters' Web of Science (which holds over 58 million items), a paper needs to have been cited over 12000 times. Many famous papers are not ranked in the top 100. The majority that are ranked, relate to what are now very frequently used experimental methods or software programs.

*****

The record holder in terms of citations (over 305,000) is held by biochemist Oliver Lowry, whose 1951 paper describes an assay to determine the amount of protein in a solution. The area that appears to get the most citations is the laboratory techniques of biologists (six in the top ten most cited papers).

*****

A survey carried out of The Top 100 Most Cited Scientific Papers in the Public, Environmental & Occupational Health Category of Web of Science by Vicenç Hernández-González et al, found that the top ten countries with the most highly-cited articles were: USA, England, Canada, Switzerland, Netherlands, Denmark, Italy, Australia, Spain, France, Sweden and Norway.

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_14

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14.1 What is an Introduction?

The Introduction of your paper is not just a historical summary. It is a constant comparison between what OTHERS have done and what YOU did or are proposing to do.

The Introduction presents the background knowledge that readers need so that they can appreciate how the findings of the paper are an advance on current knowledge in the field.

Citing key papers, but without stating how specifically you build on them, is insufficient. So you need to (i) describe, with at least one sentence, what others have done, as far as relevant for the direction of your paper, and (ii) highlight how your contribution is original and distinguishes itself from previous work. You can do this by:

- listing the shortcomings of previous approaches with a clear analysis of how your proposed approach is an improvement. Match each shortcoming with the advantage that your approach offers
- introduce a new approach, algorithm, procedure, set-up, experiment etc and validate it

A key issue is to make it clear whose work you are talking about: yours or another author's. To learn how to do this see Chapter 6.

14.2 How should I structure it? How long should it be?

An Introduction generally answers the following questions. You can use the answers to these questions to structure your Introduction.

- What is the problem?
- Are there any existing solutions (i.e. in the literature)?
- Which solution is the best?
- What is its main limitation? (i.e. What gap am I hoping to fill?)
- What do I hope to achieve?
- Have I achieved what I set out to do?

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There is no specific length for an Introduction, but I suggest that it should be no more than 20% of the paper. Readers want new information; they don't want to read a history of their topic or have to plough through a lot of detail that they already know. So try to limit what you say to what:

- will help the reader see the context of your research
- is essential for an understanding of the motivation behind your research, what you found, and what your findings mean

If your Introduction is more than a couple of pages, subheadings will make it much more 'digestible' for the reader.

14.3 How does an Introduction differ from an Abstract?

There is some overlap between an Abstract and the Introduction. However, a frequent problem is that authors may cut and paste from their Abstract into their Introduction, which can be very repetitive for readers.

Below are the first two sentences from the Abstract and Introduction from a paper (or 'Letter' as it is called in the journal where this study appeared) entitled *Fragmentation of Rods by Cascading Cracks: Why Spaghetti Does Not Break in Half* by Basile Audoly and Sébastien Neukirch. These sentences highlight the distinct ways that an Abstract and Introduction should be written.

ABSTRACT When thin brittle rods such as dry spaghetti pasta are bent beyond their limit curvature, they often break into more than two pieces, typically three or four. With the aim of understanding these multiple breakings, we study the dynamics of a bent rod that is suddenly released at one end.

INTRODUCTION The physical process of fragmentation is relevant to several areas of science and technology. Because different physical phenomena are at work during the fragmentation of a solid body, it has mainly been studied from a statistical viewpoint [1–5].

The Abstract immediately tells the readers the specific topic of the paper and then what the author's goal is. Instead, the Introduction sets the context in very general terms (Point 2).

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The abstract then continues as follows.

ABSTRACT We find that the sudden relaxation of the curvature at this end leads to a burst of flexural waves, whose dynamics are described by a self-similar solution with no adjustable parameters. These flexural waves locally increase the curvature in the rod, and we argue that this counterintuitive mechanism is responsible for the fragmentation of brittle rods under bending.

As you can see, the Abstract gives no further background information, but highlights what the authors found in their research. An absolute minimum number of words have been used. This gives the Abstract substantial impact by telling readers only what they need to know to enable them to decide whether to read the whole paper. As is standard for Abstracts, no references to the literature are made.

On the other hand about 50% of the rest of the Introduction is dedicated to helping the readers see that the general trend given in the first two sentences is being countered by another line of research. In this case, references to the literature are made. Readers are alerted to the alternative trend by the link word *nevertheless*.

INTRODUCTION Nevertheless a growing number of works have included physical considerations: surface energy contributions [6], nucleation and growth properties of the fracture process [7], elastic buckling [8, 9], and stress wave propagation [10]. Usually, in dynamic fragmentation, the abrupt application of fracturing forces (e.g. by an impact) triggers numerous elementary breaking processes, making a statistical study of the fragments sizes possible. This is opposed to quasi-static fragmentation where a solid is crushed or broken at small applied velocities [11].

The concluding sentence of the Abstract is:

ABSTRACT A simple experiment supporting the claim is presented.

This eight-word sentence is expanded considerably in the Introduction, by describing more about what the experiment consisted in, and the result it gave. Note: the text reported below is the rest of the Introduction in its entirety.

INTRODUCTION Here we consider such a quasi-static experiment whereby a dry spaghetti is bent beyond its limit curvature. This experiment is famous as, most of the time, the pasta does not break in half but typically in three to ten pieces. In this Letter, we explain this multiple failure process and point out a general mechanism of cascading failure in rods: a breaking event induces strong flexural waves which trigger other breakings, leading to an avalanche like process.

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I suggest you use a similar comparison between Abstracts and Introductions taken from your chosen journal, to see:

- what parts from Sects. 11.2 and 11.3 are covered in the Introduction. In the spaghetti paper, Parts 1–8 are condensed into eight sentences, Parts 9 and 10 are not mentioned
- how they are structured differently
- what elements from the Abstract the Introduction expands on
- how sentences from the Abstract are paraphrased in the Introduction
- what information is covered in the Abstract but not in the Introduction, and vice versa
- the relative word counts. This will give you an idea of the proportionate length of the Introduction compared to the Abstract. In the spaghetti paper the Abstract is 116 words, and the Introduction 201 words, so the Introduction is approximately twice as long. This is quite typical

14.4 How should I begin my Introduction?

This section and 14.5 provide an example of the structure of a typical Introduction. It consists of a sequence of ten parts, each with a specific role. Your Introduction will not necessarily include all ten parts and they may not be in the same order.

Your aim is to include only enough background information to allow your reader to understand why you are asking the questions you are, in what context they appear, and why your hypotheses, predictions or expected results are reasonable. It is like a preview to the rest of the paper. Thus nearly every Introduction, irrespectively of the discipline, would incorporate those parts marked with an asterisk (*).

The proportion of space given to each part (particularly with regard to the review of the literature) will obviously vary from discipline to discipline, and from paper to paper. In brackets I have suggested the typical number of sentences that would be required for each part.

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You could begin with one or more of the first four parts listed below.

PART 1 DEFINITION OF THE TOPIC PLUS BACKGROUND (1–3 sentences)

Example:

An XYZ battery is a battery that… The electrodes in an XYZ telephone battery are made of a composite of gold and silver, coated with a layer of platinum. The gold and silver provide structural support, while the platinum provides resilience.

This introductory phrase may not be necessary in your paper. Here the definition of the XYZ battery indicates to the reader that this is the background topic (i.e. the general context) of the paper. This is the place to include notations, technical definitions, and explanations of key words.

The second sentence gives information that readers should already be familiar with and suggests why the topic is important and of interest. It will help readers to understand why you are investigating this area and how you hope to extend the knowledge. It sets the context for the information that will follow in (3), which may be less familiar for your readers. Readers want to quickly learn what the specific topic of your research is, they are much less interested in being reminded how important the general area of research is.

PART 2 ACCEPTED STATE OF THE ART PLUS PROBLEM TO BE RESOLVED (2–4 sentences)*

Example:

The performance of the battery can be strongly affected by the number of times the battery is recharged and the duration of each individual recharge. The battery is subject to three possible failure modes. …

In the example text, XYZ batteries is the general context. The authors now move from this general context to the specific area of their research: XYZ batteries in telephones, and more specifically, the problems inherent in such batteries. This is the gap that the authors want to fill and that the readers should be most interested in. This part should state in simple and clear language exactly what the problem is, why you chose it, and why you claim it is important.

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PART 3 AUTHORS' OBJECTIVES (1–2 sentences)*

Example:

A research program has recently been started by the authors in collaboration with a major battery manufacturer, with the goal of developing new design models for XYZ batteries. Analytical techniques are needed that can predict …

Here the authors outline their major objectives, i.e. how they intend to fill the gap. Parts 6 and 7 (see next page) could be incorporated here. This part also serves as a transition into the review of the literature.

PART 4 INTRODUCTION TO THE LITERATURE (1-3 sentences)*

Example:

Computational techniques have been extensively applied to the study of the lifetime of XYZ batteries, in particular with regard to the number of times a battery is charged. However, little research to date has focused on the length of each individual recharge.

This introduces the background literature that the authors intend to refer to in order to motivate their particular research. It makes a reference to current insufficient knowledge of the topic.

This may be in a separate section with its own heading: Review of the Literature – see Chapter 15.

14.5 How should I structure the rest of the Introduction?

PART 5 REVIEW OF THE LITERATURE (from a few sentences to a few pages)*

Example:

Recent research has occurred in the field of laptop and jPud batteries. Evans [15] studied the lifetime in 5G jPud batteries. Smith [16] and Jones [18] found that … However their findings failed to account for ...

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This part reviews the literature in the author's precise field. As in the previous part, it often draws attention to problems that have still not been solved. For example, you may think a particular study did not investigate some necessary aspect of the area, or that the authors failed to notice some problem with their results.

You only need to describe what is necessary for the specific purposes of your paper. Much of this literature will then be used for comparative purposes in the Discussion.

The length of the literature review (i.e. Parts 4 and 5) ranges from a paragraph to several pages.

For more details on the Review of the Literature see Chapter 15.

PART 6 AUTHORS' CONTRIBUTION (1–2 sentences)*

Example:

To the best of our knowledge there are no results in the literature regarding how the length of each recharge impacts on the silver and gold in the electrodes.

Here the authors make a very clear statement of how what they describe in the paper represents an advance on current knowledge (i.e. the knowledge outlined in parts 2, 4 and 5).

PART 7 AIM OF THE PRESENT WORK (1–2 sentences)*

Example:

The aim of the present work is to construct a model to perform a comprehensive investigation of the effect of recharging on the electrodes, and to find a new proportion in the amount of metals used. The assumptions of Smith [16] and Jones [18] are used as a starting point …

This statement of the goal to be reached is essential in any Introduction. It should be in a separate paragraph and expressed so that the referee (and readers) are 100% clear about the objectives of your research and the expected outcome. You will need to tell readers what method you used and possibly why you chose this method.

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PART 8 MAIN RESULTS OF THE PRESENT WORK (1–4 sentences)

Example:

The results of the model are encouraging and show that …

Although your main results will be given in other sections of your paper (typically in your Abstract, Results, Discussion and Conclusions), many authors also announce them here to show how the background situation plus their contribution have led to particular results.

PART 9 FUTURE IMPLICATIONS OF THE WORK (1–2 sentences)

Example:

This new model will be able to …

Below is an analysis of Parts 5–11.

Some authors prefer to delay mentioning implications to the Discussion or even to the Conclusions. However, mentioning implications here gives readers an instant idea of the possible importance of your work, which may be useful for them as they read the rest of the paper.

PART 10 OUTLINE OF STRUCTURE (3–4 VERY SHORT SENTENCES)

Example:

Section 2 introduces the concept of …

This may not be necessary if the structure of your paper is completely standard for your chosen journal, and thus readers will already know in what order the various elements of your research will be presented. See 14.8 on how to write the structure.

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14.6 What tenses should I use?

The PRESENT SIMPLE is generally used to begin the Introduction in order to describe the general background context, i.e. what is known already.

The PRESENT PERFECT is then used to show how the problem has been approached from the past until the present day.

The physical process of fragmentation *is* relevant to several areas of science and technology. Because different physical phenomena are at work during the fragmentation of a solid body, it *has mainly been studied* from a statistical viewpoint [1–5].

Persistence *is* an attribute valued by many. Persistence *has most often been studied* in terms of cultural differences.

Use the PRESENT PERFECT again to refer to ongoing situations, i.e. when authors are still investigating a particular field. Even though specific past dates are mentioned in in the two examples below, these dates are part of a series of dates that describe situations that researchers are still working on today and will continue in the future.

This means that PAST SIMPLE cannot be used in any of these three cases.

Since 2016 there have been many attempts to establish an index [Mithran 2017, Smithson 2018], but until now no one has managed to solve the issue of ….

As yet, a solution to Y has not been found, although three attempts have been made [Peters 2001, Grace 2014, Gatto 2018]..

Note the underlined words. These are adverbials of time that are typically used with the PRESENT PERFECT because they indicate something that began in the past (i.e. when research first began in this area) and continues into the present. Other typical adverbials are: *to date, so far*.

The PRESENT PERFECT is also used when talking about research that was carried out at some indefinite time in the past, or when the moment it was carried out is of no relevance for the purposes of the present paper:

It *has been shown* that there is an inverse relation between the level of bureaucracy in a country and its GDP.

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Although in the example above there is no reference, the author is implying that the 'inverse relation' was not found by him/her, but by another author. In any case, it is always advisable to put a reference. If the present tense had been used ('it is shown') then the reader would think that the author is talking about the present paper.

Other research [Green, 2023] *has proved* that bureaucracy can have a negative impact on incentivizing companies to adopt environmental measures.

The sentence above indicates a case where an explicit date is given in the reference (i.e. 2023), but for the author of the present paper it is the finding (i.e. bureaucracy's negative impact) that is the key point rather than the date this finding was reported in the literature.

You must use the PAST SIMPLE when:

- The year of publication is stated within the main sentence (i.e. not just in brackets)
- You mention specific pieces of research (e.g. you talk about initial approaches and methods that have subsequently probably been abandoned)
- You state the exact date when something was written, proved etc.

In the three examples below we are talking about completely finished actions, so the PRESENT PERFECT cannot be used.

The first approaches used a manual registration of cardiac images, using anatomical markers defined by an expert operator along all images in the temporal sequence. Then in 1987, a new method was introduced which …

This problem was first analyzed in 2023 [Peters].

Various solutions were found in the late 2010s [Bernstein 2018, Schmidt 2019].

In all other cases, the simplest solution is to follow the style of the examples below.

Lindley [10] investigated the use of the genitive in French and English and his results agree with other authors' findings in this area [12, 13, 18]. He proved that …

Smith and Jones [11, 12] developed a new system of comparison. In their system two languages are / were compared from the point of view of … They found that ….

Evans [5] studied the differences between Italian and English. He provides / provided an index of.. He highlighted that …

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At the end of the Introduction, the PRESENT SIMPLE is used again when the authors state what they will do in the rest of their paper (*we explain, I hypothesize*).

In this Letter, we *explain* this multiple failure process and point out a general mechanism of cascading failure in rods: a breaking event induces strong flexural waves which trigger other breakings, leading to an avalanche like process.

Because of these findings, I *hypothesize* that subjects with internal attribution styles (as measured by the APCSS), higher levels of perfectionism, and any form of feedback *will show* greater task persistence.

In the first example above, the authors use the PRESENT SIMPLE to report their findings (see the underlined verbs). Not all authors use the PRESENT SIMPLE in this context because a general convention (but not rule) is that when you present your findings you use the past simple - the idea is to use the PRESENT SIMPLE for what is already accepted in the literature, and the PAST SIMPLE for your new contribution.

In the second example, the author uses the FUTURE SIMPLE to talk about his claim / conclusion. This usage of the future tends to be confined to where authors set out to prove a hypothesis, rather than to give hard results.

14.7 How long should the paragraphs be?

Your aim is to allow your reader to quickly digest the background information that forms the basis of your research.

An Introduction should thus not be one long paragraph or a series of very long paragraphs. This problem is typical of Introductions and Discussion, and it is vital to break the paragraph up (3.7).

Your paragraphs should range between 75 and 175 words. Try not to regularly exceed 150 words, but feel free to use fewer than 75 words if you want your paragraph to stand out in order to make a key point.

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The idea of a series of shorter paragraphs (rather than one long paragraph) is that the series highlights the logical progression (argumentation) of what you are saying - how various ideas are connected and follow on from each other.

This gradual buildup of ideas and evidence easily gets lost in a long paragraph

The moments to begin a new paragraph in the Introduction are when you:

- change topic, or you look at a different aspect of the same topic
- move from talking about one step / phase / period to another
- mention another author and this author has a slightly different take (i.e. view, perspective) on what you have been talking about so far
- want to talk about the consequences of what you have just been describing
- talk about the aim of your study / paper
- talk about the structure of your paper

If you print your Introduction you will immediately see the undesirable effect of having long paragraphs. They are not inviting for the reader.

14.8 How should I outline the structure of the rest of my paper?

Typically at the end of your introduction, you will outline how the rest of your paper is organized. Describe this structure as concisely as possible, as in the blue version below, which is half the length of the red version even though it contains the same information.

The paper is structured as follows: in Section 2 *a survey of the works related to X is provided*. In Section 3 the method that we propose for the analysis of X is shown. In Section 4 the tool that automatizes this methodology is presented and in Section 5 its components are described. In Section 6 *the experience in the application of the tool to industrial case studies is reported and discussed* and finally, *in Section 7, conclusions are provided and future works described.*

Section 2 *surveys* the works related to X. Section 3 *outlines* our method for analyzing X. In Section 4 the tool that automatizes this methodology is presented, and in Section 5 its components are described. Section 6 *discusses some industrial case studies* using the tool.

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14.9 My research area is not a 'hard' science. Are there any other ways of beginning an Introduction?

Clearly, not all disciplines would use the structure outlined in the previous subsection, though they would still cover some of the same main points. An alternative, and quite common approach, is to set the context and research goal in a series of questions.

Here is an example from a dissertation entitled *The Effects of Feedback and Attribution Style on Task Persistence* by psychology student Chris Rozek. Persistence means the ability to adhere to a task, to persevere with something rather than giving up.

Persistence is an attribute valued by many. What makes some people persist longer than others? Are internal factors, such as personality traits, or external situational factors, such as feedback, responsible for persistence? Could the answer include a combination of both? These are the questions this experiment attempted to answer.

The general topic is mentioned in the very first word (*persistence*) of a very short sentence (seven words). This enables the reader to immediately focus on and understand the context (corresponding to point 2 in the structure of an Introduction given in 14.4). Within this context, the second sentence, in the form of a question, outlines the issue that Chris plans to address in his paper (point 3). His next sentence poses the typical attributes associated with persistence (similarly to point 5). The question *Could the answer include a combination of both?* hints at what the likely findings are of his paper (point 7). The final sentence highlights that Chris will cover all the aspects he has mentioned so far. His Introduction then continues with a literature review (point 6) and concludes with his final hypothesis (similarly to point 9).

Chris has neatly covered a lot of points typically mentioned in an Introduction. He has achieved this in very few sentences and with a format (questions) that immediately involves the reader by encouraging them to formulate their own answers and thus to continue reading.

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14.10 Can a chatbot help me to write my Introduction?

You can ask a bot to:

- provide you with the key steps in writing a review of the literature
- find references for you (i.e. studies in your field that might be relevant to you). However, the references the bot provides are not always to freely available papers and are not necessarily found on Google Scholar

More usefully, a bot can also give you clues as to what areas your Introduction should cover and how the initial paragraphs could be structured.

I gave a bot the following prompt: *Write me an introduction to a paper entitled: "How perceptions of old age are frequently negatively distorted by people in their 20s."*

Old age is a natural and inevitable stage of life that affects everyone, yet it is often accompanied by negative stereotypes and biases. While many factors contribute to ageism, one of the most pervasive sources is the distorted perceptions of old age among younger individuals. People in their 20s, in particular, are prone to harbor negative views of old age, perceiving it as a time of decline, incapacity, and dependency. These negative perceptions can have serious consequences for older adults, including social isolation, discrimination, and reduced quality of life.

This paper aims to explore the prevalence and causes of negative perceptions of old age among people in their 20s. Through a review of the relevant literature, this paper will examine the ways in which ageism is perpetuated and how it impacts the lives of older adults. Additionally, this paper will explore potential strategies to mitigate ageism and promote positive perceptions of old age. By shedding light on the distorted perceptions of old age, this paper seeks to raise awareness of ageism and inspire action towards building a more equitable and inclusive society for people of all ages.

The bot has basically provided you with a structure:

Paragraph 1: General background details – the problem, the reason, the consequences

Paragraph 2: Aims of the paper and possible implications of the aims being achieved

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The bot generates a highly generic text, which you can use as template to which to add your specific details.

> Note: you should NEVER use a chatbot to generate a text for you, and then submit it to your professor or for publication without making any changes and without acknowledging that a bot was involved in creating the text.

14.11 Summary: How can I assess the quality of my Introduction?

To make a self-assessment of your Introduction, you can ask yourself the following questions.

- Does my Introduction occupy too high a proportion of the entire paper and does it contain too many general statements that are already widely known?
- Are the rationale and objectives are defined? Is it clear what problem I am addressing or trying to solve and why I chose my particular methodology?
- Is the background information all related to the objective of the paper?
- Is it clear what the reader can expect in the rest of the paper (i.e. main results and conclusions)?
- Does my Introduction act as a clear road map for understanding my paper?
- Is it sufficiently different from the Abstract, without any cut and pastes? (some overlap is fine)
- Have I mentioned only what my readers specifically need to know and what I will subsequently refer to in the Discussion?
- Have I been as concise as possible?
- Have I used tenses correctly? PRESENT SIMPLE (general background context, description of what will be done in the paper), PRESENT PERFECT (past to present solutions), PAST SIMPLE (my contribution, though this may also be expressed using the PRESENT SIMPLE or FUTURE SIMPLE).

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Chapter 15

Writing a Review of the Literature



Plagiarism and Ethics

The introduction of AI chatbots in 2022/3 led to panic in both the national and international press, with a huge number of articles about the ethics of university students using bots to write their essays, theses and papers.

*****

According to the website of plagiarism-detector software producers, iThenticate, one in three editors regularly encounters plagiarism, and nine in ten editors say that plagiarism software is effective in revealing cases of plagiarism. iThenticate was launched in 2004, when AI was still in its early stages. Twenty years' later software was being designed that would detect AI 'plagiarism', i.e. texts that had been generated by chatbots.

*****

Many famous authors, artists, musicians and politicians (in their speeches) have been involved in alleged plagiarism, including Dan Brown (author of *The Da Vinci Code*), Alex Hayley (author of *Roots*) Andy Warhol (US artist), Damien Hirst (British artist), John Lennon and George Harrison (of The Beatles), and Led Zeppelin (British rock band).

*****

Wikipedia cites around 20 famous cases of plagiarism in academia, a Rumanian mathematician and computer scientist claims to have published more than 400 papers, a number of which have been shown to be duplicates of papers previously published by other researchers.

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_15

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15.1 I need to write a Review Paper. How should I structure my Abstract? What are my aims and conclusions?

As with all abstracts of all disciplines, when you are writing a review you need to tell your audience what your primary objective is. Given that you will not have space to review every paper in the literature, you should then explain your reasons for selecting certain papers. Your 'results' are your findings drawn from analyzing the literature. Finally, for your review to have a real purpose, you need to state your conclusions and what implications they have for further research in your field.

So your structure is: aim, methodology (selection process), results, conclusions, and implications.

In addition to the guidelines given in this chapter, when you write a Review Paper, you should:

- have a clear idea of why you are writing the Review (i.e. not just to please your professor or to add to your list of publications), and how your readers will benefit
- use lots of headings within each subsection – remember you need to help your readers navigate your paper and quickly find what they are looking for
- use the tenses outlined in 14.6.
- give clear conclusions about what you have found while conducting your review and how people in your field can benefit from your observations
- suggest ways of how your review could be extended to other areas of your field

To get more detailed ideas on how to write a Review Paper see: https://www.ilovephd.com/how-to-write-a-best-review-paper-step-by-step-procedure/?utm_content=cmp-true

For some additional tips, see 15.13 and 15.14.

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15.2 How should I structure my review of the literature in my Introduction to my paper?

A review of the literature generally answers the following questions, and generally in the following order. You can use the answers to these questions to structure your Literature Review.

- What are the seminal works on my topic? Do I need to mention these?
- What progress has been made since these seminal works?
- What are the most relevant recent works? What is the best order to mention these works?
- What are the achievements and limitations of these recent works?
- What gap do these limitations reveal?
- How does my work intend to fill this gap?

Unless you are writing a review paper, then you do not need to cover absolutely all the literature. You need to cover the literature that justifies your research and relates to it - both positively and negatively.

By 'negatively' I mean any literature in your specific field that is not in agreement with your hypotheses, approach, and findings.

Sometimes in the Literature Review you want your readers to note the strong features of your work and the limitations of previous works by other authors. If what you propose has never been done before, you can begin your sentence as indicated by the words in italics below.

As far as we know, there are no studies on …

To [the best of] our knowledge, the literature has not discussed …

We believe that this is the first time that principal agent theory has been applied to …

If you want to mention the limitations of previous works you could adapt one or more of the following sentences:

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Generally speaking patients' perceptions are *seldom* considered.

Results often appear to *conflict* with each other …

So far X *has never been applied* to Y.

Moreover, no attention has been paid to …

These studies have *only* dealt with the situation in X, *whereas* our study focuses on the situation in Y.

To learn more about how to highlight your contribution and discuss the limitations of others see Chapter 6 and Chapter 7, respectively.

15.3 How should I begin my literature review? How can I structure it to show the progress through the years?

Below is an extract from the Introduction to a paper entitled *The Effects of Feedback and Attribution Style on Task Persistence* where psychology student Chris Rozek begins his review of the literature (see 14.9 for how he begins the Introduction).

Persistence has most often been studied in terms of cultural differences. Blinco (1992) found that Japanese elementary school children showed greater task persistence than their American counterparts. School type and gender were not factors in moderating task persistence. This left culture as the remaining variable.

Heine et al. (2001) furthered this idea by testing older American and Japanese subjects on responses after success or failure on task persistence. Japanese subjects were once again found to persist longer (in post-failure conditions), and this was speculated to be because they were more likely to view themselves as the cause of the problem. If they were the cause of the problem, they could also solve the problem themselves; although, this could only be accomplished through work and persistence. Americans were more likely to believe that outside factors were the cause of failure.

These cultural studies hinted that task persistence may be predictable based on attribution style. A later experiment showed that attribution style and perfectionism level can be correlated with final grades in college-level classes (Blankstein & Winkworth, 2004).

The first sentence of the first paragraph introduces the main topic (cultural differences), and the rest of the paragraph briefly reviews a major study on this topic. The implications of this study (culture as the remaining variable) are summarized at the end of the paragraph.

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The first sentence of the second paragraph then moves on to the next (in chronological terms) major study. Chris summarizes Heine's work in a way that involves the reader: he uses the verb *speculated* and then continues the next sentence using *if* which gives an example of this speculation.

The first sentence of the third paragraph summarizes the findings of the first two paragraphs in order to introduce some more recent findings.

Note also his use of tenses. In his first sentence, which is a very general overview, he uses the PRESENT PERFECT. Then when he talks about the work of specific authors and makes a summary of each step in the chronology of the literature he uses the PAST SIMPLE.

Chris's structure is thus:

1. introduction to topic
2. support from the literature
3. mini summary
4. introduction to next topic. And so on.

This technique works very well because it tells a story - it is a logical build up to the reason behind Chris's investigation that readers can easily follow. In fact, the final sentence to his Introduction begins: *Because of these findings, I hypothesize that …* Chris has gradually prepared his readers for the focus of his work: his own personal hypothesis regarding persistence.

15.4 What is the clearest way to refer to other authors? Should I focus on the authors or their ideas?

There are various styles for making reference to other authors. The four styles below contain the same information, but the focus is different.

STYLE 1 *Blinco [1992] found* that Japanese elementary school children showed …

STYLE 2 *In [5] Blinco found* that Japanese elementary school children showed …

STYLE 3 A *study* of the level of persistence in school children *is presented by Blinco* [1992].

STYLE 4 A greater level of persistence has been noticed in Japan [*5*].

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In Style 1, the author, Blinco, is given as much importance as what he (i.e. Blinco) found. You might choose this style for one of three reasons: (i) it is simply the easiest style to use and the most readable for authors, (ii) you may want to focus on the author more than what this author found, (iii) you may want to compare two authors (e.g. *While Blinco says X, Heine says Y*).

Style 2 is similar to Style 1, but in this case perhaps you are talking about more than one paper by Blinco, so the paper is the most logical first element in the sentence.

In Style 3, what Blinco found is more important than the fact that Blinco found it. This is a very typical style, but inevitably involves using the passive, which then leads to longer and heavier sentences.

In Style 4 Blinco is not mentioned at all, but only a reference to his paper in parentheses.

The style you use will depend on your journal's "Style Rules", but is likely to contain an element of flexibility. In fact, Chris Rozek's Introduction in 15.3 he uses two styles:

Heine et al. (2001) furthered this idea by testing …

… can be correlated with final grades in college-level classes (Blankstein & Winkworth, 2004)

He does this to:

- Change the focus from author to findings
- Create variety for the reader

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15.5 How can I talk about the limitations of previous work and the novelty of my work in a constructive and diplomatic way?

Sometimes in the Literature Review you want your readers to note the strong features of your work and the limitations of previous works by other authors. If what you propose has never been done before, you can begin your sentence as indicated by the words in italics below.

As far as we know, there are no studies on …

To [the best of] our knowledge, the literature has not discussed …

We believe that this is the first time that principal agent theory has been applied to …

If you want to mention the limitations of previous works you could adapt one or more of the following sentences:

Generally speaking patients' perceptions are *seldom* considered.

Results often appear to *conflict* with each other …

So far X *has never been applied* to Y.

Moreover, no attention has been paid to …

These studies have *only* dealt with the situation in X, *whereas* our study focuses on the situation in Y.

To learn more about how to highlight your contribution and discuss the limitations of others see Chapter 6 and Chapter 7, respectively.

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15.6 How can I reduce the amount I write when reporting the literature?

The phrases in italics in the red sentences below are phrases that need to be expressed more concisely. If you fill your Introduction with redundant words and phrases, the reader will quickly get bored and may even decide to stop reading.

S1 Long sentences *are known to be* characteristic of poor readability [Ref].

Long sentences are a characteristic of poor readability [Ref].

S2 *In the literature* the use of long sentences *has also been reported* in languages other than English [Ref].

Long sentences are not exclusive to English [Ref].

S3 The use of long sentences *has been ascertained* in various regions of Europe during the Roman period [Ref].

Long sentences were used during the Roman period in various regions of Europe [Ref].

S4 The concept of author-centeredness *has been suggested as playing* a role in the construction of long sentences [Ref].

Author-centeredness may play a role in the construction of long sentences [Ref].

S5 *Several authors have proposed* that in scientific writing the occurrence of a high abundance of long sentences *is* correlated to ... [Ref].

In scientific writing the occurrence of a high abundance of long sentences may be correlated to ... [Ref].

Nearly all the words in italics in the red versions could be removed. This is because the reader knows from the reference (*[Ref]*) at the end of the sentence that you are discussing another author's work or one of your previous papers.

However, if you do remove the words in italics, you still have to indicate whether something is known to be true (S1-S3), or is simply a suggestion or a proposal (S4 and S5). For things that are known to be true today (S1 and S2) you can use the PRESENT SIMPLE, and for things that are known to be true regarding the past (S3) you can use the PAST SIMPLE. To indicate that something has been suggested or proposed, you can use *may* (S4 and S5). Because you have put the reference, your use of *may* indicates a general feeling in the community and not exclusively your feeling.

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15.7 In my review I need mention what other authors have written. How should I do this?

It is perfectly normal to copy phrases from other people's papers. However, these phrases must be generic. In fact, such phrases may even help you to improve your English.

Let's look at what you can paste from another paper.

Here is an example from the literature review of a very interesting paper entitled *International scientific English: Some thoughts on science, language and ownership* by Alistair Wood of the University of Brunei Darussalam. In the extract below Wood talks about different styles of scientific writing around the world and how non-native authors may be at a disadvantage with respect to native authors.

Let's imagine that you work in the same field of research as Wood. I have highlighted phrases in italics that would be perfectly acceptable to paste into your own paper. In fact, these phrases are completely generic.

In fact there is some cross-linguistic contrastive research *to suggest that* the foreigner is *at a disadvantage*. *Even where* the grammar and vocabulary *may be perfectly adequate*, *it seems to be the case tha*t a non-native *may tend* to transfer the discourse patterns of her native language to English. *It has been suggested, for example, that* Asian languages such as Chinese, Japanese and Korean have different patterns of argument to English [3]. Thus *one study found that* those Korean academics trained in the United States wrote in an 'English' discourse style, while their colleague who had trained and worked only in Korea, with *a paper published in the same anthology*, wrote in a Korean style *with no statement of purpose* of the article and a very loose and unstructured pattern *from the English point of view* [4]. *More generally* Hinds *has put forward* a *widely discussed position that* Japanese has a different expectation *as to the degree of* involvement of the reader *compared to* English, with Japanese giving more responsibility to the reader, English to the writer [5].

It might be objected though that this is relevant only to languages and cultures which *differ greatly to* English. *However*, research on German *has shown* that German academic writing in the social sciences has a *much less linear structure* than English, *to the extent that* the English translation of a German textbook was criticized as haphazard or even chaotic by American reviewers, *whereas* the original had received no such reviews on the European continent [6]. Academic respectability in English *is evidenced by* the appropriate discourse structure but in German *by the appropriate level of abstraction* [7]. *Similarly*, academic Finnish texts *have been shown to* differ in the way they use connectors and previews and are much less explicit than English in their *drawing of conclusions*. Spanish also has *a similar pattern* [8]. English, *therefore, would seem to be a more* 'writer-responsible' language *than at least some other* European languages.

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Note how none of the phrases in italics contain unique information. The phrases could be used in many other contexts.

15.8 How can I quote directly from other papers?

If you use any of the parts of Wood's text that are not in italics without any acknowledgement you are committing plagiarism. Plagiarism in its simplest terms means cutting and pasting from other studies and papers. It also means taking credit for work that others have done. Plagiarism includes plagiarizing your own work. In fact, some journals stipulate that you cannot use more than five consecutive words from another paper that you have written.

Let's imagine you wanted to quote from the last line of Wood's paper, which concludes as follows:

The owners of international scientific English should be international scientists not Englishmen or Americans.

You can cite the exact phrase or sentence used by putting it in quotations marks. Then reference the author.

As noted by Wood [1997]: "The owners of international scientific English should be international scientists not Englishmen or Americans".

As an alternative to *As noted by Wood [1997]* you could say:

Wood [1997] concludes:

As Wood [1997] states:

As Wood states in his 1997 paper:

In his Conclusions, Wood [1997] writes:

How you make the reference to Wood's paper will obviously depend on your journal's style.

Putting quotation marks ("…") around an unaltered sentence and giving the proper citation for the origin of the work does not technically constitute plagiarism. But it may indicate to supervisors and referees that you have not actually understood what you have written – it is not your own work.

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The following comment comes from Dr Ronald K. Gratz's very useful article *Using Another's Words and Ideas*.

> It is important that you understand the work you are using in your writing. Quoting someone's sentences does not necessarily require this understanding. On the other hand, you must understand the author's meaning if you are going to be able to paraphrase correctly. This is not to say that one should NEVER quote a reference exactly. Exact quotes have value when it is important to give the precise wording used by the original author. It is unacceptable when it is used to make up the bulk of a paper, or of a part of a paper. It is also unacceptable when it is used to avoid the work of putting the ideas into your own words.

However, using quotation marks is acceptable when you are reporting another's author's definition or a philosopher's statement.

15.9 How can I use a bot to help me paraphrase?

I used QuillBot to paraphrase Gratz's comments in the previous subsection:

> It is crucial that you comprehend the source material you use in your writing. This knowledge is not always necessary when quoting someone's statements. On the other hand, in order to paraphrase effectively, you must comprehend the author's intent. This is not to argue that you should never paraphrase something. When it's crucial to provide the exact terminology used by the original author, exact quotes are valuable. When it constitutes the majority of a paper or a portion of a document, it is unsatisfactory. Also, using it as a way to avoid having to put your thoughts into your own words is wrong.

The orange text represents the changes made, the blue is phrases of four or more words that were contained in the original.

The main advantage of QuillBot over some other bots is that it presents its answer in a window next to the original version. This makes it very easy to compare versions.

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Clearly, you can learn a lot by making comparisons. For example, you may find new synonyms (i.e. ones you had not encountered before) for non-key words. ChatGPT can do something similar and is also likely to change the order in which the words appear within a sentence. So with ChatGPT, you will learn ways to restructure your sentence, convert passive sentences into active (and vice versa), how to divide up a long sentence, and how to join up two short sentences.

> Note: you should NEVER use artificial intelligence to paraphrase or transform a text for you without acknowledging the source.

15.10 How to quote from another paper by paraphrasing

Imagine you wanted to paraphrase the following sentence:

S1. It is important that you understand the work you are using in your writing.

Below are some paraphrased versions in order of increasing difference with respect to the original.

S2. It is crucial that you completely understand the works you use in your paper [Gratz 2006].

S3. You must have a clear understanding of the reference papers that you quote from in your own manuscript [Gratz 2006].

S4. If you cite any works by other authors in your own paper, it is vital that you really understand the full meaning of what the other authors have written [Gratz 2006].

S5. Researchers should ensure that they fully grasp the meaning of any of the literature that they cite in their papers [Gratz 2006].

Here is an analysis of the types of changes made in S2-S5. This should help you see the many devices that can be used in paraphrasing.

S2: *Crucial* is a synonym for *important*; *completely* is redundant but is a modification of the original; *work* (singular) vs *works* (plural); the present continuous (*are using*) vs present simple (*use*); *writing* (an – ing form used to indicate an activity) vs *your paper* (a noun). S5 is an example of what Gratz would define as 'unacceptable' (15.8) because it is essentially identical to the original. Nevertheless, the devices used (synonyms, change of tense etc.) are very useful when paraphrasing.

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S3: The concept of *important* (adjective) has been replaced by *must* (a modal verb); *understand* (verb) vs *understanding* (noun); *works you use in your paper* vs r*eference papers that you quote from in your own manuscript* (three synonyms for three nouns).

S4: The order in which the information is presented in the original is reversed.

S5: The major change here is in the way readers are addressed (*you* vs *researchers*), this factor along with the other changes make the sentences almost unrecognizable compared to Gratz's original sentence. However, Gratz is still referenced at the end of the sentence. This is because the concept contained in the sentence still 'belongs' to Gratz.

You may be thinking that paraphrasing is a pointless exercise particularly if you quote the original reference to indicate that the concepts contained are not yours. However what I have outlined above is generally considered to be good practice in the international community. In addition, to be able to paraphrase as in S4 and S5 means that you really have to understand the original sentence, which is clearly beneficial for you.

Note also that you may wish to paraphrase your own writing within the same paper, i.e. to not repeat in the Conclusions the same phrases you have used in the Abstract (19.5).

15.11 Paraphrasing: a simple example

Albert Einstein has been quoted as saying: *The true sign of intelligence is not knowledge but imagination.*

How could you paraphrase Einstein's quotation? [NB: 1935 in the examples below is just my guess as to the year when Einstein made his claim].

SYNONYMS

verbs: Einstein proposed / suggested / stated / found / revealed that … (1935).

nouns and verbs: A clear *indicator* of someone's *power of intellect* is not *how much they know* but how well their imagination functions (Einstein, 1935).

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ACTIVE TO PASSIVE

It has been claimed / proposed / suggested / stated / found / revealed that … (Einstein, 1935)

DIFFERENT WORD ORDER

According to (Einstein, 1935), it is imagination rather than knowledge that is the real sign of intelligence.

Intelligence should be judged in terms of imagination rather than knowledge (Einstein, 1935).

Note how the three key words - *intelligence, knowledge* and *imagination* - have not been paraphrased into words such as *smartness, knowhow* or *fantasy*. None of these three words are exact synonyms and they do not have the same semantic roots. It is important that key words remain as they are. However saying *power of intellect* and *how much they know* is approximately equivalent to saying *intelligence* and *knowledge* and is thus probably acceptable.

The words that can be paraphrased are the more generic words such as *indicator* for *sign*.

15.12 Paraphrasing: how it can help you write correct English

Paraphrasing avoids:

- plagiarism (at least to some extent)
- repetition of phrases within your paper (e.g. not repeating sentences in the Conclusions that you already wrote in the Abstract)

But paraphrasing is also very useful when you are not sure that a sentence you have written is correct English. You can simply paraphrase the sentence using a form that you know is correct. A great rule for writing in English is:

"Only write what you know is correct".

Let's imagine that you write S1 but you are not sure whether you can begin a sentence with such a time clause (*It is six years that ...)* or whether you have used the correct tense for the second verb (i.e. should it be *is* or something else?).

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S1. * *It is several years that* this technology *is* available on the market.

Think about how else you could write the phrase. Here are three other possibilities.

S2. This technology *was introduced* onto the market *in 2015*.

S3. This technology first *became* available *several years ago*.

S4. This technology *has been* on the market *for several years*.

The idea is that you think about various alternatives, and then choose the one that you are most confident about. S2 and S3 contain very simple constructions. S4 is more complex, as this use of the present perfect is almost unique to English.

This technique should help you not only in writing manuscripts, but also in your correspondence with editors (emails and letters), and when writing proposals and reports.

15.13 What are some more quick tips for writing a review paper?

A review paper must serve a purpose, i.e. to bring other researchers up to date on a particular topic and to highlight the implications of the results that have been obtained so far.

A review paper has a method (i.e. how and why you selected certain papers and rejected others), and results (what you found), and most importantly a discussion and conclusion (i.e. what the implications of your 'results' are).

Your review paper must be easy to navigate. This means you need a LOT of headings and subheadings, so that readers can quickly find the information they are looking for, or to help them locate later where they found some particular information.

You need to be as concise as possible when reporting what has been stated in the literature. If you continuously repeat phrases such as *it has been shown / demonstrated that* $x = y$, readers will soon get bored and may stop reading.

The implications of what you have found will be clearer if you list them in the conclusions in the form of bullets (i.e. take-home points). Basically you are telling the reader what you have learned from conducting the review, and what the implications are for the wider community.

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15.14 How should I conclude a Review Paper?

Concluding a review paper involves summarizing the key points of your paper and providing a final perspective on the topic you have covered.

Restate the reasons why you felt a review was necessary in your particular field: Remind your readers of the main focus of your review paper.

Summarize the key points: Briefly summarize the main ideas and findings of your review paper.

Make connections: Connect the main points you have discussed to the aim that you outlined at the beginning of your review paper. Show how the evidence you presented supports your argument.

Discuss the implications: Provide a final perspective on the topic by discussing the implications of your findings. What are the practical implications of your research? What are the potential future directions for this area of study?

Conclude with a final thought: End your conclusion with a final thought or message. This could be a call to action, a recommendation for future research, or a reflection on the significance of the topic.

15.15 Summary: How can I assess the quality of my Literature Review?

To make a self-assessment of your Literature Review, you can ask yourself the following questions.

- Have I shown that I am familiar with the state of the art?
- Have I mentioned only what my readers specifically need to know and what I will subsequently refer to in the Discussion?
- Have I avoided only mentioning the literature that supports my hypotheses?
- Are the papers I have mentioned in a logical order? Is it clear why I have chosen these papers and not others?
- Have I selected a disproportionate number of papers from my own country?
- Have I ensured that there are no papers cited in the bibliography that are not cited in the paper, and vice versa?

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- Have I followed my journal's instructions regarding how I make references to the literature? Where possible have I done this in a variety of ways?
- Have I removed any redundancy when reporting the literature?
- Have I used tenses correctly? PRESENT SIMPLE (descriptions of established scientific fact), PRESENT PERFECT (at the beginning of review to give general overview; for past-to-present evolutions), PAST SIMPLE (when specific dates are mentioned within a sentence; for the verbs that introduce an author's findings).
- Plagiarism is a serious issue in international science. If you use AI to help you write a paper then you should acknowledge this to the editor. NEVER use AI to paraphrase someone else's work without acknowledging this author.
- Copying phrases from other people's work is perfectly acceptable and is a good way to learn useful phrases in English that you can then use in your own work. However, such phrases must be 100% generic in the sense that they hold absolutely no hard information.
- Typical ways to paraphrase:
  - use of synonyms for non-key words (especially verbs, adverbs and adjectives)
  - change of part of speech, for example: from noun to verb, from noun to adjective, from one category of noun to another category of noun (e.g. *science* to *scientist*)
  - change of nouns and pronouns from singular to plural and vice versa
  - change of verb form, for example: from *–ing* form to infinitive, from simple to continuous, from active to passive
  - change of style from personal to impersonal
  - reversal of the order in which information is presented
- Never paraphrase technical words
- If the original contains ideas that in some sense 'belonged' to the original author, then this author should be acknowledged. This is true even if you have radically changed the original so that it is now unrecognizable
- If you are worried that you might have committed plagiarism check with your professor and co-authors

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Chapter 16

Methods



Ancient and medieval medical methods

Miscellaneous health problems: the Egyptians used the dung (feces) from various animals and insects (e.g. donkeys, dogs, gazelles and flies). Today it is known that the microflora found in some types of animal dung contain antibiotic substances.

*****

Bubonic plague (14th century): in order to avoid death, sufferers were ordered to confess their sins in the presence of a priest.

*****

Generic illnesses: a blood-sucking worm was placed on the affected part (leaching), a vein was cut in order to drain a substantial quantity of blood (venesection), or a hole was bored in the skull (trepanation).

*****

Hemorrhoids: body parts were burned (cauterization) to remove the hemorrhoid. Cauterization was also used to close an amputation and to prevent severe blood loss.

*****

Pain relief during surgery: as an anesthetic a mixture of lettuce juice with the contents of the gall bladder from a castrated boar (a wild pig) was used along with opium, henbane (a plant with psychoactive properties), hemlock juice (a highly poisonous plant), vinegar and wine.

*****

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_16

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Superhuman powers: mixtures containing mercury, sulfur and arsenic were supposed to give the patients the ability to walk on water and even to become immortal.

16.1 What is a Method's section?

This section of a paper has several different names including: 'Methods', 'Methods and Materials', 'Experimental', 'Method Description and Validation'.

In most journals the Methods section follows the Literature Review, in others it follows the Conclusions.

Your aim is to describe the materials you used in your experiments and/or the methods you used to carry out your research, in a way that is sufficiently detailed to enable others in your field to easily follow your method and, if desired, even replicate your work. Ensure the descriptions are complete and yet are also as concise as possible, for example by referring to other works in the literature, including your own, that make use of the same or similar methods.

Another key skill is to write extremely clearly, with generally not more than two steps described in one sentence, and in a logical order. This will then enable your readers to easily follow your description, and possibly replicate what you have described.

Researchers generally agree that the Methods is the easiest section to write because your methods are likely to be clear in your mind, so it may be a good point for you to begin writing your manuscript.

Look at this extract from the Methods section What other problems are there? The extract describes a methodology for defusing a bomb that is contained inside a box.

First, the lid at the top of the box should be carefully removed, provided that it has been ascertained that there is no trigger device. Second, the three wires at the side of the explosive device should be identified before proceeding with step three: the cutting of the green wire. Finally, the red wire should also be subjected to a cutting process after the blue wire has been disconnected.

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The principle problem is that some sentences contain more than one step, and that these steps are not always in chronological order - if you followed the steps correctly the bomb would explode immediately (if there was a trigger device on the lid) or just before the last step! The enumeration of the steps through the use of *first*, *second* and *finally*, is in the case not helpful as in fact there are six steps. The phrases are sometimes unnecessarily verbose (e.g. *the cutting of the green wire ... be subjected to a cutting process*).

16.2 Are there any ways that a chatbot could help me with my Methods section?

One thing you can do is to instruct the bot to rewrite a paragraph by putting the information in a more logical (e.g. chronological) order. I requested a bot to put the red text in 16.1 into chronological order. The bot returned the following:

1. Carefully remove the lid at the top of the box, provided that it has been ascertained that there is no trigger device.
2. Identify the three wires at the side of the explosive device.
3. Proceed with step three: the cutting of the green wire.
4. Disconnect the blue wire.
5. Finally, subject the red wire to a cutting process.

You now have the information in the most logical order. The next step is to get the bot to 'improve the English'. The bot returned the following:

1. Carefully remove the lid from the top of the box, making sure that there is no trigger device.
2. Identify the three wires located at the side of the explosive device.
3. Proceed to step three, which involves cutting the green wire.
4. Disconnect the blue wire.
5. Finally, cut the red wire.

Some additional notes on the improvements:

- "Provided that it has been ascertained" is a bit wordy, so I simplified it to "making sure that there is no trigger device."
- I added the verb "cut" to the final instruction to make it clear that the red wire should be cut like the green wire and blue wire.
- I also rephrased the instructions to use active voice and avoid starting each sentence with a verb.

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As you can see, the bot has not only corrected your English, but has also explained why certain improvements have been made. They are very good suggestions, which you can then implement throughout your methods.

However, the bot has a big limitation. It does not tell you that the first step should be divided into two steps:

1. Ensure that there is no trigger device.
2. Carefully remove the lid from the top of the box.

16.3 How should I structure the Methods?

The Methods section should answer most of the following questions, obviously depending on your discipline:

- What / Who did I study? What hypotheses was I testing?
- Where did I carry out this study and what characteristics did this location have?
- How did I design my experiment / sampling and what assumptions did I make?
- What variable was I measuring and why?
- How did I handle / house / treat my materials / subjects? What kind of care / precautions were taken?
- What equipment did I use (plus modifications) and where did this equipment come from (vendor source)?
- What protocol did I use for collecting my data?
- How did I analyze the data? Statistical procedures? Mathematical equations? Software?
- What probability did I use to decide significance?
- What references to the literature could I give to save me having to describe something in detail?
- What difficulties did I encounter?
- How does my methodology compare with previously reported methods, and what significant advances does it make?

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You should provide enough quantitative information (concentration, temperature, weight, size, length, time, duration etc.) so that other researchers can replicate what you did.

Describe everything in a logical order to enable readers to easily follow what you did. This will usually be chronological (but see Sect. 16.9), i.e. the order in which you conducted the phases of your tests. It may also help the reader if you use subheadings to explain the various stages of the procedure, which you can then use again (perhaps with modifications) in the Results.

Your experiments, sampling procedures, selection criteria etc. may have more than one step. It helps your readers if your description of each step follows the same logical order.

Ensure that you cover every step required. Because you are very familiar with your method, you may leave out key information either thinking that it is implicit (and thus not worth mentioning) or simply because you forget.

16.4 What style: should I use the active or passive? What tenses should I use?

The passive is good style in this part of a research paper because the focus is on what was done rather than who did it. Thus you can ignore any expert advice that tells you that the passive should always be avoided. It should be avoided, but only where it is not necessary. In the Methods the passive is both necessary and appropriate.

Most Methods sections are written in the PAST SIMPLE and/or PRESENT SIMPLE. The choice will depend on your discipline (and whether it is applied or theoretical), your chosen journal, and what actions you are describing.

The PAST SIMPLE is required when you describe actions that you did, both before and during your experiments (in the lab, in the field, while conducting surveys etc). Thus the PAST SIMPLE is likely to be used in most of this section.

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Below is an example of the PAST SIMPLE used to describe some preliminary work:

An explorative research approach *was adopted* using a seven-page survey on opinions and religious background. The findings *were collected* using an internet questionnaire survey. Six hundred religious institutions *were selected* from AMADEUS database, which *were then classified* into three groups based on …

Here are the methods used by an agronomist, again using the PAST SIMPLE:

A test bench (Fig. 1) *was used* in order to evaluate the effectiveness of the flame burner. Steel plates *were treated* with an open--flame burner (Fig. 2).

The PRESENT SIMPLE is required when you describe a standard method, i.e. not one you invented yourself for the specific purpose of the research that you are reporting in your paper.

For example, if your paper is on ways to recycle paper, when describing methods that have been reported in the literature, you will use the PRESENT. But then, when you describe the various phases of your own system for recycling, then you would probably use the PAST.

The PRESENT SIMPLE is also often used when you are presenting your procedure, model, software, device etc. In this case the active form is often used.

Firstly, we *define* x as an exogenous measure of the natural rate of longevity of people.

As in Chakraborty (2017), we *assume* that … The rule is thus given by the following formula:

Our machine *uses* diesel … It *has* a 1000 hp engine …

The application *requires* 10 TB of space ...

If you are in doubt, then look at your chosen journal and check what tense other authors use in the Methods section.

To learn about the ambiguity cause by using the passive form, see 6.4.

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16.5 How should I begin the Methods?

How you begin will very much depend on your discipline. To help you decide, take a look at the Methods section in papers from your chosen journal, and see how authors start this section.

Typical ways include:

a) Making a general statement about your method.

The method described here is simple, rapid, sensitive and …

b) Referring to another paper.

The materials used for isolation and culture *are described* elsewhere [20].

Materials *were obtained* in accordance with Burgess et al.'s method [55].

c) Stating where you obtained your materials from.

Bacterial strains …*were isolated* and kindly supplied by …

Agorose for gel electrophoresis *was purchased* from Brogdon plc (Altrincham, UK).

d) Explaining how you found your subjects, i.e. begin with the setting.

Subjects *were chosen* from a randomly selected sample of …

Participants *were selected* from patients at the Gynecology Faculty of the University of …

e) Indicating where (i.e. a geographical region) your investigation was focused.

Our empirical investigation focused on Tuscany, a central region of Italy, …

The study *was carried out* in four boulevards in Athens (Greece) and …

f) Referring the reader to a figure which shows the experimental set up.

To highlight the advantages of the system, Fig. 1 shows the …

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g) Starting directly with the first step in your procedure.

Frontal cerebral cortices *were dissected* from …

Core-cell composite materials *were prepared* by colloidal assembly of …

16.6 My methods use a standard / previous procedure. Do I need to describe the methods in detail?

No. You can alert the reader that the method is 'standard' and is described in another paper or in some manufacturer's instructions to their product (make sure you give a clear reference to the related paper or the instructions).

If you use a phrase such as *based on methods previously described*, then you need to make it clear whose these methods are: yours or someone else's. In they are yours then it would be less ambiguous to write: .... *based on methods described in our previous paper [56].*

Even though you might use a standard procedure, you will probably have adjusted it in some way and you should mention these modifications:

Our methods followed the procedures outlined in [Wallwork, 2025] *with the two minor modifications:*

Our procedure is as according to [Wallwork, 2025] *with the following exceptions:*

If the methods in the paper you are writing now are (almost) identical to the methods I published in a previous paper, you cannot just cut and paste from this previous paper as this is known as self-plagiarism.

Simply putting a reference to the original paper where your methods are described is probably not enough i) it will make your Methods section look very short, ii) it is not very helpful for readers.

A better solution is to write:

Full details of the methods used can be found in our previous paper [45]. In brief, ...

In brief alerts the reader that what they are going to read now is not the full version of your methods. Also, this gives you the opportunity to talk about any modifications you may have made to your original methods.

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16.7 Should I describe everything in chronological order?

The basic idea is present everything in your experiments, trials, procedures etc. in a way that will make best sense to your reader. The fact you did something before or after something else, may not be relevant for your reader, so in such cases chronology is not important.

However within a sentence or paragraph, readers should feel they are moving forward chronologically and logically.

The sample was filtered and acidified at pH 2 and then mixed with X.

The sample was filtered and acidified at pH 2, and then mixed with X.

The sample, which had been filtered and acidified at pH 2, was mixed with X, which enabled the resulting solution to stabilize at …

The sample was filtered and acidified at pH 2. It was then mixed with X, which enabled the resulting solution to stabilize at …

The first red sentence uses *and* twice, which means the reader may be initially confused with regard to which two items are connected with each other (*filtered* + *acidified*, or *acidified* + *mixed*). The blue version solves this problem by adding a comma after *pH 2*.

The second red sentence contains two relative clauses (i.e. a phrase introduced by *which*), whereas the blue sentence removes the first relative clause and is also split into two sentences. The blue version thus become easier to understand immediately.

16.8 How many actions / steps can I refer to in a single sentence?

A frequent problem in the Methods is that the description reads like a manual, where each individual detail or action is described in a single sentence. Given that you are describing a procedure rather than making a complex analysis, it is perfectly acceptable to have two actions in one sentence.

Below is the first paragraph from a medical paper in which the author describes how she selected the participants for her survey on depression. The word ‘practice’ means an association of medical doctors who offer a service to the public. The ‘list size’ is the number of patients the practice has.

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A first postal invitation to participate in the survey was sent to 26 practices in South Yorkshire. A total of five practices indicated their willingness to participate. Multidisciplinary focus groups in four diverse practices were purposively identified. The identification entailed using a maximum variation approach. This approach was based on socio-economic population characteristics and ethnic diversity. These characteristics were taken with reference to census data.

Following a postal invitation to participate sent to 26 practices in South Yorkshire, five responded positively. Multidisciplinary focus groups in four diverse practices were identified using a maximum variation approach, based on socio-economic population characteristics and ethnic diversity by reference to census data.

The red version is in correct English and is perfectly acceptable provided that this style is not used continuously throughout the Methods. If it is used continuously, the reader will soon find it tedious, particularly as each sentence begins in the same way (i.e. with a noun).

The technique of the blue version is to combine two steps into a single sentence, with no extra effort on the reader's part in terms of understanding. However, you do not want to have too much information in the same sentence. In the red version below, the reader would find the information much more difficult to assimilate than in the blue version, even though the information given is exactly the same.

The four practices, which had been previously identified as having list sizes between 4750 and 8200, comprised firstly an inner city practice (hereafter Type 1) with an ethnically diverse population for which the team frequently required translators for primary care consultations, secondly, two urban practices with average levels of socio- economic deprivation (Type 2), and thirdly, a mixed urban/rural practice (Type 3).

The four practices had a list size ranging between 4750 and 8200. They comprised:

- an inner city practice with an ethnically diverse population, where the team frequently required translators for primary care consultations
- two urban practices with average levels of socio-economic deprivation
- a mixed urban/rural practice

However, refer to your journal's style guide to check whether bullets are permissible.

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16.9 How can I reduce the number of words but avoid my Methods appearing like a series of lists?

It is important to be concise (Chapter 4) in the Methods. To reduce the word count you can

- assume your readers have basic knowledge of the techniques used in your field, you can thus delete any superfluous information
- cite a reference rather than detailing the procedure again if any of your methods are fully described elsewhere (in one of your papers or someone else's)
- use tables and figures to summarize information

But conciseness does not mean writing a series of lists, as in the red example below:

Processes which often occur in lipids include: oxidation, hydration, dehydration, decarboxylation, esterification, aromatisation, hydrolysis, hydrogenation and polymerization. Factors that affect the chemistry of these materials include: heat (anthropogenic transformations), humidity, pH, and microbial attacks.

This style may be appropriate on a presentation slide, but should be avoided in a paper. What you write should always sound natural if read aloud. The blue version below sounds more natural - even though it requires more words.

Several processes often occur in lipids, including oxidation, hydration, dehydration, decarboxylation, esterification, aromatisation, hydrolysis, hydrogenation, and polymerization. In addition, the chemistry of these materials can be affected, for example, by heat (anthropogenic transformations), humidity, pH, and microbial attacks.

16.10 What potential errors of grammar and syntax do I need to be careful of?

AMBIGUITY CAUSED BY BAD WORD ORDER

Can you spot the ambiguity in these two sentences'

Being over 500 years old, we handled the painting with great care.

Preserved in a frozen state, we carefully examined the samples.

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In first case it seems that *we* are 500 years old and in the second that *we* are frozen. Yes, these examples are ridiculous, but such ambiguity is common in the Methods section. Ensure that it is 100% clear what or who the subject of the verb is.

Since the painting was over 500 years old, we handled it / it was handled with great care.

The samples were preserved in a frozen state and were examined carefully.

AMBIGUITY CAUSED BY *AND* … *AND*

Be careful how you use and in a sentence. In the red version below it is impossible to understand which animal are on the endangered list. The two blue versions are alternative ways of clarifying this.

We investigated lions and tigers and elephants that are on the endangered species list.

We investigated lions and tigers, and also those elephants that are on the endangered species list.

We investigated lions, tigers, and elephants. All three types of animals are on the endangered species list.

USE OF INFINITIVE TO TALK ABOUT AIMS AND CHOICES

For validating / For to validate the results, we first had to

In order to validate the results, we first had to …

In an attempt to identify the components, it was decided to …

To provide a way of characterizing the samples, an adaptation of Smith's method [2011] was used.

Our aim was getting a general picture of …

Our aim was to get a general picture of …

This choice was aimed at getting a general picture of …

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CONSTRUCTION WITH *ALLOW*, *ENABLE* AND *PERMIT*

All three verbs require the same specific construction, which is probably different from in your language.

allow someone or something to do something

This equipment allowed to identify X.

This equipment allowed us to identify X.

allow someone or something to be + past participle

This equipment allowed X to be identified.

allow + noun

This equipment allowed the identification of X.

If you find yourself using *allow* and *enable* very frequently, then consider using the alternatives given below.

Limiting the Xs *allows* the complexity of Y *to be reduced* and permits *the user to control* the deduction process.

Limiting the Xs *reduces* the complexity of Y, and *facilitates control* of the deduction process.

The analysis *allowed the characterization of pine resin* as the main organic constituents in the sample to be achieved.

The analysis *showed that pine resin* was the main organic constituent in this ample.

This model *permits the analysis* of X.

This model *can analyze* X.

With this model *we can analyze* X.

With this model, X can be determined

The use of these substrates *enabled us to highlight* the presence of several nucleases.

The use of these substrates *highlighted* the presence of ...

The use of these substrates *meant that we were able to highlight* the presence of ...

The use of these substrates offered a means *to highlight* the presence of ...

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16.11 What other points should I include in the Methods? How should I end the Methods?

The Methods section is often the shortest section in a paper. However, in some studies the methods are the main contribution of the paper. In such cases, subsections with subheadings (e.g. *sampling procedure, experimental set up, testing the model*) may help readers to understand the various stages or various components.

Your first subsection may be a general overview of the methods chosen, how they relate to the literature and why you chose them.

Then in each subsequent subsection you:

1. preview the part of the procedure / method you are talking about
2. detail what was done and justify your choices
3. point out any precautions taken (this also helps you gain credibility as a researcher who carries out his / her work accurately and thoroughly)
4. discuss any limitations in your method or problems you encountered
5. highlight the benefits of your methods (perhaps in comparison to other authors' approaches)

If your Methods section is short and does not require any subsections, then you could end it with one or more of Points 3–5 above. If it is long, then you could end with some conclusions regarding the limitations and benefits (Points 4 and 5) of your overall methodology.

16.12 Summary: How can I assess the quality of my Methods section?

To make a self-assessment of your Methods section, you can ask yourself the following questions.

- Are enough details given? Have I really described my Methods in a way that is easy for readers to follow and which would enable them to retrace or replicate my work? Have I ensured that I have covered every step? Is my structure clear and complete?
- Have I justified the choices I made (particularly when the choices might not be obvious)?

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- Have I written everything clearly in reasonably short sentences, with no unnecessary semicolons? Have I been as concise as possible?
- Have I used references to previous works rather than repeating descriptions that readers could easily find elsewhere?
- Do the individual sentences in each paragraph contain too many, too few, or just the right manageable number of steps? Have I ensured that my sentences don't sound like lists?
- Have I thought about the way readers prefer to receive information? (no ambiguity, no back referencing, everything in chronological order.
- Have I checked my grammar (infinitive, gerund, *allow, thus* etc.) with regard to how I outline how and why I made certain choices?
- Have I checked my journal's guidelines on how to use numbers?
- Have I used tenses correctly? PAST SIMPLE (in the passive form to describe what I did), PRESENT SIMPLE (descriptions of established scientific fact or processes, software applications, standard devices etc)

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Chapter 17

Results



'Results' that were subsequently questioned

1796: Samuel Hahnemann, a German physician, creates homeopathy as a system of alternative medicine. There has never been any conclusive scientific evidence that it works other than as a placebo.

1945: Scientists warn that fluoride in drinking water is poisonous and many local governments around the world ban it. However in low concentrations (one part-per-million) fluoride reduces dental decay.

1962: Rachel Carson, a US marine biologist, forecasts that birds will die out and humans will contract cancer due to increasing exposure to the insecticide DDT. No plausible biological mechanism was identified and research failed to support the claims. DDT was nevertheless banned and millions may have died unnecessarily from malaria.

1968: Paul Ehrlich, US ecologist and demographer, publishes his book *The Population Bomb*, where he writes: 'The battle to feed humanity is over. In the 1970s, the world will undergo famines. Hundreds of millions of people are going to starve to death'.

1979: A small epidemiological study reports an association between hypothesized exposure to electromagnetic fields and childhood leukemia. Thousands of studies have failed to establish a link between actual exposure and any health effect.

1996: Scientists speculate that a variant of Creutzfeldt-Jakob disease (CJD) might be contracted from eating beef from cattle with BSE, and forecast that the disease would kill 10 million people by 2011. This led to the slaughter of 8 million cattle in Britain.

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_17

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2005: David Nabarro, Senior United Nations System Coordinator for Avian and Human Influenza, warns that an outbreak of avian influenza could kill anywhere between 5 million and 150 million people.

2015: Data apparently shows that the 'vanishing' of polar ice is not the result of runaway global warming.

17.1 What is a Results section?

Look at these football results from the football World Cup.

Vatican City 0 Vanuatu 1	Germany 7 Brazil 1
Malta 2 Liechtenstein 1	Italy 4 Senegal 4
Monaco 0 Maldives 2	Argentina 4 France 2

Think about the answers to these questions.

1. Which ones are readers most likely to be interested in? Why?
2. Which results would you just put in a table, and which in both a table and in the main text?
3. If you were Brazilian, would you avoid mentioning the result?

Don't tell the reader ALL your results – just the most relevant and / or the most unexpected. Unless you were actually born in one of the countries listed in the first column (six of the smallest countries in the world), you are highly unlikely to be interested in reading details about them. However, they could still be listed in a table, but with no need for comment in the main text.

The results in the second column represent your three big results, i.e. your three most important findings. Such results should be i) put in a table, and ii) commented about.

Even if a result goes against what was expected (e.g. the Germany vs Brazil result), you should still include it.

Not all journals require a separate Results section, often it is integrated with the Discussion, under the section title Results and Discussion.

If you have a separate Results section then the standard procedure is to present them with little or no interpretation or discussion.

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The key skill is first to decide what results are representative, and then to organize them in a sequence that highlights the answers to the aims, hypotheses or questions that you set yourself at the beginning of the paper. In many disciplines this involves the use of figures and tables, which are commented on in the text. In other disciplines, findings are only reported in text form.

You should also mention any important negative results here.

From an English point of view the key skill is in reporting your results simply and clearly. If the referees of your paper cannot understand your results, then your contribution to the current knowledge base will be lost.

Bear in mind the following comment that a reviewer wrote of one of my client's papers:

At times this paper reads like a thesis. The authors seem to have included all their results, with the consequence that I am not sure which findings are significant and which are not. However, I also suspect that some contradictory findings have not been included. So although I generally recommend brevity, this should not include leaving out key findings that do not support the authors' line of logic.

This chapter is designed to help you avoid that problem.

17.2 How should I structure the Results?

The Results should answer the following questions.

- What did I find?
- What did I not find?
- What did I find that I was not expecting to find? (e.g. that contradicts my hypotheses)

A typical structure is to follow the order you used for the protocols or procedures in your Methods. You then use figures and tables to sequence the answers to the above questions.

Alternatively, before you begin writing, arrange your figures (tables etc.) in the most logical order for your readers. This order should support the initial aim or hypothesis that you stated in your Introduction. Then associate key findings with each of your figures, excluding any results that are not relevant in supporting your research hypothesis. Note that 'not relevant' does not include results that contradict your hypothesis (Sect. 13.4).

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The rest of the section then consists in commenting on these figures one by one. Maeve O'Connor in her book *Writing Successfully in Science*, recommends the following structure.

1. Highlight those results (including those from controls) that answer your research question
2. Outline secondary results
3. Give supporting information
4. Mention any results that contradict your hypothesis and explain why they are anomalous

Very important: Whatever structure you use, throughout your results section you MUST make it clear when you are talking about your findings and not the findings of others. To learn about this extremely important aspect see Chapter 6.

17.3 How should I begin the Results?

There are two typical ways to begin the Results. The first is to give a general panorama of your surveys, experiments etc. without repeating the details you gave in the Methods section, as in the three examples below:

Overall, the results presented below show that …

The three key results of this empirical study are: …

The following emergent themes were identified from the analysis: …

The most common way is to simply go directly to your results, often by inviting readers to look at one of your figures or tables, either in the first sentence or very shortly after:

Figure 1 shows the mass spectra obtained from an analysis of the two residues. The first residue reveals a .. (Fig. 1a)

A total of 34 wheat genotypes (Table 1) were screened for … Responses to increased sunlight varied significantly (Figure 1) …

An analysis was made to look for … To do this, the average times of x and y were compared … Figures 1–3 show the differences between …

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17.4 What tenses and style should I use when reporting my Results?

Your results are things that you found before you started writing the paper. They therefore relate to past events, consequently the PAST SIMPLE is used to report them, often in a mixture of the active and passive forms.

You may occasionally wish to use the PRESENT SIMPLE. This is the case when you are taking the reader through your results as if you were a professor at the whiteboard and your reader was a student in the class. If opt for this style, which where possible I would avoid, it needs to be absolutely clear that you are talking about your own results and not someone else's (Chapter 6).

Many authors use an impersonal style. This style, in the mind of some editors, serves to add an element of objectivity to your findings.

There was a perceived failure of the NHS to provide adequate services.

But more immediate and clearer to the reader is:

We found that doctors viewed the NHS as having failed to provide adequate services.

17.5 Should I report any negative results?

Of course!

Dr Ben Goldacre, a campaigner against the suppression of negative data in medical papers, says:

> When you get a negative result, it feels as if it's all been a bit of a waste of time. It's easy to convince yourself that you found nothing, when in fact you discovered a very useful piece of information: the thing that you were testing doesn't work.

In a book published in 1988 and entitled *What do you care what other people think?* Nobel Prize winner, Richard Feynman wrote:

> If you are doing an experiment, you should report everything that you think might make it invalid - not only what you think is right about it: other causes that could possibly explain your results.

To learn how to deal with negative results see Chapter 7.

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17.6 How can I show my readers the value of my data, rather than just telling them?

Ken Lertzman, a Professor of Ecology at the Simon Fraser University, gives the following advice:

Rather than telling the reader that a result is interesting or significant, show them how it is interesting or significant … show the reader what they need to know to come to their own conclusion about the result.

Ken gives two examples to highlight the difference:

The large difference in mean size between population C and population D is particularly *interesting*.

While the mean size generally varies among populations by only a few cm, the mean size in populations C and D *differed by 25 cm*. Two hypotheses could account for this, …

The adjective *interesting* presumably means something very definite for the author, but not for the reader who has not been given the tools to assess why the *mean size* is *interesting*. Such descriptive adjectives (*interesting, intriguing, remarkable*) are rarely helpful.

You need to give your readers sufficient information for them to be able to say to themselves: "wow that is interesting!" This is what the blue version does by highlighting specific details (*differed by 25 cm*).

Adverbs such as *interestingly, intriguingly, remarkably* also suffer from the same problem. However, they can be used effectively if used at the beginning of a sentence, in order to attract attention to a key finding.

Interestingly, while the mean size generally varies among populations by only a few cm, the mean size in populations C and D *differed by 25 cm*. Two hypotheses could account for this, …

However this technique should be used only once or twice in the whole paper, otherwise it loses its effect.

If you have a Discussion section, then you do not need not to interpret your data in the Results.

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17.7 How should I comment on my tables and figures?

Dr Lertzman (17.6) has similar ideas about 'showing not telling' with regard to figures and tables:

When writing Results sections you should use the tables and figures to illustrate points in the text, rather than making them the subject of your text.

For example:

Figure 4 shows the relationship between the numbers of species A and species B.

The abundances of species A and B were inversely related (Figure 4).

In the red sentence the author is merely telling readers what they can already see in the figure. Readers are forced to make their own interpretations, which may be interpretations that you don't want them to make.

The blue version is much more helpful, because it focuses on the meaning that can be inferred from the figure. S2 saves readers from making any mental effort and at the same time guides them towards the interpretation that you want them to have.

Below, note how the blue versions don't force the reader to read the obvious.

We can see from Table 2 that in the control group, values for early adolescence (13–15) were 6.5. On the other hand, values for mid adolescence (16–17) were 6.7.

Values for early adolescence were lower than for mid adolescence: 6.5 versus 6.7 (Table 2).

Figure 1 shows that levels of intolerance are 9, 15 and 20 during early, mid and late adolescence, respectively.

Levels of intolerance are highest during late adolescence (Figure 1).

Lack of conciseness is a frequent problem when describing data in figures and tables (Sect. 5.13). Avoid phrases such *as can be seen* and *we can see*. Simply put the figure or table reference in brackets at the end of the sentence. The second red sentence also repeats information that should already be contained in the table, i.e. the respective age ranges for the three stages of adolescence.

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17.8 What more do I need to know about commenting on tables?

Below is a table from Wikipedia showing some statistics on a football World Cup semi-final in which Brazil lost to Germany.
A typical mistake when writing the main text is to repeat information from the table. For example:

Statistic	Brazil	Germany
Goals scored	1	7
Total shots	18	14
Shots on target	8	10
Ball possession	52%	48%
Corner kicks	7	5
Fouls committed	11	14
Offsides	3	0
Yellow cards	1	0
Red cards	0	0

As shown in the table, the total number of goals scored was one on the part of the Brazilian team and seven by the German team. The Brazilians achieved 18 shots, whereas the Germans accounted for a lower number of shots, namely 13.

The type of commentary in S1 adds no value to the reader. It tells them nothing that they could not have deduced for themselves. When commenting on a table, your job is to:

- interpret / discuss the results
- bring to the reader's attention anything that is particularly meaningful or significant

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- add further details that help to explain the results or which enable them to be compared with previous results

For example, you could write:

Although a close match was expected - both teams had reached the semi-final undefeated - the result was a shocking loss for Brazil (see Table 1). For what was the first time in football history, Germany scored four goals in the space of six minutes. Despite achieving a greater number of shots, having 4% more possession and committing less fouls, and having only two shots less on target, the Brazilians were humiliated. This result recalls the 1952 final when Brazil were defeated by Uruguay.

Note that given that this is the Results section, you should reserve detailed interpretations for the Discussion section.

17.9 How can I use a chatbot to help me write the Results section?

A bot can help you with the structure for your interpretation of data contained in a table. I asked a bot to: *Write a description for a results section of an academic paper based on the information in this table*. The table I inserted is the one from the previous section on the World Cup semi-final in which Brazil lost to Germany. Below is the bot's output. Note that it is in blue because the text is correct from an English point of view; however the content is not optimal, as explained below.

The results show that Germany scored seven goals, while Brazil scored only one. Germany's dominance in the match is also evident in their higher number of shots on target (10) and a greater percentage of ball possession (48%). In comparison, Brazil had eight shots on target and 52% of ball possession.

In terms of set pieces, Brazil had more corner kicks than Germany, with a total of seven, while Germany had five. However, Germany committed fewer fouls than Brazil, with 14 and 11, respectively. Moreover, Brazil had three offsides, whereas Germany had none. Finally, Brazil received one yellow card, and neither team received any red cards during the match.

Overall, the statistical data suggests that Germany played a more dominant game, with superior attacking and defensive performances. Brazil, on the other hand, had a more balanced game, with more ball possession and attempts at goal, but with fewer conversions. These findings can provide insights into the teams' strengths and weaknesses and inform future strategies for improvement.

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What the above highlights is that there is very little information in the text that cannot be more immediately understood from the table itself. So there is really no added value for the reader. As an author you need to focus on those results that are particularly interesting in some way, e.g. they are counterintuitive, they confirm a hypothesis you made earlier in the paper (or equally they go against your hypothesis), they have an important implication.

On the positive side, the bot's texts provides a lot of examples of how to make comparisons (*fewer, whereas*) and also gives you a possible structure:

- summary of the key findings, highlighting the significant differences between the two teams
- detailed description of the statistical data
- possible implications

I also asked the bot to provide a legend for the table. However, first it listed the meanings of each individual column and row. In my second attempt, it explained the meaning of the terms column, row, data and legend. Finally, I asked it to provide in a single sentence a legend for the entire table, which resulted in:

This table presents a comparison of various statistical measures between Brazil and Germany during a football match, with columns representing the values recorded for each team, and rows representing metrics such as goals scored, total shots, shots on target, ball possession, corner kicks, fouls committed, offsides, yellow cards, and red cards.

Clearly, the legend is much too detailed, but it gives you an idea of what you could include and what you could leave out. You do NOT need to begin the legend with *This table presents*, you can just start with: *A comparison of ...*; or given that tables are always a means of comparison you could start with: *Statistical measures ...*

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17.10 What about legends and captions?

A typical mistake is to repeat word for word the caption / legend to your figures and tables within the main text.

Legends should have a number. They should be as short as possible and be sufficiently detailed to enable your readers to understand the figure or table without having to read your text. It is vital that you pay attention to legends as some readers may only look at your figures and tables, without even reading the paper itself!

Note that when referring to a figure in the text, the word "Figure" is abbreviated as "Fig.", while "Table" is not abbreviated. Both words are spelled out completely in descriptive legends (see 1.8 in *English Grammar, Usage and Style*).

The rest of this subsection is taken directly from the biology website at Bates College - a special thanks to Greg Anderson for allowing me to reproduce it. Although Greg's advice relates to biologists, much of it is true for other hard sciences as well.

Every figure and table included in the paper MUST be referred to from the text. Use sentences that draw the reader's attention to the relationship or trend you wish to highlight, referring to the appropriate figure or table only parenthetically:

Germination rates were significantly higher after 24 h in running water than in controls (Fig. 4).

DNA sequence homologies for the *purple* gene from the four congeners (Table 1) show high similarity, differing by at most 4 base pairs.

Avoid sentences that give no information other than directing the reader to the figure or table.

Like the title of the paper itself, each legend should convey as much information as possible about what the table or figure tells the reader:

- what results are being shown in the graph(s) including the summary statistics plotted
- the organism studied in the experiment (if applicable),

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- context for the results: the treatment applied or the relationship displayed, etc.
- location (ONLY if a field experiment),
- specific explanatory information needed to interpret the results shown (in tables, this is frequently done as footnotes)
- culture parameters or conditions if applicable (temperature, media, etc) as applicable, and,
- sample sizes and statistical test summaries as they apply.

How much methodology and results are reported in the legends is journal specific. Hot-off-the-press journals, such as *Science* and *Nature*, limit the body text so that virtually all of the Methods are presented in the figure and table legends or in the footnotes. Many of the results are also reported in the legends.

17.11 My research was based on various surveys and interviews. How should I report quotations from the people we interviewed?

Generally speaking:

- there is no need to translate / report word for word what your interviewees said
- your transcript should simply enable the reader to understand the core points the interviewee made
- even if the sentence uttered by the interviewee was not complete, if appropriate you can complete it if it was obvious what the interviewee wanted to say
- remove any tangential / irrelevant phrases
- remove any filler words - *I mean, in other words, that is to say, you know, um, er*

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The following sentence comes from an interview with a business manager talking about the actions her company is taking to be more sustainable. In the red version there are a lot of brackets and dots (...). Readers are NOT interested in knowing the exact words that the manager used, they just want to know the substance of what the manager said.

We realized that if we continued to do what we were doing, as we used to, it would be problematic, precisely because [...] legislation [...] [is changing], in favor of preserving our planet.

We realized that if we continued to do what we were doing, in the way we have always done it, would be problematic, precisely because legislation is becoming stricter in favor of preserving our planet

However, you should ignore all the above points if the whole purpose of the interviewee was to report the exact words that were said, e.g. for some linguistics research.

The quotes you provide need further discussion - otherwise the reader is left to make their own interpretations and to try to make sense of the quotation.

17.12 Summary: How can I assess the quality of my Results section?

To make a self-assessment of your Results section, you can ask yourself the following questions.

- Have I expressed myself as clearly as possible, so that the contribution that my results give stands out for the referees and readers?
- Have I limited myself to only reporting the key result or trends that each figure and table conveys, rather than reiterating each value?
- Have I avoided drawing conclusions? (this is only true when the Results is an independent section)
- Have I chosen the best format to present my data (e.g. figure or table)? Have I ensured that there is no redundancy between the various figures and tables?
- Have I ensured that my tables of results are comprehensive in the sense that they do not exclusively include points that prove my point?

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- Have I mentioned only what my readers specifically need to know and what I will subsequently refer to in the Discussion?
- Have I mentioned any parts of my methodology (e.g. selection and sampling procedures) that could have affected my results?
- Have I used tenses correctly? PAST SIMPLE for your findings (in the passive form), PRESENT SIMPLE (descriptions of established scientific fact)

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Chapter 18

Discussion



Albert Einstein's top quotations

If you can't explain it simply, you don't understand it well enough.

To raise new questions, new possibilities, to regard old problems from a new angle, requires creative imagination and marks real advance in science.

Look deep into nature, and then you will understand everything better.

Learn from yesterday, live for today, hope for tomorrow. The important thing is not to stop questioning.

The difference between stupidity and genius is that genius has its limits.

Education is what remains after one has forgotten what one has learned in school.

The only reason for time is so that everything doesn't happen at once.

Most people say that it is the intellect which makes a great scientist. They are wrong: it is character.

There are two ways to live: you can live as if nothing is a miracle; you can live as if everything is a miracle.

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_18

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18.1 What is a Discussion?

Here are two discussions of the same findings. Note how in the blue version the key result is highlighted more effectively.

In an in-depth study (Smith et al (2024) of the intelligence quotas of doctoral students researching in the fields of engineering, robotics, biosciences, agriculture and veterinary sciences revealed that such students had a level of intelligence, which on average, was equal to a value of 8.115% above the norm. In the present work, with a larger sample (i.e. 5000 students as opposed to the 500 students analysed in the study by Smith et al) it was found that students in the same disciplines (i.e. engineering, robotics, biosciences, agriculture and veterinary sciences) had an intelligence quota of 9.996% below the norm.

Smith et al (2024) found that PhD students in engineering, robotics, biosciences, agriculture and veterinary sciences had above average intelligence (just over 8%). Our study totally contradicts Smith's finding. Using a sample that was ten times larger, our experiments proved that such PhD students have very limited intelligence (a surprising 10% below the norm), and would in fact be more suited to cleaning toilets than carrying out research. This radical finding may help governments reduce the amount of funding given to post-graduate university education.

The blue version reveals some important points:

- important findings get lost in long sentences (the sentence length in Discussion A is around 50 words, in B less than 20)
- a short sentence attracts attention (e.g. the 6-word sentence in B)
- you should make clear comparisons and contrasts with literature data
- readers need to be alerted when you are about to announce something important (e.g. *Our study totally contradicts Smith's finding*)
- clear writing with no redundancy is easier to read, easier to immediately understand the importance, and easier to remember
- the occasional use of emotive words (*surprisingly, radical*) may help to give more impact
- using the active and personal forms is more dynamic (e.g. *Our experiments proved* versus *it was found that*)

The primary aim of a discussion is to highlight the level of innovation of your research. Basically, it justifies why you wrote the paper. Thus, along with the Abstract, it is generally the most important section in the paper.

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In the Discussion / Conclusions it is essential to:

- be clear what YOU did and what other authors have done
- highlight your UNIQUE contribution
- discuss LIMITATIONS of your findings
- state what the applications and implications of your research are

You always need to have the following questions in your head.

1. What is your most important finding?
2. Why is it so fantastic?
3. How does it compare with similar findings made by other researchers? What are its advantages and disadvantages?

18.2 How should I structure the Discussion?

The Discussion should answer the following questions, and possibly in the following order. You can thus use the answers to structure your Discussion. This gives you a relatively easy template to follow.

- What are my most important findings?
- Do these findings support what I set out to demonstrate at the beginning of the paper?
- How do my findings compare with what others have found? How consistent are they?
- What is my personal interpretation of my findings?
- What other possible interpretations are there?
- What are the limitations of my study? What other factors could have influenced my findings? Have I reported everything that could make my findings invalid?
- Do any of the interpretations reveal a possible flaw (i.e. defect, error) in my experiment?
- Do my interpretations contribute some new understanding of the problem that I have investigated? In which case do they suggest a shortcoming in, or an advance on, the work of others?
- What external validity do my findings have? How could my findings be generalized to other areas?

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- What possible implications or applications do my findings have? What support can I give for such implications?
- What further research would be needed to explain the issues raised by my findings? Will I do this research myself or do I want to throw it open to the community?

Whatever your discipline you will need to answer all the questions above, with the possible exception of Question 8 (your findings may only be very preliminary). Whether you answer Questions 8–10 will depend on whether you have a separate Conclusions section, if so, the Conclusions may be a more appropriate place.

It may make sense for you to organize your Discussion following the same sequence as you presented your findings in the Results section. In this case, you discuss each survey, study or experiment, and interpret it within the overall scenario of the problem.

18.3 Active or passive? What kind of writing style should I use?

In the Discussion you will constantly be comparing your work with other authors' work. In your head you know what you did, and you know what other authors have done. But the reader doesn't. You need to make a very clear distinction, so that in every sentence the reader is 100% clear about whose work you are referring to (Chapter 6).

Passive sentences do not reveal the author of the action and so the reader will not understand if you are referring to your findings or another person. So, to avoid ambiguity, where possible use active sentences.

The table below shows five examples. The first two make it 100% clear to the reader whose work is being talked about. The other three are in order of decreasing clarity.

In 2024, *we confirmed* that complex sentences reduce readability [25].

We clearly indicates that you are referring to your own work.

In 2024, *Carter suggested* that complex sentences could also lead to high levels of stress for the reader [36].

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Carter, who is another author, is the subject of the verb. Thus it is clear to the reader that this is not your work.

In 2024, *it was suggested* that complex sentences could also lead to high levels of stress for the reader [Carter, 36].

The passive form means that the reader is not sure until the end of the sentence if it was you or another author. A long literature review or Discussion full of sentences like this is very heavy and annoying for the reader.

In 2024, *it was suggested* that complex sentences could also lead to high levels of stress for the reader [25].

Readers cannot know who made the suggestion unless they go to Ref. 25 and see if it was you or someone else.

In 2024, *it was suggested* that complex sentences could also lead to high levels of stress for the reader.

There is no reference. Readers cannot be sure if you made the suggestion or someone else. This is a very typical mistake in papers and is a very dangerous way of referring to the literature.

18.4 How should I begin the Discussion?

Below is a possible way to begin your discussion.

First, remind readers of your goals, preferably in a single sentence:

One of the main goals of this experiment was to attempt to find a way to predict who shows more task persistence.

Second, refer back to the questions (hypotheses, predictions etc.) that you posed in your Introduction:

These results both negate and support some of the hypotheses. It was predicted that greater perfectionism scores would result in greater task persistence, but this turned out not to be the case.

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Third, refer back to papers you cited in your Review of the Literature:

Previous studies conflict with the data presented in the Results: it was more common for any type of feedback to impact participants than no feedback (Shanab et al., 1981; Elawar & Corno, 1985).

Fourth, briefly restate the most important points from your Results:

While not all of the results were significant, the overall direction of results showed trends that could be helpful to learning about who is more likely to persist and what could influence persistence.

You could begin with any of 1–4 above, or perhaps use them all in combination. Next, you give readers a very brief statement of what you can conclude from your findings. You can then use this statement as a starting point for interpreting your findings and comparing them to what is already known in the literature.

18.5 Why and how should I compare my work with that of others?

A good structure for comparing your work with others is:

1. Make a general statement regarding your findings
2. Mention another author's work that relates directly to your findings
3. Make a link between her/his work and your work
4. Clearly state how your work differs from her/his work
5. State the conclusions that can be drawn from your results in light of these considerations

The text on the next page comes from a paper entitled *Exploring Stock Managers' Perceptions of the Human Animal Relationship on Dairy Farms and an Association with Milk Production* by Catherine Bertenshaw and Peter Rowlinson.

The authors did a postal survey of 516 UK dairy (i.e. milk) stockmanagers (i.e. farmers) about how they believed humans could affect the productivity, behavior and welfare of cows and heifers (young female calves that have not given birth). Nearly half said they called their cows by name – such cows had a 258 liter higher milk yield than those who that were not called by their name. About 10% said that a fear of humans resulted in a poor milking temperament.

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Below is the beginning of the Discussion section (the numbers in brackets are mine):

(1) Our data suggests that UK dairy farmers largely regard their cows as intelligent beings, capable of experiencing a range of emotions. Placing importance on knowing the individual animal and calling them by name was associated with higher milk yields. (2) Fraser and Broom [1997] define the predominant relationship between farm animals and their stock managers as fear. (3) Seventy-two percent of our commercial respondents thought that cows were not fearful of humans, although their reports of response to an approaching human suggest some level of fear, particularly for the heifers. With both cows and heifers this would appear to be greater in response to an unfamiliar human. Respondents also acknowledged that negative experiences of humans can result in poor behavior in the parlor. (4) Hemsworth et al. [1995] found that 30–50% of the variation in farm milk yield could be explained by the cow's fear of the stockperson, therefore recognizing that fear is important for animal welfare, safety, and production.

In (1), Catherine begins with an overall summary of her key finding and its implications. In (2) she mentions a previous study (by Fraser) in the same topic area and thus connects her findings with the literature.

Fraser's study gave contrasting results to what Catherine reveals in (3). However, in (3) Catherine also tries to account for some of what Fraser's found (*although … heifers*) and in (4) finds further confirmation of Fraser's findings in another study.

Catherine thus adopts a diplomatic approach in which she questions the findings of other authors in a constructive way. She uses their results either to corroborate her own results, or to put her results and their results in a new light.

Another useful skill that Catherine uses throughout her Discussion, is that she constantly clarifies for the reader between whether she is talking about her findings or those of other author, or whether she is just talking in general,

(5) The elaborated responses reported in *our postal survey* contribute some examples of the capacities of cattle, and this contextual human insight may be useful for developing hypotheses for further study.

(6) Most respondents (78%) thought that cows were intelligent. (7) However, a *study by Davis and Cheek* (1998) found cattle were rated fairly low in intelligence. *They* suggested that the ratings reflected the respondents' familiarity with the animals. (8) The stock managers in *our survey* were very familiar with their cattle and had a great understanding of the species' capabilities, through working with them daily. (9) *Stockpersons' opinions offer* valuable insight into this subject, which could enable more accurate intelligence tests to be devised; for example, to test whether cows can count in order to stand at the feed hopper that delivers the most feed.

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(10) Hemsworth and Gonyou (1997) doubt the reliability of an inexperienced stockperson's attitudes towards farm animals. *Our* survey found an experienced workforce (89.5% > 15 years).

In (5) Catherine concludes a paragraph by suggesting a future course of action.

(6) is the first line of the next paragraph, so it is clear that the *respondents* are her respondents and not another author's.

In (7) she uses *however* to indicate that she is going to give some contrasting information. Her use of *they* clearly refers back to Davis and Cheek.

In (8) Catherine then clarifies for the reader that she is now focusing on her study. She does this again using *our*. If she had not inserted the phrase "in our survey", the reader would not know which stock managers she was talking about. Not making this distinction is an incredibly common error in Discussions and leads to total confusion for the referee and readers. In the literature *our* is often used, even if the style of the rest of the paper is impersonal (i.e. the passive is used, rather than *we*). Using *our* can be crucial in differentiating your work from others.

In (9), like she does in (5), Catherine makes a mini summary of what she has said in the rest of the paragraph. Her use of the SIMPLE PRESENT (*offer*) shows that she is talking about all stockpersons – not just those in her study or in Davis and Cheek's study. She also recommends a course for future action.

In (10) Catherine begins a new paragraph to indicate that she is now going to cover another subtopic. Good use of paragraphs is essential in signaling to readers that you are moving on to discuss something different. Catherine begins with a reference to the literature to establish the new subtopic, and then immediately moves on to her findings to make a contrast between inexperienced and experienced workers.

The rest of her Discussion is structured in a similar manner, in which she provides more conclusive evidence that calling a cow by its name, rather than problems connected with fear, is more likely to affect milk production. In each case, she makes it 100% clear to her readers why she has mentioned another person's work and how it relates to her work.

For more on this critical point see Chapter 6.

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18.6 How can I give my interpretation of my data while taking into account other possible interpretations that I do not agree with?

In a paper that won him an Ignobel Prize, Magnus Enquist made a case for the fact that chickens are able to discriminate between good looking and ugly human beings. Here is an extract of the Discussion section of his paper, *Chickens prefer beautiful humans*. The numbers in brackets are mine.

(1) We cannot of course be sure that chickens and humans processed the face images in exactly the same way. (2) This leaves open the possibility that, while chickens use some general mechanism, humans possess instead a specially evolved mechanism for processing faces. (3) We cannot reject this hypothesis based on our data. (4) However, there are at least two reasons why we do not endorse this argument. First, it is not needed to account for the data. We believe that the existence of a task-specific adaptation can be supported only with proofs for it, rather than with absence of proofs against. Second, the evolutionary logic of the argument is weak. (5) From observed chicken behaviour and knowledge of general behaviour mechanisms we must in fact conclude that humans would behave the same way with or without the hypothesised adaptation. There would thus be no selection pressure for developing one.

His strategy for anticipating possible objections to his argument is to:

1. admit that he might be wrong - sentence (1)
2. put forward an alternative interpretation (2)
3. reiterate that his data could be used to confirm this alternative interpretation (3)
4. give reasons for not agreeing with this alternative interpretation (4)
5. propose his own conclusion (5)

18.7 How can I use *seems* and *appears* to admit that I have not investigated all possible cases?

It is crucial to be totally honest and non-misleading as to the status of results.

Let's take the example of a mathematical proof. There may be some cases that you have not checked, i.e. you are making an intuitive claim or guess based on what you have checked so far.

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In such cases you can use *it appear*s to be or *it seems*. Such phrases say exactly the truth, i.e. that something is true for the cases you have checked. You are telling the reader that you intuitively suspect or expect that it could be always true, but you don't claim it. That is what 'appears' means. You make no assertion as to the probability because you have not computed or assessed a probability.

It appears that stochastic processes for which x = y can produce finite dimension values.

This completes the proof of Theorem 1. Note how this enables us to determine all the Xs and Ys at the same time. Thus *it seems that* some natural hypotheses can be formulated as ..

However, you must make it 100% clear to the reader that, for example, you have not checked all cases, that your sample size was small, and that some external factors may have influenced you results.

18.8 What about the literature that does not support my findings - should I mention it?

Yes. Your aim is to be transparent. And don't forget that the reviewers will quickly spot that you have only mentioned other works that support your own - this is one of the main jobs of a reviewer.

18.9 How can I show the pitfalls of other works in the literature?

There are three areas to call into question regarding the work of other authors.

1. Hypotheses that have never really been tested. You want to test them.
2. Other studies have only been conducted very generally or in one specific field. You want to apply this research to a new area.
3. Other studies have limitations. You are trying to overcome these limitations.

The important thing when criticizing other people's work is not to undermine their credibility. The idea is that if you treat others with respect, they will treat you with respect.

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18.10 Should I discuss the limitations of my research?

It is essential that you inform readers of any limitations to your research or any failures or contradicting data. There is no need to consider these aspects of your research to be totally negative. Your readers will appreciate learning about what went wrong, as this may help them with their own research.

However, don't end the Discussion or Conclusions with your limitations. You want your paper to end on a positive note, so in your final paragraph(s) talk about the benefits and wider applications of your research.

To learn how to describe and discuss your limitations, see Chapter 7.

18.11 How can I be more concise?

After the Abstract, the Discussion is generally the most important section in the paper. This is where you highlight for the reader what you have achieved and what it all means in the context of the state of the art. So it is very important that you present this information as concisely as possible. Compare these two versions of the same sentence:

Furthermore, PCB 180 *has been reported* to share several toxicological targets with dioxin-like compounds [Ref. 1]. Hence, *it appears reasonable to assume* that PCB 180 may affect the AhR pathway in pituitary apoptosis. In fact, the involvement of the AhR pathway in the regulation of apoptosis *has been recently reported* [Ref. 2]. The contents of the PCB were in agreement with *the results of* Chad et al [Ref. 3] and similar to *those reported by* Jones [Ref. 4]. (78 words)

Furthermore, PCB 180 *shares* several toxicological targets with dioxin-like compounds [Ref. 1]. Hence, PCB 180 *may* affect the AhR pathway in pituitary apoptosis. In fact, the AhR pathway *may be* involved in the regulation of apoptosis [Ref. 2]. The PCB contents *were in agreement with* Chad et al [Ref. 3] and similar to Jones [Ref. 4]. (56 words)

The red version forces readers to read the same phrase (*has been reported*) or a similar phrase (*reported by*, *the results of*) again and again. Other phrases that you may be able to remove are: *it has been suggested / proposed that ..., it is well known that ...*

Such redundancy may cause the reader to read with less attention, and thus they may miss the important points you are trying to make.

The blue version is one third shorter than the red version, but with no loss of detail. If your Discussion was originally three pages long, then in theory you could save one page by removing redundancy (Chapter 4).

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18.12 How can I use a chatbot to help me with my Discussion?

You can ask the bot to answer specific questions (i.e. the questions outlined in the subsections of this chapter). Below are some examples. In *italics* is my prompt, and in blue is the bot's answer.

I am writing the Discussion section. I have this negative result: Given the small sample size we are unable to draw any certain conclusions. How could I express this negative result in a more positive way?

One way to express a negative result in a more positive way is to focus on the potential implications of your findings and suggest avenues for future research. Here is an example:

"While our study's small sample size precludes drawing definitive conclusions, our findings suggest interesting directions for future research. Our results highlight the need for larger, more diverse samples to better understand the phenomenon under investigation. In addition, future studies could explore alternative methods for data collection or analysis to better capture the complexities of the phenomenon."

In this way, you are acknowledging the limitations of your study, but also emphasizing the value of your findings and offering constructive suggestions for future research.

As you can see from the blue text above, the bot has provided useful advice and a possible construction for a sentence justifying your negative result. As usual, the bot's sentence is very generic. You would need to tailor it to your specific case by adding relevant details.

Below is another example:

This text comes from a Discussion. Please make it more concise:

PCB 180 has been shown to share toxicological targets with dioxin-like compounds [Ref. 1], suggesting a potential effect on the AhR pathway in pituitary apoptosis, which is regulated by the AhR pathway [Ref. 2]. The PCB contents were consistent with previous studies [Ref. 3, 4].

The bot's version is even shorter than the blue version I presented in the previous subsection (18.11): 45 vs 56 words. It has achieved this by not putting the names of the author and by combining the last two references together. Clearly, you need to decide whether the bot's methods of being concise are suitable for you and your journal.

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You can also use a bot to check for any potentially controversial or biased statements in your Discussion (or other parts of the paper). I inserted some sample texts into ChatGPT, which returned comments such as:

This statement may be seen as divisive and biased towards one political party over another, and lacks evidence to support its claim.

This statement may be considered overgeneralizing and lacking in evidence.

It may be beneficial to rephrase some statements and provide more evidence to support the claims made.

The language used in this text may be considered offensive to xxx, and the portrayal of xxx as unethical and unprofessional could be harmful to their reputation.

Furthermore, the text contains multiple ethical violations, such as ...

Therefore, the text should not be used in academic or professional settings as it may be considered offensive and inappropriate.

Basically, the bot has acted as a sensitivity editor. This might be helpful to you if your research is in any way controversial.

18.13 What are the dangers of using a chatbot to help me write my Discussion?

The bot's ability to do what you request is severely limited by the information you give it. I asked a bot to rewrite the red text from 18.1 "so that the key results are highlighted more clearly". Below in italics is the bot's response.

A recent study by Smith et al (2024) examined the intelligence quotas of 500 doctoral students in engineering, robotics, biosciences, agriculture, and veterinary sciences. Their findings showed that on average, these students had a level of intelligence 8.115% above the norm. However, in a larger study with a sample of 5000 students in the same disciplines, it was discovered that these students had an intelligence quota of 9.996% below the norm. This indicates a significant difference between the two studies, with the latter showing a surprising and potentially concerning result.

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If you compare the bot's output with my suggested improvement in 18.1 and reported again below, you will see that the bot can only work with the info that you give it. It may suggest additional sentences or different ways of phrasing something, but it cannot search your brain to find the additional important info (e.g. government spending on education) that needs to better highlight your key findings.

Smith et al (2024) found that PhD students in engineering, robotics, biosciences, agriculture and veterinary sciences had above average intelligence (just over 8%). Our study totally contradicts Smith's finding. Using a sample that was ten times larger, our experiments proved that such PhD students have very limited intelligence (a surprising 10% below the norm), and would in fact be more suited to cleaning toilets than carrying out research. This radical finding may help governments reduce the amount of funding given to post-graduate university education.

The bot has also made serious mistakes.

In the blue version above you can see that there is a comparison of OUR study with Smith's. The bot's version refers to 'a larger study', with no reference to the fact that it was our study. So the whole point of the comparison is lost.

Be careful: A bot may mistakenly transform your meaning.

Note: you should NEVER use a chatbot to generate a text for you, and then submit it to your professor or for publication without making any changes and without acknowledging that a bot was involved in creating the text.

18.14 How long should the paragraphs be?

Your aim is to allow your reader to quickly understand how your results add to the current state of the art.

Your Discussion should thus not be one long paragraph or a series of very long paragraphs.

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The moments to begin a new paragraph in the Discussion are when you:

- change topic, or you look at a different aspect of the same topic
- move from talking about one result to another result
- mention another author with similar or different results
- justify any differences between your work and the literature
- want to talk about the consequences of what you have just been describing
- talk about the limitations of your work
- talk about the implications of your study and any future research lines
- draw conclusions

However, for some of the points above you will certainly need more than one paragraph. Consider having a series of sub-headings within the Discussion, and under each heading you can have one or more paragraphs.

If you print your Discussion you will immediately see the undesirable effect of having long paragraphs. They are not inviting for the reader.

18.15 How should I end the Discussion if I have a Conclusions section?

If your journal requires both a Discussion and a Conclusions then you can:

1) Tell your readers if and how your findings could be extended to other areas. But you must provide evidence of this. If you repeated your experiment in a different context, would you get the same result?

We only used a limited number of samples. A greater number of samples could lead to a higher generalization of our results …

Although this is a small study, the results can be generalized to …

Our results may hold true for other countries in Asia.

2) Suggest ways that your hypothesis (model, device etc.) could be improved on.

We have not been able to explain whether x = y. A larger sample would be able to make more accurate predictions.

A greater understanding of our findings could lead to a theoretical improvement in …

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3) Say if and / or why you ignored some specific areas.

Our research only focuses on x, whereas it might be important to include y as well. In fact, the inclusion of y would enable us to …

We did not pay much attention to… The reason for this was …

4) Admit what you have not been able to do and as a consequence cannot provide conclusions on.

Unfortunately, our database cannot tell the exact scale of Chinese overseas R&D investment. Consequently we cannot conclude that …

5) Reiterate your reasons for choosing your topic of investigation in order to convince your readers of the validity of what you have said in the Discussion.

As mentioned in the Introduction, so far no one appears to have applied current knowledge of neural networks to the field of mass marketing fraud. The importance of our results using such networks thus lies both in their generality and their relative ease of application to new areas, such as counterfeit products.

18.16 How should I end the Discussion if I do not have a Conclusions section?

Whether or not you have a Conclusions section, your Discussion should end with a summary of the main points you want your readers to remember.

Catherine Bertenshaw concludes her Discussion (18.5) in the classic way by stating:

1) what her findings imply

The attitudinal information from our survey shows that farmers hold cows in very high regard.

2) what her recommendations are

These results create a positive profile of the caring and respectful attitudes of UK farmers to their stock, and this image should be promoted to the public.

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3) how her research could be continued

A 56% response rate suggests the respondents are a good representation of UK stock managers. Further on-farm interviews, observations, and animal-centered tests are needed to confirm the inferences made from the data collected in this postal survey.

Many Discussions end in the same way as Catherine's, particularly those that have no Conclusions section. Catherine's paper does in fact have a Conclusions section, but it is only 70 words long and provides an overall summary of her data, and what she thought the implications of her findings might be.

18.17 Summary: How can I assess the quality of my Discussion?

When you have finished writing your Discussion, it is a good idea to make sure you can honestly answer 'yes' to all the questions below. This will enable your peers to make a critical assessment with regard to the strengths and weaknesses of (a) how you carried out your research (b) and how you analyzed your findings. The result will be that you will be seen as a credible researcher.

- Is my contribution to the knowledge gap clear? Have I underlined the significance of my findings? Have I related my findings and observations to other relevant studies?
- Have I explained what I believe to be new and important very clearly but without exaggerating? Have I ensured that I have not over-interpreted my results (i.e. attributed interpretations to them that cannot actually be supported)?
- Have I truly interpreted my results, rather than just reiterating them? Have I shown the relationship (confirmation or rejection) between my results and my original hypothesis? Have I generated new theory rather than simply giving descriptions?
- Is there a good balance, or is it a one-sided version? Have I really offered alternative explanations?
- Have I clearly distinguished fact from speculation? Will the reader easily be able to understand when I am merely suggesting a possible interpretation rather than providing conclusive evidence for something?

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- Have I ensured that there is no bias in my research? (i.e. I have not hidden any of my data or any unexpected results, simply because they do not confirm what I was hoping to find)
- Have I included those works in the literature that do not corroborate my findings? Likewise, have I avoided distorting the magnitude or direction of the data of the literature that I have selected? (i.e. I have made sure that I have not committed publication bias)
- Have I discussed my findings in the context of what I said in the Introduction? Have I exploited my Review of the Literature?
- Have I integrated my results with previous research (including my own) in order to explain what I observed or found?
- Have my criticisms of the literature been justified and constructive?
- Have I ensured that I have not introduced any new findings (i.e. findings not mentioned in the Results)?
- Are all the statements I have made in the text supported by the data contained in my figures and tables?
- Have I removed any trivial information? Have I been as concise as possible?

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Chapter 19

Conclusions



Back to the future

The Year 2000 – A Framework for Speculation on the Next Thirty-Three Years written in 1967 by staff members of the Hudson Institute listed 'One hundred technical innovations very likely in the last third of the Twentieth Century', including:

- New methods of water transportation (e.g. large submarines and special purpose container ships)
- Major reduction in hereditary and congenital defects
- Extensive use of cyborg techniques (mechanical aids or substitutes for human organs, senses, limbs, or other components)
- New and useful plant and animal species
- Pervasive techniques for surveillance, monitoring, and control of individuals and organizations
- Some control of weather and/or climate
- Human hibernation for relatively extensive periods (months to years)
- General and substantial increase in life expectancy, postponement of ageing, and limited rejuvenation
- Permanent inhabited undersea installations and perhaps even colonies
- Personal "pagers" (perhaps even two-way pocket phones) and other personal electronic equipment for communication, computing, and data processing

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19.1 What is the purpose of the Conclusions section?

One of my PhD students once remarked to me: *I find the conclusions quite difficult to write, even in my own language. If I wrote everything in the paper, what should I add at the end?* Her question sums up the dilemma that authors have with the Conclusions. It's not that the Conclusions section is difficult to write, it's just that authors don't know what to write. In fact, several journals do not even have a separate Conclusions sections, authors simply write a concluding paragraph in their Discussion.

Although the Conclusions may not be the last section that readers read, there is a strong probability that they will be the last thing that the referee reads. Consequently, the Conclusions must be clear and concise, and leave the referee with a good impression. If your structure and English are poor this will have a negative impact on the referees and may affect their final decision as to whether to accept your paper or not.

The key skills are in knowing what referees and readers expect to find in Conclusions, not repeating exactly the same phrases and information from your Abstract and Introduction, and in providing a clear and high-impact take-home message for readers.

As clearly stated in an editorial in *Nature Physics* (https://www.nature.com/articles/nphys724)

> Conclusions are not mandatory, and those that merely summarize the preceding results and discussion are unnecessary (and, for publication in *Nature Physics*, will be edited out). Rather, the concluding paragraphs should offer something new to the reader.

However, many readers do expect a Conclusions section or your journal may require one. Bear in mind the words of wisdom of Jonathan Shewchuk, professor of computer science at Berkeley:

> Conclusions should synthesize the results of your paper and separate what is significant from what is not. Ideally, they should add new information and observations that put your results in perspective. Here's a simple test: if somebody reads your conclusions before reading the rest of your paper, will they fully understand them? If the answer is *yes*, there's probably something wrong. A good conclusion says things that become significant after the paper has been read. A good conclusion gives perspective to sights that haven't yet been seen at the introduction. A conclusion is about the implications of what the reader has learned. Of course, a conclusion is also an excellent place for conjectures, wish lists, and open problems.

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19.2 What tenses should I use?

Many tenses and constructions are used in the Conclusions – the future, conditionals, modal verbs etc. For details on how to use these forms see Chapter 14 in *English for Academic Research: Grammar, Usage and Style*.

One distinction that many authors make is between what they did during the research (SIMPLE PAST) and what they did during the writing process of the manuscript (PRESENT PERFECT).

We *have described* a method to extract gold from plastic. We *used* this method to extract 5 kg of gold from 50 kg of plastic. We *found* that the optimal conditions for this process were …

The first verb (*have described*) says what the authors have done in the paper, whereas the second and third verbs (*used, found*) say what they did in the laboratory (i.e. a finished action).

The following two sentences are incorrect in the Conclusions because they use the present simple instead of the PRESENT PERFECT:

In this paper we *consider* the robust design of an extractor for removing gold from plastic.

In this study, it *is demonstrated* that by using an ad hoc extractor gold can be easily removed from plastic.

However, the above two sentences would be correct in the Abstract or Introduction.

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19.3 I have no idea how to write a Conclusions section. Is there a quick way that a chatbot could help me?

If you insert your Abstract into the chatbot, you can tell it to generate a Conclusions section on the basis of the info in your Abstract.

Below is an existing Abstract that I instructed a chatbot to use as a basis to generate a Conclusions section. Below is my prompt.
Write a conclusions section to an academic paper on university students' behavior, based on the following abstract:

"In this work, PHDI, a tool for the prediction of the behavior of students at university is developed. PHDI is based on a lumped and distributed parameters approach and is capable of describing both the social and in-lecture behavior of first-year students. On the basis of the boundary conditions applied to the model, it is possible to obtain the operating map of changes in behavior. Particular care was devoted to the analysis of the number of hours spent using social media and of its influence on exam outcomes. The predictive capabilities of our tool were evaluated by simulating a reference case: first-year students from the 1950s. The most important parameters for the description of the behavior are detailed. In addition, a set of these parameters was found, in order to accurately simulate the complete operating map. Finally, numerical results were compared to measurements and a good agreement between experimental values and numerical predictions is found. The study highlights that excessive social media use is responsible for some divergent behavior amongst university students."

The bot generated the Conclusions below (in blue).

In this work, we have presented PHDI, a novel tool for predicting the behavior of first-year university students. Our model is based on a lumped and distributed parameters approach, which enables us to describe both the social and in-lecture behavior of students. By applying boundary conditions to our model, we *have obtained* the operating map of changes in behavior.

Our analysis *has focused* on alcohol consumption and its influence on exam outcomes. We *have found* that social media use are responsible for the inappropriate behavior of many university students. Our tool *has been evaluated* by simulating a reference case: first-year students from the 1950s. We *have identified* the most important parameters for the description of behavior, and found a set of these parameters that accurately simulate the complete operating map.

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Our numerical results *have been compared* to measurements, and we *have found* a good agreement between experimental values and numerical predictions. Our study has *important implications* for understanding the behavior of university students, and provides a valuable tool for predicting and mitigating problematic behavior. By identifying the *key parameters* that influence student behavior, our work can help universities to develop *targeted interventions* to improve student outcomes and reduce negative behaviors. Overall, our study represents an important step forward in understanding the complex social dynamics of university life, and has *important implications* for policy and practice in higher education.

The bot provides a very good template for your to work with.

However, the usage of tenses (19.2) is not always correct. The verbs highlighted in red would be better in the simple past tense (*we obtained, we focused* etc), as they describe what you did in the lab rather than relating to the process of writing the paper.

Moreover, there is some repetition ('important implications').

But the most significant failure of a chatbot is its inability to be specific. This is the result of it working with very limited input on your part. The last three sentences are very vague, though they are typical of the way researchers end their papers. Instead, readers need to be:

- reminded what the key parameters are
- given examples of targeted interventions
- told what the important implications are

Note: you should NEVER use a chatbot to generate a text for you, and then submit it to your professor or for publication without making any changes and without acknowledging that a bot was involved in creating the text.

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19.4 How should I structure the Conclusions?

The Conclusions section is not just a summary. Don't merely repeat what you said in the Abstract and Introduction. It is generally not more than one or two paragraphs long. A Conclusions section typically incorporates most or all of the following:

1. Statement of your principal findings – a very brief revisit of the most important findings pointing out how these advance your field from the present state of knowledge
2. Strengths and weaknesses of your study – an indication of the limitations of your study (though the Discussion may be a more appropriate place to do this)
3. Strengths and weaknesses in relation to other studies, discussing important differences in results
4. Meaning of the study: possible explanations and implications for researchers and also practical suggestions for those who in the public and private sector might usefully exploit your findings
5. Unanswered questions and recommendations for future research

Unlike the Abstract, the Conclusions:

- does not provide background details
- gives more emphasis to the findings (Point 1)
- talks about limitations, which are not normally mentioned outside the Discussion and Conclusions (point 2)
- covers three additional aspects (Points 3–5)

19.5 How can I differentiate my Conclusions from my Abstract?

These two sections have completely different purposes. The Abstract is like an advertisement for your paper – it has to attract the reader's attention. On the other hand, the Conclusions section is designed to remind readers of the most salient points of your paper. However, the Conclusions also have to add value. This added value is typically contained in the recommendations, implications and areas for future research.

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In any case, it is a good idea to revise the Abstract and Conclusions together, and even shift information from one to the other.

There will be some overlap between the two sections, but this is both accepted practice and inevitable.

The example below shows you how to avoid a 'cut&paste' from your Abstract. It comes from a fictitious paper entitled: *Six key strategies to a meaningful life: the non-believer's worldview*

As you read, compare the structure and the information given. The numbers in the Conclusions are just for the purposes of this book.

ABSTRACT With no hope of an afterlife, atheists may have difficulty rationalizing their purpose on earth. With the aim of understanding the coping mechanisms of non-believers, we interviewed 150 UK-born couples (125 mixed, 25 same sex; average age 46) who had happily cohabited for more than 15 years. Interviewees were asked ten simple questions regarding their attitudes to the meaning of life. Our results revealed that there are six key strategies in an atheist's pursuit of a happy and meaningful existence: keep everything simple, have fun, cultivate a sense of community, delight in the wonder of nature, find time for creativity, and help other people through frequent acts of kindness. Atheists that implement a combination of these six strategies were found to be more equipped than other non-believers to deal with the death of close ones, health problems, financial difficulties, and bad luck.

CONCLUSIONS 1) We found that six strategies are key to atheists having a satisfying life: simplicity, fun, community, a love of nature, and the importance of creativity and of helping others. 2) An additional but not unexpected finding, not considered in the original research aim, was that an unbridled respect for one's partner is fundamental for a long-lasting relationship. 3) In the light of the vacuous and aimless nature of Western society, our findings suggest that the six strategies should be taught in schools as part of children's philosophy or religious education lessons. 4) Comparisons with traditional religions revealed no substantial differences in approach, apart from a believer's blind faith in a benevolent omniscient overlord and the promise of an afterlife (or reincarnation). These commonalities indicate that traditional religions should attempt to be more sympathetic to atheists, and vice versa. 5) Future work will investigate how the promise of an afterlife may undermine the fulfillment of one's true potential on earth.

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Here is the structure of the Conclusions:

1. Repetition of key findings, paraphrased from the Abstract – this is the only overlap between the Abstract and the Conclusions
2. Additional relevant findings
3. Recommendations for policy makers
4. Implications
5. Areas for future research

Not all Conclusions will contain all of the above five points.

19.6 How can I differentiate my Conclusions from my Introduction and from the last paragraph of my Discussion?

The same comments made in 19.5 regarding the difference between the Abstract and the Conclusions, are also substantially the same as for the Introduction, so they are not worth repeating.

If your journal has a separate section for Conclusions, i.e. the conclusions are not included in the Discussion, then it may be best to shift any overall conclusions you may have made in your Discussion into your Conclusions. This means that the final paragraph of your Discussion may just be a conclusion regarding one specific point, rather than an overall summary of the whole paper.

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19.7 How can I increase the impact of the first sentence of my Conclusions?

Just as professional copy editors advise against beginning a paper with *This paper describes*, they also suggest avoiding ending the paper in the same way (*This paper has described*).

In this study it is concluded that compression plays an important part in ... It was found that ...

Compression plays an important part in ... In fact, it was found that ...

This work has demonstrated that a number of compounds present in X are responsible for delaying the onset of ...

A number of compounds present in X are responsible for delaying the onset of ...

We have shown that the crystal structure of X reveals that ...

The crystal structure of X reveals that ...

It has been suggested in this paper that the localization of X in neurons is a good marker for neuronal viability.

The localization of X in neurons suggests that it is a good marker for neuronal viability.

It is not difficult to be more direct and concise. In fact, the blue versions have simply removed the initial 5–8 words of the red versions.

This is because such phrases:

- waste a lot of words (5–7 words that tell the reader nothing)
- delay the main topic
- are not memorable for the reader and have no impact

The result is that the main topic of the paper now appears in the first two to four words of the Conclusions. The result is a Conclusions section that is more concise and has more impact.

The RVs versions are considerably more direct and are found in many disciplines, particularly in medicine and biology related disciplines. If you are worried that they are too direct, then you can make them 'softer' by writing, for example, *could be responsible* (second blue sentence) and *seems to reveal* (third blue sentence).

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19.8 I don't have any clear Conclusions, what can I do? Should I mention my limitations?

Sometimes it is impossible to leave the reader with clear conclusions regarding the contribution of your work – maybe your method turned out to be inappropriate and your results were not as brilliant as you were hoping for! In such cases simply say what you have learned about the problem and then suggest possible lines of future research. Such a final section is generally entitled Concluding Remarks.

If you don't have any clear conclusions, it is important not to present your findings in an exaggerated light or to say something uninteresting or irrelevant. Readers may still be able to benefit from what you found, or equally important, did not find (Chapter 7).

Here are some examples of authors admitting that their work did not achieve all that they had hoped for. In some cases readers are immediately warned of this 'failure' through the use of the words highlighted in italics.

Unfortunately, we could not assess how much of the difference in outcome was due to …

When results are compared across different components, the confidence intervals overlap, and we have no conclusive evidence of differences in …

Although some progress has been made using our model, this incremental approach provides only a partial answer

Unfortunately this trial had too few subjects to achieve sufficient power and had a low …

It is also unclear what conclusion should be drawn …

Regrettably, we did not have the means to …

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To make your Conclusions not sound too negative, you can add some hope for the future.

Although it is too early to draw statistically significant conclusions, two patterns seem to be emerging …

However, more definite conclusions will be possible when …

Nevertheless, our study confirms recent anecdotal reports of …

Despite this, our work provides support for …

In any case, we believe that these preliminary results indicate that …

Again, the first words of the sentence alert the reader that you are now going to qualify the negative stuff you said before by offering some optimism. You could also use some conditional sentences to show what might have been possible if you had had different circumstances, or what might be possible in the future.

If we had managed to … then we might have been able to …

If we manage to … then we might be able to.

19.9 How can I end my Conclusions?

You can use one or more of these two ways.

The first is to show how your work could be applied in another area.

Our findings could be applied quite reliably in other engineering contexts without a significant degradation in performance.

These findings could be exploited in any situation where predictions of outcomes are needed.

Our results could be applied with caution to other devices that …

Note how the above phrases all make use of *could* as a way not to make your statement too strong.

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You might however like to say where they could not be applied for the moment.

However, it remains to be further clarified whether our findings could be applied to …

Further studies are needed to determine whether these findings could be applied to components other than those used for …

The second typical ending is to suggest future work.

In the future, we plan to develop additional features for the software, such as …

In the future, additional features for the software could be developed, such as …

In the future, we suggest that additional features for the software could be developed, such as …

Finally, don't attempt to lessen the negative impact of the limitations of your research (see Chapter 7) by simply claiming that these limitations could be solved in 'future work'. Referees and editors can quickly see through this strategy and thus dismiss your claims as being unfounded or vague. Instead, you need to give some details regarding how they could be solved.

19.10 To be honest, I don't really know what the implications of my research are or what possible avenues for future research there are. How could a chatbot help me?

Sometimes, you may be unable to see how your work fits into a bigger picture and how it might possibly inspire future research.

The bot can make suggestions for you. Here is an example of an instruction that you could use:

The text is from a Conclusion section to an academic paper. Please tell me i) what the implications of the findings are, ii) what possible avenues of future research could be.

The bot will generate some interesting ideas. And you can use the text generated by the bot as a template for writing the final sentences of your Conclusions – but ONLY as a template.

> You CANNOT simply accept what the bot produces as being accurate and reliable. You must consult with your professor or fellow researchers.

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19.11 How should I write the Acknowledgements?

The Acknowledgements generally include one or more of the following.

1. Sources of funds.
2. People who gave significant technical help (e.g. in the design of your experiment, in providing materials).
3. People who gave ideas, suggestions, interpretations etc.
4. The anonymous reviewers

It is a good idea to let the people that you wish to acknowledge see the exact wording of how you want to acknowledge them – they might think it is too effusive (or occasionally, insufficient).

The style of giving acknowledgements may be quite different from the style of the rest of the paper. For example, you can use the first person (*I, we*).

Keep your acknowledgements as short as possible, they are generally of little interest to anyone apart from those mentioned.

19.12 Summary: How can I assess the quality of my Conclusions?

To make a self-assessment of your Conclusions, you can ask yourself the following questions.

- Is what I have written really a Conclusions section? (If it is more than 200–250 words, then it probably isn't – it needs to be much shorter)
- If the conclusions are included in the Discussion, have I clearly signaled to the reader that I am about to discuss my conclusions (e.g. by writing *In conclusion …*)?
- Have I given a maximum of one line to comments related to descriptions of procedures, methodology, interviews etc.? (Generally such comments are not needed at all, unless the primary topic of your paper is the methodology itself)
- Have I avoided cut and pastes from earlier sections? Do my Conclusions differ appropriately from my Abstract, Introduction and final paragraph of my Discussion?

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- Are my Conclusions interesting and relevant?
- Have I given my Conclusions as much impact as possible and have I avoided any redundant expressions?
- Have I avoided any unqualified statements and conclusions that are not completely supported?
- Is my work as complete as I say it is? (i.e. I am not trying to get priority over other authors by claiming inferences that cannot really be drawn at this stage)
- Have I introduced new avenues of potential study or explained the potential impact of my conclusions? Have I ensured that I have only briefly described these future avenues rather than getting lost in detail?
- Are the possible applications I have suggested really feasible? Are my recommendations appropriate?
- Have I used tenses correctly? PRESENT PERFECT (to describe what you have done during the writing process), PAST SIMPLE (what you did in the lab, in the field, in your surveys etc.)

In addition, you should look at the summary questions for the Discussion (18.17), as these may also be helpful in deciding whether your Conclusions will have the necessary impact on your readers.

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Chapter 20

The Final Check



Referees' reports

The following are exact quotes from editors and reviewers of papers submitted by Prof. Charles Fox of the University of Kentucky.

- Fox seems to have an enlarged view of the significance of his work.
- The explanation is interesting, and worth stating [but] I didn't need to read the paper to get the point.
- This is still a most unexciting paper, but it is probably useful to confirm experimentally what everyone knows intuitively.
- Fox's productivity in terms of number of papers is impressive [but] there is nothing new in any of his work.
- The point is so elementary that it does not require a manuscript of this length to develop it.
- It is with great regret that I must inform you that your submission … will be accepted, pending revision. My regrets are motivated, of course, by your excessive profile in the literature. Nothing would have given more pleasure than to reject the paper, but the excellent reviews, and my own opinion of the work, make this impossible.
- I went through the paper, trying to find something wrong with the manuscript, because that is what editors do. Then I tried to think of changes I could ask you to make because the other thing editors do is slow up the publication process by requesting revisions. I failed at both of these … Consequently, I have violated all traditions [and recommend] that your manuscript be published exactly as is.
- The results are not earth-shattering, but the paper is beautifully written … In fact, the paper is so well-crafted that it may become a classic. … You have done such a good job that I now want to go back and read all your other papers.

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0_20

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20.1 How good is my paper?

At http://shitmyreviewerssay.tumblr.com/ researchers can leave the negative comments made by reviewers of their papers (the site is wonderful, I recommend you visit it). Some reviewers can be very insulting. For example:

It is clear that this manuscript will not win a beauty contest. There are still many awkward sentences that make me feel like listening to someone that scratches a glass plate with an iron nail.

The biggest problem with this manuscript, which has nearly sucked the will to live out of me, is the terrible writing style.

However many comments may be true of your manuscript. You can use the ones below as a checklist to judge the quality of your work.

1. I'm sorry, but I'm reading this thinking, we've already heard about that, you're telling us again and now you're telling me I'm going to hear about it again in the next chapter.
2. Considering by the end of the paper I had no idea what was being said, I don't believe the paper was successful in its argument. In fact, it felt like the author was merely saying the same thing over and over again but with synonyms.
3. I could not find any passage in the MS that would explain to me what is the exact novel idea, proposal, argument, or hypothesis.
4. The interchangeable use of evaluate and validate is a concern because it is not clear if authors know the difference between these two verbs.
5. There are not enough headings.
6. I counted 15 uses of 'clear' or 'clearly.' That's one per page. 'Clearly' the results aren't as clear as the author would like them to be.
7. The authors presents a flurry of statistics, but they do not explain why or how those are relevant to the study.
8. It is not clear what the author wants to accomplish.
9. Overall, this paper goes to significant amounts of trouble to accomplish something that is already well established as a technique in a less elegant manner than is currently employed. This paper should not be published.
10. The paper brings to mind the Mark Twain quote: 'I didn't have time to write a short letter, so I wrote a long one instead.'
11. The paper is grossly over referenced, and reads a little like a student trying to impress a supervisor that a lot has been read, rather than a mature and parsimonious use of citations. The conclusions are hardly world shattering.

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12. It was agonizing for this reviewer to read a total of nine pages describing the overall methods.
13. Alarming errors, even if they could be easily corrected, gave the manuscript a sense of carelessness that makes me hesitant to recommend it for revisions.
14. This would have been a question of high interest 10 years back
15. I also would add that while I don't have anything serious to complain about, I also see this work as being pretty straightforward – there is no "big idea" that I see as especially imaginative or creative. Again, it is extremely solid, just not necessarily sexy.
16. There are FAR too many analyses and results. The reader is swamped. It's simply not possible to take it all in. It needs to be pruned.
17. So, what is the point of this?
18. Words are used inappropriately—I count, for example, 13 instances of "unique", but it is used correctly only once.
19. Not sure how to say this diplomatically, but the manuscript is really dull.
20. The abstract says absolutely nothing, and I mean this literally.

According to Professor David Dunning, professor of psychology at Cornell University:

A full 94% of college professors state that they do 'above average' work, although it is statistically impossible for virtually everybody to be above average.

You too may consider your paper to be above average work, but it is worth checking the coverage (i.e. what referees expect to find) and quality of each section by referring to the final subsection in each of Chapters 11 to 19.

If you have time it is a good idea is to get colleagues to review your manuscript (including the title), and you review their work. Often it is much easier to spot mistakes (grammatical, stylistic, structural etc.) in other people's work than in your own. But you can improve your critical skills of your own work if you become accustomed to critically evaluating other people's papers.

Referees are famous for asking for revisions before accepting a paper. These revisions often involve what you might consider as trivial details, such as typos and spelling mistakes. Such delays cost you time and money and may also mean that another paper on the same topic gets published before yours.

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20.2 Use AI to check your paper, then print it out

You can use various AI tools to automatically check the accuracy of your paper. These are described in Chapter 9.

When you have checked everything (including the spelling – see 20.9), reformat your document into a font that you find easy to read. Use 'double space' line spacing and larger margins. Then print it.

You will

- be more likely to find mistakes connected with grammar, word order, and structure
- notice how your paper looks visually and be able to break up long paragraphs into shorter ones that are easier on the eye. Breaking up paragraphs is quick and easy to do (3.7)

Bear in mind the following:

- Cut as much as you can. On the basis of identical content, very few referees would prefer to review a paper of twenty pages rather than fifteen (4.10).
- Judge your paper with the same criteria as you would if you had written it in your own native language.
- Double check you have followed the journal's guidelines/instructions for authors.
- Ensure everything is accurate (data, dates, references, bibliography).
- Ensure everything is consistent (US vs GB spelling, punctuation, capitalization).
- It takes much longer for editors and reviewers to read badly-written work than well-presented work. They may not react well.
- Rewriting (which includes cutting) can be a very satisfying as well as being an essential process.
- After weeks or months of working on your paper, you will find it hard to spot your own errors – ask a colleague to help you.
- Consider using a professional editing agency to edit and proofread your work – they will also act as a pre-review by highlighting aspects of the paper that may need reworking. However, if you are not in a hurry to have your paper published, it might be worth waiting to hear the referees' comments before submitting your paper to a professional agency, who can then work on your final version.
- Never make last minute changes without checking that you have not inadvertently introduce a mistake – typo or grammar error.

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Also, ask a colleague to read your printed version. Your colleagues will very likely find mistakes that you have overlooked – in fact, your familiarity with your own work makes it quite difficult to spot errors. But don't ask the colleague to read the whole work – that might be asking too much. Just highlight in the margin, some parts of the paper that you think are particularly important.

20.3 Check your paper for readability

Website designers follow the principle of 'don't make me think'. This means that everything should be so clear to visitors to their websites, that these visitors intuitively know where to find the information they need. The visitors are not required to think.

You do not want referees and readers to consider your work wordy, unclear, pompous, or dull, so when you make the final check of you manuscript, ask yourself the following questions:

- are my sentences reasonably short? (sentences longer than 30 words are generally hard to assimilate without having to be read twice)
- are my paragraphs reasonably short?
- have I only written what adds value, have I ensured there is no redundancy?
- have I clearly differentiated my work from the work of others so that the referees can understand what I did in relation to what others have done before me?
- have I highlighted my contribution and the gap it fills so that the referees can judge whether my paper is suitable for my chosen journal?

Readability is also affected by the following factors (these are all covered in Part I of this book):

- poor layout: large blocks of text are hard to read, whereas short paragraphs with white space in between them are much easier
- ambiguity and lack of clarity: the reader is not sure how to interpret a phrase
- lack of structure: within a sentence, paragraph or section
- too much abstraction: the reader is not given concrete explanations or examples
- lack of consistency

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20.4 Always have the referee in mind

The key factor when revising your paper is to have the referee in mind. Here are two quite typical comments made by referees and which are related to poor writing skills.

I often had to defer my interpretation of the meaning of a sentence until I had read it in its entirety. Frequently I got lost in a series of subordinate clauses. The paper would thus benefit from a major revision from a language point of view.

This paper could be improved considerably if the authors gave more consideration to their readers. At times it was difficult to follow the logical connection of the authors' ideas, and on several occasions I was tempted to stop reading completely.

Referees often make a direct connection between the time and effort that an author makes in presenting information, and how much time and effort the author has spent in doing their research. If the information is presented badly, then the implication is that the research may have been conducted badly too. Also it helps to remember that referees make reports on manuscripts in their free time for no financial reward – they are of much more benefit to you, than you are to them! To learn about how reviewers write their reports see Chapter 11 in *English for Academic Correspondence*.

20.5 What to do if your paper is subject to a 'blind' review

Before you submit your paper, find out if the paper will be subjected to a blind review or not. A blind review is when the referees do not know the author of the paper that they are reading. This means that the editor will delete your names and institutes from the top of the manuscript. The idea is to enable referees to be totally subjective in their recommendations.

Consequently, you should avoid giving any clues as to who you are.

So if your name is John Doe, in your draft version you should not write a sentence such as:

In a previous paper (Doe et al, 2024) we demonstrated that …

The red sentence makes it clear to the referees that you are John Doe and thus defeat the objective of a blind review. Instead you could write:

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Doe et al (2024) demonstrated that …

However, when the paper has been accepted for publication, you should change all such sentences to the personal form (S1) so that you enable the reader to understand that when you write *Doe et al* you are in fact referring to your own work.

20.6 Be careful with cut and pastes

If you write your paper in conjunction with other authors you multiply the chances of mistakes and ambiguity. Words such as *it, that, this, one, former, latter* and *which* are potentially dangerous if the words they refer to are subsequently changed by another author. For example, imagine Author 1 writes

… Russia, Canada and the United States. In the former …

Then, in order to put the countries in alphabetical order, Author 2 modifies it as follows:

… Canada, Russia and the United States. In the former …

The problem is that *the former* in Author 1's sentence refers to Russia. But in Author 2's sentence *the former* refers to Canada. To avoid such mistakes it is always best to repeat the key word rather than using *it, that, this, one, former, latter* and *which.* In any case, if it is your job to read the final version of the manuscript it is worth taking such problems into consideration.

For more on sources of ambiguity, see Chapter 5.

20.7 Make sure everything is consistent

Most papers are a collaborative effort. Maybe you write the Abstract and Introduction, some else writes the Methods, and another person writes the rest of the paper. You need to check:

- consistency – same key words
- that there are no issues with cut&pastes that have been carried out incorrectly
- spelling – maybe some parts of the paper have the spell check on and others don't. You need to highlight the entire paper and re-set the spelling
- references within the paper are all in the same format

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Referees will suggest a delay in the publication if they find inconsistency in your paper. Here is a genuine example from a referee's report. The only thing I have changed is the key words (X and Y).

1. "Figure 1" on page 4, yet "fig 5a" on page 8.
2. Page 4: "Figure 1 shows an example of an X graph," yet page 5: Figure 1 caption states "Example of Y". So is it a Y or an X graph?
3. Commas after some equations like on page 10, but not on all equations.
4. Caption to Fig 4 states "Initial Size Distribution," yet the illustration is of a graph not a size function.
5. Sometimes comma after i.e. e.g., and other times not

Here is an extract from another referee's report, which again highlights the importance of what you may consider to be fairly marginal issues:

This work is novel and is worthy of publication. However, the presentation of the work is, quite frankly, unprofessional. There are many sloppy mistakes like spelling mistakes and incorrect references, as well as inconsistency such as changing terminology and differences between captions and inline text.

20.8 Check that your English is suitably formal

There are certain words and expressions that are considered by most journals to be too informal. Check that your manuscript doesn't contain any of the following (note these are just examples and do not represent a comprehensive list):

- contracted forms: *doesn't, can't, we'll* etc
- informal nouns: *kids* (rather than *children*)
- informal adjectives: *trendy*; prefer *topical*
- informal expressions of quantity / size / appearance: *a lot, big, tiny, nice*
- informal conjunctions and adverbs: *so, till, like*; prefer *thus, until, such as*
- informal phrasal verbs: *check out, get around, work out*; prefer *examine, avoid, resolve*
- *going to* when *will* or the present tense could be used
- use of *you*

See 13.12 in *English for Research: Grammar, Usage and Style.*

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20.9 Don't underestimate the importance of spelling mistakes

I cannot overestimate the importance of doing a final spell check as the very last thing you do before submitting your manuscript.

Spelling checkers only pick up words that are not contained in their dictionaries. Mistakes and typos like the ones below would not normally be found because they are words that are in the dictionary (though not with the meaning that the author intended).

The company was *funded* in 2011. (founded)

The samples were *weighted* and *founded* to be 100 g. (weighed, found)

It was different *form* what was expected. (from)

We *asses* the values as being … (assess)

Be careful of: *choose / chose / choice, filed / field / filled, from/form, there / their / they're, then / than, through / trough, use / sue, were / where, with / whit*.

Remember that spell checkers tend to ignore words in CAPITALS. Given that sometimes titles of papers and other headings are in 'all caps', you need to double check them.

An additional problem is that your paper may have been written by various people and the language set for the spell checker may vary throughout the text. It should be the responsibility of the person who sends the paper to the editor to ensure that the language is set on English throughout the paper, and that American or British English have been chosen as appropriate.

Regarding US vs GB spelling, don't worry about *-ize* vs *-ise*, both forms are used by US and GB authors indifferently (it is only Word that has decided that *-ize* is US spelling). However, differences such as *color /colour, modelled / modeled* should be taken into account.

To learn more about spelling see 11.7–11.9 in *English for Academic Research: Grammar, Usage and Style*.

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So, as the very last thing you do before you send your manuscript to the journal, use the Editor function of your word processing application to check for spelling and grammar mistakes. Do not make any last-minute changes after this final check – Wallwork's law states that 'Any last-minute modification to your text will invariably contain a spelling mistake'.

20.10 Write a good letter / email to accompany your manuscript

If your English is poor in your email, the editor may suspect that the English will be poor in the manuscript too. This is not a good start.

To learn how to write effective emails, see Chapter 13 in *English for Academic Correspondence*.

20.11 Keep an open mind and deal with rejections in a positive way

Most journals reject large numbers of papers. In general, the higher the impact factor of a journal, the higher the risk of rejection. Don't be demoralized – yours is certainly not the first paper to have been rejected! See 11.2 in *English for Academic Correspondence.*

The highest ranked journals also tend to have the fastest turnaround and may thus return your rejected paper quite quickly. The benefit to you is that you are likely to be given a peer review of an excellent standard, which should help you to revise your paper before submitting it elsewhere. See rejection as an opportunity for making your paper even better.

To give you an idea of how difficult it is to publish a paper in a top ranking journal, here are some statistics from the 'Welcome to resources for authors' page of the website of the British Medical Journal (BMJ), one of the world's most prestigious journals.

We can publish only about 7% of the 7,000–8,000 articles we receive each year.

We reject about two thirds of all submissions without sending them for external review.

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However there are still advantages of sending your paper to such a journal, even if there is a very high chance of rejection. The BMJ makes very quick decisions (2–3 weeks) so you don't really delay your chances of publishing elsewhere. If they don't even send your paper for external review, it either means your paper is outside the scope of the journal, or that it has some serious flaws in terms of science and/or structure and language. This is a clear indicator that you need to seriously revise your paper. If the BMJ does decide to submit your paper to peer review, the reports you will receive from the reviewers will be very helpful in indicating how your paper can be improved.

See Chapter 12 in *English for Academic Correspondence* to learn how to write to editors.

20.12 Take the editor's and reviewers' comments seriously

There is a tendency to only take into account the editor's and referees' comments that you agree with and to discount everything else. However, if a referee says that they cannot understand what you mean, there is a very good chance that readers will have the same problem.

Don't underestimate editors and reviewers when they ask for a 'linguistic revision'. Here is what one editor wrote after the authors revised the technical aspects of their paper but failed to address English language issues:

This new revision does address many of the concerns regarding the technical substance of the manuscript. Unfortunately, the English writing (which the reviewers raised, and which was explicitly listed as requirement #2 in the review summary) continues to be an issue (in fact, the newly revised portions have the most language issues). There are problems with word order, commas, (missing or incorrect) articles, duplicated or missing words, logical inconsistencies, and general grammar issues throughout.

There will therefore need to be yet another minor revision, after which point I hope the manuscript will be in an acceptable state.

Let me stress the fact that improving the writing is *not* optional. If the manuscript comes back with significant remaining language issues, then it will unfortunately have to be rejected.

See Chapter 13 in *English for Academic Correspondence* to learn how to communicate with editors.

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20.13 A final word from the author: Let's put a bit of fun into scientific writing!

There's a tendency in the academic world to take oneself and one's discipline very seriously. This seriousness often leads to extremely tedious presentations at conferences where presenters feel obliged to be mega formal. An over inflated sense of importance also leads to papers that lack any sign of the presumed enthusiasm of the researcher for his / her work.

The British Medical Journal is one of the most respected journals in the world, not just in the field of medicine but in science in general. Yet they felt it was perfectly acceptable to publish the following paper: *Orthopaedic Surgeons: As Strong As an Ox and Almost Twice As Clever?*

The objective of the research was *To compare the intelligence and grip strength of orthopaedic surgeons and anaesthetist*s, and its Conclusions (as reported in their structured abstract) were: *Male orthopaedic surgeons have greater intelligence and grip strength than their male anaesthetic colleagues, who should find new ways to make fun of their orthopaedic friends*.

The Introduction to the paper reads as follows (I am very grateful to Paddy – Padmanabhan Subramanian – first author of the paper and an Orthopaedic Registrar, in London, for allowing me to quote from his paper):

A humorous anaesthetic colleague recently repeated the following popular saying while an operating table was being repaired with a mallet: "typical orthopaedic surgeon—as strong as an ox but half as bright." Making fun of orthopaedic surgeons is a popular pastime in operating theatres throughout the country. This pursuit has recently spread to the internet; a humorous animation entitled "orthopedia vs anesthesia" had received more than half a million hits at the time of writing. [1] Several comparisons of orthopaedic surgeons to primates have been published, and the medical literature contains suggestions that orthopaedic surgery requires brute force and ignorance. [2–4]

The stereotypical image of the strong but stupid orthopaedic surgeon has not been subject to scientific scrutiny. Previous studies have shown that the average hand size of orthopaedic surgeons is larger than that of general surgeons. [2, 3] However, a search of the worldwide scientific literature found no studies assessing the strength or intelligence of orthopaedic surgeons. In the absence of a cohort of willing oxen as a control group, and given that the phrase is popular with anaesthetists, we designed this study to compare the mean grip strength of the dominant hand and the intelligence test score of orthopaedic surgeons and anaesthetists.

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I am not advocating that everyone should adopt Paddy's humorous approach, but I am suggesting that now and again we should all try to inject a bit of fun into our writing and presentations.

People tend to remember and implement what they have enjoyed reading and learning, but quickly forget torrid tedium.

20.14 Summary of this chapter

- Respect the referee. Don't waste his or her time by submitting a poorly written manuscript.
- Use AI to check the accuracy of your English.
- Get a colleague to read through your paper or use a professional editing service.
- Print a hard copy of your manuscript. Don't rely on reading it on screen.
- Check for all types of mistakes in English: grammar, vocabulary and spelling.
- Apply the same standards as if you had written your manuscript in your own mother tongue.
- Cut as much as you can.
- Check your manuscript for readability and logic.
- Be careful with problems caused by multiple authors, e.g. cut and pastes.
- Ensure you have followed the journal's style guide, e.g. for citing the literature
- Check for accuracy and consistency.
- Take editorial comments seriously.
- As your last task before sending the manuscript to the journal, do a spell check. Don't rely 100% on automatic spell checkers. Spell checkers do not know the difference between *witch* and *which*, *asses* and *assets,* or *tanks* and *thanks.*

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20.15 Summary of the entire book: 10 key concepts

1. Preparation is vital. Read as many papers as possible on similar topics from your chosen journal to learn about the expected style, length of the various sections etc.
2. Always have the referees and readers in mind. You are writing for them, not for yourself.
3. Don't equate the length of a paper with its importance. Do everything you can to reduce redundancy (words, sentences, paragraphs, even sections).
4. Keep it simple. Write reasonably short sentences using grammatical structures and vocabulary that you are familiar with. Your aim is to add to the state of the art of your discipline, to communicate your results in the clearest and most accessible way possible. Your aim is not to impress your reader with your wonderful writing style.
5. Don't underestimate the visual impact that your paper has on the readers' eyes. Avoid long blocks of text and very long sentences. Use headings to help readers navigate. Place tables and figures strategically to break up blocks of text.
6. Be very careful when using pronouns (if possible replace them with the noun they refer to); likewise avoid synonyms for key words – the reader may not know what the pronoun or synonym refer to.
7. Ensure the reader can see your key findings (don't bury them in the middle of a long paragraph) and understand the gap you are filling and the level of innovation. And make it clear whose findings you are talking about – yours or another research group's.
8. Always mention any limitations.
9. Where possible, show how your work could be applied in other fields. Remember that your work is often funded by public money, and the public needs to feel that their money has been invested well.
10. Although AI can help you considerably to remove errors from your use of English, you should still triple check everything.

And if you can … Find ways to enjoy writing your manuscripts. If you don't find it a pleasurable experience, how can you expect your readers to enjoy or at least appreciate what you have written?

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Additional information

Other books in this series

English for Academic Research: Grammar Exercises

English for Academic Research: Writing Exercises

English for Academic Research: Vocabulary Exercises

The 2024 editions of the above exercise books practice the rules and guidelines given in this manual. The latest book in this series is: *AI-Assisted Writing and Presenting in English* - covers ChatGPT and automatic translators, with chapters for non-native English-speaking researchers, as well as native English-speaking translators, scientific editors, and EAP teachers.

English for Research: Usage, Style, and Grammar – covers those areas of English grammar that non-native academics tend to have difficulty with. It also contains chapters on how to use machine translation and ChatGPT to translate and improve your texts. There is also a chapter on the new uses of personal pronouns.

100 Tips to Avoid Mistakes in Academic Writing and Presenting – outlines the typical mistakes made in each section of a research paper; explains how to avoid redundancy and ambiguity; lists typical mistakes in proposals, journal submissions and emails; also tells you how to improve your presentation skills.

English for Presentations at International Conferences – all the tricks for overcoming your fears of presenting in English at a conference.

English for Academic Correspondence – tips for responding to editors and referees, networking at conferences, understanding fast-talking native English speakers, using Google Translate, and much more. No other book like this exists on the market.

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English for Interacting on Campus – covers your daily activities as a university student: interacting with professors and other students, going to lectures, socializing, overcoming difficulties of living in a new country.

English for Academic CVs, Resumes, and Online Profiles – your CV / resume is probably the most important document that you write as an academic – it is the one that gets you jobs and funds.

English for Academic Research: A Guide for Teachers – written for EAP teachers. Tips on how to prepare English courses, plus insights into the academic world of research, congresses and publishing.

I am a trainer in EAP and EFL. How should I use this book?

If you are a teacher of English for Academic Purposes or English as a Foreign Language you will learn about all the typical problems that non-native researchers have in the world of academia. You will be able to give your students advice on writing quality research papers and getting referees and editors to accept their papers.

You can also use the three exercise books (writing, grammar, vocabulary) that are part of this *English for Academic Research* series, plus the teacher's book that contains notes on how to exploit all the books: *English for Academic Research: A Guide for Teachers.*

If you are teaching intermediate level students, then use the following books:

Writing an Academic Paper in English – Intermediate Level

Essential English Grammar and Communication Strategies – Intermediate Level

Giving an Academic Presentation in English – Intermediate Level

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Download a free compendium of useful phrases for a research paper

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https://extras.springer.com/?query=978-3-319-26092-1

About the author

Adrian Wallwork runs an agency specialized in editing and revising scientific papers. He also teaches English for Academic Purposes (EAP) to PhD students. In addition to his many books for Springer, he has written course books for Oxford University Press, and discussion books for Cambridge University Press.

His passion is teaching PhD students and researchers how to write and present their research. He is also particularly interested in using the amazing advances in artificial intelligence to help students translate, write and correct / edit their work.

Are you an ESL, ELT or EAP teacher? Are you looking for new discussion topics with audio support?

You may be interested in Adrian Wallwork's other work – check out tefldiscussions.com. Adrian and team have developed a huge number of materials (written and audio) related to interesting discussions designed to stimulate even the dullest of groups and the most burnt out teachers! There is a blog for teachers (tefldiscussions.com/blog/), covering not just grammar and vocabulary issues, but also how to deal with challenging students, how to teach diversity and inclusivity, and how to branch out of teaching into more remunerative language-related areas.

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I would like to thank those readers who contacted me with errors that they had found in the original version of this book and others in the series. Thank you so much!

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Finally, thanks to Anna Southern for reading through the manuscript and weeding out the nonsense!

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Sources

Note: .0 refers to the factoids/quotations at the beginning of each chapter. So 1.0 means the factoids presented in Chapter 1. If no references are given for the factoids, it means that the facts are in the public domain.

1.0 https://www.universityworldnews.com/post.php?story=20180905095203579; https://en.wikipedia.org/wiki/List_of_countries_by_number_of_scientific_and_technical_journal_articles; https://www.findaphd.com/blog/5403/how-rare-or-common-is-it-to-have-a-phd; https://onlinelibrary.wiley.com/doi/full/11.1111/ejed.12514#:~:text=According%20to%20Eurostat%20estimates%2C%20in,current%20value%20of%20a%20doctorate%3F **1.1** *Writing for Science*, R Goldbort, Yale University Press, 2006; *How to Write and Publish a Scientific Paper*, R Day, Cambridge University Press, 2006; *Handbook of Writing for the Mathematical Sciences*, N Highman, SIAM, 1998. Highman's book is one of the best books I have read on scientific writing. Any researcher in mathematics should seek out a copy. **1.10** Translated by me from the original Italian *I compagni sono pregati di scrivere in modo semplice … che a un tempo stimoli le energie mentali dei lettori senza richiedere loro uno sforzo inadeguato*. (https://marxpedia.org/il-psi-e-il-biennio-rosso/) **2.0** 4) John Adair's "The Effective Communicator" (The Industrial Society, 1989 – also available on Google Books). **3.0** Chapter 2 *English for Interacting on Campus*, Adrian Wallwork, Springer 2016 **3.1** John Kirkman, *Good Style – Writing for Science and Technology*, Routlegede, 2005 **3.8** *Clarity in Technical Reporting*, S Katzoff, NASA Scientific and Technical Information Division. Free download at: http://courses.media.mit.edu/2010spring/mas111/NASA-64-sp7011.pdf **5.0** First three quotations – The Penguin Dictionary of Twentieth Century Quotations (1996) edited by M J & J M Cohen. Barbara Kingsolver – a BBC interview with her on June 9, 2011. Bruce Cooper *Writing Technical Reports*, Penguin UK, 1999. **5.1** https://www.nature.com/nature-portfolio/for-authors/write **6.0** 1) *The Mother Tongue*, Bill Bryson, HarperCollins, 1990; (2) http://www.mfa.gov.il/mfa/foreign-policy/peace/guide/pages/un%20security%20council%20resolution%20242.aspx **7.0** *I wish I hadn't said that: Experts speak and get it wrong*, C Cerf & V Navasky, HarperCollinsPublishers, 2000: (**7.1** *Brilliant Blunders: From Darwin to Einstein – Colossal Mistakes by Great Scientists That Changed Our Understanding of Life and the Universe*, Mario Livio, Simon & Schuster, 2014; BMJ site: https://www.bmj.com/about-bmj/resources-authors/article-types **7.2** *Why People Believe Weird Things*, Michael Shermer, Holt Paperbacks, 2002: http://abacus.bates.edu/~ganderso/biology/resources/writing/HTWtablefigs.html; Pauling quoted in Livio. **7.3** For more on this topic, see Dr Maggie Charles's very useful article "Revealing and obscuring the writer's identity: evidence from a corpus of theses" in Chap. 9 of "Language, Culture and Identity in Applied Linguistics", a book by the British Association of Applied Linguistics. **8.0** https://gobraithwaite.com/thinking/the-business-story-behind-the-kiss-principle/#:~:text=That%20was%20the%20genius%20of,for%20more%20than%2040%20years; *Consequences of Erudite Vernacular Utilized Irrespective of Necessity: Problems with Using Long Words Needlessly*, by Daniel Oppenheimer, available at: http://web.princeton.edu/sites/opplab/papers/Opp%20Consequences%20of%20Erudite%20Vernacular.pdf; https://www.theguardian.com/books/2010/jul/15/slow-reading **8.1** http://wattandedison.com/Prof_Leggett_Notes_on_the_Writing_of_Scientific_English.pdf; https://ibadanbookclub.wordpress.com/2012/04/11/the-art-of-slow-reading/ **9.0** https://www.fastcompany.com/90820090/the-internet-loves-chatgpt-but-theres-a-dark-side-to-the-tech; https://www.telegraph.co.uk/business/2022/12/05/meet-scarily-intelligent-robot-who-can-do-job-better/; https://www.theatlantic.com/technology/archive/2022/12/chatgpt-ai-writing-college-student-essays/672371/;https://gizmodo.com/chatgpt-free-ai-10-fails-nyc-schools-college-essay-porn-1849956907; https://www.forbes.com/sites/lanceeliot/2023/02/26/legal-doomsday-for-generative-ai-chatgpt-if-caught-plagiarizing-or-infringing-warns-ai-ethics-and-ai-law/?sh=95976e2122bd; https://www.positive.news/society/the-robots-are-coming-its-time-to-be-more-human-chatgpt/; https://www.keranews.org/education/2023-02-23/chatgpt-wont-ruin-education-texas-academic-says **9.11** *Clarity in Technical Reporting*, S Katzoff, NASA Scientific and Technical Information Division. Free download at: http://courses.media.mit.edu/2010spring/mas111/NASA-64-sp7011.pdf **9.15** *7 Common Types of Plagiarism, With Examples* https://www.grammarly.com/blog/types-of-plagiarism/#:~:text=Plagiarism%

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20is%20the%20act%20of,likely%20has%20a%20plagiarism%20policy. **11.0** (1–4) winners of Ignobel prizes (www.improbable.com/ig/winners/), (5–6) arxiv.org, 7–12 http://mathoverflow.net/questions/44326/most-memorable-titles **11.15** This section was taken from the website of the American Journal Experts (AJE) – a very useful website both for researchers and scientific editors, and was written by Michaela Panter: https://www.aje.com/en/author-resources/articles/editing-tip-crafting-appropriate-running-title **12.0** (1) http://arxiv.org/ftp/arxiv/papers/1110/1111.2832.pdf; (2) The Journal of Emergency Medicine Vol 8 Issue 3 May-June 1990; (3) http://scholar.harvard.edu/files/joshuagoodman/files/goodmans.pdf; (4) http://arxiv.org/abs/0711.4114; (5) http://arxiv.org/abs/1004.2003v2; (6) http://link.springer.com/article/11.1007/s00300-003-0563-3?no-access=true (7) EOS Trans. AGU Vol 72, No 27-53, p456 **12.5** Coates article: https://pubmed.ncbi.nlm.nih.gov/11827675/; U.S. National Library of Medicine: http://www.nlm.nih.gov/bsd/policy/structured_abstracts.html **12.6** Style 1) http://www.tribunes.com/tribune/art97/wooda.htm (note, no longer available); Style 2) R A J Matthews Tumbling toast, Murphy's Law and the fundamental constants, 1995 Eur. J. Phys. 16 172–176, available at: http://www.iop.org/EJ/journal/EJP; Style 4) Copyright © 1999 by the American Psychological Association. Reproduced with permission. Kruger, Justin; Dunning, David, Unskilled and unaware of it: How difficulties in recognizing one's own incompetence lead to inflated self-assessments, Journal of Personality and Social Psychology. Vol 77(6), Dec 1999, 1121–1134. The use of APA information does not imply endorsement by APA. **13.0** originally published in *English for Interacting on Campus*, Adrian Wallwork, Springer, 2016 **13.6** http://www.sigsoft.org/resources/pughadvice.htm **13.9** www.nature.com/nature/authors/gta/Letter_bold_para.doc **14.0** https://www.genscript.com/top-100-most-cited-publications.html; https://www.mdpi.com/1660-4601/19/15/9645/pdf **14.3** Fragmentation of Rods by Cascading Cracks: Why Spaghetti Does Not Break in Half, was published in Physical Review Letters Vol. 95, 095505 (2005). The full version available at: http://www.lmm.jussieu.fr/spaghetti/audoly_neukirch_fragmentation.pdf **14.9** For the full version of Chris Rozek's paper "The Effects of Feedback and Attribution Style on Task Persistence" see: http://gustavus.edu/psychology/files/Rozek.pdf **15.3** For the full version of Chris Rozek's paper "The Effects of Feedback and Attribution Style on Task Persistence" see: http://gustavus.edu/psychology/files/Rozek.pdf **15.7, 15.8** Alistair Wood's article was originally published in Science Tribune in April 1997 **15.8** "Using Another's Words and Ideas" is no longer available online. **17.0** (2–6) https://www.ipa.org.au/publications/1964/a-history-of-scientific-alarms (7) https://en.wikipedia.org/wiki/Influenza_A_virus_subtype_H5N1 (8) http://www.telegraph.co.uk/news/earth/environment/globalwarming/11395516/The-fiddling-with-temperature-data-is-the-biggest-science-scandal-ever.html **17.5** *Bad Science*, Ben Goldacre, Harper Collins, 2008. See also videos on Goldacre's website: www.badscience.net; *What do you care what other people think?* Richard Feynman, W. W. Norton & Company, 2001 **17.6, 17.7** Ken Lertzman's "Notes on Writing Papers and Theses" are available for free download at: http://www.jstor.org/stable/20167913?seq=1#page_scan_tab_contents; **17.8** http://en.wikipedia.org/wiki/Brazil_v_Germany_%282014_FIFA_World_Cup%29 **17.10** Bates College site: https://www.bates.edu **18.5** Catherine Bertenshaw and Peter Rowlinson's article, "Exploring Stock Managers: Perceptions of the Human-Animal Relationship on Dairy Farms and an Association with Milk Production," appeared in Anthrozoos: A Multidisciplinary Journal of The Interactions of People & Animals, Volume 22, Number 1, March 2009, pp. 59–69(11), Berg Publishers, an imprint of A&C Black Publishers Ltd. **18.6** "Chickens prefer beautiful humans" originally appeared in Human Nature Volume 13, Number 3, 383–389. **18.7** Based on Professor Shahn Majid's notes for math students, "Hints for New PhD students on How to Write Papers". **19.0** *The Year 2000 – A Framework for Speculation on the Next Thirty-Three Years*, Herman Kahn and Anthony J. Weiner, Macmillan, 1967. **19.1** http://www.nature.com/nphys/journal/v3/n9/full/nphys724.html; http://www.cs.cmu.edu/~jrs/sins.html = **19.4** The University of Toronto's excellent website on writing skills can be found at: https://ecp.engineering.utoronto.ca/resources/online-handbook/components-of-documents/ **20.0** http://www.uky.edu/~cfox/PeerReview/Index.htm **20.1** The reviews come from: http://shitmyreviewerssay.tumblr.com/ Big thanks to Rogier Kievit. The Dunning quote comes from *Ignobel Prizes – The Annals of Improbable Research by* **20.14** http://www.bmj.com/content/343/bmj.d7506

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A. Wallwork, *English for Writing Research Papers*, English for Academic Research, https://doi.org/10.1007/978-3-031-31072-0

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